

The Elementary Education System in India
*A field based investigation into institutional structures, processes and
dynamics*

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GLOSSARY

AIE: Alternative and Innovative Education
AIES: All India Education Survey
AP: Andhra Pradesh
APC: Assistant Project Coordinator
APPEP: Andhra Pradesh Primary Education Project
APSES: Andhra Pradesh School Education Society
AS: Alternative School
BC: Backward Caste
BEO: Block Education Officer
BRC: Block Resource Centre
BRCF: Block Resource Centre Facilitator
BSS: Balika Shikshan Shivir
BSTC: Basic School Training Certificate
CABE: Central Advisory Board of Education
CBSE: Central Board for Secondary Education
CEO: Chief Executive Officer
CIE: Central Institute of Education
CRC: Cluster Resource Centre
CRCF: Cluster Resource Centre Facilitator
DEO: District Education Office
DEO: District Education Officer
DIETs: District Institutes of Education and Training
DISE: District Information Statistics for Education
DM: District Magistrate
DPEP: District Primary Education Programme
Dy Eos: Deputy Education Officers
EGS: Education Guarantee Scheme
GCDO: add to chap 5
GER: Gross Enrolment Ratio
GoAP: Government of Andhra Pradesh
GoI: Government of India
GoR: Government of Rajasthan
GP: Gram Panchayat
GPS: Government Primary School
GSDP: Gross State Domestic Product
GSDP: Gross State Domestic Product
IAS: Indian Administrative Service
IASEs: Advanced Studies in Education
IIEDC: Add to chap 3
IT: Information Technology
JSN/PL&CE: Add to chap 3
KSPS: Khand Stariya Prabandan Samiti
MEO: Mandal Education Officer
MHRD: Ministry of Human Resource Development
MLLs: Minimum Levels of Learning
MP: Mandal Panchayat
MRP: Mandal Resource Person
MSV: Mahila Shikshan Vihar
MTA: Mother Teachers Association
NCAER: National Council for Applied Economic Research
NCERT: National Council for Educational Research and Training
NCF: National Curriculum Framework

NCTE: National Council for Teacher Education
NER: Net Enrolment Ratio
NFE: Non-formal Education
NIEPA: National Institute for Educational Planning and Administration
NPE: National Policy on Education
NPEGEL: National Programme for Education of Girls at Elementary Level
NREP: National Rural Employment Programme
NSDP: Net State Domestic Product
OB: Operation Blackboard
ODA: Overseas Development Assistance
PLC: add to chap 3
PRIs: Panchayati Raj Institutions
PRTU: Progressive Recognized Teachers' Union
PS: Primary School
PTA: Parent Teacher Association
PTR: Pupil-Teacher Ratio
RCPE: Rajasthan Council for Primary Education
RFLP: Rural Functional Literacy Projects
RGSJP: Rajiv Gandhi Swarna Jayanti Pathshala
RPSC: add to chap 5
SA: School Assistant
SAS: add to chap 3
SC: Scheduled Caste
SCERT: State Council for Educational Research and Training
SDC: School Development Committee
SDMC: School Development and Monitoring Committee
SDP: State Domestic Product
SEC: School Education Committee
SGT: Secondary Grade Teacher
SIDA: Swedish International Development Authority
SIERT: State Institute of Educational Research and Training
SKP: Shiksha Karmi Project
SOPT: Special Orientation for Primary Teachers
SPD: State Project Director
SRG: State Resource Group
SSA: Sarva Shiksha Abhiyan
SSK: Sahaj Shiksha Kendra
ST: Scheduled Tribe
OBC: Other Backward Caste
SY: Saraswati Yojna
TLM: Teaching-Learning Material
TPR: Teacher-Pupil Ratio
UEE: Universalisation of Elementary Education
UPE: Universal Primary Education
UPS: Upper Primary School
UT: Union Territory
VCA: Village Education Committee
VDO: Village Development Officer

CHAPTER 1: INTRODUCTION

Vimala Ramachandran and Rashmi Sharma

The state shall endeavour to provide, within a period of ten years from the commencement of this Constitution, free and compulsory education for all children until they complete the age of fourteen years (Article 45, Directive Principles of State Policy, Constitution of India, 1950).

The State shall provide free and compulsory education to all children of the age of six to fourteen years in such manner as the State may, by law, determine (86th Amendment, Article 21 A, Constitution of India, December 2002).

A span of more than half a century separates the Directive Principles of the Indian Constitution that foresaw elementary education as *a moral obligation that the state would endeavour towards*, and the Fundamental Right to Education that would enable *citizens to demand it of the state*. This shift encapsulates the story of post independence state action in elementary education in the country. At the time of India's independence, less than a fifth of its population was literate and the task of making elementary education universal was truly monumental for a newly independent country. Six decades later, in the next century, the task was still looming as the approach paper to the Tenth Plan (2002-2007) pointed out: 'Out of approximately 200 million children in the age group 6-14 years, only 120 million are in schools and net attendance in the primary level is only 66 per cent of enrolment'. But in 2002, unlike immediately after independence, the inability of India, a country that was celebrating its accelerating economic growth and racing ahead in information technology, to make elementary education universal, seemed both incomprehensible and shameful. Six years later in 2008, as this book is written, the economy remains buoyant and the Indian pride in succeeding in the high-tech field of information technology is still rising, but elementary education is not yet universal.

These limitations of elementary education in India form the point of departure for this book. We begin with the assumption that this problem needs to be understood in its complexity, rather than in terms of simplified phrases such as 'lack of political will', 'because of poverty' or 'grand collusive designs of the national elites with the forces of international capital'. We do not deny that all these, and many other such phrases, do contain several grains of truth. But they can also keep at bay the vibrant and intricate dynamic that underlies the problems of elementary education in India, which this book attempts to articulate.

The difficulties and dilemmas of elementary education need to be seen against the fact that they have persisted across six decades of varying contexts in terms of economic scenarios, changing political regimes (at least since 1977) and shifting overarching policy frameworks. In other words, they are not simply a function of the level of economic growth or political ideology, though these can provide important basis for addressing various issues. This is visible in the relative success in elementary education achieved by various states. For instance, Punjab and Harayana have witnessed a high degree of economic growth, but are not leaders in education. In contrast, Kerala is well known for its success in elementary education and literacy despite modest economic growth. Equally, West Bengal has had some success with land reforms and a range of pro-poor economic policies, but not with elementary education.

At the same time, the problems of elementary education have persisted not only across the changing larger context, but, as the discussion in the following sections of this chapter shows, also across a *fairly wide range of various types of policy thrusts and programmes*. For instance, a fairly wide set of strategies and programmes have been initiated by the central

government vis-à-vis elementary education, addressing the expansion of schooling, upgradation of teacher qualifications and salaries (followed by reversals during the 1990s), provision of infrastructure, creation of an institutional structure for teacher education, non-formal and alternative schooling, institutionalization of community participation, curricular and textbook reform, and so on. Yet, all these programmes, from the simplest to the relatively complex, from the most rigid to the fairly flexible, have been bogged by a common problem of 'poor implementation'. While some policy thrusts and programmes have been better conceptualized than others, and consequently have had somewhat greater impact, the commonality of the problems that confront all policy thrusts and programmes are striking.

The fact that the successes and failures of elementary education in India are not fully explicable in terms of levels of economic growth, ideologies of political regimes and even policy thrusts, though these are, of course, important, *points to the need to examine issues that are embedded more deeply in the 'system' or the institutional structures, processes and dynamics*. Two other scenarios point in the same direction. First, there is increasing evidence in recent studies that school functioning itself is a major problem area (Probe Report, 1999). A fairly large part of the 'school problem' comprises poor motivation and skill of teachers and supervisors, inadequate time spent on classroom teaching and unsatisfactory classroom processes. Second, the elementary education scenario has been characterized by a paucity of administrative and institutional reforms. While policies have been formulated to address a range of concerns such as out-of-school children and curricula, no policy thrusts have comprehensively addressed the reform of an administrative system set up in the colonial era, or addressed the lack of vibrancy and productivity of various institutions. In this book, we explore this silence, i.e., make an attempt to comprehend 'the system', or the institutional set up of elementary education in India.

But before exploring the institutional structures, processes and dynamics, it is important to be familiar with the current situation in, and the main problems of, elementary education in India, as well as the policy and programmatic thrusts. In this chapter we shall briefly sketch the national picture regarding the same, and while doing so, also substantiate the case for the need to examine the 'system' i.e., the institutional structures, processes and dynamics. In the first section of this chapter, we highlight the major milestones of national policy development; in the second section the main problems of elementary education in India are outlined; in the third section, major national level programmatic thrusts have been described; and finally, the last section explains the structure of the rest of the book.

The Overall National Policy Context

As per the Indian Constitution as originally enacted after independence, education was a 'state' subject, i.e., the responsibility of state governments, rather than the national government. Yet, in the post-independence scenario, the central government played a critical role in developing a fairly similar system across the country. One way in which it did so was by creating national institutions that could advise, play a leadership role and set standards. The Central Advisory Board of Education (CABE)¹ was in existence since the pre-independence times, and played a role in the formulation of policy. The National Council for Educational Research and Training (NCERT)² was set up in 1961 and the Central Board for Secondary

1 CABE was set up in 1921 to facilitate coordination after education was transferred to Provincial Legislatures under 'dyarchy' (whereby a few subjects were transferred to elected state legislatures), but was abolished two years later. In 1935, CABE was revived, after the diarchy was replaced with 'Provincial Autonomy', whereby the whole field of provincial administration was placed under a ministry responsible to a legislature (Ghosh, 1995).

2 Originally, the Central Institute of Education (CIE), set up to train secondary teachers for Delhi, was proposed to be developed as a high level institute for educational research, training and extension. Several central government institutes were set up subsequently between 1954 and 1959: Central Bureau

Education (CBSE)³ in 1962. NCERT's mandate includes a range of activities to improve school education including research, curriculum, textbook production and training. The most important role of CBSE is to conduct examinations and grant certificates and diplomas for secondary schools (Shukla, 1989). In the same year (1962) an Institute of Educational Planning and Administration was established in collaboration with UNESCO, of which the central government took over responsibility in 1970. In 1979, the institute was renamed the National Institute for Educational Planning and Administration (NIEPA).⁴ Its mandate includes training of educational planners, administrators and teacher educators, facilitation of seminars and discussions, research in planning and administration and provision of consultancy services to state and local governments as well as other countries on request (Sharma, 1989). Much later, the National Council for Teacher Education (NCTE) was set up in 1995 with the goal of accrediting teacher education institutes and providing guidance regarding curricula and teaching methods in teacher education (GoI, 1996).

An important landmark in the early years was the setting up of the Education Commission, also known as the Kothari Commission, (1964-1966) (at this time the Congress formed the government at the centre) and the subsequent passing of a resolution on the National Policy on Education (NPE) in 1968. As per NPE 1968, a radical reconstruction of education was essential for the economic and cultural development of the country, for national integration and for realizing the ideal of a socialist pattern of society (NPE, 1968: 1). The policy laid emphasis on the provision of free and compulsory education as per the Directive Principles of the Constitution; enhancement of teachers' status through an increase in emoluments and improvement of other service conditions as well as teacher education; the development of regional languages and their use as media of education; equalization of opportunity of education through a common school system,⁵ emphasis on the education of girls, Scheduled Caste/Scheduled Tribe (SC/ST) children as well as physically and mentally challenged

of Textbook Research, Central Bureau of Educational and Vocational Guidance, All India Council for Secondary Education, National Institute of Basic Education, National Fundamental Education Centre and the National Institute of Audio-Visual Education. In 1960, as the work of these institutes was reviewed, they were found to be too small and isolated, and without the necessary autonomy to be effective. These institutes were then converted into a single, autonomous organization, i.e. NCERT (NCERT, 1969).

3 The first school education board to conduct public examinations at the end of the school stage was set up in 1921, with jurisdiction over United Provinces (later Uttar Pradesh), Rajputana, Central India and Gwalior. Subsequently, a separate board was set up in Rajputana in 1929 and in 1950-51, Madhya Bharat formed its own board. In 1952, the Government of India named this board the Central Board for Secondary Education (CBSE) and extended its jurisdiction to ten states. It was thought that Indian schools should not be affiliated to a foreign body, i.e., the School Syndicate of Cambridge University, and the demand for centrally managed secondary education was increasing from various groups such as the armed forces and central government servants. CBSE was subsequently reconstructed in 1962 to cover the whole country (Shukla, 1989).

4 In 1958, in a general council of UNESCO, India moved a resolution for initiating a major programme for primary education in Asia. It was thought that the planning skills needed for such a programme had not developed. Consequently, in 1962, the UNESCO Regional Centre for Training of Educational Planners, Administrators and Supervisors in Asia was set up in Delhi, subsequently renamed the Asian Institute of Educational Planning and Administration, serving 18 Asian countries. In 1970, as per the recommendations of the Kothari Commission, it was made an autonomous national institute, called the National Staff College for Educational Planners and Administrators, and renamed the National Institute of Educational Planning and Administration in 1979 (Sharma, 1989).

5 The Education Commission noted that schools run by the government and local bodies were generally of poor quality. High fee charging private schools were better, but were generally availed of by only the middle and higher classes. The result was that education itself increased social segregation. The Kothari Commission emphasized the need for a common school system of public education, open to all children, with an adequate standard in all schools, no tuition fee, and access to education dependent upon talent and not wealth, so that the average parents would not feel the need to send their children to private schools (GoI, 1966: 10).

children and an emphasis on work experience to bring the school closer to society. A common educational structure of education, i.e., ten years of school education followed by two years of higher secondary education (and another three years of education for the first college degree) was seen as advantageous and attempts were to be made to increase the expenditure on education to the level of 6 per cent of the national income.

In 1976, through a constitutional amendment, education became a concurrent rather than a state subject, i.e., both the central and state governments could legislate on it. The next major policy statement came in 1986 (during Congress rule again).⁶ A document, 'Challenge of Education', was prepared and used to initiate a nation-wide debate. On the basis of this document and the debates that followed, the National Policy on Education, 1986 (NPE, 1986) was adopted. In 1990, as the United Front government was formed at the centre, a committee set up under the chairmanship of Acharya Rammurthy reviewed NPE 1986. The report of this committee was tabled in Parliament, but the government lost its mandate before it could be approved. The revision of the policy was completed in 1992-93, and the NPE 1986 was substantially endorsed. A modified and National Policy on Education, 1986 and Programme of Action were introduced in 1992.

As with NPE 1968, NPE 1986 emphasized economic and technical development and reiterated several of the policy thrusts of the previous policy such as provision of access to education of comparable quality to all children and better service conditions and training for teachers. NPE 1986 laid a special emphasis on universal education for all children up to the age of 14 years, removal of disparities across gender and various social groups (see Chapter 7 in this volume) and on teacher education. It also sought to create special opportunities for talented rural children and envisaged decentralization and the creation of District Boards of Education for the management of school education at the district level. Important centrally sponsored schemes followed NPE 1986, including Operation Blackboard and the Scheme for Restructuring and Reorganization of Teacher Education.

There has been no separate overarching national policy statement on education subsequently, but through the 1990s, there were important developments that had a policy-like impact. These are discussed in detail below, but the background to these developments needs to be kept in view. During the 1990s, important shifts took place in the country's economic policy. In 1991, India liberalized its economy, lifting controls on domestic industry and reducing external tariff barriers. A parallel important development was that the pressure to make elementary education universal increased during the decade. In 1993, the Supreme Court of India delivered a landmark judgment, (*Unnikrishnan vs. State of Andhra Pradesh*), pronouncing the Right to Education a Fundamental Right.⁷ Subsequently, the 86th Constitutional Amendment Act of 2002 led to the new Article 21A, which has been quoted in the beginning of this chapter. A corresponding 'Free and Compulsory Education Bill, 2004' was drafted by the Ministry of Human Resource Development for consideration by Parliament, but has not yet been passed. A nation-wide debate on whether 'transitional arrangements' (short-term camps/non-formal learning arrangements) do justice to the spirit of equality, coupled with lack of political consensus on the extent of investment that may be required, has led to the bill being kept in abeyance. A draft bill has been posted on the GoI website⁸ and state governments have been advised to enact suitable legislation.

⁶ An effort at formulating policy was made in 1979 by the Janata government, but the government fell before the policy could be passed by Parliament.

⁷ Before the 86th amendment to the Constitution, compulsory education acts had existed in 19 states and union territories, but these lacked teeth and were rarely implemented (Juneja, 2003).

⁸ Government of India, July 2004. Free and Compulsory Education Bill, 2003 (Revised). Posted for comments on the official website of the Ministry of Human Resources Development, New Delhi.

Additionally, from the 1990s onwards, international pressure to make elementary education universal also increased as India became a signatory to several international agreements (Jomtien, 1990; Dakar, 2000; MDGs, 2001). Importantly, before the 1990s, external funding for primary education existed on a small scale and was financed in the form of grants. The first few externally financed projects included the Andhra Pradesh Primary Education Project (1987) with ODA assistance, the Shiksha Karmi Project (1987) with assistance from the Swedish International Development Authority (SIDA), the Mahila Samakhya Programme supported by the Dutch (1989), the Bihar Education Project (1991) and the Lok Jumbish Project in Rajasthan (1992). But in March 1991, CAGE endorsed the idea of tapping external funds for basic education from multilateral as well as bilateral sources⁹ (GoI, 1991). In the post-liberalization scenario, the Social Safety Net Credit became available and loans were taken for primary education from the World Bank¹⁰, and the quantum of external finance for primary education increased significantly.

In 1994, the District Primary Education Programme (DPEP) was conceived as an umbrella programme to channel support from various funding agencies¹¹ (Aiyer, 1995; GoI, 1996). DPEP, which spread to nearly 60 per cent of the country, aimed at increasing enrolment, retention and achievement at the primary stage and also at reducing gender and social disparities in districts with low female literacy rates. Several observers have seen developments in DPEP being influenced significantly by international funding agencies (Kumar et al., 2001; Sadgopal, 2004, Sarangapani and Vasavi, 2004). Subsequently, in 2000, the central government took up the Sarva Shiksha Abhiyan (SSA), across the country, covering upper primary as well as primary education. SSA aims to universalize elementary education by 2010, and is based on decentralized, consultative planning.

The Elementary Education Scenario

We now outline briefly the problems and issues of elementary education in India, which show a school system that expanded rapidly after independence, but does not provide education of high quality, and does not treat all Indian children equitably.

Access and enrolment

In the post-independence scenario, the elementary education system in India has been characterized by an exponential growth of government and private schools in rural as well as urban areas. Between 1950-51 and 2003-04, the number of primary schools in the country increased over three-fold, from 209,671 to 712,239 and the number of upper-primary schools, 19 times from 13,596 to 262,286 (GoI, 2006).¹² This expansion of schooling led to a steady increase in enrolment and a reduction in the percentage of children who had never enrolled in schools. Between 1950-51 and 2004-05, enrolment in the primary cycle went up from 19.2 million to 130.8 million and at the upper primary stage from 3.1 million to 51.2 million¹³. There was an appreciable acceleration in enrolments after 1986. (see [Table 1.1](#)).

⁹ CAGE recommended that external funds be tapped in addition to domestic resources and that all programmes taken up be in conformity with national policies (GoI, 1991).

¹⁰ A test case for such assistance was launched through the UP Basic Education Project.

¹¹ Programme funds were obtained from the World Bank, the European Commission and DFID. State governments contributed 15 per cent of the total cost.

¹² Initially, as education was a state subject, the expansion of schooling was led by state governments, while after 1978 the central government also began to play an important role, and after the 1990s, many new schools were started under DPEP and SSA.

¹³ Select Education Statistics, Department of Education, MHRD, GoI, 2007.

Table 1.1: Percentage increase in schools, enrolment and teachers

INDIA		
Percentage increase in the number of schools	1986-93	1993-2003
Primary	7.89	14.18
Upper Primary	17.11	50.65
Percentage increase in enrolment	1986-93	1993-2003
Classes 1-5	12.94	26.15
Classes 6-8	24.93	37.49

Source: 6th All India Educational Survey (1998) and 7th All India Educational Survey (2004).

But the increase in the number of schools did not address the needs of *all* children of the school going age. Though access to education improved a great deal, it did not become universal (see Table 1.2 below). Notably, state policies for providing new government schools were based on providing schools within a certain distance of all habitation i.e. within one kilometre for primary schools and three kilometres of upper primary schools provided there were a certain number of children or a certain population available within the habitations. Children in very tiny habitations that did not qualify for a proper school, continued to lack access to elementary education. Even more important was the fact that schools could not be provided as per the stated norms, and in many states, attempts were made to provide schools as per existing norms only in the 1990s. In 1993, 83.36 per cent eligible habitations had a recognised primary school¹⁴ within one kilometre and 76.15 per cent had an upper primary school within 3 kilometres (see Table 1.2). This went up marginally to 86.96 per cent and 78.11 per cent for primary and upper primary schools respectively in 2003. This data does not include “alternative schools¹⁵”. In 1999, the Tapas Mazumdar Committee projected a need for an additional 400,000 schools to achieve universal elementary education. These figures indicate inadequate provision of funds for stated policy goals. As can be seen, there are significant inter-state differentials in terms of access to education, as the quantum of funding that states provided for elementary education varied.

¹⁴ A recognised school as defined by NCERT for the purpose of this survey is “A recognized school is that in which the course(s) of study followed is/are prescribed or recognized by the government (central/state) or university or a Board constituted by law or by any other agency authorized in this behalf by the central or state government and which satisfies one or more of the authorities e.g. Directorate of Education, Municipal Corporation/Committee, Board, etc., with regard to its standard of efficiency.”(Source: NCERT 7th Educational Survey Research Tools - <http://7thsurvey.ncert.nic.in/VIF.htm>)

¹⁵ An alternative school as defined by NCERT for the purpose of this survey is “Schools set up in unserved habitations (with no schooling facilities within one Km) under the Education Guarantee Scheme (EGS) component of the EGS&AIE Scheme to provide education to out of schoolchildren are termed as Alternative Schools. EGS schools in the States of Madhya Pradesh, Orissa, Uttar Pradesh; Maavadi in Andhra Pradesh; Multi- grade learning centers in Kerala; Shishu Shiksha Karamsuchi Kendras in West Bengal; Contract schools in Maharashtra; Rajiv Gandhi Swarna Jayanti Pathshalas in Rajasthan are some of the examples of Alternative Schools.”

Table 1.2: Percentage rural habitations with primary and upper primary schooling facilities 1993 and 2003, Select states

Sl. No	State / UT	Primary Schools				Upper Primary Schools			
		Within habitation ¹⁶		Within 1 km of habitation		Within habitation		Within 3 km of habitation	
		1993	2003	1993	2003	1993	2003	1993	2003
1	Andhra Pradesh	69.73	78.49	88.57	93.91	13.82	24.49	65.40	74.73
2	Bihar	58.20	56.71	92.33	88.92	13.48	13.15	87.04	81.10
3	Karnataka	60.36	67.44	83.75	88.41	24.71	30.66	85.32	88.26
4	Madhya Pradesh	58.35	69.03	81.13	89.20	11.37	22.52	61.69	79.56
5	Himachal Pradesh	20.99	28.82	59.44	74.98	5.69	9.40	64.75	77.60
6	Rajasthan	51.11	53.41	74.58	79.84	14.59	21.07	64.43	78.26
7	Tamil Nadu	53.46	53.93	98.62	88.48	14.73	15.89	80.79	74.31
INDIA		49.79	53.04	83.36	86.96	13.87	18.45	76.15	78.11

Source: 6th and 7th Educational Surveys (2005).

Along with uneven physical access across states and habitations, the number of children who enrolled in the school system varied across gender and social groups too, with girls as well as SC and ST children lagging behind. However, recent studies indicate that these groups have been catching up as far as enrolment at the primary school level is concerned, though as shown below, the drop out rates tell another story. Though the enrolment of girls remains below those of boys, it has risen steadily and faster than that of boys over the years. The enrolment of SC and ST children has shown the same growth trend, and is now in proportion to their population (Arun C. Mehta, 2006). However, as Box 1.1 shows, within the SC and ST categories, the enrolment of girls remains less than those of boys, indicating their extreme marginalization. Moreover, disaggregated data on out-of-school children who have never enrolled in schools reveal a disproportionate presence of special groups like working children, those in urban slums, residents of far flung habitations, SCs/STs and nomadic groups.

¹⁶ A habitation as defined by NCERT for the purpose of this survey is “A habitation is a distinct cluster of houses existing in a compact and contiguous manner; with a local name; and its population should not be less than 25 in plain areas and not less than 10 in hilly/desert/sparsely populated area”

Box 1.1: Enrolment among SC and ST girls

In 2002-03 there were 21.67 million SC children (44.9% of these were girls) and 11.83 million ST children (45.7% girls) in primary school. In upper primary school there were only 7.49 million SC children (40.9% girls) and 3.25 million ST children (40.6% girls). As [Tables A and B](#) below show, SC and ST girls' enrolment at the primary and upper primary stages has increased quite sharply over the years, but remains less than that of boys, particularly for ST. However, if one were to look at Gross Enrolment Ratios (GER), which reveal a lot more than absolute numbers, the ratios of SC and ST children to their total population remains far lower than the national average. Only 48.6 per cent SC girls and 40.6 per cent ST girls are enrolled at the elementary level as compared to the national average of 56.22 per cent for all girls. Another significant point to note is that there is a drastic fall in GER rates from Primary to Upper Primary – even in 2004-05 only 61.5 percent SC girls 59.9 per cent ST girls reach upper primary school. Equally significant is that the gender gap among both SC and SC students is high at both primary and upper primary levels.

Table A: Gross Enrolment Ratios of Scheduled Caste children from 1986 to 2005

Year	Primary				Upper Primary			
	Boys	Girls	Total	% Girls	Boys	Girls	Total	% Girls
1986-87	103.8	64.8	84.8	37.31	52.7	26.6	40.4	29.90
1990-91	125.5	86.2	106.4	38.35	68.7	35.8	52.7	33.97
1995-96	109.9	83.2	97.1	41.16	71.4	44.5	58.5	36.58
2000-01	107.3	85.8	96.8	43.10	76.2	53.3	65.3	39.26
2004-05	123.3	106.6	115.3	44.81	77.9	61.5	70.2	41.39

Source: Select Education Statistics, Department of Education, MHRD, GoI, 2007.

Table B: Gross Enrolment Ratios of Scheduled Tribe children from 1986 to 2005

Year	Primary				Upper Primary			
	Boys	Girls	Total	% Girls	Boys	Girls	Total	% Girls
1986-87	111.0	68.8	90.1	32.77	45.6	21.9	34.1	27.63
1990-91	125.4	81.4	104.0	36.57	53.9	26.7	40.7	30.40
1995-96	115.0	80.2	96.9	36.99	57.3	35.0	46.5	33.74
2000-01	116.9	85.5	101.1	40.64	72.5	47.7	60.2	36.63
2004-05	128.1	115.5	121.9	42.43	73.9	59.5	67.0	39.07

Source: Select Education Statistics, Department of Education, MHRD, GoI, 2003-04.

Hierarchies of access

An important dimension of access to elementary education in India is that children belonging to different social categories access different types of schools (Ramachandran, 2003; Sadgopal, 2006). The existing school types may be classified into three major categories: one, 'formal' government schools that provide free (or nearly free) education, with reasonably well-qualified and paid teachers; two, various types of non-formal centres for education run or supported by the government, with locally recruited teachers who are paid a very small salary or honorarium; and three, fee charging private schools that may or may not receive government funds. Recent research reveals a hierarchy of access whereby alternative or non formal centres have a high proportion of the less well off, usually girls and the socially marginalized, while private fee charging schools have a high proportion of with the better-off, especially upper caste boys (Ramachandran, 2003).

Notably, in the elementary school system, not only is there a distinction between government and private schools, but government schools are also divided in the 'formal' and 'non formal'

schools, with the latter lacking trained teachers and basic facilities and catering to the poorest children. Additionally, 'formal' government schools too are of several types. There are well-staffed and well-equipped schools run by the central government called 'central schools', for children of government employees and Navodaya Vidyalayas which are residential schools for a very small number of rural children selected on the basis of merit. The bulk of government schools are formal schools run by state governments, are accessed by a vast majority of the children. These schools have qualified teachers and some basic infrastructure. But there are large variations among these schools. Generally, larger villages and areas with greater clout tend to have schools with more teachers and facilities (Govinda and Verghese, 1993; Mehta 2005; Sharma 1997).

Private schools are of two types in terms of source of funding: aided schools, i.e., schools that are privately managed but receive financial support from state governments (state funds may comprise as much as 95 per cent of their expenses) and unaided schools that are privately managed and financed. While in the 1970s and 1980s, private unaided schools catered to a small section of the privileged classes, in the 1990s the number of such schools grew considerably, though a vast majority of the children continue to access state government schools¹⁷. Equally, the number of private aided schools has been shrinking. The new unaided private schools have a high proportion of higher caste children and a low proportion of SC and ST children (Tilak and Sudarshan, 2001). Micro-studies show that these are frequently located in prosperous localities and have a preponderance of boys. These schools are also highly diverse and set up for varied social, linguistic, religious or commercial motives (De et al., 2001).

Dropout and completion

Serious, systemic problems with the elementary school system are visible in the high dropout and low completion rates. In the 6 to 11 age group, while close to 90 per cent children are formally enrolled in primary schools and alternative centres of various types, nearly 40 per cent dropout at before completing class 5. There is a further drop in the number of children enrolling in upper primary and those completing the cycle. The National Family Health Survey (1998-99), reported that while there was a significant increase in the number of children in the 6-14 years age group attending school (73.7 per cent girls, 83.1 per cent boys), only 34.2 per cent girls and 53.3 per cent boys completed primary education. The median number of years of schooling was 5.5 for men and only 1.8 for women; only 35.6 per cent boys and 19.7 per cent girls had completed middle school in 1998. As [Table 1.3](#) shows, dropout rates are the highest among socially disadvantaged children: girls, SC and ST children.

17 A National Council for Applied Economic Research (NCAER) survey in 1993 showed that in elementary education, 75 per cent children were enrolled in government schools, 16 per cent in private aided schools and a little less than 10 per cent in private unaided schools (Tilak and Sudarshan, 2001).

Table 1.3: Dropout Rates at Primary and Elementary Stages

Year	Primary (1 to 5)						
	All Boys	SC Boys	ST Boys	All Girls	SC Girls	ST Girls	Total
1990-91	40.1	46.3	60.3	46.0	54.0	66.1	42.6
1995-96	41.4	43.7	55.0	43.0	48.5	58.9	42.0
2001-02*	38.4	43.7	51.0	39.9	47.1	54.1	39.0
2004-05*	31.81	32.7	42.6	25.42	36.1	42.0	29.0
Year	Elementary (1 to 8)						
	All Boys	SC Boys	ST Boys	All Girls	SC Girls	ST Girls	Total
1990-91	59.1	64.3	75.7	65.1	73.2	82.2	42.6
1995-96	56.6	64.7	62.3	61.7	70.5	71.2	42.1
2001-02*	50.3	58.6	67.3	57.7	63.6	72.7	39.0
2004-04*	50.49	55.2	65.0	51.28	60.0	67.1	61.92

Source: Select Education Statistics, Department of Education, MHRD, GoI, 2007.

* Provisional

The DISE data analysis shows that ‘67 children out of 100 reach grade V... the corresponding figure for rural and urban areas are 64 and 83. More boys survived up to grade V (68) compared to girls (66). Rural urban and regional differences are marked—there is 17 percentage point gap between rural and urban boys and a 22 percentage gap between rural and urban girls...’ (Arun Mehta, 2006: 126-128).

Severe social barriers to meaningful participation of children from some communities continue to exist. For example, children of communities engaged in scavenging, disposal of dead animals and other ‘impure’ occupations are shunned by their classmates. Teachers are also not very sympathetic to their predicament. Recent qualitative micro-studies conducted in six states have captured a range of discriminatory practices in schools (Ramachandran, 2004). The combination of gender, social and location disadvantages combine to make some groups of children extremely vulnerable. For instance, 71 out of 100 tribal girls dropout before completing elementary education, and the situation of SC girls is not significantly better (Table 1.3). Unfortunately, data on children with special needs—physically and mentally challenged—remains scanty. An educational environment remiss at handling ‘normal’ children has a long way to go before it can be expected to respond satisfactorily to the needs of challenged children.

There are important inter-state differences here too. A recent analysis of the District Information Statistics for Education (DISE) data¹⁸ from 581 districts revealed that a majority of the children in Andhra Pradesh, Chandigarh, Himachal Pradesh, Karnataka, Kerala, Pondicherry, Punjab and Tamil Nadu survived up to Class 5. On the other hand, a majority of the children in Arunachal Pradesh (44.2 per cent survival rate), Bihar (39.3 per cent survival rate) and Rajasthan (50.85 per cent survival rate) did not survive up to Class 5 (Arun Mehta, 2006: 126-128). In Karnataka, Tamil Nadu and Himachal Pradesh the gap between Gross Enrolment Ratio (GER) and Net Enrolment Ratio (NER)¹⁹ at the primary and upper primary

18 DISE is a computerized district based system of educational statistics of the Government of India.

19 GER is simply the ratio of the total number of children enrolled to the total number of children in the relevant age group. NER is the ratio of the number of children enrolled from within the relevant age group to the total number of children in the age group. GER at the primary stages often exceeds 100 per cent (this phenomenon is fairly common in India because children below and above the 6-14 years age group enroll in primary classes), though the net enrolment ratio is about 20 per cent lower (Selected Educational Statistics 1999-2000 and 2004, GoI).

stages was not so marked, while in Bihar and Rajasthan it was substantial. Andhra Pradesh and Madhya Pradesh fell somewhere in between. A large gap essentially implies that the dropout in the primary cycle and from primary to upper primary is very high.

School quality

Studies and surveys of various types show that the quality of schools is not satisfactory. Recent studies (Agarwal, 2000, 2001; Jha and Jhingran, 2002; Nair et al., 2001; Ramachandran 2003) argue that poor quality education effectively pushes children out. The Probe Report (1998), based on a survey in five north Indian states, concluded that parental demand for education existed and was often quite high, but that the poor quality of schools discouraged both parents and children. The various dimensions of poor school quality, i.e. inadequate infrastructure, poor school functioning and inadequate teaching and learning indicate inadequate resources as well as problems in management and pedagogic processes.

In general, the availability of teachers and infrastructure in government elementary schools has remained poor and some commentators have perceived a decline as schooling has acquired a more mass character. Numerous studies of the 1990s showed government schools to be characterized by poor buildings, lack of teachers, inadequate teaching materials, books and other facilities such as drinking water and toilets (MHRD, 2001; Gopinathan Nair et al., 2003; Probe, 1996). The official policy is for primary schools to have at the minimum two rooms, two teachers and a pupil teacher ratio of 40:1. However, the DISE data shows that as many as 1,07,842 schools of all categories had only one room in 2004-05 and 94.73 per cent of such schools were located in rural areas (Table 1.5) (Arun Mehta, 2006).

Table 1.5: Key indicators across seven states

	AP	TN	Kar	HP	MP	Bihar	Raj
% Single Teacher Schools (P)	8.50	11.20	17.10	11.50	26.70	14.20	40.00
% Single Teacher Schools (P +UP)	0.20	1.20	0.40	0.40	4.10	3.20	2.30
% Single Classroom Schools (P)	34.50	11.10	23.60	5.20	11.10	14.40	5.50
% Single Classroom Schools (P+UP)	3.40	0.70	0.50	1.10	3.70	3.30	1.00
% Schools with ≤ 50 children (P)	48.90	31.30	65.30	60.60	27.60	3.90	32.50
% Schools with ≤ 50 children (P + UP)	3.60	2.30	7.50	33.50	21.00	2.20	7.10
% Schools with PTR (Pupil-teacher ratio) > 100 (P)	0.50	1.30	0.50	0.10	3.40	26.30	3.40
% Schools with PTR > 100 (P + UP)	0.50	3.60	0.50	0.80	4.70	35.70	3.60

Source: Arun C Mehta: Elementary Education in India: Where do we stand; State Report Cards based on DISE data, NIEPA, 2006.

There is a major shortage of teachers, resulting in teacher pupil ratios going as high as 1:80 in select districts of some states (Tables 1.6 and 1.7) (7th All India Educational Survey, 2003). In addition, there is a perennial problem multi-grade classrooms, i.e. where there are fewer teachers than there are grades, so that one teacher teaches more than one (and sometimes as many as five) classes at one time. An estimated 14.66 percent rural schools are single teacher schools. Forty per cent of rural primary schools in Rajasthan had only one teacher, while the percentage was 8.50 in Andhra Pradesh (Arun Mehta, 2006). The situation at the upper

primary level was slightly better with both teacher pupil ratios as well as student classroom ratios being marginally better—with the all-India teacher-pupil ratio (TPR) at the upper primary level being 1:34 (all) and 1:35 (rural).

Table 1.6: Teacher-pupil ratio in different states

Sl. No.	State/UT	Primary Schools			Upper Primary Schools		
		Rural	Urban	Total	Rural	Urban	Total
1	Andhra Pradesh	32	35	33	31	27	30
2	Bihar	85	60	83	77	51	73
3	Himachal Pradesh	22	23	22	15	13	15
4	Karnataka	26	28	26	37	37	37
5	Madhya Pradesh	38	29	36	31	22	28
6	Rajasthan	42	33	41	34	25	31
7	Tamil Nadu	35	33	34	42	38	40
INDIA		44	36	42	35	32	34

Source: 7th All India Educational Survey done in 2003, NCERT, 2005

Table 1.7: Teacher-pupil ratio

Year	Primary	Upper Primary
1950-51	01:24	01:20
1970-71	01:39	01:32
1980-81	01:38	01:33
1990-91	01:43	01:37
1999-00*	01:43	01:38
2000-01*	01:43	01:38

Source: Select Education Statistics, Department of Education, MHRD, GoI, 2002-03.

* Provisional

There were important differences across states. For instance, Bihar recorded a rural TPR at 1:77 at the upper primary level followed by Jharkhand at 1:60 while Tamil Nadu recorded a TPR of 1:40.

A major problem of school quality is that many schools simply do not function as educational establishments, where the main goal is teaching and learning. The Probe Report found critical problems in school functioning in north Indian states: the absolute teaching time was quite low, teachers often arrived late and left early and casual absenteeism was common. Even when teachers were in school, they spent a large part of their time in activities other than teaching. There are some differences among state here, with a better performance of the southern states as compared the northern states. Within north India, the Probe Survey found that Himachal Pradesh presented a far better picture than the other states, in the greater commitment of its teachers, better-maintained infrastructure and a friendlier school environment (Probe 1998, Manabi Mazumdar in Gopinathan Nair et al, 2003).

Several studies also confirm ineffective teaching learning processes in schools (e.g. DPEP 2001 and 2003, Probe 1998), though systematic all-India data is not available. The concrete outcome of poor school quality is inadequate pupil achievement, and that learning outcomes in schools are alarmingly poor has been confirmed by several studies. For instance, in 1994 the National Council of Educational Research and Training (Delhi) made a country-wide assessment of children's basic reading, writing and numerical skills. The results of the study presented a dismal picture. On two basic tests, arithmetic and reading-comprehension in the mother tongue, the scores showed that many children were not capable of answering even half

the questions correctly, i.e., they were nowhere near mastering basic reading and numerical skills (Shukla et al., 1994). Studies undertaken after this survey (Bashir, 1994; Sharma, 1997) showed a similar pattern.

Programmatic Thrusts

We now discuss the major initiatives of the central government to address the critical problems that the elementary education school system in India is confronted with. This discussion illustrates that while the central government initiatives span a range of strategies, they share the common problem of ‘poor implementation’ (which the later chapters of this book will show to be a highly complex dynamic). Additionally, many strategies, especially of the 1990s, openly or tacitly admit this problem, and attempt to redress it.

Alternative centres

As noted above, the spread of schooling did not lead to universal enrolment and less so to every child remaining in school up to the age of 14 years because of lack of physical access in some habitations and because of the special problems faced by socially marginalized children. Non-formal or alternative schooling was one strategy adopted by the national government as education became a concurrent subject. In 1981-82, the scheme for Non-formal Education²⁰ (NFE) was introduced in nine educationally backward states to supplement the formal school system and provide children who could not attend formal schools with an alternative mode of education, and extended to all the states in 1987-88. The NFE programme provided funds to state governments and to NGOs for running community based schools with flexible timings. The centres were expected to run for one and a half to two hours every day for six days a week. The teachers in NFE were local persons, who were paid a tiny honorarium.

However, as government documents admit, NFE did not emerge as a credible solution. The problem lay with the design of the programme as well as in the unrealistic expectations that working children could be provided primary schooling in two years through part-time instruction provided by poorly qualified and paid para-teachers. Also, ‘earnest and effective implementation by states of NFE (was) also lacking’ (GoI 1996: 44). The NFE programme was withdrawn in 2000 and a revised programme, the Education Guarantee Scheme and Alternative and Innovative Education (EGS and AIE) was launched instead.

In parallel to the NFE programme, several important initiatives were pioneered in the Shiksha Karmi and Lok Jumbish programmes in Rajasthan. These were in partnership between the state and central governments and NGOs with financial aid from SIDA. These initiatives differed from the NFE programme as they developed a range of strategies for children who faced different types of problems in attending school as against a single strategy (Box 1.2), in their capacity to develop processes which allowed for real engagement with the community, and their success in actually implementing stated strategies.

Box 1.2: Strategies to enroll out-of-school children in Lok Jumbish, Rajasthan

- i. **Muktangan**—A flexible programme for the education of tribal children.
- ii. **Madrassa strengthening programmes** to reach out to children from minority communities.
- iii. **Hostels for tribal children** (short- and long-stay) to cater to the educational needs of children of seasonal migrants as well as those living in remote/scattered habitations.
- iv. **Mahila Shikshan Vihar (MSV)**—Residential condensed courses for out-of-school adolescent girls/young women in the 15+ years age group.
- v. **Balika Shikshan Shivar (BSS)**—A residential camp of 6 to 8 months for out-of-school

²⁰ NFE was started in 1977 on a pilot basis.

girls in the 9+ years age group, to serve as a bridge programme to facilitate re-entry into the formal school system.

- vi. Institutionalization of **Sahaj Shiksha Kendra (SSK)**, a rigorous non-formal education programme to cater to the needs of working children.
- vii. Creation of **Village Education Registers** (maintained by the VEC) and **Retention Registers** (maintained by the primary school/SSK) to track children's participation/attendance.

Source: Ramachandran (2003).

The success of these initiatives had an impact on national programmes. For instance, the Working Group Report for the Ninth Plan points to 'NFE in Lok Jumbish and Shiksha Karmi Projects' of Rajasthan (pp42-43). Large-scale interventions on a more flexible pattern were tried in the mid-1990s in DPEP, wherein access and enrolment of children in the 6-11 years age group received unprecedented attention. A host of initiatives were developed across various states and different types of alternative education centres were set up in very small habitations and for children who had dropped out. Other strategies such as bridge courses to enroll out-of-school children were also taken up.

A major initiative that emerged at this point was the Education Guarantee Scheme in Madhya Pradesh (Box 1.3).

Box 1.3: Madhya Pradesh Education Guarantee Scheme

The Education Guarantee Scheme (EGS) in Madhya Pradesh (MP) was introduced in January 1997 with the objective of providing minimum education facilities to socially, economically and geographically marginal groups through schools that were sensitive to the needs of local communities and were also cost-effective. The process involved in setting up an EGS school was simple. A settlement with a population of 40 students in the age group of 6 to 14 years (the number was reduced to 25 in the case of socially marginal groups or sparsely populated areas) and no school in the radius of 1 kilometer could demand a primary education facility from the state. The government was mandated to respond with a fully functional EGS school within 90 days of receiving the demand. Teachers for EGS were recruited locally and hired on a yearly contract by the panchayat with salaries much lower than regular teachers. The expansion of EGS led to an unprecedented surge in enrolment, especially of girls and children from disadvantaged communities.

An important shift visible here was that EGS centres were established *in lieu* of formal government schools, as against providing education to children who could not access formal education. Community control was stressed and in the early years of the programme it was claimed that these schools functioned regularly and provided good quality education (Gopalakrishnan and Sharma, 1998). Several states across the country adapted the EGS idea—for example, Rajasthan created Rajiv Gandhi Pathshalas along the lines of the MP EGS programme. It is interesting to note that several states in northern India adopted the model of alternative schools, while most southern and western states focused on enhancing access to formal primary/upper primary schools, using the alternative route only in specific situations or as a time-bound intervention to enable out-of-school children to return to the formal stream. In other words, such schools were more popular in states where the problems were severe.

In fact, an interesting set of needs and motives seems to underlie the growth of such schools in the 1990s. As noted above, non-formal or alternative schools were originally meant to provide education to children who could not attend formal schools and the efforts in this direction increased during the 1990s as the pressures to make elementary education universal heightened. However the establishment of such schools *in lieu of* formal schools needs to be viewed against the fact that these were low-cost schools in a scenario where state governments

were increasingly cash-strapped. Additionally, some state governments even considered these schools to be better than formal schools, particularly as they were seen to be closer to the community. This last logic is especially significant, as it illustrates the huge problems that the formal elementary school system had run into and the emergence of the need to evolve new type of schools that were managed differently. The problem of systemic functioning clearly contributed to the popularity of these schools (which does not imply that this was, in fact the right solution).

Privatization

We have noted above the growth of private schools since the late 1980s and the 1990s. As in the case of alternative education centres, the poor quality of the government school system was one reason for this growth (Probe Report, 1999). Notably, during the 1990s state policy towards private schools grew more liberal. The Working Group Report for the Ninth Plan stated that:

‘.... It cannot be denied that a large number of private institutions are providing quality education... While States should curb the tendency of commercialisation of education, they should provide adequate space and opportunity to deserving private schools to spread elementary education in rural areas’ (MHRD, 1996: 25).

To some extent this may also be seen as an attempt to counter the problems of the government school system.

Community engagement

The concept of informal schools or alternative education centres has been tied up with the idea of community participation from the time of the NFE schemes. The premise has been that community engagement would motivate parents to send children to school regularly. In Lok Jumbish, engagement of the community was central to enrolling out-of-school children through processes such as ‘school mapping’ and ‘micro-planning’ (GoR, 1991). The Shiksha Karmi Project (SKP) in Rajasthan, launched in 1987, also emphasized community engagement and provided locally recruited teachers in dysfunctional primary schools (Box 1.4).

Box 1.4: Shiksha Karmi Project, Rajasthan

Responding to widespread teacher absenteeism and non-availability of teachers in remote rural habitations, and consequent poor enrolment and high dropout rates, the Government of Rajasthan and the Government of India jointly launched the Shiksha Karmi Project (SKP) in 1987. The concept of Shiksha Karmi was based on a supposition that a change agent, especially in the field of education, can work effectively if he/she belongs to the same locality even though he/she is lesser qualified than a regular teacher. The social background and location of the teacher was seen to be of primary importance. The local community was involved in identifying potential candidates and the education department/Shiksha Karmi Board selected the most appropriate person through an intensive selection-cum-training process. Young men and women with basic high school education were identified and put through intensive induction training. These teachers were then given continuous academic support through a local resource agency (in many cases a local NGO with a proven track record in elementary education). The local community provided support and also monitored the school and hence this was conceived as a ‘people’s programme’ to educate children.

During the 1990s, in DPEP the government had engaged with the community on a large scale. Community based bodies such as Village Education Committees (VECs), Parent Teacher Associations (PTAs) and Mother Teachers Associations (MTAs) etc. were set up and trained to monitor school enrolment and attendance and take up enrolment drives (DPEP, 1999).

Notably, community participation was further identified with attempts to improve the quality of schools. The community-based bodies mentioned above were expected not only to increase enrolment and reduce dropouts, but also improve school functioning by participating in their management. This was another attempt to address the problems of school functioning. The SSA document made community ownership of schools a key cornerstone of the programme (GoI 2001).

Incentives and nutritional support

Given the high drop out rate of children groups marginalized groups, state governments had also put in place various incentive schemes—scholarships as well as uniforms and textbooks were provided to children who were at a high risk of dropping out, i.e., SC and ST children and girls (see Chapter 2 for more details). However, the Probe Survey (1998) found the coverage of these schemes to be very limited with incentives being supplied in an erratic manner. During the 1990s, the central government supplemented this effort, beginning with DPEP, which provided for textbooks for girls and SC/ST primary school students and expanding the scope in SSA up to the upper primary level. The National Programme for Education of Girls at Elementary Level (NPEGEL), implemented under the SSA umbrella, provides additional support for the education of underprivileged/disadvantaged girls at the elementary level in selected educationally backward blocks.

A major initiative of the central government of the 1990s was the Midday Meal Scheme²¹. During the 1990s, as the stocks of food available with the central government became substantial, several activists began to renew protests about the persistence of hunger in this context. Hence, in 1995, through the National Programme for Nutritional Support for Primary Education, the central government provided food grains free of cost to the state governments, who were expected to make additional investments and provide cooked midday meals to the children. However, in most states, dry monthly grain rations were distributed to children (Dreze and Goyal, 2003). In November 2001, the Supreme Court directed all state governments to introduce cooked midday meals in schools and increasingly the state governments began to comply.

Up-gradation of school infrastructure

The rapid expansion of schooling in independent India meant that state governments could often not keep up with the infrastructure requirements. The central government first attempted to redress this situation through the Operation Blackboard (OB) Scheme of 1987, which sought to provide a basic minimum number of teachers and infrastructure to all primary schools, and was subsequently expanded to upper primary schools in 1992. In terms of infrastructure the goal was to provide two all-weather rooms and a covered veranda in all primary schools changed to three rooms for primary schools if enrolment exceeded 100, and one room per classroom for upper primary schools in 1992; separate toilets for boys and girls; and teaching-learning materials.

An evaluation of this scheme conducted in 2000 showed that though the scheme reduced the number of single teacher schools, there were still nearly 15 per cent schools with single teachers continued to exist. Similarly, while infrastructure had improved, a substantial number of single classroom schools continued to exist. Some observations of the evaluation report of the OB scheme are interesting. The report observed wide differences among states in implementing the scheme, observing that, ‘the states which have been found to be efficient in implementing the on-going state funded primary education development activities, also happened to be states which have been utilizing the funds provided under the OB scheme’

21 During the 1980s, a few states had started providing free noon meals. Tamil Nadu is a much admired case.

(NIEPA, 2000: 80). It also noted that in several cases, additional classrooms had not been constructed where they were required and sites for additional classroom were decided under local pressure (p. 76). Moreover, an overwhelming majority of the new classrooms constructed were 'inappropriate from the point of view of design and inadequate from the point of view of quality of construction' (p. 80).

Substantial school infrastructure was subsequently created in DPEP from the mid-1990s onwards and continues to be created in SSA, as these programmes allow substantial percentage of funds to be used for creation of infrastructure. Additionally, in DPEP for the first time teachers were allowed a grant of Rs. 500 per year to purchase teaching-learning materials and each school was given a grant of Rs. 2,000 per year for repair of buildings etc. These features have been enhanced in SSA.

Teachers

Operation Blackboard attempted to address the problems of high teacher pupil ratios as well as multi-grade and single teacher schools by providing extra teachers. The goal was to provide two teachers, one of them preferably a woman, for primary schools, changed to three teachers if enrolment exceeded 100, and at least one teacher per class in upper primary schools in 1992. This trend of the central government stepping in to provide funds for extra teachers continued in DPEP and subsequently SSA.

An important policy thrust in the early years was the enhancement of the status of teachers, emphasized in both NPE 1968 as well as NPE 1986. Teacher salaries and other benefits improved significantly across the states in the post-independence scenario. However, during the 1990s, several states began to hire para-teachers, who had low salaries and no security of tenure. Notable, para-teachers were hired not only in alternative education centres, EGS and EGS-like education centres but also in formal schools. As was the case for alternative or non formal education centres, northern states tended to hire para-teachers far more than the southern states.

Para-teachers reduced the financial investment needed to reach required teacher-pupil ratios in cash-strapped states. Moreover, the unsatisfactory performance of 'regular' teachers lent support to this initiative, and para-teachers, as local and rooted in the community, were seen as an appropriate option. For instance, the Probe Survey, while weighing the pros and cons of hiring para-teachers, noted the negatives of the growing dualism i.e., existence of teachers who received different salaries for the same work within the system, and lower teacher qualifications; yet it also admitted that several para-teachers worked distinctly harder than regular teachers (pp. 98-99). The para-teacher remained a subject of intense debate through the 1990s (Kumar et. al., 2001; Sharma, 1999)

Teacher education has been consistently emphasized in policy documents and today teachers in most states tend to be fairly well qualified. An important initiative in this direction was the Scheme for Restructuring and Reorganization of Teacher Education, begun in 1987. For elementary education, the most important feature of this scheme was the establishment of District Institutes of Education and Training (DIETs), envisaged as institutions to provide academic and resource support. Till this scheme was made operational, academic and resource support for elementary education was provided only at the national and state levels with elementary teacher training colleges providing pre-service training to teachers. Through DIETs, academic support was sought to be extended to the district level (GoI, 1989). However, by 2001, though 461 DIETs had been set up across the country, they had not become the pace-setting institutions that they were envisaged to be and had addressed the role visualized for them in a very limited way. By and large, the institutes were involved with pre-service and some in-service training of teachers, but not activities such as developing curriculum and teaching-learning materials. Moreover, even the training provided by the

institutes tended to ignore ground realities. The Working Group Report for the Tenth Plan noted that the development of DIETs across different states had been uneven, and the southern states had developed these institutions better than the northern and eastern states (DPEP, 1999; GoI, 2001; Probe, 1998).

In DPEP, the teacher support structure was extended still further and grassroots resource centres at sub-district level were established, which have been continued in SSA. The Teacher Education Scheme had also made provisions for large-scale training for primary teachers. In DPEP, the scope of teacher training was expanded to a very large scale with the help of grassroots resource centres (DPEP, 1999a). SSA too allows for very substantial in service training of primary as well as upper primary school teachers.

Curriculum and textbook reform

One way in which the central government sought to ensure quality of education was by developing uniform patterns of education and standards across the country. The recommendations for a common pattern of schooling,²² formulation of curricula and textbooks by NCERT²³ and State Council for Educational Research and Training (SCERTs) and public examinations were the main methods of doing so. A much debated step in this direction was the articulation of Minimum Levels of Learning (MLLs) for the primary stage, which were followed by large-scale teacher training programmes. MLLs emphasized the concept of ‘competency’, laying out the competencies to be acquired by children in each class for the primary school cycle, and attempted to shift the common concern of teachers regarding completion of syllabus to what children had actually learned (NCERT, 1998). Several state governments formulated textbooks on the basis of MLLs and trained teachers, but there is little evidence that MLLs resulted in better classroom teaching (Probe, 1999). At the same time, MLLs have been critiqued by several educationists as being narrowly and rigidly defined, not taking the nature of children’s learning into consideration, not taking into account the multi-grade teaching situation in the country etc. (e.g. NCERT 1998, Probe 1999).

NPE 1986 had also stressed the need to make primary education a joyful experience for children by making it more child-centred. Various small initiatives were taken by the government and by non-government agencies in this direction. In DPEP, such initiatives were attempted on a much larger scale. Several states revised their textbooks and took up massive teacher training programmes to change pedagogic practices, with varying success. Available studies indicate that while these initiatives brought about some changes in the classroom, only in a few instances, e.g. Kerala, (Rampal, 2001) were the changes substantial, and these too were followed by reversals (DPEP 1999, 2001, 2003; World Bank, 2003). A major curricula and textbook reform process has been initiated recently by NCERT, and is discussed in greater detail in Chapter 6.

The paucity of administrative and management reforms

In contrast to the various programmatic initiatives highlighted above, i.e. to increase the outreach of the school system, midday meals, incentives, improvement of school infrastructure, setting up an institutional structure to improve school quality, curricular and textbook reforms etc., attempts to reform practices of administration and management of schools have been few (Sharma, 2000). The basic departmental structure for the administration

22 The Secondary Education Commission recommended an 11-year cycle for school education, while the Education Commission of 1966 recommended a 10+2 pattern, which was reiterated by NPE 1968.

23 In 1975, NCERT formulated the ‘Curriculum for the Ten Year School: A Framework’, and prepared curricula and syllabi for the primary and upper primary stages. The implementation of this curriculum remained uneven across the country. In 1986, a National Curriculum Framework was developed by NCERT after NPE 1986 (NCERT, 1998).

of school education has continued in the inspectorial pattern established during colonial rule²⁴. Policy advisors and makers have been aware of the problems posed by this structure, and the 1968 as well as the 1986 policies recommended several reforms vis-à-vis the management of school education such as strengthening the field offices, better recruitment policies and training. However, there have been no major central schemes that have focused specifically on the reform of the existing administrative and management structure and processes. At the same time, several recent studies show this area to be the most problematic. For instance, the Probe Report brought out sharply a supervision system that was indifferent to these all too apparent, and major, problems (Probe report. 1999)

While this has been the general trend, some initiatives have been visible during the 1990s. First, large scale 'project structures' were created in DPEP, and the programmes was administered through a registered society at the state level, and district project offices as well as sub-district structures were established to implement the programme. This project structure functioned in parallel to the regular structure, and has been retained in SSA. Second, in both these programmes, instead of pre determined schemes, the central government channelled its funds on the basis of need based district and state plans, which covered a wide range of activities, selected by the districts and states, to achieve overall programme goals.

In Chapter 5, as we look at the internal dynamic of the system, we illustrate why such reforms are especially difficult to initiate, and what the attempted changes lead to. For the present, it is important to note that this underlines further the need to examine systemic and institutional functioning.

About This Book

In this book we explore the 'system' and its institutions, the manner in which they are structured and function, their capacities, processes and dynamics. This exploration is based on an in-depth field study of several types of institutions across various levels of government, situated in two states, Rajasthan and Andhra Pradesh. Rajasthan and Andhra Pradesh are selected, as these are low and middle income states respectively, and between them, provide contexts that are common in large parts of India. At the same time, they lie in north and south of India respectively, and consequently provided some idea of the variations that exist in these two regions. However, these differences are much less marked than would be in the case of say, Tamil Nadu and Uttar Pradesh. In the selection of states, preference has been given to bringing out a picture that is common to much of India, rather than bringing out differences across states and regions.

The reasons for studying states that are not sharp contrasts are threefold. First, key, *similar* systemic problems exist across the country, and this book attempts to highlight these, and this best done by selecting states that are representative of much of the country, rather than 'outlier' states. Second, as the detailed study of institutions attempted in this book is the first of its kind in India, it is important to understand general trends. Later studies may bring out inter-state differences more vividly. Finally, inter-state comparisons pose considerable difficulties, involving as they do varying socioeconomic contexts, political dynamics and policies, and also

24 'School administration' was initiated in the pre colonial era as the East Indian Company established a General Committee of Public Instruction in 1823 in Calcutta to disburse a grant of Rs. 1 lakh made to education. In 1842, this Committee was replaced by a Council of Education, which advised the government on all matters relating to education. Similar, but subordinate boards and councils were also established in the Bombay and Madras presidencies and north-western provinces. This Council drew the government's attention to the necessity of providing for proper inspection and supervision of government institutions, and in 1844 an Inspector was appointed. After the Education Despatch of 1854, these boards and councils were replaced by departments of public instruction, headed by a Director of Public Instruction in each province, assisted by inspectors (Ghosh, 1995).

huge within-state variations. Consequently, the main goal in this book is to identify common trends in these two states, which are in any case much greater than the differences, and which can be assumed to hold across much of the country. At the same time, the differences that exist are noted and analyzed.

To understand institutional structures and processes, it is important to be aware of the educational scenario and various policy thrusts, which articulate the expectations from the institutions, and form the context in which the activities within these institutions can be understood. The national level picture of the status of elementary education and policy and programmatic thrusts has already been described in this chapter. A similar picture for Andhra Pradesh and Rajasthan is provided in Chapter 2, which also describes the institutional structure in the two states. Subsequently more details about the study undertaken are provided. These should ideally have been provided here, but to understand the study design, it is important to first have a clear picture of the institutional structure. An analysis of the financing of education is provided in Chapter 3. This chapter highlights the relative priority that is accorded to elementary education vis-à-vis other sectors, as well as future needs.

The analysis of the 'system' begins with a brief glimpse of a small number of schools and villages in Chapter 4. This is meant to enable the reader to get an idea of the social (village level) context in which elementary schools function, as well as the characteristics of these schools. A large-scale survey of schools has not been undertaken. Such a survey would have required resources that were not available, and notably, broad features of this situation are already known, and studies exploring the situation of schools have already been quoted in this chapter. In stead, Chapter 4 attempts to bring out the nuances of this picture, by enabling the reader to look in greater depth at a small number of villages and schools. Importantly, we do not attempt to compare the two states through this picture of a very small number of schools, but in stead, attempt to bring out a dynamic and detailed picture of the villages, the educational situation within them, and school functioning. In subsequent chapters, we relate this picture to institutional functioning in the two states.

The next three chapters examine the various institutions of the system in detail. In this scrutiny, we draw upon recent scholarship on the state and its institutions (e.g. Evans 1995, Migdal 1995) that argues that states are not unitary actors, but complex organizations. As Peter Evans (1995) has shown, states vary dramatically in their internal structures and relations to society, and these variations in turn produce vastly different outcomes in formulating as well as pushing through developmental as well as re-distributive agendas. The broad approach to the study of state institutions followed here is that suggested by Peter Evans (1995), of viewing state institutions and their capacity to carry out various agendas, in terms of their internal structure and dynamics on the one hand, and their relation to various societal groups on the other. Additionally, the 'system' is seen as comprising not just a few policy making apex institutions, but also of its implementing, field level institutions. To quote, 'as in ideology and policy formulation, policy implementation also reflects the state's engagement with other social forces' (Migdal 1994: 14). These ideas, i.e. viewing the state as a complex organization, comprising a range of institutional mechanisms at various levels, to be understood in terms of their internal organization and relation to society, form the basis of analysis.

Systemic practices are further examined from three points of view, in terms of the play of politics and the administrative and management ethos and style (examined in Chapter 5), pedagogic beliefs and practices (Chapter 6) and the approach to and impact on social equity (Chapter 7). Each of these perspectives adds depth to the understanding of the system as a whole, and provide a picture in which salient issues can be separated from more mundane ones. These three aspects are of course, interrelated and their relationship is also highlighted across the chapters. It is important to note here that the focus of this book is on government

institutions, and, given the huge scope of this task, it does not tackle private initiative in elementary education in any significant way.

In the last chapter, we sum up our findings and look at the future and attempt to delineate what the story told in this book implies in terms of what should form the focus of deliberations and action among all those interested in equity and quality in elementary education in India, and all the rewards that it might bring.

CHAPTER 2: STATE EDUCATION POLICY AND INSTITUTIONS

Rashmi Sharma and Vimala Ramachandran

To explore the dynamics and processes of the various institutions elementary education in Andhra Pradesh and Rajasthan, one needs to understand the socio-political context, the elementary education scenario, state policies as well as the structure of the elementary education system in the two states. This chapter provides this background and is organized into five sections. Section 1 provides a brief introduction to the socio-political context in the two states. In the three sections that follow, we examine the educational scenario and indicators, policy framework and major programmes and the structure of the elementary education system in the two states. In the last section, the detailed study design for examining the dynamics of the education system in the two states is provided.

The Socio-Political Context of Andhra Pradesh and Rajasthan

Andhra Pradesh²⁵ and Rajasthan²⁶ are characterized by several pan-Indian features—a caste based, hierarchical social structure; a high degree of gender inequality; the engagement of a majority of the population in an agrarian economy and the simultaneous existence of a vibrant ‘modern’ sector comprising manufacturing and service industries; inadequate physical and social infrastructure; a high degree of poverty; and low literacy and high infant mortality rates. But Andhra Pradesh and Rajasthan lie in south and north India respectively and historically, important differences have existed between these two regions. There are also several points of differences (along with many similarities) in the post independence trajectory of these two regions. The sway of feudal forces is far greater in north India than it is in the south. The latter region has done better than the former in terms of economic performance, and the difference has widened in the last one and a half decade (Joseph, 2004). While the northern region has remained an area of highest population growth in the country, the reduction of poverty has been the sharpest in the south. There are differences too in terms of social development. The average literacy rates in the north are much lower than in the south as is life expectancy, while infant mortality is higher.

Andhra Pradesh and Rajasthan exhibit several pan Indian characteristics, and some regional differences, though these are less marked than in the case of say, Tamil Nadu and Uttar Pradesh, as Andhra Pradesh is less 'advanced' among south Indian states on many indicators. And, both these states also have several characteristics that are unique to them.

25 The state of Andhra Pradesh was formed on 1 November 1956, by merging the nine Telegu speaking districts of the old Hyderabad state with 11 districts of Andhra state formed in 1953 by separating them from the Madras state. It was the first linguistic state in India. AP is conventionally classified into three broad regions—Coastal Andhra, Rayalaseema and Telangana, though there is a great deal of heterogeneity in Coastal Andhra and Telangana (Sarma, J.V.M. (2003)).

26 Rajasthan, geographically the largest Indian state, was formed by the amalgamation of 19 princely states and three principalities between 1947 and 1950, and the former British territory of Ajmer-Merwara was added to it in 1956 (Hooja and Hooja, 2000). The Aravali hill ranges from the north-west and south-west divide the state approximately into the western and eastern semi-arid region. The western part, that comprises one-third of the state is covered by the Thar desert and characterized by extreme temperature and semi-arid conditions, with a paucity of water (Choudhry, 1999).

Table 2.1: Some key indicators of AP and Rajasthan

	India	Andhra Pradesh	Rajasthan
Basic demographic information, Census 2001			
Population	102.70 crores	7.57 crores	5.65 crores
Decadal population growth rate 1991-2001	21.34	13.86	28.33
Male-Female ratio	933	978	922
Population density	324	275	165
Child health indicators, NFHS-3²⁷			
Children 12-23 months fully immunised	43.5 %	46.0%	26.5%
Infant Mortality Rate	57.0	53.5	65.3
Under-5 Mortality Rate	74.3	63.2	85.4
% children 0-71 months receiving any service from ICDS Centre	32.9	30.5	21.1
% children (36-71 months) who went to any pre-school / Child care centre	18.1	22.0	9.6
Literacy rate from 2001 Census			
Total	65.38	61.11	61.03
Male	75.85	70.85	76.46
Female	54.16	51.17	44.34
From NSSO 61st Round 2004-05²⁸			
Proportion Per 1000 of households in Rural areas without any literate adult (15 years and above)	261	364	326
Proportion Per 1000 of households in Rural areas without any literate female member (15 years and above)	500	577	703
Proportion per 1000 population currently attending any educational institution 5-14 years, male	889	917	889
Proportion per 1000 population currently attending any educational institution 5-14 years, female	867	902	842

Source: Census of India, 2001 and NFHS-3 2005-06 and NSSO 61st Round 2004-05

As Table 2.1 shows, Andhra Pradesh and Rajasthan do not differ a great deal in terms of overall literacy levels, and are below the all India average, though in both these states, the growth of literacy has been higher than the all-India average in the last decade (Reddy and Rao, 1999). In fact, in the last decade (1991-2001), Rajasthan recorded the highest growth of literacy rate, which jumped to 61.03 per cent in 2001 from 38.55 per cent in 1991, among all Indian states. As is evident in Table 2.1 female literacy levels in Rajasthan are much lower than in Andhra Pradesh. In parallel, the male to female ratio in Andhra Pradesh is higher than the all-India average; in Rajasthan, it is much lower. The 'southern' character of Andhra Pradesh and the 'northern' character of Rajasthan are also demonstrated by the fact that while in Andhra Pradesh the growth rate of population was high between 1951-91, there was a sharp decline in 1991-2001, and a replacement level of fertility is likely to be achieved in the near future (Rao and Dev, 2003). In Rajasthan in contrast, during the last decade, the rate of growth of population was 2.5 per cent as against the all-India average of 1.9 per cent (Joseph, 2004).²⁹

²⁷ International Institute for Population Sciences and Marco International. 2007, National Family Health Survey (NFHS-3) 2005-06: India, Volume I, Mumbai IIPA

²⁸ NSSO, Government of India, 2006. Employment and Unemployment Situation Among Social Groups in India 2004-05, Report No 516(61/10/2), New Delhi

²⁹ With a population of over 5.6 crores (2001) Rajasthan recorded a decadal growth rate of 28.33 (1991-2001) (GoR, 2001). Andhra Pradesh witnessed a dramatic decrease in the decadal population growth rate—from 24.20 in 1981-91 to 13.86 in 1991-2001.

Similarly, the infant mortality rate for Rajasthan is higher than the all-India average, while for Andhra Pradesh it is lower.

Rajasthan has the highest proportion of Scheduled Caste (SC) population in India (averaging over 17 per cent) and a high proportion of Scheduled Tribe (ST) population (12.4 per cent) (GoR, 2001). In Andhra Pradesh 16.2 per cent of the population are SCs, while 6.6 per cent are STs. Notably, the two states differ substantially in terms of the residential pattern of their populations,³⁰ a fact that has important repercussions for elementary education. In Andhra Pradesh, 70 per cent of the rural population lives in 32 per cent of the large size (population above 2,000) villages. Only 10.5 per cent of the population lives in villages of 1,000 population. This has made it easier to provide 'supply side' factors such as access to education (Reddy and Rao, 2003). Rajasthan, on the other hand, is sparsely populated, with many very tiny villages.

Andhra Pradesh is a middle-income state, while Rajasthan qualifies as a low-income state (see Chapter 3). In both these states, a majority of the population depends on agriculture. In Rajasthan, only around a third of the cropped area is irrigated and agricultural productivity is lower than the all-India average, with food security and subsistence the primary concern of farmers. Given the high levels of ecological stress upon land, water and forests, the state has witnessed a great deal of conflict with respect to control over these resources. After agriculture, particularly in the western arid and semi-arid regions of the state, where the potential for farming is limited, cattle and other livestock are the most important source of livelihood, especially for poor and nomadic groups (GoR, 2001). In contrast, Andhra Pradesh has been a participant in the Green Revolution, especially for rice. This has been facilitated by the availability of a good irrigation infrastructure (Sarma, J.V.M. (2003)). In the last ten years, however, there has been a shift from food to cash crops, followed by significant agrarian distress, which has also received a great deal of media attention.³¹

Andhra Pradesh is also more industrialized than Rajasthan. In 1995-96, while Andhra Pradesh's share in India's manufacturing sector was 6.9 per cent (up from 5.0 per cent in 1979-82), Rajasthan's share was a mere 3.1 per cent (up from 2.9 per cent in 1979-82) (Thomas, 2002). In recent years, Andhra Pradesh has had considerable success in the vibrant information technology sector in India, and among Indian states and union territories (UTs), Andhra Pradesh is the sixth largest exporter of software, while exports from Rajasthan remain small. There have been attempts to speed up the pace of industrialization in Rajasthan too in recent years and the share of industry in the state SDP has increased from one-fourth at the end of the 1980s to nearly a third in recent years. However, employment generation has continued predominantly in agriculture rather than in industry (Joseph, 2004).

The per capita income in Rajasthan continued to be way below the national average through the 1980s and the 1990s (Joseph, 2004). Periodic droughts have a significant impact on poverty in the state. While the severity of a drought's impact lasts for a short duration, there are persistent adverse implications for poor people's livelihood and survival strategies such as loss of land, cattle and household goods and valuables. Debt and bondage are often the result of external shocks to the peasant economy. During periods of crisis and under-employment, migration to cities, both within the state and outside, is very widespread—entire families migrate for several months at a time withdrawing their children from schools. Under-employment is widespread, especially in the rural areas (GoR, 2001).

30 At the all-India level, every two out of three villages have a population of less than 500 and a quarter of the rural population lives in these small villages. Only a sixth of the total villages were in the category above 2,000 population. (Reddy and Rao, 2003).

31 According to official statistics, there were 8,000 suicides by farmers between 2001-2006 in the states of AP, Karnataka, Kerala and Maharashtra (Suri, 2006).

The level of per capita income in the AP too has always been lower than that at the all-India level and the gap appears to have widened in the 1990s. Poverty estimates for Andhra Pradesh vary. While some estimates show the number of people living below the poverty line to be much lower than the all-India average, more recent estimates show poverty rates in the state to be closer to the all-India pattern (26 per cent for rural areas and 12 per cent for urban areas) (Rao and Dev, 2003). There are also wide regional differences within the state. For example, it is estimated that 48 per cent of the population of Anantapur and 40 per cent of the population of Srikakulam live in poverty (Anirudh Krishna, 2006).

Educational Indicators in Andhra Pradesh and Rajasthan

Access

As table 1.2 in Chapter 1 shows, in Andhra Pradesh,, there are fewer habitations without access to primary schooling compared to the all India average, while for Rajasthan, this figure is higher. In 2003, while 93.91 per cent habitations in Andhra Pradesh had a primary school within 1 kilometer, in Rajasthan only 79.86 per cent habitations did. For the upper primary stage, both the states had poorer access than the all India average in 1993, but in Rajasthan, this situation changed significantly in 2003, when the access to upper primary schooling actually became a little better than the all-India average. In 2003, 78.26 per cent habitations in Rajasthan and 74.73 per cent habitations in Andhra Pradesh had an upper primary school within 3 kilometers. Notably, in Rajasthan, schooling facilities started at a very low base at independence and increased slowly for the next four decades³². But after 1990, a concerted effort was made to enhance access to schools (Table 2.2). For primary schools, the major increase in access has come in the last five years, with the government starting 21,306 Rajiv Gandhi Pathshalas (as of December 2004) in the state (Ramachandran, 2005), though the number of regular primary schools seems to have actually come down (Table 2.3). As can be seen, increase in access to schooling has been such slower in Andhra Pradesh in recent years, though the state made steadier progress in previous decades.

Table 2.2: Percentage increase in rural habitations having schooling facility between 1992-93 and 2002-03

State	Primary schooling facility up to 1 km	Upper Primary schooling facility up to 3 km
Andhra Pradesh	12.13%	20.84%
Rajasthan	35.56%	53.81%
All India*	21.12%	19.09%

Source: Govinda and Biswal using flash report of the 7th All India Educational Survey (2002).

Note: * Includes 35 states and UTs.

Table 2.3: Number of schools in 1993 and 2003, Rajasthan

	Number of regular schools in 1993	Number of regular schools in 2003
Primary (regular)	33349	32953
Upper Primary	10176	23181
Secondary	3330	5616
Higher Secondary	1204	2915

Source: 6th All India Educational Survey (1998) and 7th All India Educational Survey (2004).

32 In 1949 when Rajasthan was formed, there were only 3,195 primary schools (2,864 for boys and 331 for girls). This went up sharply to 23,125 in 1981; and had gone up to 39,335 by 1991 and 55,757 in 2004.

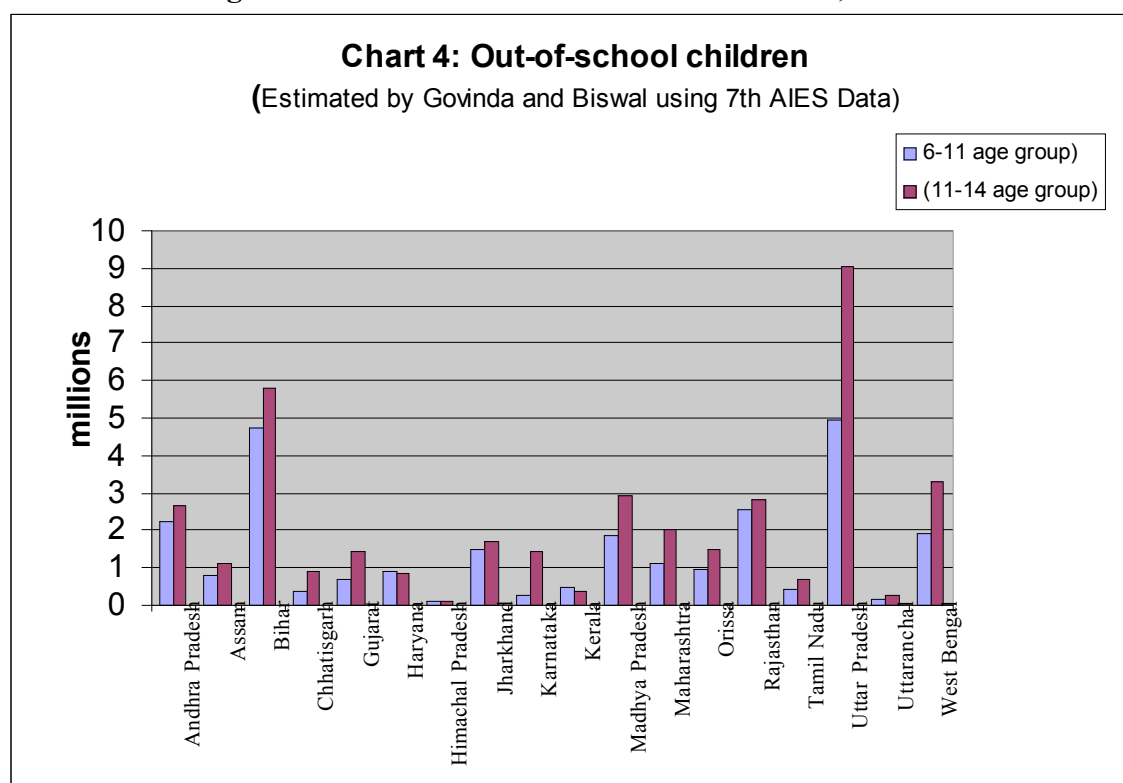
Enrolment

While discussing enrolment rates, out of school children and retention rates, the problem of the lack of a reliable data set needs to be kept in view. According to Govinda and Biswal, (2004), ‘there seems to be total confusion on the number of out-of-school children; there is no consistent data on number of children dropping out without completing the elementary cycle; also, no information is available on net enrolment ratios which are considered vital for international comparisons and for computing other indices such as gender parity index and education development index’. According to them, huge gaps exist in data and official statistics often under-report the actual achievements in the elementary education sub-sector.

Data shows a large gap between gross enrolment ratios at the primary level (classes I to IV or V) and upper primary level (classes V or VI to VIII). According to the DISE data – the gap between GER and NER is also fairly significant. The Net Enrolment Ratio (considered a far more accurate indicator of educational access) stands at 75.28 at primary for Andhra Pradesh and 81.52 in Rajasthan. At the upper primary level the NER for the two states are 53.02 for AP and 44.66 for Rajasthan (Arun C Mehta, 2007). The high GER in Rajasthan indicates the presence of a large number of overage and underage children in the system.

There is some difference between the District Information Statistics for Education (DISE) data (which shows a very low NER rate) and the 7th All India Educational Survey Data, which shows a far less proportion of children who are out-of-school. In the last five years, and more so after the launching of Sarva Shiksha Abhiyan (SSA) in 2000-01 all the states are expected to conduct household surveys to identify out-of-school children and enroll them. Additionally the NCERT All India Education Survey (AIES) has also estimated the number of out-of-school children. As is evident from Figure 2.1, Rajasthan and Andhra Pradesh are fairly close.

Figure 2.1: Out of School Children – state-wise, 2003



Source: Govinda and Biswal (2004).

Enrolment rates, however, are not reliable indicators of educational participation. Rajasthan has consistently recorded very high enrolment rates — often higher than the national average. Assessments and reviews conducted under the aegis of the District Primary Education Programme (DPEP) (1994-2004) and now under SSA regularly comment on the huge gap between enrolment figures and the number of children who actually attend school. (DPEP and SSA Joint Review Mission Reports, various years)

The percentage of students enrolled in private schools is not dramatically different across the two states with 22 per cent children in Andhra Pradesh studying in private schools as opposed to 18.62 per cent children in Rajasthan. Government schools continue to attract a majority of the children in both the states (Table 2.4).

Table 2.4: Percentage of children enrolled in government and private schools

	Rural and Urban		Rural only	
	Government	Private	Government	Private
Andhra Pradesh				
Primary	85.66	14.34	92.95	7.05
Primary + Upper Primary	77.20	22.80	85.82	14.18
Primary + UPS + HS	45.15	54.85	45.66	54.34
UPS	62.94	37.06	89.30	10.70
UPS + HS	72.73	27.27	87.87	12.13
Total	77.93	22.07	89.93	10.07
Rajasthan				
Primary	93.88	6.12	94.93	5.07
Primary + Upper Primary	70.94	29.06	75.26	24.74
Primary + UPS + HS	36.64	63.36	45.93	54.07
UPS	86.67	13.33	90.45	9.55
UPS + HS	91.95	8.05	96.54	3.46
Total	81.38	18.62	85.20	14.80

Source: NIEPA (2004).

Retention

When it comes to retention, according to GOI, SES data of 2004-05 drop out rate at the primary level is 57.9 in Rajasthan and 42.6 in Andhra Pradesh. As per this data, a significant proportion of the children seem to dropout of the school system after the primary stage in both the states, and the situation is worse in Rajasthan. However, the DISE data (Table 2.6) shows the survival rate up to class 5 to be much higher in Andhra Pradesh (90%) than Rajasthan (54%). There is not much difference in transition rates — the children who do survive up to class 5 largely make it to make it to Class 6 (Table 2.6). As Table 2.5 shows, for the elementary cycle as a whole the drop out rate is higher than the national average in both the states, and higher in Rajasthan than in Andhra Pradesh.

Table 2.5: Enrolment and Dropout rates, AP and Rajasthan by social group and gender: 2003-04

	All			ST			SC		
	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total
Rajasthan									
GER (classes 1-5)	120.18	109.41	115.07	94.89	87.84	91.61	92.10	82.00	87.33

Dropout rate (Classes 1-5)	59.29	55.83	57.94	52.19	38.31	47.80	44.09	46.12	45.09
GER (classes 6-8)	74.30	47.22	61.54	82.03	48.42	66.88	81.13	46.60	65.29
Dropout rate (classes 6-8)	64.64	73.87	68.50	70.42	79.63	74.00	63.41	68.87	66.05
Andhra Pradesh									
GER (classes 1-5)	87.42	88.03	87.72	81.85	77.20	79.57	91.41	90.92	91.17
Dropout rate (Classes 1-5)	42.42	42.80	42.61	63.29	68.47	65.76	53.07	36.29	47.69
GER (classes 6-8)	67.10	62.49	64.86	87.94	62.17	75.71	96.51	88.21	92.55
Dropout rate (classes 6-8)	57.93	61.78	59.79	76.80	82.49	79.33	69.65	80.70	73.87
INDIA									
GER (classes 1-5)	100.63	95.58	98.20	94.66	87.77	91.37	93.12	83.00	88.30
Dropout rate (Classes 1-5)	33.74	28.57	31.47	49.13	48.67	48.93	36.83	36.19	36.56
GER (classes 6-8)	66.76	57.62	62.40	84.00	66.62	75.76	79.39	63.35	71.86
Dropout rate (classes 6-8)	51.85	52.92	52.32	69.04	71.43	70.05	57.33	62.19	59.42

Source: Select Educational Statistics, GOI, 2006.

Inequity

Enrolment rates as well as drop out rates in both states indicate the persistence of social inequalities— participation of children from socially disadvantaged groups and minority communities (Muslims) has been far lower than the forward social groups. Equally, rural-urban differences are also marked. The dropout rates for SC and ST groups remains unacceptably high with a significant proportion of those who dropout belonging to these social groups. Perhaps the most glaring inequality in Rajasthan has to do with gender. The differential access and opportunities available to boys and girls is marked in the state. As Table 2.5 shows, the differentials between the GERs of girls and boys at the primary as well as the upper primary stages are much higher in Rajasthan than in Andhra Pradesh or at the all-India level.

School facilities, teachers and infrastructure in the two states

As far as the availability of teachers and school facilities are concerned, there are some interesting differences between the two states. While the availability of schools with respect to population size in the two states is quite comparable, Table 2.6 shows that the percentage of single teacher primary schools in Rajasthan (38.6 per cent) is much higher than in Andhra Pradesh (15.7 per cent). However, when it comes to single classroom schools these trends are reversed—34.3 primary in Andhra Pradesh and 4.4 per cent in Rajasthan. For upper primary schools the percentage of single classroom schools is much lower at 4.3 per cent in Andhra Pradesh and 0.7 per cent in Rajasthan. Similarly, the number of single teacher schools at the upper primary stage is very small (0.7 per cent in Andhra Pradesh and 1.3 per cent in Rajasthan). The average pupil- teacher ratio (PTR) in both the states is comfortable at 32 for primary schools and 31 for upper primary in Andhra Pradesh and 38 for primary and 32 for upper primary in Rajasthan. In both the states only a very small number of schools have a PTR higher than 100 (below 3 per cent).

Table 2.6: Select educational indicators AP and Rajasthan

Education indicators	Andhra Pradesh	Rajasthan
Government Primary schools	54,008 49,747 Rural	49,621 47,451 Rural
Private Primary Schools	3,882 2,067 Rural	2,991 2,412 Rural
Government: Primary with upper primary	10,882 9,757 Rural	11,395 10,491 Rural
Private: Primary with upper primary	2,981 1,443 Rural	5,856 4,317 Rural
Gross Enrolment Ratio (Primary) ³³	86.6	87.5
Gross Enrolment Ratio (Upper Primary)	48.4	36.9
Net Enrolment Ratio (Primary)	68.7	61.9
Net Enrolment Ratio (Upper Primary)	35.1	25.0
Ratio of primary schools to upper primary	3.10	3.07
% Single classroom schools (primary)	34.3	4.4
% Single teacher schools (primary)	15.7	38.6
% Enrolment in government schools (primary)	85.7	93.9
% Enrolment in government schools (primary + upper primary)	77.2	70.9
Regular teachers (primary)	80,203 (M) 54,122 (F)	62,802 (M) 22,906 (F)
Para-teachers (primary)	6,414 (M) 4,783 (F)	19,000 (M) 3,827 (F)
Regular teachers (primary + upper primary)	46,270 (M) 31,015 (F)	74,974 (M) 21,070 (F)
Para-teachers (primary + upper primary)	3,346 (M) 2,699 (F)	2,276 (M) 4,152 (F)
Pupil Teacher Ratio (PTR)—primary	32	38
% Schools with < 50 students (primary)	46.3	32.9
% Schools with < 50 students (primary + upper primary)	4.4	9.8
% Schools with PTR > 100 (primary)	1.1	2.7
% Schools with PTR > 100 (primary + upper primary)	1.5	1.8
Apparent survival rate up to Class 5	89.8	42.5
Repetition rate in Class 1	11.3	18.5
Repetition rate in Class 2	6.7	10.2
Transition rate from primary to upper primary	62.91	67.87

Source: MHRD, GoI (2004-05) and NIEPA (2004).

Infrastructure availability (at least on paper) is quite good in Rajasthan but teacher shortage was a big issue till October 2005. On the other hand, infrastructure remains a big problem in many areas of Andhra Pradesh. Teacher availability is generally better in Andhra Pradesh, though in recent years because of a complex litigation, vidya volunteers have increasingly substituted 'regular' teachers.

³³ This figure differs from the figure in table 2.5 because it is for a different year and also derived from a different data set.

Education Policies and Programmes in Andhra Pradesh and Rajasthan

Major programmes

When it comes to state education policies and programmes it needs to be kept in mind that unlike at the national level, state level comprehensive policy statements are rare,³⁴ and were not in existence in either of the two states at the time of this study. Rather than following a stated policy agenda, state action in elementary education tends to be based on national policies and programmes, with some state specific programmes. However, though state level thrusts draw a great deal from national policy, they are not identical to it as state governments implement various national policies with different degrees of vigour. State policies are implicit rather than explicit, and are an amalgam of national policies, the state government's approach to these policies and various programmes and activities initiated by the state government. It is from these that the 'policy environment' of the state has to be discerned.³⁵ Generally speaking, there tend to be broad similarities in policy across states, but some differences do exist.

The national thrusts identified in the previous chapter form an important ingredient of policies of both Andhra Pradesh and Rajasthan. Both these states implemented central government schemes and programmes—NFE, OB and Teacher Education during the 1970s and the 1980s, and DPEP, introduced in 1995 in Andhra Pradesh and in 1999, in each state spreading to 19 districts in two phases. At the time of this study, in Andhra Pradesh, DPEP was coming to an end, while in Rajasthan it was still on-going.³⁶ At present, SSA is the most significant programme in elementary education in both these states.

However, before DPEP, state specific projects with external assistance with varying focus, i.e., Andhra Pradesh Primary Education Project (APPEP) in Andhra Pradesh and Lok Jumbish and the Shiksha Karmi Project (SKP) in Rajasthan had been initiated. APPEP focused on quality of schools and the aim of the project was to enable teachers to handle primary classes and to acquire necessary skills for activity-based teaching and learning in classrooms for better pupil achievements. The main activities in the project included teacher training, establishing Teacher Centres to support teachers academically, strengthening DIETs and supply of teaching-learning materials to schools. APPEP was merged in DPEP after 1995 (GoAP, 1994; Reddy and Rao, 2003).

In contrast, in Lok Jumbish as well as SKP in Rajasthan, an important focus area was enrolment of out-of-school children, especially in remote habitations and of girls, by involving the community and setting up alternative education centres outside the formal school system.³⁷ SKP involved activating dysfunctional schools in remote villages through the appointment of

³⁴ Of the major Indian states, Karnataka, Maharashtra and West Bengal have formulated their own policies (Premi, 2001).

³⁵ The vision 20/20 document in AP, an overall vision of the state by 2020, promoted by the previous government, but no longer the official statement of the present one, contained a chapter on education. The policy thrusts of this document for elementary education included ensuring full literacy, strengthening programmes for out-of-school children, non-formal education, providing more schools, classrooms, teachers, focused strategies for disadvantaged groups and girls, teacher training, redesigning curricula to make it more activity based, involving the community and so on (GoAP, 1999).

³⁶ In AP, DPEP was introduced in five districts (phase 1) in 1995 and then expanded to another 14 in 1997 (phase 2). There has been a UNDP assisted programme in selected mandals of the remaining four districts. In Rajasthan, DPEP was launched in 10 districts in 1999 and extended to another nine in 2001.

³⁷ SKP was started in 1974 through the efforts of the NGO SWRC (Social Work and Research Centre) and was supported by NCERT. In 1984, a fresh project was formulated and 13 remote villages taken up in 1987. SKP was launched by the state government in partnership with the national government and SIDA.

local teachers.³⁸ Rigorous processes of selection, training and career advancement were followed and the Rajasthan Shiksha Karmi Board was set up to administer its planning, implementation, coordination and management (Choudhary, 2000). Lok Jumbish attempted to make education a people's movement by ensuring active and sustained participation of people at every level. The main innovations of Lok Jumbish included school mapping, micro-planning and special initiatives to enroll out-of-school children and girls. The project was managed through the 'Lok Jumbish Parishad' set up at the state level and a new school management system was developed where the block was the main unit of programme delivery (Choudhary, 2000). At the time of the study, the project had been merged in SSA.

Free and compulsory education

Both the states had liberal admission policies in government schools to encourage children to enroll. In primary schools, children could be admitted any time during the academic year. Additionally, in Rajasthan, in class 1 there was no age limit for admissions (GoR, 2004-05). In Andhra Pradesh, no fee was charged from elementary school children. However, in Rajasthan there was a complex fee structure. While no fee was charged from girls up to class 8 and from boys up to class 3, boys paid a monthly fee of Rs. 5 in classes 4 and 5 and Rs. 10 in classes 6 to 8. Students from SC and ST categories as well as from other backward castes paid 50 per cent of these amounts (GoR, 2004-05).

However, neither state had attempted to make education compulsory with any vigour. Rajasthan had no act for free and compulsory education. In Andhra Pradesh, a Compulsory Primary Education Act³⁹ was introduced in 1961 for children in the age group 6-7 years, extended to all children in the 6-11 years age group in 1965-66 (NIEPA, 1979). However, the act lacked teeth and had not been implemented with any seriousness. In fact, several government officials were not clear about the status of the act. At the time of this study, a new act was being considered, but no final decisions had been taken. Notably, in Andhra Pradesh, an NGO/social activist led movement for the elimination of all forms of child labour had led to increasing pressure to enrol all children. The MV Foundation had led this movement since 1991 and was subsequently joined by several other voluntary organizations. With increasing pressure on the government to eliminate child labour and provide full time education for all children, in March 2001, the government passed a resolution to end all forms of child labour and achieve Universal Elementary Education (UEE) in the state by 2004 (Venkateshwarlu, 2004).

Both Rajasthan and Andhra Pradesh provided several types of benefits to students. In Rajasthan, free textbooks were provided to all children in classes 1 to 12 (GoR, 2005-06, 2004-05). In Andhra Pradesh, free textbooks were provided to all children in classes 1 to 5 and also to SC/ST/BC/physically handicapped/minority children in classes 6-10. Additionally, with SSA, all girls in classes 6-8 were provided free textbooks. In Andhra Pradesh, scholarships for SC/ST children were provided by the Social Welfare Department, and there

³⁸ In SKP, earlier the minimum qualifications used to be middle school for men and primary school for women. Now these are secondary school for men and middle school for women. The selection is done by the Shiksha Karmi Committee. The salary is Rs. 1,800. After eight years of work the Shiksha Karmi is made a Grade 3 teacher.

³⁹ As per the existing AP Education Act, 1982 any local authority, if called upon by the government to do so, shall make a scheme for compulsory primary education. The government will then scrutinize this scheme and sanction it by declaring primary education compulsory in that area. The local authority may appoint an 'attendance authority' who can put questions to guardians. It is the duty of the guardian to get a child to attend school. Wherever the guardian fails, the attendance authority may hold an inquiry and pass an order directing the guardian to get the child to attend school.

were also merit scholarships. Rajasthan had a fairly complex structure of scholarships (see Box 2.1).

Box 2.1: Scholarships to SC and ST students in Rajasthan

For SC/ST/migratory groups and other backward caste students of classes 6-8	Boys: Rs.15 per month. Girls: Rs. 20 per month.
For students of other backward castes, students from families below the poverty line (annual income less than Rs. 4,500)	Day scholars: Classes 1 to 5: Rs. 25 per month. Classes 6 to 8: Rs. 40 per month and an annual grant of Rs. 500.
For children from families that collect garbage etc.	Classes 1 to 5: Rs. 40 per month. Classes 6 to 8: Rs. 60 per month and an annual grant of Rs. 550.
For students from extremely poor families (income less than Rs. 2,500 per year)	Rs. 10 per month for Classes 6 to 8.
Students whose fathers died in Kargil	Classes 1 to 8: Rs. 1,800 per year.

Source: GoR (2004-05).

The Rajasthan state government had also started an insurance scheme for students (GoR, 2004-05). Moreover, 32 upper primary schools had been identified in which handicapped children were paid special attention. These students were entitled to various grants, including aids for physical handicap, medical examinations, stationery, uniforms and transport.⁴⁰ In Andhra Pradesh, a 'Student Safety Insurance Scheme' for death or disability due to accident was at an inception stage at the time of the fieldwork (GoAP, 2005b).

Both the states had implemented the national midday meal programme. Since 1995 uncooked food grains were distributed as part of the National Programme for Nutritional Support to Children in both the states.⁴¹ From 2003 onwards, both the states began to provide cooked midday meals to primary school students (GoAP 2005b; GoR, 2005a; GoR, 2004-05). In the midday meal scheme, food grains were provided free of cost by the central government, along with an additional Re.1 per child per day for conversion to cooked meals. The Rajasthan state government provided an additional Re. 0.50 per day per child. In Andhra Pradesh, a higher amount was made available: in schools with less than 50 children, the total conversion cost provided is Rs. 2.00 per child per day, in schools with 50-100 children Rs. 1.75 per child per day, and in schools with more than 150 children, Rs. 1.50 per child per day.

Privatization

The two states differed in their policies regarding the regulation of private schools. In Andhra Pradesh, all persons or organizations interested in setting up such schools had to apply to the state government for recognition and deposit Rs. 25,000 as an endowment fund, to be utilized for purchasing necessary furniture etc. for the school. There were detailed criteria for recognition including qualifications of the staff and availability of library and sports equipment.⁴² Moreover, private schools had to follow the syllabus and textbooks prescribed by the government. The government also reserved the right to prescribe reservations for social categories and no student could be denied admission on the grounds of caste, religion, language etc. All schools had to advertise posts before recruitment (GoAP, 1994).

In contrast, in Rajasthan, though similar conditions existed for setting up upper primary schools, government recognition was not necessary for private primary schools since March 1995 (GoR,1995). Additionally, Rajasthan had promulgated the 'Bhamashah scheme' to

⁴⁰ The entitlement was for Rs. 400 for stationery, Rs. 200 for uniforms, Rs. 500 for transport, Rs. 75 for escort of blind students, Rs. 2,000 for aids and Rs. 150 for a medical examination.

⁴¹ During the 1960s, in a CARE project milk was provided to school children in Rajasthan (GoR 1966).

⁴² The recognition is valid for five years after which the school has to apply again.

encourage private donations to education. Under this scheme, those who donated 50 to 75 per cent of the expenditure for establishing a school, were entitled to have the school named as per their wishes, with the word 'government' added. If the donation covered more than 75 per cent of the expenses, the school was simply named after the donor. If the donor funded a part of the building, then that part could be named as per his/her wishes. Donors who contributed more than Rs. 5 lakh were honoured at the state level, and those who contributed Rs. 5 lakh at the divisional level (GoR, 2004-05).

School types

In both the states the primary school cycle was from class 1 to 5, but while Rajasthan the upper primary school cycle was from class 6 to 8 as per the nationally prescribed pattern, in Andhra Pradesh it is from classes 6 to 7. Board (public) examinations are conducted after class 8 and class 7 and in Rajasthan and Andhra Pradesh respectively. In both the states, there were three types of 'formal' schools, i.e. primary schools (PS) for classes 1 to 5, upper primary schools (UPS) for classes 1 to 7 in Andhra Pradesh and 1 to 8 in Rajasthan, and high schools for classes 6 to 10 (in Rajasthan there were also higher secondary schools for Classes 6 to 12). In both the states, a large majority of the schools were primary schools, i.e., from classes 1 to 5, while upper primary classes were usually clubbed with high schools, though there were also some schools that combined primary and upper primary classes, i.e., classes 1 to 7 or 8. In parallel, Rajasthan also had a fairly complex system of alternative education centres which functioned in lieu of formal schools, i.e., the Shiksha Karmi Pathshalas (SKPs) and the Rajiv Gandhi Swarna Jayanti Pathshalas (RGSPs). Andhra Pradesh had a fairly successful residential school system⁴³ for classes 5 to 10, managed by a Residential Education Society, in which tribal children could study. Both the states had alternative education centres too.

In Rajasthan, SKPs were started under the Shiksha Karmi Project, whereby a teacher from within the community was selected through a rigorous process and trained intensively. Subsequently, RGSPs were started in 1999, on the pattern of the Education Guarantee Scheme in Madhya Pradesh. Under this scheme, any human settlement that did not have an educational facility within a radius of one kilometre and had 25 children between 6-11 years of age was entitled to a Rajiv Gandhi Pathshala. A Rajiv Gandhi Pathshala was set up when a proposal was received from the ward/gram sabha and approved by the gram panchayat. Most RGSPs were up to class 3 and some up to Class 4 (Choudhary, 2000), and 21,077 such schools were started. At the time of this study, there had been some backtracking on this policy and since 2003, a total of 3,994 RGSPs had been converted into regular government schools.

In both the states, alternative education centres, which were not in lieu of formal schools but meant for children from very tiny habitations or those who could not attend government schools, had also been started. In Rajasthan, these included full time (6 hourly) alternative schools in habitations of population exceeding 100, with 30 non-school going children available and no schooling facility within 1 kilometer (there were 1,560 such schools with 79,263 children enrolled) and evening/part time alternative schools (4 hourly) where there were 15 non-school going children (there were 525 such schools with 39,978 children). Additionally, selected *madrassas* were given full time alternative school status and provided with a para-teacher to teach Hindi, math and EVS (246 such *madrassas* had been strengthened). Bridge courses for girls and mobile schools in buses for cover slum areas had also been started.

⁴³ The first residential school was started in 1971. In 1972 the AP Residential Educational Institutions Society, an autonomous body with a Board of Governors and 22 members, was set up. The Minister for School Education was the chairman and Secretary Education, the vice chairman. Funds are provided by the state government. In 1983 the government decided to establish one residential school each for boys and girls in every district. There are 120 such institutions in existence (GoAP, 2004).

In Andhra Pradesh, alternative and ‘Mabaadi’ schools were set up in habitations that did not qualify for a formal school, including tribal areas where the population of school-going children was less than 15 (2,955 such schools had been started). In Andhra Pradesh, there has also been an emphasis on long-term bridge course camps as well as non-residential bridge course centres to mainstream out-of-school children, i.e., working, migratory and street children etc. In fact, Andhra Pradesh pioneered the ‘Back to School’ programme in residential and non-residential bridge courses, a model that was subsequently adopted by the central government under the aegis of the SSA programme. Andhra Pradesh differed from Rajasthan in the fact that its alternative centres were not in lieu of formal schools, and were confined to habitations that were too small to qualify for a formal school and to children who for various reasons could not attend formal schools.

Teachers

Both the states had two types of teachers—‘regular teachers’, who enjoyed good salaries and security of tenure and ‘para-teachers’, who had small salaries and no security of tenure.

Among regular teachers, two grades of teachers were recruited in Andhra Pradesh—secondary grade teachers (SGTs) who taught elementary classes and school assistants (SAs) who taught in high schools. Rajasthan had three types of regular teachers—grade 3 teachers who taught in primary as well as upper primary schools, grade 2 teachers who taught in high and higher secondary schools and acted as headmasters in upper primary schools and grade 1 teachers who taught in high schools. In both the states, teachers with the lowest qualifications and salaries taught at the primary school level, while high and higher secondary school teachers had the highest salary. However, in Andhra Pradesh upper primary teachers had college degrees in their subject of specialization, while in Rajasthan primary and upper primary teachers were drawn from a common pool and a college degree was not a mandatory qualification. The mandatory qualifications and salary structure of elementary school teachers in the two states are given in Box 2.2.

Box 2.2: Regular teachers in Andhra Pradesh and Rajasthan

AP			Rajasthan		
Post	Qualifications	Basic scale	Post	Qualifications	Basic scale
School Assistant for Classes 6-10	Graduation & B. Ed, recruited directly for separate subjects	Rs. 4,850-10,250 with DA @ 34.45%	Grade 1 for secondary school, recruited for separate subjects	Post graduate and B. Ed	
Secondary Graduate Teacher for Classes 1-5	Class 12 and Teacher Training Certificate or BA and B.Ed.	Rs. 3,750-7,650 with DA @ 34.45%	Grade 2 for classes 6-10, posted in secondary schools and also as headmasters in upper primary schools	BA/B.Sc and B.Ed	Rs. 5,500-8,000
			Grade 3 to teach all subjects in classes 1-8.	Class 12 and BSTC/ B.Ed	Rs. 4,500.

Source: GoAP (2001) and information (on salaries) from the directorate.

In Andhra Pradesh there were two types of teacher cadres among regular teachers—a small percentage (7 per cent) of teachers belonged to the education department and worked in education department schools located mainly in urban areas, while a large majority (93 per cent) belonged to the zila parishads and panchayat samitis and worked in panchayat schools spread across rural areas. In the pre-independence period, the panchayat cadre teachers had much lower salaries and benefits than the government cadre teachers. However, subsequently

the salaries and other emoluments of the teachers of the two cadres were made equal⁴⁴. But teachers in the state government cadre continued to enjoy better promotion opportunities. In 1992, the government merged the two teacher cadres with the intention of bringing about uniformity in their service conditions. This led to a long process of litigation (GoAP, 2005).

In Andhra Pradesh, all regular teachers were selected on the basis of a written test and academic and professional qualifications and record. The examination was organized by the Commissioner and Director of School Education, question papers were prepared by the State Council for Education Research and Training (SCERT) and actual recruitment took place at the district level (GoAP, 2001).⁴⁵ In Rajasthan, before 1998, grade 3 teachers were recruited through panchayats and grade 2 teachers through the education department. However, the recruitment of regular primary teachers was stopped in 1998. It was subsequently resumed in 2003 and appointments made through the Rajasthan Public Service Commission.

State governments of both the states had gone in for large-scale recruitment of ‘para-teachers’. In Andhra Pradesh, the state government decided to provide financial assistance to School Education Committees (SECs) to enlist ‘*vidya volunteers*’ on a contract basis in regular schools. This was viewed as a temporary intervention to bring down the teacher-pupil ratio. The number of vidya volunteers recruited was substantial, e.g. 62,685 in 2004-05 (GoAP, 2004). Para teachers were appointed in alternative schools of DPEP too. In Rajasthan, para teachers had been recruited in SKP, RGSP schools and alternative centres of DPEP, and also in other schemes from time to time too. For instance, a woman para-teacher had been provided for girls’ education in every gram panchayat and in the Saraswati Yojna (SY) local women (8th pass) were given training and financial assistance to run a courtyard school in their homes. RGSP para-teachers were the largest in number.

Notably, the policy vis-a vis teacher recruitment in Rajasthan had shifted often in recent years. For instance, between 1998 to 2003, para-teachers were appointed through the gram panchayats in RGSP schools⁴⁶, and there was no recruitment of ‘regular’ teachers for the primary level. At the same time, severe problems with respect to para-teachers arose, including court cases, as they started demanding that they be made regular (for more details see Chapter 5). As the government changed in 2003, recruitment of parateachers to Rajiv Gandhi schools was stopped and that of ‘regular’ teachers resumed.

All the regular teachers in Andhra Pradesh, and all but a small minority in Rajasthan were trained (Firdausi et al., 1999), but this was not the case for para-teachers (Bhargava and Sharma, 2004). In both the states, regular teachers could be transferred from one place to another, while para-teachers are recruited for a particular school. In Andhra Pradesh, teacher transfers had been rationalized and made completely transparent (see Chapter 5 for details). (Go AP 2005a). The transfer policy in Rajasthan, on the other hand, had been the subject of a great deal of controversy in recent years with allegations of political favouritism as well as bribery, though in June 2004, a new transfer policy for teachers had been announced (GoR, 2005).

⁴⁴ Andhra Pradesh was one of the earliest states to accept the principle of parity in pay and benefits across various types of teacher cadres (GoI, 1966).

⁴⁵ In AP, the District Selection Committee comprises the District Collector as the chairperson and the DEO as the secretary, the Chief Executive Officer, zila parishad, one of the commissioners of the municipalities/ municipal corporations nominated by the collector, and the Project Officer Tribal Development Agency as members (GoAP, 2001).

⁴⁶ Originally, these teachers belonged to the same gender and social category as their panchayat ward representative and were paid Rs. 1,200 per month with an additional Rs. 200 added every year. Recruitment was made through a committee comprising the sarpanch, upper primary school headmaster of the school at the panchayat headquarters or in the panchayat area, a representative of BEO and gram sewak. Subsequently, between April 2001 and August 2002, the government made various changes regarding recruitment processes and qualifications.

The Institutional Structure

The institutional structure for elementary education in both the states was similar. Education ministries, headed by elected ministers, were responsible for policy level initiatives and decisions. The administrative head was a senior official (called the principal secretary or the secretary), assisted by a team of additional secretaries, deputy secretaries and other officials. Several departmental organizations at various levels were responsible for implementing government policy and managing the school system. These fell broadly into three categories: the administrative and regulatory system, the academic system and the programme structure. At the state level, these were headed by the directorates of education, state councils/institutes for research and training and education societies created specifically in the context of DPEP and SSA respectively.

Administratively, the states in India are divided into districts and further into blocks or mandals (see Table 2.7). Mandals in Andhra Pradesh comprise of 25 to 50 villages, while blocks in Rajasthan comprise around 200 villages. All three types of the sub-systems mentioned above had offices or institutions at the district level. In Rajasthan, administrative as well as programme offices also existed at the block level, while in Andhra Pradesh, these had been combined into a single office at the Mandal level. Running parallel to the administrative system was the system of the elected local government, i.e., Panchayati Raj Institutions (PRIs) at three tiers.

Table 2.7 Administrative units in AP and Rajasthan

	Andhra Pradesh	Rajasthan
Number of districts	23	32
Number of mandals/blocks	1127	237
Number of gram panchayats	21,176	9189
Number of villages	28,123	
Number of habitations	72,154	
Number of municipalities		183

Source: Census of India, 2001.

State level institutions

In Andhra Pradesh, most state level institutes were located in the state capital Hyderabad, while in Rajasthan they were dispersed in different cities across the state, capitals of the erstwhile princely states. The directorates were the oldest structures of the departments, established in 1949 and 1956 in Rajasthan and Andhra Pradesh respectively. The directorate for school education was common for elementary and secondary education in Andhra Pradesh, but in Rajasthan a separate directorate of elementary education was formed in 1997, and the director's post was subsequently elevated to that of a 'commissioner'.

The directorates were responsible for the overall administration of schools. The functions of the directorates comprised personnel management including teacher recruitment; planning, budget formulation, disbursement of funds to field offices, audit etc; overall supervision and monitoring; educational development such as starting of new schools; ensuring smooth implementation of government schemes such as supply of free textbooks, disbursement of scholarships, etc. The directorate for school education was headed by a director in Andhra Pradesh, while that of Rajasthan by the commissioner for elementary education. The basic unit in both the directorates was a 'section', i.e., a small unit that handled one or two schemes or areas such as departmental enquiries, recruitment of teachers etc.. A section was headed by an assistant director in Andhra Pradesh and a section officer in Rajasthan, who were assisted by clerks and ministerial staff. The work of these officials was supervised at two levels by

officials such as deputy directors, joint directors and additional directors, who reported to the director or the commissioner. The building and facilities were adequate (see annexure A for more details on all institutions).

SCERT in Andhra Pradesh, set up in 1967, and the State Institute for Educational Research and Training (SIERT) at Udaipur in Rajasthan, set up in 1978, were the academic wings of the departments and played a leading role in setting curricula, making textbooks, teacher training and other academic activities. The Andhra Pradesh SCERT had 14 departments, i.e. Elementary Education, ECE and Research Studies, Teacher Education, Curriculum and Textbooks, Planning and Statistics, Population Education, Science Education, Math Education, Educational Technology, Vocational Education and Tribal Education, Language Education, Social Science, Innovations, Research Studies, Computer Education and Evaluations. Each department was headed by a professor who worked with a team of 2-3 lecturers. The SCERT was not autonomous, and had no Advisory Committee. As annexure A shows, the building and other facilities were very inadequate.

In SIERT Rajasthan there were nine departments i.e. Humanities and Social Science, Science and Mathematics, Psychology and Vocational Education, Teacher Education, Educational Planning and Administration, Educational Production, Educational Research and Communication, Non-formal and Deprived Groups and Curriculum and Evaluation. Each department was headed by a joint director or deputy director who worked with lecturers, research officers and teachers. The SIERT in Rajasthan had an Advisory Council. Building and other infrastructure available were adequate.

During the 1990s new organizations, which may be called 'programme implementation structures', were set up in DPEP and subsequently implemented SSA in the two states. Both the 'Rajasthan Council for Primary Education (RCPE) at Jaipur and the Andhra Pradesh School Education Society (APSES) at Hyderabad were registered societies⁴⁷, and enjoyed greater autonomy than the directorates. Personnel in these societies were drawn from various organizations. RCPE was housed in a very fine building and infrastructure in RCPE as well as APSES was adequate.

The APSES had a General Council with the chief minister as president and the minister for school education as vice president. Its Executive Committee was chaired by the secretary of the education department. The State Project Director was the administrative head. There were a total of 18 units in the office:: Research and Evaluation, Alternative Innovative Education, Girls Education and NPGEL, Early Childhood Education, Capacity Building, Focus Areas/Groups, Management (Administration), Planning and Management, Community Mobilization and Participation, Pedagogy and School Improvement, Distance Education Programme, Children with Special Educational Needs, Media and Documentation, Management Information Systems, Kasturba Gandhi Balika Vidyalaya, Civil Works, Finance and Administrative Office, Janshala (APDPEP 2003). These were headed by joint and deputy directors, who worked with teams of assistant directors. In RCPE there were six units: Planning, Primary Formal Education, SSA/Research and Monitoring, Community Mobilization, Senior Training Coordinator and Administration. Each was headed by a deputy director who was assisted by one or two assistant directors.

In addition to these three main organizations, both states had separate organizations for printing textbooks. Moreover, in Andhra Pradesh, there was a directorate of government examinations that mainly handled the secondary school examinations, but was also formally in

⁴⁷ They are registered under the Societies Registration Act 1864, which is a mechanism open to anyone to create a formal organization that comes under the broad jurisdiction of the Registrar of Societies. Elected or selected executive committees are empowered to take all decisions regarding the conduct of these societies. Almost all NGOs, research institutions and even universities are registered as societies.

charge of the class 7 examinations. There were also a few other organizations in each state, charged with specific programmes or special types of schools such as the Residential Schools Society in Andhra Pradesh and the Shiksha Karmi Board and the Rajasthan Madrassa Board in Rajasthan.

District level institutions

District education offices were critical nodal points in school administration in both the states. In Rajasthan, there were separate offices for elementary and secondary education,⁴⁸ while in Andhra Pradesh there was a single office for school education up to Class 10. While the district office in Andhra Pradesh was more than 50 years old, the one in Rajasthan was established in 1991 when the Directorate of Elementary Education was separated from the Directorate of Secondary Education. The combined district office in Andhra Pradesh was far better staffed and equipped in terms of personnel and infrastructure than the district office for elementary education in Rajasthan. The Andhra Pradesh offices were staffed by a District Education Officer (DEO), five deputy education officers and a few other officials for accounts. The Rajasthan district office had a DEO, an assistant DEO and another junior official along with staff. The district offices were linked to the directorates through intermediate offices, six ranges headed by the Deputy Director Education in Rajasthan and Regional Joint Directors of Education in Andhra Pradesh (APSES, 2005-06).

In the academic sub-system, at the district level there were District Institutes of Education and Training (DIETs) that conducted pre-service training for elementary school teachers and were also responsible for other academic activities such as in-service training and action research in both the states. In Andhra Pradesh, between 1987 and 1990, 23 teacher-training institutes were upgraded to DIETs and one tribal teacher-training institute as a sub-DIET (GoAP, 2004). Rajasthan had established 30 DIETs. In Andhra Pradesh, posts sanctioned for a DIET included a principal, seven senior lecturers and 14 lecturers. In Rajasthan DIETs, the teaching staff strength sanctioned was half of this.

There were also district project offices. In both the states the DEO was designated as the district project coordinator. But in fact, the day-to-day work of the project office was managed by an Assistant Project Coordinator (APC) in Andhra Pradesh and an Assistant District Project Coordinator (ADPC) in Rajasthan. In Andhra Pradesh, the APC was assisted by four programme officers as well as engineering staff and in Rajasthan, the ADPC was assisted by three programme officers for academic activities, community mobilization and women's development.

Sub-district institutions

While the institutional structure at the state and district levels was similar in the two states, their sub-district structures differed. Rajasthan followed a pattern that is prevalent in most Indian states, i.e., there was an administrative office at the block level, i.e. the Block Education Office with one Block Education Officer (BEO) and three officials to assist along with clerks etc., and a Block Resource Centre, essentially an extension of the programme structure, with a Block Resource Centre Facilitator (BRCF) and two resource persons. The facilities available to the block level offices were very limited. The programme structure extended further through one Cluster Resource Centre Facilitator (CRCF) for every 12-15 villages.

In Andhra Pradesh, the sub-district structure for education existed at the level of the Mandal, where a single centre had both administrative and programme personnel. The staff available at the mandal office included one Mandal Education Officer (MEO) and three Mandal Resource

⁴⁸ At one time, Rajasthan had separate district level set ups for the education of boys and girls, but this was subsequently restructured.

Persons (MRPs). The support staff was negligible. Additionally, 'Teacher Centres' were created for 15-20 schools in Andhra Pradesh in the APPEP programme in 1987 (Rao, 1999) and have continued since in DPEP (1994) and SSA (2001). TC activities consisted of planning for lesson preparation, group discussion, use of teaching aids etc. (Firdausi et al., 1999).⁴⁹

Local elected bodies and school committees

Along with the departmental structures, local governments or PRIs also take up various aspects of school management. Local governments existed in both the states at the district level (called zila parishads), at the block level in Rajasthan (called panchayat samitis), at the mandal level in Andhra Pradesh and at the level of one or more villages (called gram panchayats) in both the states. In Rajasthan, the zila parishads had been made responsible for the management of rural primary schools in 1959, and in 1998, all rural schools up to the upper primary level were also transferred to the panchayats. The DEO, elementary education was made Additional Chief Executive Officer (CEO) of the zila parishad and the BEO, additional CEO of the block panchayat (GoR, 2003a). In Andhra Pradesh, a large number of primary and secondary schools were transferred to local bodies in 1961. The administration of schools was entrusted to local bodies, while the task of academic inspections and examinations was retained by the education department (Kumar, 1998).

Parallel to the PRIs, there were also other community-based bodies specifically for elementary education. A Village Education Committee that was in existence in Rajasthan earlier had been replaced by a School Development and Monitoring Committee (SDMC) in DPEP. To begin with the village panchayat ward member was the chairperson of the SDMC, but in 2002, the composition of SDMC was changed, and the school headmaster was made its chairperson, with a teacher functioning as the secretary. Three parents from different social groups, three elected representatives, a good student, a Bhamashah (a person making monetary contributions to the school), two retired persons from the education department, a representative of school teachers and one representative of alternative schools of various types were nominated members. The SDMC's role was to undertake enrolment drives, ensure children's attendance, create a development fund for the school (to include funds for various government schemes as well as contributions from the community) (Bhargava and Sharma, 2004: GoR 2003). There was also a parent teacher association in each school in which all parents could participate (GoR 2004-05).

The parallel structure in Andhra Pradesh at the time of the fieldwork was the School Education Committee (SEC), governed by the Andhra Pradesh School Education (Community Participation Act), 1998. A Parent Teacher Association (PTA), comprising the teachers and the parents of All the children studying in the school, formed the General Body of the SEC and. Four persons from among the parents, including at least two women and one person of SC/ST/backward caste/minority, were elected to the SEC by the PTA. The SEC was elected every two years and at the time of this study there had been three batches of the SEC. Andhra Pradesh had also attempted to link SEC to the PRI structure. In every village panchayat, a Panchayat Education Committee was constituted with the sarpanch as chairperson, headmaster of high school/upper primary school as convenor, two members from each SEC (the SEC chairperson and a woman) and NGO members. Donors who donated Rs. 50,000 or more were life members. In a similar fashion, the Mandal Education Committee and Zila Education Committee, with members elected from among the membership of lower tier committees were established.

⁴⁹ Teachers meet once a month in these centres to discuss various pedagogic issues. TC activities slowed down under DPEP – especially as compared to APPEP (1987-1993). TC meetings were not effective because of inadequate planning, lack of proper facilities, delays in payment of TA/DA, contingency grants (Rao, 1999).

About the Study

The above description of the socioeconomic context, the education scenario, policies and programmes and the institutional structure of the two states prepares the background for understanding their processes and dynamics. This was done through an intensive field study, conducted between September 2005 to March 2006, which encompassed all the major institutions for elementary education. These included in each state, at the state level, the directorate, SCERT /SIERT and programme offices; at the district level one District Education Office, one DIET and one District Programme Office; at the sub district level the Block Education Office and Block Resource Centre or two Mandal Resource Centres; and a small number of schools and villages in each state.

To study the institutional structures and processes below the state level, one district was selected in each state. This was an 'average' district, i.e., representative of the state on the basis of the literacy rate and the percentage of SC and ST population. In Rajasthan, within the district, one block and within the block two clusters were selected. The block selected was an average type for the district, while the two clusters represented varied, i.e., more and less developed areas. The more and less developed clusters were selected in terms of infrastructure and educational development as perceived by the education officials of the block. In Andhra Pradesh, two mandals were selected to represent the diversity of the district (see Box 2.3).

Box 2.3: Districts, Blocks and Mandals Studied

A. Key socio-economic indicators of sample states, districts and sub-district units

	All India	AP	Selected sample in AP			Rajasthan	Selected sample in Rajasthan	
			District	Mandal 1 (M1)	Mandal 2 (M2)		District	Block
Population (in millions)	1027.0	76.21	3.238	0.034	0.066	56.5	1.107	0.0203
% Rural population		72.7%	86.74	100.0	100.0	76.6%	78.81	81.30
% Urban population		27.30%	13.26	0	0	23.40	11.19	18.70
% SC population		16.2%	17.78	13.85	19.80	17.2%	17.7%	20.50
% ST population		6.6%	10.58	23.61	12.26	12.6%	10.6%	15.90
Female-Male Ratio	933	978	967	968	957	922	936	928
Decadal population growth (1991-2001)	21.34%	13.86%	13.55%	NA	NA	28.33%	24.24%	31.43%
Population density	324	275	190	NA	NA	165	168	201
Literacy rate (total)	65.38%	61.11%	57.84	41.61	60.37	61.03%	52.39	55.80
Literacy rate (male)	75.85%	70.85%	70.19	63.57	72.21	76.46%	71.25	73.60
Literacy rate (female)	54.15%	51.17%	45.07	36.43	48.00	44.34%	32.30	37.00

Source: Census of India, 2001.

B. Basic information about the district and sub-district units

	AP			Rajasthan	
	District	M1	M2	District	Block
Distance from district HQ		100 km.	65 km		30km
Number of blocks/mandal	59			6	
Number of villages	1161	25	27	1106	210
Number of village Panchayats	1160	25	27	231	41
Number of habitations	3349	31	74	1480	420
Number of teacher centres/clusters	316	4	5	98	16

Source: From respective offices in the two states

Within the clusters and mandals selected, schools of various types were selected (Box 2.4). These included six formal schools in each state, encompassing the three types, i.e., high schools with upper primary classes, upper primary schools and primary schools as well as one private unaided school. In terms of alternative educational centres, the clusters in Rajasthan had three types of schools—Rajiv Gandhi Pathshalas, Shiksha Karmi schools and DPEP alternative schools. One of each type was selected for study. In Andhra Pradesh, there was only one type of alternative school, which was selected for study.

Box 2.4: Number of schools studied in each state

School type	Andhra Pradesh	Rajasthan
Formal government schools		
High school/higher secondary school	2	1
Upper primary school	2	3
Primary school	1	1
Alternative centres (government)		
DPEP alternative centre	1	1
Rajiv Gandhi school		1
Shiksha Karmi school		1
Private schools		
Unaided private school	1	1
Total number of schools	8	10

In the study of state, district and sub-district institutions, detailed, semi-structured interviews with a large number of actors from these institutions provided the most critical input (see Box 2.5 for details). These interviews explored in detail the interviewees' professional experience including training, postings and transfers, problems faced at work, achievements, autonomy, supervision, the management practices they followed, interactions with community, non government organizations, politicians etc., perceptions about the problems and issues in elementary education and possible strategies to address these, opinions about schools, teachers, perceptions and experience of making textbooks, teacher training, school examinations, perceptions about children from deprived groups, and other issues that emerged during the interviews. The interviews were supplemented by a study of existing documents and records as well as observation of the institutions concerned.

Box 2.5: Interviews conducted at the state, district and sub-district level

	Andhra Pradesh		Rajasthan		Total for the two states
	Name of organization	Number of interviewees	Name of organization	Number of interviewees	
State level	Directorate of school education	4	Directorate/Commissionerate of school education	4	8
	SCERT	4	SIERT	5	9
	State Project Office	4	State Project Office	3	7
	NGO representative	1	NGO representative	1	2
	Teacher Union Leader	1	Teacher Union Leader	1	2
<i>Sub-total</i>		<i>14</i>		<i>14</i>	<i>28</i>
District level	District Education Officer	1	District Education Officer	1	2
	DIET faculty	2	DIET faculty	2	4
	DPO office	3	DPO	3	6
	District Panchayat	1	District Panchayat	1	2
	NGO	1	NGO	1	2
	Teacher Union Leader	1	Teacher Union Leader	1	2
<i>Sub-total</i>		<i>9</i>		<i>9</i>	<i>18</i>
Sub-district level	Mandal Education Office (2)	4	Block Education Office	1	
	Mandal Panchayat	2	BRCF	1	

	(2)				
			Block Panchayat	1	
			CRCF	2	
<i>Sub-total</i>		6		6	12
Total		27		27	58

At the grassroots, a complete picture of the village as well as the school was developed. The socio-economic profile, educational facilities, access and the profile of children in an out of school in the village were obtained with the help of records and key informants. In the sample schools the enrolment profile was obtained from records and infrastructure and facilities available were observed during visits. In each school, the head teacher was interviewed, as well as the class 2 and class 6 teachers (where available) in detail. The interviews explored the teachers' work responsibilities and working conditions, motivation, ability, opinions and experiences about textbooks, teacher training, examinations, perceptions about pupils including those from deprived groups, experience with supervisors and resource persons as well as the parents, village panchayats and community and SEC/SDMC. Interviews were also held with village panchayats as well as SEC and SDMC presidents or members about their perceptions of and relationship to the schools and teachers, the activities of the panchayat and SEC/SDMC, etc. In addition, the teaching learning process in class 2 and class 6 classrooms were observed and focus group discussions were conducted with students of class 6 to get an idea about their home life. Achievement tests in math were also conducted for class 2 and class 6 students.

Table 2.8: Number of interviews conducted at the village and school level

Interviewees	Number of interviewees		
	AP	Rajasthan	Total
Headmasters/head teachers	7 (including one private school)	7 (including one private school)	14
Class 2 teachers	4	6	10
Class 6 teachers	5 (including one private school)	5 (including one private school)	10
Alternative centre teachers	1	3	4
Total teachers	17	21	38
Village panchayat representatives	6	3	9
SEC/SDMC chairpersons/members	7	5	12
Total public representatives	13	8	21
Total interviews	30	29	59

Annexure A: Personnel and infrastructure in the institutions for elementary education in Andhra Pradesh and Rajasthan

	Andhra Pradesh	Rajasthan
State level		
Directorate	<i>Personnel available:</i> one director, an additional director, 3 joint directors, 7 deputy directors, 11 assistant directors, along with ministerial staff. <i>Building and facilities:</i> There were three blocks in the building. Block A was built more than a 100 years ago, Block B, 40 years ago and Block C is around 20 years old. Up to the Assistant Director level people had independent rooms. The rest sat in halls. Almost all officers had computers as well as air conditioners. Every section had a computer and was connected by telephone. Through a Closed User Group (CEG) system all officers were connected through mobile telephones. There were several faxes and photocopiers in the office. There were four cars, used by senior officials and available to staff as per need, and also an auto rickshaw.	<i>Personnel available:</i> One commissioner, one additional director, a chief accounts officer, 2 joint directors, 3 deputy directors, 3 DEOs or junior deputy directors, 5 assistant directors, a senior accounts officer, 4 assistant accounts officers, a chief legal advisor, an assistant legal officer, an additional block officer and an administrative officer, along with ministerial staff <i>Building and facilities:</i> A large campus housed the Directorate of Elementary Education, Office of the Registrar as well as the Directorate of Secondary Education. The Director, Additional Director, Joint Director and the Chief Accounts Officer had single rooms. The rest of the officials shared rooms. Sometimes two officers were squeezed into a very small room. The ministerial staff were seated section-wise. This building was supposed to be air-conditioned, but air conditioners were not provided and now there is lack of ventilation. Telephones, faxes, photocopiers and vehicles were adequate.
SCERT	<i>Personnel:</i> Each of the 9 units was headed by one professor and has 1-3 lecturers. <i>Building and facilities:</i> SCERT was situated in two buildings that also housed the Open School, Jan Shikshan and the Hindi Academy. There were 44 rooms including 3 conference halls spread across the two buildings. The buildings were not in a good condition. There are no toilets in one of the buildings and on two floors. There was a shortage of space as several rooms of SCERT had been allotted to other offices. Professors and lecturers shared rooms, in some cases there are as many as 5 persons to one room. There was one old jeep, provided by UNICEF and a car for the Director, and only one telephone and one fax. To make a call outside SCERT, the faculty has to access the telephone of the Director's PA.	<i>Personnel:</i> The total sanctioned staff strength of SIERT was 208. <i>Building and facilities:</i> SIERT was housed in two large adjoining buildings in a pleasant campus. It had its own hostels, separate for men and women. Office rooms were large, but some were unkempt. Five to seven academic staff had individual rooms and others shared rooms. There were three meeting halls. There were two cars. There used to be 6 but several cars were taken from the SCERT and put in a common pool with the District Collector. Each department head had a telephone, and there were two faxes in SIERT.
State Programme Offices	<i>Personnel:</i> There are 34 officers, including 27 clerks, 10 computer operators, one driver and one PA for SPD and 19 contingent workers. <i>Building and facilities:</i> Situated in the new wing of the DSE building. Five senior officials had independent rooms, the rest shared. Space is short. In some rooms there were as many as five officers. All the support staff sat in one large hall. There was a great deal of noise and disturbance in the office. There was a computer in each room and 13 hired cars. Only additional directors had independent telephones on their desks, but all officers had cell phones.	<i>Personnel:</i> There was a Director, Additional Director, Chief Engineer, Controller Finance, Joint Director and 6 Deputy Directors, each working with one or two Assistant Directors <i>Building and facilities:</i> The SSA state office was housed on two floors of one block a huge building known as Shiksha Sankul, designed on the pattern of a Rajasthan palace. The Director and Additional Directors had large rooms, while officers in charge of functional areas sit in small cabins. Telephones, faxes, photocopiers and vehicles were adequate.
District		

level		
DEO Office	<i>Personnel:</i> one DEO, 5 deputy education officers, an accounts officer, an auditor a statistical officer, 3 computer personnel, 35 clerks and typists, one driver and 9 helpers. <i>Buildings and facilities:</i> There were 13 rooms and one hall, and also a telephone, fax, photocopier, 6 computers and a jeep that was however, not road worthy.	<i>Personnel:</i> one DEO, an Assistant DEO and an SDI; there were 9 posts of superintendents/clerks of which three were vacant; there were 4 helpers. <i>Buildings and facilities:</i> There were 6 rooms and 2 big halls. There was a telephone but no fax, computer, photocopier or vehicle.
DIET	<i>Personnel:</i> There was a principal, 4 senior lecturers and 11 lecturers. <i>Buildings and facilities:</i> There were 6 classrooms, 6 activity rooms, laboratories, a library, a principal's room and a staff room, and a few other offices and storerooms. There was one telephone, 11 computers, recording equipment such as television, tape recorder and video recorder, photocopier (that has been out of order for several years), but no vehicle.	<i>Personnel:</i> The principal's post is vacant. There was a vice principal, 5 lecturers posted against 7 posts, a librarian and 5 helpers. <i>Buildings and facilities:</i> The DIET building is located 3 km outside the town. The space available is adequate, but the upkeep is not very good. There is a telephone, fax, photocopier, computer, communication equipment such as projector and audio system, but no vehicle.
DPO Office	<i>Personnel:</i> Assistant Project Officer, 4 programme officers, 3 deputy engineers, 9 assistant engineers, an Accounts Officer, 3 accountants, one statistical officer, 7 clerks and data entry operators; and 3 helpers. <i>Buildings and facilities:</i> It has 4 rooms in the DEO's building. There is a telephone, fax, photocopier, computer, television and video player. Jeeps are hired as per need.	<i>Personnel:</i> Assistant District Project Coordinator, 3 programme officers for academic activities, community mobilization and women's development, an accountant, 3 clerks and 2 helpers. <i>Buildings and facilities:</i> The building was big, with sufficient space. There was a telephone, fax, a computer, a photocopier (not in working condition) and two jeeps.
	Mandal Office 1. <i>Personnel:</i> one MEO and 3 Mandal Resource Persons. A clerk from the MDO office is expected to do the work of the MEO office but did not. A cleaner had been tapped from a school. <i>Buildings and facilities:</i> 4 rooms and a training hall. There is a telephone and a computer, but no fax, photocopier or vehicle.	BEO office. <i>Personnel:</i> one BEO, 3 Assistant BEOs, one junior accountant, 4 clerks' post but only one clerk (3 posts are vacant) and one peon. <i>Buildings and facilities:</i> The building comprised one room and one large hall. There was no telephone, computer, photocopier or vehicle.
	Mandal Office 2 <i>Personnel:</i> There is one MEO and 3 Mandal Resource Persons. There are no clerks or attendants. The building has 4 rooms and a training hall. <i>Buildings and facilities:</i> There is a telephone and a computer and a cyclostyling machine, but no fax, photocopier or vehicle.	<i>Personnel:</i> a BRCF, two resource persons, a junior accountant and 2 clerical staff. <i>Buildings and facilities:</i> 2 halls, 3 stores and 2 toilets. There is no telephone, fax or computer. There is a photocopier that is out of order and though a jeep is supposed to be available for 3 days a week, it is not.

CHAPTER 3: PUBLIC EXPENDITURE IN EDUCATION IN TWO EDUCATIONALLY BACKWARD STATES

Jandhyala B G Tilak^{*}

“... it is precisely in the area of finance that so many of the nobler aims of education are defeated.” Mark Blaug (1972).

Introduction

According to the human investment revolution in economic thought (Schultz, 1961) and in India the Education Commission (1966) and more particularly the National Policy on Education 1968, expenditure on education is an investment in human beings, an important investment for development, yielding high individual and social returns—economic, social, cultural and political etc., some of which are tangible, and many of which are also regarded as externalities, deserving high priority and an activity on which we need to spend more and more. It follows that expenditure on education is to be treated not as a burden on the public exchequer and as an activity for which resources have to be saved, but as priority investment. Education is widely perceived as a crucial vehicle for the upward mobility of individuals—socially, occupationally and economically—and therefore is an important instrument of equity in society. In addition, education is also regarded as an important component of human development and a fundamental right, having both intrinsic and instrumental values. Therefore, public policy towards education is universally one of the most critical areas of modern development strategies and public expenditure on education is one of the the most significant aspects of such a policy. Public expenditure on education assumes greater significance, especially in developing societies like India, in ensuring equity and quality in education and at the same time in promoting it as a public good and as a human right.

It appears that this view is not ingrained in the minds of our union or state government functionaries, i.e., the economic and educational policy makers and planners. As a result, the education system in India suffers from a severe inadequacy of resources and is in a deep financial crisis. This is a problem of resource scarcity assuming serious proportions while at the same time, the need for additional resources is rapidly increasing. The financial crisis includes: inadequacy in the allocation of resources towards education, relatively low fiscal priority for education, and improper allocation of resources within the education sector between different levels and items of expenditure.

The crisis can be traced to certain faulty assumptions such as

- Public expenditure on education is immaterial for education development, or for overall development,
- Economically poor societies cannot afford to give due priority to education, and

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- Economic reform policies do not affect social sectors, as they get protected.

All these are untenable as our analysis now illustrates.

The education systems in Andhra Pradesh and Rajasthan testify to all the above important dimensions of the financial crisis. During the 1990s, Andhra Pradesh adopted the new economic reform policies somewhat aggressively and was labelled a 'progressive' state by neo-liberals and as a 'pioneering state to earnestly initiate economic and fiscal reforms at the state level in India' by many others,⁵⁰ while Rajasthan has over a long period been considered a *bimaru*⁵¹ state. A comparison between the two reveals interesting aspects of education development and will hopefully yield some useful lessons for other states.

As Chapter 2 shows, Andhra Pradesh and Rajasthan can be regarded as educationally backward states with literacy falling below the national average. In addition, in terms of public spending on education, both the states belong to the group of 'low performing' states along with Bihar, Uttar Pradesh and Madhya Pradesh.⁵² Notably, post independence, Rajasthan lagged behind Andhra Pradesh for a long period in terms of most indicators of education development. In recent years, however, as Chapter 2 shows, Rajasthan has made important strides in improving adult literacy and access to school and has overtaken Andhra Pradesh, where the improvement has been much slower. This is the case in spite of the fact that Rajasthan continues to trail Andhra Pradesh in terms of per capita income and poverty (Planning Commission, 2007). It can be hypothesized that much of the improvement in education in Rajasthan may be directly related to the increase in public spending on education and the relative deterioration in Andhra Pradesh to the reduced levels of public expenditure on education. However, it also needs to be noted that non formal or alternative centres of education, especially the RGSP schools account for a great deal of the new expansion in Rajasthan, and the efficacy of these is discussed in later chapters (especially Chapter 7). With this background, this chapter presents an analysis of trends in public expenditure on education in the two states.

In the rest of the chapter, an attempt is made to (a) review trends in public expenditure on education during the last two and a half decades, focusing on total and per capita expenditure in current and real prices made by the Departments of Education and 'other' departments, central and state, and also analyzes plan and non-plan expenditures, (b) examine the relative priority given to education in the fiscal framework—in state income, state budget and Five Year Plans (c) briefly analyse the pattern of intra-sectoral allocation of resources, i.e., allocation of resources to different levels of education, concentrating on elementary education, and focusing on total expenditure per student and the relative share of elementary education in state income, state budget and total education budgets, (d) have a glimpse at the inter-functional allocation of resources in elementary education and (e) analyse the sources of funds for elementary education and describes the changing centre-state responsibilities in financing education and the contribution of external aid. The chapter ends with a short summary and a few concluding observations.

A point that the reader may note here is that though the scope of this chapter is exhaustive in its coverage of issues relating to public expenditure on education, it does not examine other

⁵⁰ See Hanumantha Rao (2003) for a detailed discussion on economic reform policies in Andhra Pradesh. See also Sarma (2003) and Narasimhulu (2006) for a brief account of fiscal reforms attempted in the state.

⁵¹ 'Bimaru' (in Hindi) means sick, and this term is used symbolically to refer to a few North Indian backward states and the acronym BiMaRU refers to *Bihar, Madhya Pradesh, Rajasthan and Uttar Pradesh*.

⁵² 'Medium' performance states include Orissa, Assam, Haryana, Karnataka and West Bengal and the 'high' performance states are Punjab, Gujarat, Tamil Nadu, Maharashtra, Himachal Pradesh and Kerala (Srivastava, 2005: 136).

important dimensions of education development like the actual functioning of the schools, the quantity and quality of teaching-learning processes, the achievement levels of pupils, etc., the degree of efficiency in the utilization of funds. Some of these aspects are analysed in the following chapters of the book. The purpose here is to present an understanding of the issues relating to financing of education in the two states.

In both Andhra Pradesh and Rajasthan, like in many other states in India, public expenditure on education has increased remarkably since independence. We concentrate here on the trends over the last 25 years, i.e., from 1980-81 to 2004-05/2005-06. This period falls about five years prior to the National Policy on Education 1986 (GoI, 1986) being adopted by the Government of India and the post-Policy period including the pre-*adjustment* and *adjustment* periods. Economic reform policies were introduced in India in the early half of the 1990s; and foreign aid for education started being accepted for the first time in a big way in 1993-94. So this 25-year period is likely to capture some interesting developments in these two states.

Trends in Public Expenditure on Education

Economically, Andhra Pradesh is a middle income state with a per capita net state domestic product (NSDP) of Rs.26,211, ranking 16th among 29 major states and union territories; and Rajasthan a low income state with a NSDP of Rs.17,863, ranking 25th in 2005-06 (MOF, 2008). However, the relative economic position of the states may not necessarily explain their spending levels on education.

As table 3.1 shows, government expenditure on education, inclusive of expenditure by the Department of Education and other departments, has increased by more than 20 times between 1980-81 and 2004-05 in both the states. It increased at an annual rate of growth⁵³ of 13.6 per cent in Andhra Pradesh and at a rate of 15.5 per cent in Rajasthan. But in real prices and in per capita terms, the growth is not all that impressive.

Table 3.1: Growth in government expenditure on education

	Total (Rs. Crores)				Per Capita (Rs.)			
	In current prices		In 1993-94 prices		In current prices		In 1993-94 prices	
	AP	Rajasthan	AP	Rajasthan	AP	Rajasthan	AP	Rajasthan
1980-81	291.3	153.7	914.0	463.2	55.11	45.16	172.90	136.74
1985-86	687.2	360.6	1481.3	726.2	116.64	92.33	251.43	185.94
1990-91	1231.4	890.2	1742.5	1226.8	187.54	202.57	265.36	279.17
1995-96	2206.2	1830.6	1788.3	1556.4	307.84	371.79	249.53	316.09
2000-01	4012.9	3337.6	2472.3	2211.9	527.97	623.03	325.29	404.83
2003-04+	5965.9	4017.5	3193.8	2150.7	699.45	611.19	389.75	340.66
2004-05*	6236.4	4230.4						
Growth	13.6	15.5	4.6	7.3	11.9	13.6	2.7	5.0

Source: based on Selected *Educational Statistics* (relevant years). New Delhi: Ministry of Human Resource Development

Analysis of Budgeted Expenditure on Education (relevant years) New Delhi: Ministry of Human Resource Development

Economic Survey 2006-07. New Delhi: Ministry of Finance,

National Accounts Statistics 2005. New Delhi: Central Statistical Organisation, 2005.

Note: + revised estimate; * Budget estimate; per capita figures shown against 2003-04 refer to 2002-03 (actual), Growth: Annual rate of growth (%).

⁵³ All annual rates of growth in this chapter are estimated by using the semi-log regression equation on continuous annual data.

At real (1993-94) prices⁵⁴ the total government expenditure on education increased by only 3.5 times in Andhra Pradesh (at a rate of growth of 4.6 per cent per annum) and 4.6 times (or at a rate of growth of 7.3 per cent) in Rajasthan between 1980-81 and 2003-04. The government expenditure on education per capita (per head of population) increased by more than 13 times in both the states during this period. But at 1993-94 prices, it has in fact increased only by 2.3-2.5 times. While in per capita terms the real growth has been around 5 per cent on an average in Rajasthan, it is only 2.7 per cent in Andhra Pradesh.

The relative position of both the states is not satisfactory in relation to other states when it comes to public spending on education per head of population. Andhra Pradesh and Rajasthan respectively rank 25th and 22nd among the 28 states on which such data is available for 2004-05. They are ahead of only Bihar, Uttar Pradesh, Orissa and Madhya Pradesh. Thus both states are characterized by backwardness in public spending on education or in effect by chronic under-investment in education. Notably, up to the end of 1980s, Andhra Pradesh was spending more on education than Rajasthan but in recent years, the rate of growth of expenditure has been higher in Rajasthan than in Andhra Pradesh. In fact, in per capita terms, Rajasthan, which was behind Andhra Pradesh in 1980-81, seemed to be spending more than Andhra Pradesh in recent years.

Plan and non-plan expenditure

The total revenue expenditure⁵⁵ on education incurred by the Department of Education and/or by other departments is composed of 'plan' and 'non-plan' expenditure. Plan expenditure is generally meant for development activities (new programmes, projects, schemes, new schools, new buildings, new teachers, etc.), while non-plan expenditure is meant for maintenance of the system. The latter is regarded as a committed liability of the government on the existing system that includes expenses on salaries, allowances, wages, maintenance of existing assets in schools, supervision and administration etc. As the education system expands, non-plan expenditure increases rapidly. At present, a large bulk of the expenditure on education in India is of the non-plan category and in Andhra Pradesh and Rajasthan forms **nearly 90** per cent of the total expenditure on education. In Andhra Pradesh, the share of plan expenditure came down to five per cent in 2000-01, and it has increased only in the last few years. In Rajasthan it has been above ten per cent, and for a few years, e.g., in 1995-96 it was much above ten per cent (Table 3.2).

Table 3.2: Plan and non-plan expenditure on education (% Distribution)

	Andhra Pradesh			Rajasthan		
	Plan	Non-Plan	Total	Plan	Non-Plan	Total
1990-91	6	94	100	13	87	100
1995-96	8	92	100	22	78	100
2000-01	5	95	100	13	87	100
2004-05*	11	89	100	10	90	100

⁵⁴ State income deflators based on 1993-94 national income (GDP) and state income (SDP) series are used throughout this chapter to convert all the figures from current prices into constant (real) prices, based on data drawn from *Economic Survey 2004-05*, and *National Accounts Statistics 2005*. Thus, figures in real prices are adjusted for increase in prices.

⁵⁵ Total expenditure on education consists of revenue expenditure and capital expenditure. The share of education in capital expenditure is negligible in most states and in the country as a whole; and hence it is not included in many analyses. The capital expenditure on education forms less than 1 per cent of the total expenditure on education and the revenue expenditure forms more than 99 per cent. It is increasingly noted that the classification of expenditure into revenue and capital accounts in education in India does not serve any meaningful purpose, as non-recurring expenditure needs are met out of revenue account and recurring expenditure out of capital accounts.

2005-06**	12	88	100	14	86	100
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Source: Based on *Analysis of Budget Expenditure on Education* (relevant years) New Delhi: Ministry of Human Resource Development.

Note: * revised estimate; ** Budget estimate.

While both plan and non-plan expenditure are important, plan expenditure can set new directions for development. Plan resources provide planners freedom to exercise additional leverage as these resources can be increased or cut without affecting the existing system. This cannot be done with non-plan resources, as these are committed expenditures. For this reason, one does not find a steady growth in plan expenditures on education, as one finds in case of non-plan expenditure.

As the system of education expands, the relative share of plan expenditure can be expected to decline and the share of non-plan expenditure to increase. But as long as there are substantial unfinished tasks in terms of a need for say additional school buildings and classrooms to be constructed, additional teachers to be recruited, or additional programmes to be launched for the improvement in quality, equity etc., the financial needs of the plan category are very important. Therefore, plan expenditure should not be subject to serious fluctuations every year but instead, should increase steadily. Both Andhra Pradesh and Rajasthan, as Chapters 2 and 4 show, have accessless habitations, inadequate school buildings, multi-grade teaching and para teachers in lieu of regular teachers, and new schools, buildings, teachers etc., are required. Consequently, there is need for increasing plan investments in education.

Relative Fiscal Priorities: Inter-Sectoral Allocation of Resources

The priority given to education in the framework of the total state economy can be analysed in terms of a variety of indicators. Some of the important ones are: share of education in state income measured in terms of net state domestic product (NSDP), or gross state domestic product (GSDP) or simply state domestic product (SDP); share of education in total government expenditure; and share of education in Five Year Plan outlays/expenditures. These are briefly examined here.

Share of education in state income

The proportion of state income spent on education, one of the most standard indicators, shows the priority given by society to education vis-à-vis other sectors. On the recommendation of the Education Commission (1966), the Government of India (1968) fixed a target of and investment of six per cent of the national income from the public exchequer on education by 1986; but the goal remains elusive so far, and, recent estimates show that a substantial increase in allocation of resources would be required to reach this goal (MHRD, 2005a).

Both Andhra Pradesh and Rajasthan are far below the national target, though between the two, Rajasthan's performance is somewhat better. As Table 3.3 shows, while in 1980-81 Andhra Pradesh was spending a higher proportion of its SDP toward education than Rajasthan by 1990-91, Rajasthan had overtaken Andhra Pradesh and was allocating a higher proportion, and the trend continued in the later period. In 2004-05, Rajasthan allocated 4.5 per cent, while Andhra Pradesh allocated 2.6 per cent of its SDP for education.

Table 3.3: Changing relative fiscal priority for education, government expenditure on education

	% of GSDP		% of Total Budget Expenditure	
	Andhra Pradesh	Rajasthan	Andhra Pradesh	Rajasthan
1980-81	3.35	2.87	25.09	24.92
1985-86	4.24	3.55	24.71	23.92

1990-91	3.48	3.72	22.37	25.58
1995-96	2.76	3.87	20.79	21.97
2000-01	2.92	4.37	17.39	22.20
2004-05	2.61	4.45	14.30	21.08

Source: Based on *Analysis of Budget Expenditure on Education* (relevant years). New Delhi: Ministry of Human Resource Development.

Notably, among the major states, Himachal Pradesh spends as much as 6.1 per cent of its SDP on education (see Table 3.5 for details). Kerala earlier spent above 6 per cent but in recent years subsequent to its success in universal school education, has managed to reduce the proportion to 4.4 per cent. Both Andhra Pradesh and Rajasthan have to make special efforts to raise their allocations to education as a proportion of their SDP to reach the national norm of six per cent. In fact, both states may have to allocate much above six per cent of their SDP to provide reasonable quality education. This has also been seen as necessary for the country as a whole (see MHRD, 2005a).

Share of education in the government budget

A more important indicator of how much priority a government gives to education is the expenditure on education as a percentage share of the government's total expenditure or the budget. This is more important because the government has greater direct control on expenditure from its own revenues as compared to the SDP. Andhra Pradesh and Rajasthan were spending about 25 per cent of their respective revenue budgets on education in 1980-81. But Andhra Pradesh progressively reduced this share in subsequent years and stood at 14 per cent in 2004-05. Rajasthan also reduced the relative priority given to education in the revenue budget, but the decline was marginally less and it still spends a little above one-fifth of its revenue budget on education.

The percentage share of plan expenditure on education reveals the relative priority given to education in the plan framework. Expenditure trends show that plan priorities have not been systematic in the country, and particularly so in case of Andhra Pradesh and Rajasthan. Between 1990-91 and 2004-05, the percentage of plan funds spent on education fluctuated a good deal in Andhra Pradesh: 8.4 per cent in 1990-91, 11.9 per cent in 1995-96, declining to below 5 per cent in 2000-01 and then marginally increasing to 6 per cent in 2004-05. Similar fluctuations can be seen in Rajasthan: 25 per cent in 1995-96 and only 14 per cent in 2004-05. In case of non-plan expenditure, there is a somewhat steady decline in the share of education in both the states (Table 3.4).

Table 3.4: Share of education expenditure in total plan and non-plan expenditure (%)

	Plan Expenditure		Non-Plan Expenditure	
	Andhra Pradesh	Rajasthan	Andhra Pradesh	Rajasthan
1990-91	8.35	18.45	25.26	27.18
1995-96	11.86	25.61	22.16	21.10
2000-01	4.67	23.17	20.25	22.06
2004-05*	6.25	14.28	17.23	22.30
2005-06**	5.96	18.73	16.62	24.67

Source: Based on *Analysis of Budget Expenditure on Education* (relevant years) New Delhi: Ministry of Human Resource Development.

Note: * revised estimate; ** budget estimate.

With respect to percentage of NSDP allocated to education, both Andhra Pradesh and Rajasthan rank poorly, i.e. 28th and 14th respectively, as compared to other states (Table 3.5). However, both states fare relatively better in case of the share of budget allocated to education: Rajasthan ranks fairly high at 5th, and Andhra Pradesh at a very low level – 24th. Table 3.5 also gives data on per capita expenditure on education. In general, it can be concluded that with respect to all the three indicators, but particularly with respect to percentage share of SDP and expenditure on education per capita, Andhra Pradesh and Rajasthan lag far behind many other states. This may at least partly explain the overall educational backwardness of these two states.

Table 3.5: Expenditure on education in India: inter-state comparisons

	% of NSDP (2003-04/ 2004-05)	% of Total Budget (2004-05)	Per capita Rs. (2004-05)
Andhra Pradesh	2.61	14.30	605.80
Arunachal Pradesh	8.78	11.43	1539.53
Assam	8.00	19.82	1108.97
Bihar	7.64	25.01	488.45
Chhattisgarh	4.37	22.17	796.18
Goa	3.64	18.34	2619.43
Gujarat	3.18	18.95	856.32
Haryana	2.73	17.28	893.93
Himachal Pradesh	6.13	19.51	1738.74
Jammu & Kashmir	5.92	12.98	1000.40
Jharkhand	4.74	20.90	619.88
Karnataka	3.38	19.48	880.22
Kerala	4.42	20.96	1209.68
Madhya Pradesh	3.89	19.46	577.46
Maharashtra	3.68	22.65	1161.02
Manipur	7.46	19.48	1448.56
Meghalaya	6.30	18.53	1243.28
Mizoram	10.03	15.99	2658.51
Nagaland	4.79	14.55	1282.30
Orissa	4.28	15.63	561.89
Punjab	3.06	13.25	952.09
Rajasthan	4.45	21.08	707.35
Sikkim	11.95	9.18	2927.81
Tamil Nadu	3.29	19.16	859.16
Tripura	8.20	25.63	1698.53
Uttar Pradesh	4.32	19.45	503.10
Uttaranchal	7.36	21.01	1417.65
West Bengal	2.75	18.84	663.07

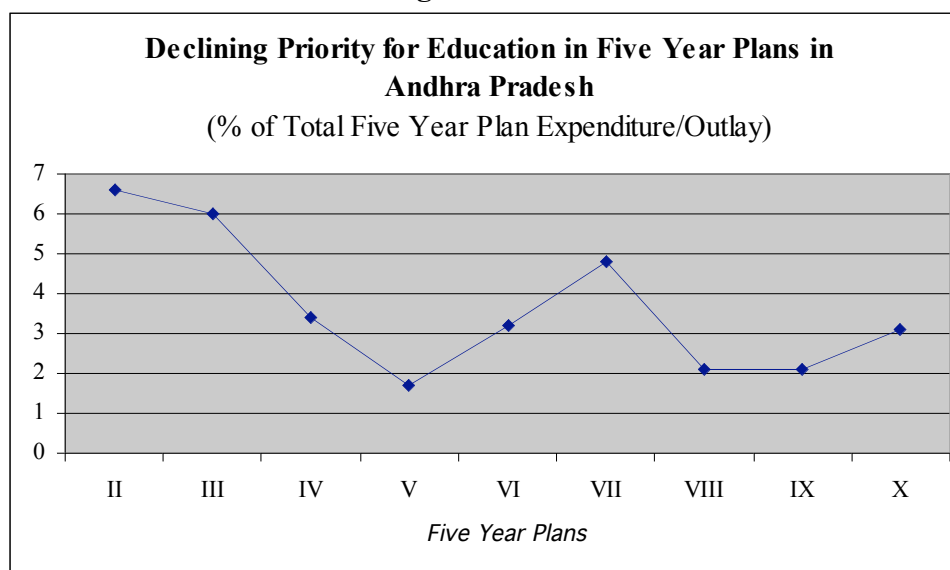
Source: *Analysis of Budgeted Expenditure on Education 2003-04—2005-06* (New Delhi: Ministry of Human Resource Development) for % of NSDP; and *Selected Educational Statistics 2004-05* (New Delhi: Ministry of Human Resource Development) for others.

Expenditure on education in five year plans

The Five Year Plan outlays provide important directions for development, setting targets to be achieved and indicating the sectoral distribution of plan resources. The relative priority given to education in the Five Year Plans in Andhra Pradesh has been quite erratic. In the Second and Third Five Year Plans (1951-56 and 1961-66), the state had allocated more than six per cent of

the total plan outlay/expenditure to education, which declined to 3.4 per cent in the Fourth Five Year Plan (1969-74). There has been a continuous decline in the proportion allocated to education in the subsequent Five Year Plans. There was a special effort to increase the priority in the Seventh Five Year Plan (1985-89) and the share of education went up to 4.8 per cent., probably because of special programmes launched after the National Policy on Education in 1986. However, the tempo could not be maintained in the following period. In the Eighth and the Ninth Five Year Plans (1992-97 and 1997-2002) as little as 2.1 per cent of the total plan expenditure was spent on education. In the Tenth Plan (2002-07) it is proposed that this be raised marginally (Figure 3.2). Even in absolute terms, the growth in expenditure on education in the Five Year Plans has not been smooth in Andhra Pradesh (Tilak, 1998).

Figure 3. 2



Source: Tilak (1998) Public Expenditures on Education in Andhra Pradesh: A Review of Trends, Issues and Problems. Government of Andhra Pradesh, District Primary Education Programme, Hyderabad; and Planning Commission (2002) Financial Progress of Education Sector during Ninth (1997-2002) & Tenth Plan Period. Education Division, Planning Commission, Government of India, New Delhi (Unpublished).

Even in the case of Rajasthan, trends have not been smooth with regard to allocations in the last three Five Year Plans (Eighth to Tenth), even though the state allocates a higher proportion of the total plan outlay/expenditure to education. In the Ninth Plan (1997-2002) the education sector received 10 per cent of the total; but according to estimates available, the corresponding proportion seems to have declined to 5.4 per cent in the Tenth Plan. Even in absolute amounts there is a big fall (Table 3.6).

Table 3.6: Expenditure on education in five year plans (in Rs. crores)

Five Year Plan	Andhra Pradesh				Rajasthan			
	VIII	IX*	IX	X*	VIII	IX*	IX	X*
General Education	175.48	491.23	364.33	1417.54	1099.27	2195.38	1691.82	1427.26
Technical Education	60.67	38.14	199.64	37.55	88.08	97.66	72.46	40.56
Total	236.15	529.37	563.97	1455.09	1187.35	2293.04	1764.28	1467.82
% of the Total of All Sectors	..	2.10	..	3.12	..	10.18	..	5.37

Source: Education Division, Planning Commission Financial Progress of Education Sector during Ninth (1997-2002) & Tenth Plan Period. Education Division, Planning Commission, Government of India, New Delhi (Unpublished); and *Mid-Term Appraisal of 10th Five Year Plan 2002-07*. New Delhi: Planning Commission, 2005.

Note: * approved outlay not available

Such zigzag trends in plan priorities do not ensure the healthy growth of the education system. The importance of Five Year Plan in setting new directions for development, has been noted, but it has to be concluded that neither state has given laid sufficient emphasis on providing a steady flow of funds to education in its Five Year Plans.

On the whole, trends in total government spending on education in Andhra Pradesh and Rajasthan reveal some interesting developments. Today, in terms of total and per capita expenditure, and relative priorities—percentage of state income and total revenue expenditure of the state—Andhra Pradesh spends less than Rajasthan on education. Noatbly, this was not the situation in the 1980s. The economic reform policies, including stabilization and structural adjustment policies involving steep cuts in public expenditure, along with liberalization, privatization and globalization of all sectors including education,⁵⁶ were found to have a significant effect on budgetary allocations for education in Andhra Pradesh, while the reform period had no adverse effect in Rajasthan. In fact, expenditure on education and also the relative priority given to this sector increased in Rajasthan during this period.⁵⁷ It is well known that Andhra Pradesh has pursued the economic reform policies quite vigorously from the beginning of the 1990s, and Rajasthan has not, though it also started implementing the reforms much later.⁵⁸ Statistically also, we find in Table 3.7 a significant negative impact of the reform period on expenditure on education as a proportion of SDP in Andhra Pradesh. The value of the coefficient of the dummy variable for the reform period used in the regression equation is negative and statistically significant.⁵⁹

Further, the general level of economic development, measured in terms of SDP per capita has no relation to the level of relative priority given to education in either of the states, confirming the general finding that level of public spending on education is not necessarily influenced by levels of economic development (Tilak, 1987). It may be felt that the influence of the per capita GSDP is not seen in the equation possibly because of the high correlation of it with the dummy variable (D). But as the results in Eqn.3 indicate, this is not the case. The level of economic development has no significant effect on the relative priority given to education. In both states, an increase (in principle also a fall) in SDP per capita is not followed by a proportionate increase (or a decrease) in the proportion of NSDP spent on education. Further, it is also well noted some of the economically better-off states (e.g., Punjab and Haryana) do not necessarily spend more on education than some of the economically poorer states (like Kerala and Himachal Pradesh) in terms of relative proportions and also in terms of absolute amounts – per capita or per student expenditure [see also Tilak, 1987].

⁵⁶ For an elaborate discussion on the impact of economic reform policies on education in India, see among many, Tilak (1996b) and Jha (2005) for analyses of impact of the adjustment policies on education.

⁵⁷ For example, Srivastava (2005: 149) also estimated that between 1975 and 2000, the decade of the 1990s marks the lowest rate of growth in expenditure on education in Andhra Pradesh (3.2 per cent between 1990 and 2000, compared to 7.8 per cent between 1975 and 1990). In contrast, the trends were the reverse in Rajasthan: expenditure on education increased at a rate of growth of above 7 per cent between 1990 and 2000, compared to 6.7 per cent in the 15 years preceding this period (see also Ray, 2005: 175).

⁵⁸ See, for example, World Bank (1996b, 2006) that describes the implementation of the reforms in Andhra Pradesh and Rajasthan respectively.

⁵⁹ The total period considered here covers 1980-81 to 2000-01/2001-02. The reform period is defined to include the period beginning from 1992-93 and the period before 1992-93 is considered as pre-reform period.

Table 3.7: Regression results of expenditure on education (*ln*EDEX/SDP) in Andhra Pradesh and Rajasthan

	Andhra Pradesh (1980-81-2000-01)			Rajasthan (1980-81-2001-02)		
	Eqn.1	Eqn.2	Eqn.3	Eqn.1	Eqn.2	Eqn.3
GSDP/pc	0.000012 (0.653)		-0.000018 (0.983)	0.000008 (0.651)		0.0000275 (2.345)
D(YEAR)	-0.092 (3.171)	-0.082 (3.377)		0.0543 (2.559)	0.6239 (3.681)	
Intercept	0.4952 (4.791)	0.5619 (35.342)	0.637 (5.627)	0.4873 (7.236)	0.5315 (46.405)	0.404 (6.061)
Adjusted R ²	0.3217	0.3422	-0.0016	0.3553	0.3739	0.1764
F-Value	5.742	11.403	0.967	6.787	13.539	5.497
N	21	21	21	22	22	22
Notation: EDEX/SDP: Expenditure on education as % of SDP GSDP/pc: Gross State Domestic Product per capita D(YEAR): Dummy for Economic reform period; year =1, if adjustment period, 0 otherwise t-values in parentheses.						

Intra-Sectoral Allocation of Resources and Expenditure on Elementary Education

The distribution of resources between different levels of education shows the relative priority that the government attaches to various levels of education. Of the total resources available to education, what have been the priorities given to different levels of education?

Given the inadequate progress with respect to universal elementary education, and following the emphasis laid on elementary education in the National Policy on Education (1986), the union government had preferred that at least 50 per cent of the total expenditure on education be allocated to elementary education, and the remaining 50 per cent be distributed among the other levels. Accordingly, many state governments made necessary changes in the pattern of allocation of resources within education. While many allocated nearly or about 50 per cent of the total to elementary education, there are some states which are far off the mark.

According to latest statistics (Table 3.8), in 2004-05, elementary education in Andhra Pradesh received only 46 per cent of the total education expenditure, while in Rajasthan the corresponding proportion was 57 per cent.. Andhra Pradesh has been spending less than 50 per cent of its education budget on elementary education: it has varied between 40 and 50 per cent. Rajasthan has been spending above 50 per cent (54-58 per cent). Secondary education is accorded a similar priority in both the states with an allocation of about one-third of the total education expenditure. Rajasthan is consistent in allocating about 30 per cent of its education budget to secondary education: the variations among years being very small, Andhra Pradesh is not so consistent: its allocation to secondary education varied between 28 and 33 per cent.

But there are significant differences in spending on higher education. While Andhra Pradesh spends about one-fifth of the total education budget on higher education, Rajasthan spends less than one-tenth. Moreover, expenditure on higher education in Rajasthan shows a decline from 9.2

per cent to 6.7 per cent of the total between 1990-91 and 2004-05. Both states spend very small amounts on technical education: Andhra Pradesh about 3 per cent and Rajasthan much less, even below 1 per cent in 2004-05. Other levels and types of education receive negligible proportions in both states.

Table 3.8: Distribution of expenditure on education, by levels (%)

	1990-91		1995-96		2000-01		2002-03		2004-05*	
	AP	Raj	AP	Raj	AP	Raj	AP	Raj	AP	Raj
Elementary	46.0	54.4	43.1	56.0	40.0	57.8	42.1	57.3	45.6	57.6
Secondary	28.2	32.1	31.8	32.6	31.6	32.6	32.3	33.3	30.2	33.1
Adult	0.7	1.0	0.2	0.8	0.3	0.1	0.3	0.1	0.3	0.4
Higher	21.3	9.2	20.8	7.6	24.6	7.2	20.9	6.9	19.0	6.7
Technical	2.8	1.7	2.9	1.4	2.5	1.0	3.3	1.1	3.5	0.9
Physical	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1
Other General	0.6	0.7	0.7	0.6	0.7	0.3	0.8	0.2	0.8	0.2
Language Development	0.4	0.9	0.4	1.0	0.4	1.0	0.3	1.0	0.3	1.0
Total	100.0	100.0	99.9	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Source: Based on *Analysis of Budget Expenditure on Education* (relevant years). New Delhi: Ministry of Human Resource Development.

Note: * revised estimate.

Public expenditure on elementary education

Though universalization of elementary education has been an important objective of educational planning in India, it suffers from under-investment of public funds, resulting in under-achievement towards realizing goals. Table 3.9 gives trends in public expenditure on elementary education in Andhra Pradesh and Rajasthan in current and constant prices.

Table 3.9: Expenditure on elementary education (Rs. crores)

	In current prices		In 1993-94 prices	
	Andhra Pradesh	Rajasthan	Andhra Pradesh	Rajasthan
1980-81	101.4	77.6	318.1	233.8
1985-86	243.1	172.9	524.0	348.2
1990-91	454.5	433.4	643.1	597.3
1995-96	724.7	939.6	587.4	798.9
2000-01	1392.7	1818.3	858.1	1205.0
2004-05*	2225.5	2284.5	..	..
Growth	14.6	20.3	3.8	8.0

Source: Based on *Analysis of Budget Expenditure on Education* (various years) New Delhi: Ministry of Human Resource Development.

Note: Growth – Annual rate of growth of the expenditure in 1993-94 prices refers to the period from 1980-81 to 2002-03 only.

* Budget estimates.

While at current prices there has been a significant increase in public expenditure on education in both the states, the real increase is very small in Andhra Pradesh, the rate of growth being 3.8 per cent per annum, while in Rajasthan the rate of growth has been reasonably satisfactory at 8 per cent. What is interesting is that Andhra Pradesh was spending higher amounts from the beginning of the 1980s until 1990-91. But the relative position changed from 1993-94 onwards,

and in 2000-01, Rajasthan was spending about 40 per cent more than Andhra Pradesh.⁶⁰ Along with this, as Andhra Pradesh has a larger number of enrolments than Rajasthan, as will be shown later, the differences in per pupil expenditures are higher.

The relative share of plan expenditure on elementary education increased from 10 per cent in 1990-91 to 21 per cent in 1995-96 in Andhra Pradesh. In contrast, in Rajasthan more than 90 per cent of the expenditure remained in the non-plan category (Table 3.10). What probably led to this situation was the fact that the assistance from the union government to plan expenditure on elementary education in the 1990s was higher in Andhra Pradesh than in Rajasthan. Even in 2004-05, one-fifth of the total expenditure in Andhra Pradesh was in the plan category. In Rajasthan, the relative proportion of plan expenditure was expected to suddenly increase to 14 per cent in 2004-05 from 2 per cent in 2000-01. These sudden increases and cuts cannot be easily explained, though in general one may find that in the last year or two of every Five Year Plan there could be a steep increase in plan allocations to exhaust the total allocations made in the plan (Tilak, 1987). But any way, these are only relative proportions of plan expenditures to total expenditures.

Table 3.10: Plan and non-plan expenditure on elementary education (%)

	Andhra Pradesh		Rajasthan	
	Plan	Non-Plan	Plan	Non-Plan
1980-81	6	94	3	97
1985-86	9	91	7	93
1990-91	10	90	7	93
1995-96	21	79	5	95
2000-01	14	86	2	98
2004-05*	18	82	14	86
2005-06**	16	84	16	84

Source: Based on *Analysis of Budget Expenditure on Education* (various years) New Delhi: Ministry of Human Resource Development.

Note: * revised estimate; ** budget estimate.

Share of elementary education in state income and government budget

The relative priority accorded to elementary education can be analysed in terms of the share of expenditure on elementary education as a proportion of (a) SDP, (b) total government budget, and (c) total budget expenditure on education (Table 3.12).

Table 3.11: Indicators of relative priority for elementary education

	% of SDP		% of Total Government Budget		% of Total Education Expenditure	
	Andhra Pradesh	Rajasthan	Andhra Pradesh	Rajasthan	Andhra Pradesh	Rajasthan
1980-81	1.17	1.45	8.73	12.58	35	50
1985-86	1.50	1.70	8.74	11.47	35	48
1990-91	1.28	1.81	8.26	12.45	37	49
1995-96	0.91	1.99	6.83	11.28	33	51
2000-01	1.01	2.38	6.04	12.09	35	54
2004-05*	1.37	2.56	6.52	11.67	46	58

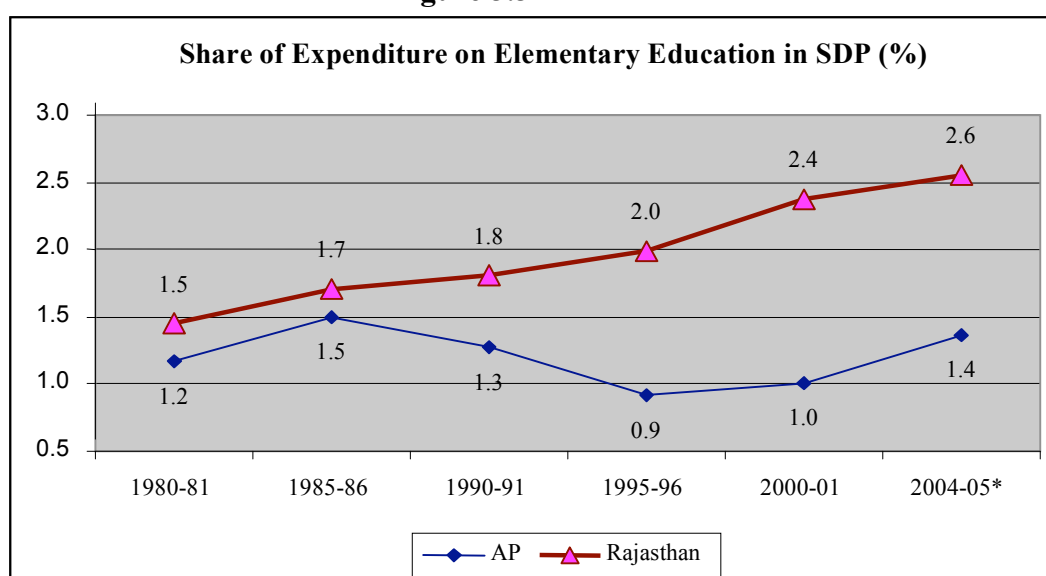
⁶⁰ Srivastava (2005) also estimated that the overall growth in real expenditure on elementary education during 1975-2000 in Andhra Pradesh had been the lowest, and that in Rajasthan had been the highest among the major states in India.

Source: Based on Selected *Educational Statistics and Analysis of Budget Expenditure on Education* (various years). (both by the Ministry of Human Resource Development, New Delhi)

Note: * Budget estimate.

All the three indicators clearly reveal that Rajasthan has accorded a higher priority to elementary education than Andhra Pradesh throughout the period examined here. The share of expenditure on elementary education in SDP declined in Andhra Pradesh from 1.5 per cent in 1985-86 to 0.9 per cent in 1995-96. Though it increased marginally later, it remained much below the 1985-86 level of 1.5 per cent (Figure 3.3). The gap between the two states on this measure has increased during the last 20 years, i.e., between 1985-86 and 2004-05. In 2004-05, Rajasthan spent nearly double the ratio that Andhra Pradesh allocated towards elementary education. Notably, while the difference between the two states was of the order of only two decimal points in 1980-81 and 1985-86, but increased to more than one per cent point by 1995-96 and to 1.2 per cent points by 2000-01.

Figure 3.3



Source: Based on *Analysis of Budget Expenditure on Education* (various years). New Delhi: Ministry of Human Resource Development; and *Domestic Product of States of India* (EPW Research Foundation) Sameeksha Trust, Mumbai, 2004.

While in Rajasthan more than ten per cent of the revenue budget was consistently allocated to elementary education, in Andhra Pradesh, it was 8.7 per cent in 1980-81 and declined to 6.5 per cent in 2004-05. This discussion shows that elementary education received better treatment in Rajasthan than it did in Andhra Pradesh during the last 25 years in the overall fiscal framework and also within the education budget. Moreover, Rajasthan spends nearly or more than half of the total education expenditure on elementary education; but in Andhra Pradesh it has remained around one-third. According to latest budget estimates, it may be above 40 per cent in 2004-05.

Rajasthan has also made significant improvements over the years in terms of expenditure per student (Table 3.12) as it spent consistently higher amounts than Andhra Pradesh. In real prices, per student expenditure on elementary education increased at a rate of growth of 2.9 per cent in Rajasthan between 1980-81 and 2000-01, while it only increased at a 1.5 per cent rate of growth per annum in Andhra Pradesh. However, the annual rate of growth has not been consistent in either of the two states. While the highest rate of increase was noticed in Andhra Pradesh during

1980-81 and 1984-85, it was only from 1985-86 to 1989-90 that Rajasthan could experience such a high rate of growth.

Table 3.12: Public expenditure on elementary education per student (Rs. per annum)

	Andhra Pradesh		Rajasthan	
	in current Prices	in 1993-94 prices	in current prices	in 1993-94 Prices
1980-81	189.72	595.21	244.33	736.54
1985-86	347.36	748.78	373.23	751.60
1990-91	581.68	823.08	835.94	1152.07
1995-96	968.40	784.98	1309.79	1113.57
2000-01	1602.66	987.40	1816.53	1203.84
2002-03*	1901.38	1059.51	2222.33	1238.68
<i>Annual Growth Rates</i>				
1980-81 to 1984-85	17.02	8.06	9.16	1.15
1985-86 to 1989-90	13.44	5.28	15.50	8.36
1990-91 to 1994-95	15.00	2.62	8.98	-0.76
1995-96 to 1999-2000	10.54	4.05	8.46	2.49
1980-81 to 2000-01	10.82	1.46	11.57	2.93

Source: Based on *Analysis of Budget Expenditure on Education* (various years); *Education in India* (Various years) and *Selected Educational Statistics* (various years) all by the Ministry of Human Resource Development, New Delhi; Note: * Revised estimate.

As in the case of total expenditure on education, expenditure on elementary education also suffered during the reform period in Andhra Pradesh. Similar equations are estimated and the results are shown in Table 3.13. The results show that the reform period has had a severe adverse effect on revenue expenditure on elementary education in Andhra Pradesh. But the expenditure on elementary education in Rajasthan was not adversely affected.⁶¹ This is despite the fact that the District Primary Education Programme (DPEP), was adopted in Andhra Pradesh in the early 1990s and aid had flown in through the union government, while this project was not adopted in Rajasthan until 1999.⁶² An important point that emerges here is that in those two states, the level of economic development has no relation to the level of spending on elementary education.

Table 3.13: Regression results of expenditure on elementary education (*ln*ELEYDEX/SDP) in Andhra Pradesh and Rajasthan

	Andhra Pradesh (1980-81-2000-01)			Rajasthan (1980-81-2001-02)		
	Eqn.1	Eqn.2	Eqn.3	Eqn.1	Eqn.2	Eqn.3
GSDP/pc	-0.0000087 (0.546)		-0.000047 (2.416)	-0.0000014 (0.097)		-0.1026 (7.834)
D(YEAR)	-0.1153 (4.503)	-0.1226 (5.741)		0.8818 (3.534)	-0.0805 (4.399)	

⁶¹ Srivastava (2005: 149) has estimated that the growth in expenditure on elementary education in Andhra Pradesh was only 4.3 per cent between 1990 and 2000, compared to 8.6 per cent between 1975 and 1990 and in the case of Rajasthan the 1990s experienced a rate of growth of 7.6 per cent compared to 5.6 per cent in the earlier period. In fact, Srivastava (2005) found that expenditure on elementary education in all other states was adversely impacted.

⁶² Details of DPEP and other externally aided projects are discussed later in this chapter.

Intercept	0.1847 (2.026)	1.136 (9.70)	0.3626 (3.106)	0.2268 (3.086)	0.2198 (17.813)	0.3567 (24.959)
Adjusted R ²	0.6403	0.6151	0.1947	0.4386	0.4664	0.7419
F-Value	16.018	32.955	5.836	9.202	19.352	61.379
n	21	21	21	22	22	22
Notation EDEX/SDP: Expenditure on education as % of SDP ELEYEDEX/SDP: Expenditure on Elementary Education as % of SDP GSDP/pc: Gross State Domestic Product per capita D(YEAR): Dummy for Economic reform period; year =1, if adjustment period, 0 otherwise Figures in parentheses are t-values.						

Inter-functional allocation of resources

An equally important question is allocation of resources within education between different items and objects. Expenditure on education can be classified into two categories: recurring and non-recurring. Recurring expenditure is presently classified in official records into only two groups: expenditure on salaries and allowances, and 'others'. A substantial part of the expenditure on school education is of a recurring nature and most of the recurring expenditure is on salaries of teachers and other staff. More than 95 per cent of the total⁶³ recurring expenditure on primary and upper primary schools in Andhra Pradesh and Rajasthan was incurred on salaries and allowances of teaching and other staff in 1996-97, the latest year for which such data are available (Table 3.14). All other items including scholarships, incentives and materials received negligible amounts. In fact, available data do not provide any details on other expenditure.

Table 3.14: Expenditure on primary and upper primary education, 1996-97

	Primary		Upper Primary	
	Andhra Pradesh	Rajasthan	Andhra Pradesh	Rajasthan
<i>Recurring Expenditure</i>				
Salaries & Allowances	97.0	98.6	96.8	95.4
Others	3.0	1.4	3.2	4.6
Total	100.0	100.0	100.0	100.0
Total Recurring Expenditure	98.9	98.2	99.3	97.8
Non-Recurring Expenditure*	1.1	1.8	0.7	2.2
<i>Grand Total</i>	100.0	100.0	100.0	100.0
Grand Total (Rs. crores)	608.16	536.28	302.74	167.93

Source: *Education in India 1996-97* New Delhi: Ministry of Human Resource Development.

Note: * No expenditure on libraries, buildings; it is categorised as 'Others'.

We have somewhat different but elaborate details for a more recent period (2000-01) on government expenditure.⁶⁴ These confirm that expenditure on non-salary items is miniscule.

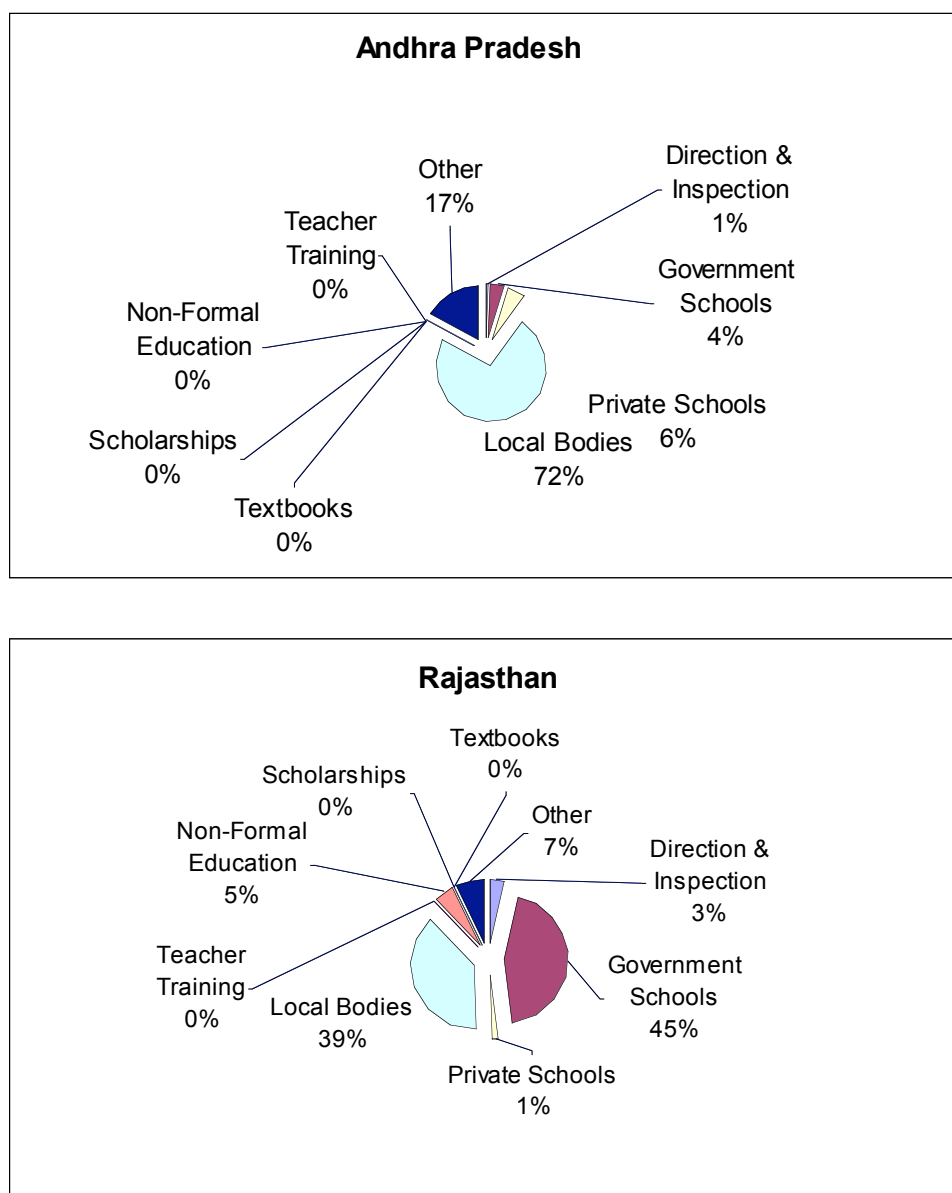
⁶³ The *total* includes government expenditure, expenditure incurred by local bodies, contributions from fees, endowments, etc.

⁶⁴ Two different sources are used in this context and some problems of comparison arise. *Education in India* (MHRD-b) an annual statistical publication of the Ministry of Human Resource Development forms the basis for 1996-97. The other source used is *Analysis of Budget Expenditure on Education* (MHRD-c) that forms the basis for the somewhat detailed break-up. The latter adopts a different classification of objects of expenditure. The latest volume of *Education in India* (on expenditure related aspects) refers to 1996-97, that too for school education; for higher education the latest one is as old as 1983-84.

Though they do not explicitly mention salary expenditure, most of the allocations to government schools, local body schools and private schools can be considered as expenditure on salaries of teaching and other staff. This forms about 90 per cent of the total expenditure. According to Figure 3.4, nil or negligible amounts are allocated to teacher training, scholarships and textbooks in both the states. Negligible amounts are also allocated to non-formal education. Andhra Pradesh spends insignificant amounts on direction, supervision and administration, while Rajasthan spent 3.4 per cent in 2004-05. Expenditures on these items positively influence the quality of education and also access to education.

A further 5.6 per cent of government expenditure on elementary education in Andhra Pradesh is given as grants-in-aid to private aided schools. The corresponding ratio is only 1.4 per cent in Rajasthan. There are a larger number of private schools in Andhra Pradesh than in Rajasthan: 3.7 per cent of the primary schools and 2.9 per cent of upper primary schools in Andhra Pradesh were government-aided private schools in 2004-05; in contrast, the corresponding ratios in Rajasthan were only 0.5 and 0.9 per cent respectively

Figure 3. 4: Distribution of expenditure on elementary education, by items, 2004-05



Source: Based on *Analysis of Budget Expenditure on Education 2003-04—2005-06*. New Delhi: Ministry of Human Resource Development, Government of India.

It is important here to note that available evidence from developing countries shows that investment in items such as textbooks, teaching-learning material and teacher training that have a close relationship with the quality of education, yields higher returns than investment in other inputs. For example, Behrman and Birdsall (1983), Fuller et al (1986) and Fuller (1990) have shown that investment in quality related inputs in education pays substantially and that unless schools have attained a threshold level of quality they cannot yield long-term economic benefits. Secondly, to improve equity in education, besides providing 'free' education it is also necessary that allocations to equity oriented items like scholarships at all levels, free textbooks, noon meals and uniforms etc., at the school level are increased. But both Andhra Pradesh and Rajasthan have been spending very little on these items in elementary education. This is not surprising as the situation is similar in many other states in the country (Tilak, 2003b). When it comes to

elementary education, the union and state governments seem to be concentrating all their expenditure mainly on teachers' salaries.

Sources of Funds for Education

The sources of funds for education essentially include the

- Governmental sector (union government, state government and local bodies such as zila parishads, municipalities, mandals and panchayats),
- Non-governmental sector (households: students/parents, fees, etc., and rest of the community in terms of endowments and donations), and
- Foreign aid, consisting of bilateral and multilateral aid.

Detailed data are available only on the first two sources. Among the contributions from the household sector, information from official sources is available only for fee income accruing to educational institutions (details about some other aspects of household expenditure are discussed later in this chapter). Focusing on the information available, one might note that like in most other developing and developed societies, the government is the most important financier of education at all levels. The government—both central and state and local bodies—has been financing 95 per cent of the total expenditure on elementary education in many states in India.⁶⁵ Of course, local bodies also receive aid from the state governments (see Tilak, 2003b for details).

The situation is identical in Rajasthan and Andhra Pradesh (Table 3.16). Government grants account for more than 95 per cent of the total income of elementary education. But this is true with respect only to recurring income. Endowments and other funds, which may be regarded as non-government or private sources of funds for education, constitute a huge proportion of non-recurring income. But it has to be noted that the total non-recurring income is very small in quantum. For the same reason, in the totals (recurring plus non-recurring), the government, including local bodies accounts for 95 per cent and above. Non-governmental contribution to the financing of education is very limited—negligible in Rajasthan and 3-5 per cent in Andhra Pradesh in primary and upper primary education. In recent years, however, efforts have been made to mobilize resources, both in cash and kind, from communities and they are believed to be sizeable, though no quantitative information on these is available. They include both voluntary and non-voluntary contributions from the community (see, for example the proposals made by GoAP, 1995).

Earlier, income from student fees was mentioned as a separate source, though it accounted for a small proportion at the primary and upper primary levels. However, in recent years in response to controversies on the appropriateness of fees in 'free' elementary education, fee income is clubbed along with endowment and other sources. So the data in Table 3.16 does not necessarily mean that there is no fee at the primary and upper primary levels of education in India or specifically in either of the two states. Several studies (e.g., Tilak, 1996a, 2002a, 2002b) have documented the prevalence of the phenomenon of fees in 'free' elementary education.⁶⁶ It is hoped that with the 86th amendment of the Constitution of India and the passage of the Free and Compulsory Education Bill (MHRD (c) and (i) 2005), all fees will soon be abolished at the elementary level of education in all states.

⁶⁵ The income received from the local bodies or the expenditure incurred by local bodies includes the grants received by the local bodies from the state governments and the resources generated by the local bodies themselves may be very small, if not nil.

⁶⁶ Among others, Majumdar (2001) also finds that children pay fees in government schools in Rajasthan and Tamil Nadu.

Table 3.16: Sources of income of primary and upper primary education, 1996-97

	Primary		Upper Primary	
	Andhra Pradesh	Rajasthan	Andhra Pradesh	Rajasthan
<i>Recurring</i>				
Government	96.9	88.6	95.0	100.0
Local Bodies	0.0	11.2	0.0	0.0
Endowments & Others	3.1	0.2	5.0	0.0
Total	100.0	100.0	100.0	100.0
<i>Non-Recurring</i>				
Government	72.6	78.8	32.4	95.2
Local Bodies	0.0	0.0	0.0	0.0
Endowments & Others	27.4	21.2	67.6	4.8
Total	100.0	100.0	100.0	100.0
<i>Total</i>				
Government	96.6	88.5	94.6	99.9
Local Bodies	0.0	11.1	0.0	0.0
Endowments & Others	3.4	0.4	5.4	0.1
Total	100.0	100.0	100.0	100.0
Total (Rs. crores)	607.65	502.10	302.31	188.66

Source: *Education in India 1996-97* New Delhi: Ministry of Human Resource Development.

Centre-state shares in financing education

The union government began taking a serious interest in elementary education after the formulation of the National Policy on Education in 1986. The 73rd and the 74th amendments to the Constitution in 1993 enlarged the powers of local bodies in school administration. However, local bodies rely on grants from state governments, as they do not have much of a role in generating funds for education on their own. In a federal system, the financial responsibilities of the union and state governments are important.⁶⁷ However, education continues to be largely funded out of state budgets, supplemented by assistance from the union government. The states meet about 90 per cent of the total plan and non-plan expenditure on education with the union government meeting the rest. The union government meets a large proportion of plan expenditure, sometimes even about half in case of elementary education; but plan expenditure itself is a very small amount in relation to non-plan expenditure⁶⁸.

Union government finances education (and a few other sectors) in the states through schemes, which can be totally or only partially financed by the union government with the state government meeting the balance. During 1992-93 and 1998-99, in all Rs.438 crores was allocated by the union government to Andhra Pradesh and Rs.493 crores to Rajasthan for centrally sponsored schemes in education (Table 3.16).

⁶⁷ See Tilak (1989) for an elaborate discussion on centre-state relations in financing education in India.

⁶⁸ This is true of the state government expenditure, after the central transfers to the states through the Finance Commission.

Table 3.16: Expenditure on centrally sponsored schemes (total expenditure incurred during 1992-93 to 1998-99)

	Andhra Pradesh		Rajasthan	
	Rs. (Lakhs)	%	Rs. (Lakhs)	%
Operation Blackboard	12444.99	28.40	11160.17	22.62
Non-Formal Education	9794.07	22.35	7569.59	15.34
Teacher Education	4256.41	9.71	6890.42	13.97
Vocational Education	2554.36	5.83	2172.26	4.40
Science Education	593.12	1.35	412.17	0.84
Education Technology	1073.81	2.45	457.03	0.93
Environment Education	15.13	0.03	37.56	0.08
Integrated Education for Disabled Children (IEDC)	213.68	0.49	321.65	0.65
Computer Literacy and Studies in Schools (CLASS)	373.81	0.85	418.17	0.85
Yoga	26.00	0.06	0.00	0.00
Rural Functional Literacy Project (RFLP)	0.00	0.00	150.00	0.30
<i>Jana Shikshan Nilayam/Post-Literacy & Continuing Education (JSN/PL&CE)</i>	1641.49	3.75	570.59	1.16
Special Assistance Scheme (SAS)	330.32	0.75	549.99	1.11
Appointment of Hindi Teachers	2.00	0.00	0.00	0.00
Development of Sanskrit	11.91	0.03	70.49	0.14
National Scholarships	0.00	0.00	0.90	0.00
Scholarships for Talented Children	105.98	0.24	1.68	0.00
Upgradation of facilities for Scheduled Castes/Scheduled Tribes	0.97	0.00	4.11	0.01
Total Literacy Campaign/ Post-Literacy Campaign	4453.25	10.16	5433.45	11.01
Madrassa Education	30.55	0.07	31.55	0.06
Minorities Education	42.90	0.10	55.60	0.11
<i>External Aided Projects</i>				
DPEP	5850.00	13.35	1968.21	3.99
Lok Jumbish			11062.00	22.42
Total	43814.75	100.00	49337.59	100.00

Source: Based on *Education Profile of States and Union Territories 1997* and *Education Profile of States and Union Territories 1999*. New Delhi: Ministry of Human Resource Development.

Of the several schemes financed by the union government, four schemes accounted for a greater part of the share i.e. the Operation Blackboard Scheme, Non-formal education, Teacher Education Scheme and Total literacy and Post Literacy Campaigns. However, the importance given to elementary education schemes like Operation Blackboard, non-formal education and teacher training had declined understandably by the end of the 1990s. The Midday Meal Scheme that was started in 1995-96 now accounts for 40-50 per cent of the total central assistance.⁶⁹ Notably, central assistance for quite a few schemes flows somewhat erratically, a huge amount in one year, and zero amounts in others, or merely token amounts in some years.

⁶⁹ Mid-day meal scheme is not listed in Tables 16 and 17, as detailed information by states is not available.

Andhra Pradesh and Rajasthan, as two major states, each account for about one-tenth of the total expenditure of central government schemes on education (Table 3.17). It means that all other states together accounted for the remaining 76 per cent of the total expenditure incurred by the union government on these centrally sponsored schemes. Rajasthan accounted for one-fourth of the total central government allocation for literacy and post-literacy programmes. Rajasthan received 83 per cent of the total aid for Lok Jumbish. If we ignore Lok Jumbish for a while, one can state that Andhra Pradesh received substantial external aid, 18 per cent of the DPEP funds, and 19 per cent of the funds meant for Mahila Samakhya.

Table 3.17: Share of Andhra Pradesh and Rajasthan in some centrally sponsored schemes 1996-2000 (% of all States)

	Andhra Pradesh	Rajasthan
Operation Blackboard	7.30	6.10
Non-Formal Education	12.10	10.40
Teacher Training	5.80	14.90
TLC/PLC	4.60	25.00
<i>External Aided Projects</i>		
DPEP	17.80	8.90
Lok Jumbish	0.00	83.0
Mahila Samakhya	18.60	..
Total	12.60	11.10

Source: Srivastava, Ravi. (2005). Public Expenditure on Elementary Education, in Mehrotra, Santosh, P.R. Panchamukhi, Ranjana Srivastava, and Ravi Srivastava, (Eds) *Universalising Elementary Education in India: Uncaging the 'Tiger' Economy*, Oxford, New Delhi, p. 164.

In general, the union government's assistance to elementary education under the plan category has been increasing since the beginning of the 1990s, with the share of the states declining relatively. The increase in central expenditure on elementary education in the states in the 1990s was partly possible due to external borrowing and hence the growth in central expenditure on education can be regarded as 'aid-led growth' (Tilak, 1999a). Now in the recent years, most of the union government's allocations to elementary education under Sarva Shiksha Abhiyan (SSA) are accounted by the revenues received from education cess, and hence they may be regarded as 'cess-driven allocations' [Tilak, 2006c]. Further, while there are some important centrally sponsored schemes such as education for disabled children and mid-day meals, which aim at promoting equity in the system, the central transfers for education have not been found to be contributing to reduction in inter-state disparities. On the whole, central transfers for education have neither been equity-oriented nor have they promoted resource mobilization by the states (Tilak, 1989; see also Dev and Mooji, 2002).

External assistance for education

Since the early 1990s, foreign aid began to flow for the first time into primary education in India in sizeable quantities. Most of the external assistance received for elementary education has been routed through the union government. Hence external aided projects viz., the District Primary Education Programme (DPEP), Lok Jumbish and Mahila Samakhya figure as centrally sponsored schemes in the tables in this chapter. Foreign aid for elementary education, though continues, is now on a decline.

Though foreign aid was very small in quantum, it exerted considerable influence on the functioning of the school system in terms of strengthening planning and monitoring mechanisms, district level planning and decentralized administration including the setting up of village level education management committees. However, foreign aid also created different kinds of

problems—instead of being additional, to some extent it substituted government efforts at mobilizing domestic resources; it led to donor dependency that viewed foreign aid as critical for the development of the system and; since the aid was not well spread geographically, different kinds of regional inequalities arose (see Tilak, 1999b, 2008). On the whole, the external aid for primary education has been uneven. Bashir (2000) found that educationally developed states received higher per capita resources from external aid as compared to educationally poor states, though of the total central assistance (or external aid for elementary education) educationally backward states accounted for a larger proportion than the others.

Sarva Shiksha Abhiyan

Since 2000-01, the union government has started an umbrella scheme called the Sarva Shiksha Abhiyan (SSA). Aimed at rapid universalization of elementary education in the country, SSA proposed a new pattern of sharing financial responsibilities of elementary education between the union and the state governments. During the Ninth Five Year Plan period, the union government met 85 per cent of the plan expenditure on elementary education with the state governments meeting the remaining 15 per cent. However, the proportion changed to 75:25 during the Tenth Five Year Plan and it will further change to 50:50 by the end of the Eleventh Plan period.

The SSA also provides, drawing from the experience of the DPEP, for a new pattern of devolution of resources from the centre -- to the autonomous bodies/societies at state and district levels, in stead of channelling the resources through the government departments. While claims are being made on the efficiency and 'grand success' of the SSA, it is yet to be seen how far the marginalisation of the government departments would be efficient and desirable. Besides, the SSA is designed to be an umbrella programme covering all central schemes, including externally aided projects in such a way that the states have flexibility in utilizing the assistance provided for various schemes. It was also intended to provide safeguards for non-relapse of unutilized funds. The unutilised funds are saved in the *Prathamik Shiksha Kosh* (elementary education fund) set up recently by the union government as a non-relapsable fund for elementary education, along with other allocations under SSA. The Prathamik Shiksha Kosh mainly consists of the revenues received through education cess.

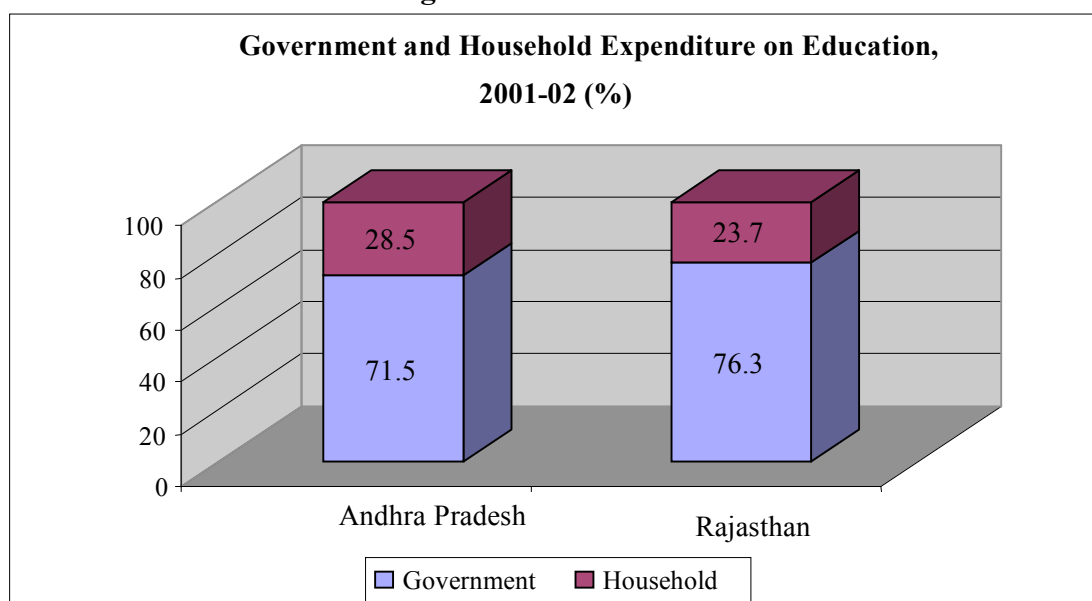
Household Expenditure on Education

Total expenditure on education consists of public expenditure as well as expenditure incurred by households. For a long period it was believed that families did not spend much on education and that the government met almost the entire total expenditure. But increasing evidence shows that households do spend significant amounts of money on acquiring education for their children, in addition to incurring opportunity costs. Family expenditure on education complements government expenditure, raising the total expenditure or alternatively, it substitutes government expenditure on education (Tilak, 2000a).

If the total expenditure on education is composed of government expenditure and household expenditure, according to the latest available estimates, published by the MHRD-a (2002-03), family expenditure constitutes 29 per cent of the total in Andhra Pradesh, and 24 per cent in Rajasthan; the balance – over 70 per cent- being government expenditure (Figure 3.5).⁷⁰ These figures relating to family expenditure on education seem to be rather high as compared to other countries.

⁷⁰ Based on 57th round of the NSSO (2001-02), the MHRD (a) presented these figures in their volume relating to 2003-04.

Figure 3.5



Source: *Selected Educational Statistics 2002-03*. New Delhi: Ministry of Human Resource Development.

In both Rajasthan and Andhra Pradesh, household expenditure on education is substantial. Families annually spend about Rs.800 per student on education. The expenditure per student increases at higher levels of education. For example, families in Andhra Pradesh spend on average Rs.430 per student per year on primary education, Rs.1,540 on secondary education and Rs.2,080 on higher education (Table 3.18).⁷¹ Between the two states, family expenditure on primary and upper primary education per student in Rajasthan is higher than in Andhra Pradesh, the difference being larger in upper primary education. But in the case of secondary and higher education, families in Andhra Pradesh incur higher levels of expenditure than those in Rajasthan, spending 50 per cent more..

Table 3.18: Average household expenditure (Rs.) per student in general education, 1995-96

	Primary	Middle	Secondary	Higher	All
Andhra Pradesh	430	820	1541	3081	825
Rajasthan	518	831	1400	2016	778

Source: Tilak (2000). Household Expenditure on Education in India: A Preliminary Examination of the 52nd Round of the National Sample Survey. National Institute of Educational Planning and Administration (July), New Delhi

While the increasing level of household expenditure on education is viewed by some as 'willingness to pay' for education, it is more likely that families feel compelled to spend on education because public expenditure is not adequate. Further, it was found that poor households spend higher proportions of their income than rich households on education, which

⁷¹ In fact, more recent studies (e.g., Panchamukhi, 2005) show that households spend a much higher level on primary and upper primary education both in rural and urban areas. This is based on a UNICEF survey in eight states conducted in 1999. See also Krishnaji (2001) and Majumdar (2001) for details on household expenditure on education based on surveys at grassroots level. Tilak (2000a, 2002a, 2002b) provides details based on NSS data and Tilak (2002c) based on NCAER (National Council of Applied Economic Research) survey.

creates new inequalities. The ‘wealth effect’, i.e., the difference in expenditure on education between the richest and poorest groups of population in public (government) schools among major states in India is found to be the highest in Rajasthan and Andhra Pradesh, after Gujarat (Tilak, 2002c). All this shows that higher levels of household expenditure on education would restrict access to education for the weaker sections. In this context, it is widely hoped that the Free Compulsory Education Bill (MHRD, 2005c) will ensure that families do not feel the need to spend on their children’s elementary education.

Requirement of Resources for Elementary Education

The requirements of the education sector in both Andhra Pradesh and Rajasthan are huge. Earlier Tilak (1994) estimated the requirement of resources for education in different states based on cost functions – costs regressed on the enrolments in and unit costs of education during the 1980s. There have been several other attempts at estimating the resources required for education (see, for example, MHRD, 1997 and Srivastava 2005). Based on a detailed estimation of costs of each item/programme, an expert group constituted by the Ministry of Human Resource Development (1999) estimated that an additional financial requirement of Rs. 4,201 crores would be required for primary education and Rs. 4,496 crores for the upper primary level in Andhra Pradesh for a ten-year period. Similarly, Rajasthan’s requirement was Rs. 6,508 crores for provision of universal elementary education— Rs.4,273 crores for primary education and Rs.2,235 crores for upper primary education Table 3.20 presents the details. The estimates included requirement of resources for improvement in access to education, improvement in quality of education, improvement in equity and improvement in processes— planning, management and the functioning of the school. Important line items of costs were construction of school buildings, buildings for district institutes of training (DIETs), block resource centres (BRCs) and teacher salaries. The next most important group of items included equity and quality related items such as school uniforms, scholarships, midday meals and teaching-learning equipment.

Table 3.19: Additional Requirement of Resources for Universal Elementary Education in Andhra Pradesh and Rajasthan (Rs. Crores)

Andhra Pradesh and Rajasthan (Rs. Crores)							
		Andhra Pradesh			Rajasthan		
		Primary	Middle	Elementary	Primary	Middle	Elementary
Access and retention: Non-recurring costs							
A1	Construction of schools with community supervision	1346.8	1391.3	2738.1	949.8	610.3	1560.1
A2	Provision of school equipments by decentralized procurement	7.2	102.3	109.5	16.3	60.6	76.9
A3	Establishment of new DIETs and up gradation of DIETs	1080.0		1080.0	1770.0		1770.0
A4	Establishment of cluster centres	3.5	4.2	7.7	7.1	3.7	10.8
A5	Est. of Block Resource Centres	23.1	1497.8	1520.9	18.6	874.6	893.2
Access and retention: Recurring costs							
A6	Teachers' salaries	1060.5	1056.3	2116.9	940.4	382.2	1322.6
A7	Teachers' support material	14.2	13.8	28.0	12.6	10.1	22.7
A8	Maintenance and repair of school infrastructure with community support	16.9	14.1	31.0	14.9	12.3	27.1
A9	Provision of sustainable replacement/ repair maintenance of school equipment	16.9	14.1	31.0	14.9	12.4	27.2
A10	Salaries of DIET staff	81.4		81.4	58.4		58.4

A11	Salaries of block level institutions	5.7		5.7	4.1		4.1
Access and retention: Special Needs: Non-recurring costs							
B1	Integrated education for disabled children	99.6	70.9	170.5	84.9	52.1	1376.0
Access and retention: Special Needs: Recurring costs							
B2	Teachers for disabled children	33.8	16.9	50.7	29.7	14.9	44.6
Access and Retention: Incentives: Recurrent Costs							
C1	Free uniforms	103.7	73.9	177.6	98.4	54.3	152.7
C2	Midday meals	93.0	59.1	152.1	70.7	13.4	84.2
C3	Scholarships	103.7	73.9	177.6	83.4	54.3	137.7
C4	Teaching-learning equipment	66.4	88.7	155.0	56.6	65.1	121.7
Curriculum and text books: Non-recurring costs							
D1	Curriculum and textbook	0.15		0.2	0.2		0.2
Curriculum and Textbooks: Recurring costs							
D2-	Curriculum and textbooks	25.5	13.8	39.3	22.6	10.1	32.7
D4	Community based monitoring, supervision and research	4.6		4.6	5.4		5.4
D5	Advocacy and environment building	4.6		4.6	5.4		5.4
D6	Classroom observation by resource persons	10.2	5.1	15.2	8.9	4.5	13.4
Total		4201.3	4496.1	8697.4	4273.1	2234.9	6508.1

Source: Report of the Expert Group on Financial Requirements for Making Elementary Education a Fundamental Right. New Delhi: Ministry of Human Resource Development, 1999.

These estimates are already dated, and the additional needs might indeed have arisen. Given the present levels of spending, and the growth rates, it is clear that both Andhra Pradesh and Rajasthan need special efforts to generate the required amount of resources for elementary education. Though some efforts are being initiated to generate resources through mobilization from the community etc., a bulk of the resources have to come from the state exchequer, and/or from the union government, at least as far as elementary education is concerned. Moreover, a strong rationale also exists for public financing of secondary as well as higher education (Tilak, 2004).

Concluding Observations

Despite some significant achievements, particularly in terms of quantitative expansion, the education situation in India, and particularly in Andhra Pradesh and Rajasthan, is dogged by major problems and even some conspicuous failures. Both these states rank poorly among the major states in the country with respect to any indicator of educational development. Many of their problems can be traced to inadequate investments, though adequate investments in education would not necessarily solve all problems automatically and satisfactorily. In this sense, finances form a *necessary*, but not a *sufficient* condition for an improvement of the education situation.

Both Andhra Pradesh and Rajasthan are characterized by backwardness in public spending or by chronic under-investment in education. Relatively speaking, Andhra Pradesh is economically an advanced state with a higher per capita income than Rajasthan. But Rajasthan could get ahead of Andhra Pradesh by an increased outlay on education. In terms of all the indicators— absolute levels of expenditure, per capita expenditure, percentage shares in SDP and in the budgets, Rajasthan now ranks ahead of Andhra Pradesh. Not only with respect to the total education sector, but also with respect to relative priority given to elementary education, Andhra Pradesh has lost its comparatively advantageous position to Rajasthan over the last 10-15 years. Andhra

Pradesh was spending higher amounts in the 1980s, from the beginning of the 1980s until 1990-91. But from 1993-94 onwards the relative positions changed. The economic reform policies were found to have a severer effect on budgetary allocations to education in Andhra Pradesh than in Rajasthan. By 2000-01 Rajasthan was spending about 40 per cent more than Andhra Pradesh.

It is possible that, as a result of all this, Rajasthan will improve its relative position in literacy and education development and march ahead of Andhra Pradesh even with respect to the human development index and reduction in poverty. For the same reason, some also portray Rajasthan as the 'most successful state of the past decade'. The improvement is not confined to literacy and elementary education; it extends to other levels of education also. As a result, the mean years of schooling of the population, a summary statistic of education development, has also improved in Rajasthan to exceed the levels attained in Andhra Pradesh.⁷² If these trends continue in both the states, then Rajasthan can hope to come of the infamous *bimaru* fraternity and Andhra Pradesh may take its place, at least in the field of education.

The review attempted in this chapter also confirms that

- Economically less advanced states can also spend a greater level of funds on education,
- Public expenditure on education matters significantly for the development of education; and
- Economic reform policies in general may harm public spending on education and impact even a priority sector like elementary education that is expected to be protected from the adverse impact of reforms, and also harm overall education development.

Towards improvement

There is a national goal of spending 6 per cent of the national income on education from the public exchequer. Both Rajasthan and Andhra Pradesh are much below the national target, ranking respectively 17th and 23rd among the 28 states, Rajasthan's performance being somewhat better. In 2004-05, Rajasthan allocated 4.5 per cent, while Andhra Pradesh allocated 3.6 per cent of SDP for education. To reach the target of allocating 6 per cent of SDP in a meaningful way, both the states, like many others in the country, have to step up allocations to education very substantially. In fact, given the magnitude of the unaccomplished task, it is felt that both states should aim at allocating much above 6 per cent of their respective SDP to education.

Allocations to education in the Five Year Plans in the two states do not show any systematic pattern. Since frequent and violent fluctuations in allocations to education and to various levels of education cause serious damage to its planned progress, it is necessary that such fluctuations be avoided. As a thumb rule, it has to be seen that the allocations to any sector of education—total and per student—do not decline over time in real prices. After all, decrease in expenditure per student in real terms, *ceteris paribus*, might suggest a decline in the quality of education.

Equally important is the issue of intra-sectoral allocation. Andhra Pradesh has been spending less than 50 per cent of its education budget on elementary education, while Rajasthan has been spending above 50 per cent. With respect to elementary education in state income, the gap between the two states has increased during the last 20 years, i.e., between 1985-86 and 2004-05. Rajasthan spends nearly double the ratio that Andhra Pradesh allocates to elementary education. Consequently, there is also much scope for improvement in intra-sectoral allocation of resources in both the states. However, given the inter-sectoral links between elementary, secondary and higher levels of education, it is also important to see that there is smooth and steady progress in

⁷² The mean years of schooling of population was 3.4 in Andhra Pradesh and 2.8 in Rajasthan in 1992-93 (for the whole population). But in 1998-99, the corresponding ratio for males was 4.7 in Andhra Pradesh and 4.9 in Rajasthan; for females the ratios were 2.8 and 2.2 respectively for Andhra Pradesh and Rajasthan (author's calculations based on *National Family and Health Survey*).

secondary and higher education as well. Sudden and significant reductions in allocations to secondary and higher education may inflict severe damage to balanced education development.

Further, there seems to be much scope for improving the inter-functional allocation of resources. Very meagre resources are allocated to non-salary items in school education. Since items like textbooks and other teaching-learning material and teacher training are found to have significant impact on the quality of education, allocations to these items have to be increased. Tentatively it may be suggested that at least 5 per cent of the elementary education budget be allocated to these items and another 5 per cent to equity related items.

Trends in public spending on education have not been smooth either in Andhra Pradesh or in Rajasthan. Since frequent and violent fluctuations in allocations to education and to various levels of education cause serious damage to its planned progress, it is necessary that such fluctuations be avoided. As a thumb rule, it has to be seen that the allocations to any sector of education—total and per student—do not decline over time in real prices. After all, decrease in expenditure per student in real terms, *ceteris paribus*, might suggest a decline in the quality of education.

The education systems in both the states requires a huge amount of additional resources. In 1999 it was estimated that Andhra Pradesh required Rs.8,697 crores and Rajasthan Rs.6,508 crores for the universalization of elementary education alone. Since elementary education is a 'fundamental right', the government has to finance elementary education 100 per cent without relying on any non-governmental resources. Voluntary community contributions, if any, would only supplement the supplementary activities or programmes of additional quality improvement. In short, the state has to come forward to liberally finance education, as after all, 'no nation goes bankrupt by investing in education'.

CHAPTER 4: GLIMPSES FROM THE VILLAGES AND SCHOOLS

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In this chapter, we describe the educational scenario in the sample villages. This purpose is to familiarize the reader with the socio-political context and characteristics of schools. For each state, we begin by describing one village in detail as a case study, and then describe the general socio-economic scenario, access to education and the schools in all the sample villages.

Rajasthan

Rajvillage – a profile

Rajvillage, situated about 75 kilometers from a big city, surrounded by eight small habitations or hamlets, was the largest village within a radius of 10 kilometers. The total population of Rajvillage as per the 2001 census was more than 4,000, and included OBCs (Cheepas), SCs and STs (Bairwa, Meena), and upper caste families of Rajputs, Brahmins, Sonis and Mahajans. People lived in separate caste-wise settlements. Surrounding habitations comprised mainly of SC and ST communities.

Despite proximity to the state capital, Rajvillage betrayed little urban influence. The chief occupation of the people was agriculture, but it was rain-dependent, and there was no irrigation or rainwater harvesting. People also reared livestock. Inadequate rainfall for several consecutive years had depleted the water resources and the economic situation of many families was precarious. Many families migrated to Jaipur and Delhi in search of work, returning just before the monsoon. A few people had shops for groceries, clothes, cycle repair etc. Some families from the Bairwa community in the village as well as in nearby habitations, owned power looms and wove carpets. A few upper caste people held government or other jobs.

A metal road connected Rajvillage with the surrounding areas. Buses, taxis and jeeps were readily available. The village had a small market of 20-30 shops, a post office, a bank, a health centre and a veterinary centre. It had electricity, but power was supplied for only a few hours a day and the lone telephone was usually out of order. Hand pumps were the only source of drinking water. The fluoride content in the water was way above stipulated levels; both the people and the government were aware of this. As a consequence, some families had left the village moved to the nearby town, commuting everyday to tend to their farms, while children returned to the village during holidays.

Politically, Rajvillage was rather inert. A member of the Meena community was elected as the sarpanch (president of the village panchayat) for five years, as the seat was reserved for STs. Subsequently, in the previous year, the seat was declared reserved for the OBC community, and a 21 year old young man from the Soni community was elected as sarpanch. Investigations revealed that the lone Brahmin (Chaturvedi) family of the village called the shots, the unwritten rule being that every decision of the village panchayat was routed through this family. Both the former and the current sarpanch were elected only on the approval of this family. One member of the family was a teacher in the Rajiv Gandhi Pathshala in the village, but rarely went to teach. Another taught regularly in a Shiksha Karmi school.

According to data collected under Sarva Shiksha Abhiyan (SSA), nearly 95 per cent children were enrolled in various schools. The village had three 'formal' government schools, four unaided private schools, one school for working children (run by an NGO and funded by the labour department of the central government) and one madrasa supported by DPEP. Notably, the nearby habitations had various types of alternative centres that imparted primary education. For upper primary education, children from these habitations had to come to Raj village. However, since the habitations were around 3 kilometers away from the main village, girls were not able to attend upper primary schools. The government schools in Raj village included a co-educational senior secondary school, a girls' upper primary school and a co-educational primary school.

The government primary school was located one kilometer outside the village and presented a curious scenario. This two-teacher school had a substantial building, a huge boundary wall and a playground. The 50 children who were enrolled belonged to SCs, STs and OBCs. Not a single child from the upper castes was enrolled in this school. During the ten visits that the research team made to the village, one of the two teachers was found missing, sometimes reportedly attending a block-level meeting or training, or conducting a survey. This school also served as the nodal centre (salary centre) for neighbouring schools and one teacher was usually found to be engaged in administrative paperwork during the visits of the research team. The school appeared to have no actual teaching activity, and people who lived nearby were not even aware that it was a school. They believed it was some wing of the education department!

We learned that eight years ago, some 200 students were enrolled in this school, and it was functional. However, subsequently, Shiksha Karmi and RGSP schools came up in the neighbouring hamlets, and the enrolment went down to 50 students. People living around the school said that persistent absence of one of the two teachers and the lackadaisical attitude of the school authorities effectively pushed the children out. The tragedy was that when this school became dysfunctional, the boys in the village were left with no government primary school. As a result, most of them attend a private school in the village.

The upper primary school for girls was located in the middle of the village and functioned regularly. This school had recently been upgraded from a primary school to an upper primary school. Classes 6 and 7 are already under way, and class 8 was expected to start in the next academic year. The school had 164 students. There were five classrooms, one of which served both as a store and for teaching. Two classes were taught in the veranda. Two of the rooms were so small that children were packed in like sardines. A hand pump had been installed next to the school. The school did not have a playground. There were six teachers, including the headmaster. During the last visit that the research team made to the school, three teachers had gone to participate in a block-level workshop and one was busy filling information forms along with an official from the department of literacy. Only two classes had teachers. The remaining children had been left to their own devices.

There was one coeducational senior secondary school with classes 6-12 in the village. An old school, it used to function as an upper primary school between 1947 and 1985, and was subsequently upgraded to a secondary school. Girls were admitted in class 9, as there was a separate government school for them up to class 8. Boys moved from the private school to this one in class 6. Children from neighbouring villages also came to this school as it enjoyed a good reputation. As of November 2005, 650 children were enrolled. Only 10 of the 17 posts of teachers had been filled but the school functioned regularly. Parents were happy with the quality of education and said that 89 per cent of the students who appeared for the terminal board examination (Class 12) cleared it. The headmaster told us that the school had all 17 teachers the year before last and the results were 100 per cent. However, when a class 6

mathematics class was observed, teachers taught conventionally. Interviews revealed that they had not attended any training or refresher courses. Inspectors never visited the school.

One madrasa, that was given academic and teacher salary support by DPEP, was dysfunctional, and people nearby did not even know of its existence. Its two teachers, appointed through a process, but practically by the CRCF, held jobs elsewhere (one had been working in Bombay for the previous six months!). Though people had complained to the CRCF about the school, he had taken no action. Similarly, the special child labourers' school supported by the central government reportedly catered to 50 working children, but in fact around 15-20 children attended the school. The headmaster informed us that the 'contract' to run the school was awarded to an NGO and admitted that it functioned as per the personal whims of the NGO leader. Five teachers received a salary much lower than due and were made to sign on blank receipts. Some of them were happy to get the reduced amount for doing nothing. In the event of an inspection, some teaching material was requisitioned. Officials at the district level were aware of the situation but chose to ignore it. The village panchayat also did not intervene on behalf of either of these schools.

The number of private schools in the village had increased over the years. The Saraswati Secondary School came up about 20 years ago, and at the time of the field-work, there were four private schools, two of which had come up in the last five years. While the Saraswati Secondary School had classes from 6 to 10, two schools had classes from one to eight and one school from one to five. The fee charged in these schools varied between Rs. 50 to Rs. 100 per month, the fee for higher classes being more. Irregular payment of fee was a recurrent problem, especially in times of economic hardship such as drought. Contact with parents was extremely limited.

The teachers in private schools were under-qualified. Only in one school, i.e. the Saraswati Secondary School, 4 (out of 11) teachers were trained. In other schools, teachers did not have teacher training degrees. Several were studying for college degrees. In one school, most of the teachers belonged to the same family. Teachers were paid between Rs. 800-1,500 a month, and the turnover was high. Teachers often left in the middle of the academic session. In terms of infrastructure, except for one school, all had adequate classrooms, and had toilets and drinking water facilities. However, none had a library or a playground.

The village panchayat has little say in the education system of the village. Our investigations included a School Development and Monitoring Committee (SDMC) of the government upper primary school for girls. The headmistress headed the committee and one of the teachers was the secretary. However, other members were not even aware that they were on the committee. No meetings were held. Apparently, the headmistress asked for signatures as and when needed from a few parents, who obliged. People appeared to have resigned themselves to the dismal state of education in the village.

In theory midday meals were served in all the government schools (except the higher secondary school) and the madrasa. However, since the government primary school and the madrasa were dysfunctional, with no students, there were not midday meals in practice, and it was difficult to ascertain how the allotted funds were used. The midday meal was served in the girls' upper primary school. '*Ghugur'i*' and '*dal-bat*' were served three times a week each. Children ate *dal bati*, but not *ghughri*, some taking it home to feed the cattle.

Sample villages in Rajasthan

The research team visited eight villages/habitations in the sample district in Rajasthan, in a block situated 30 km from the district headquarters. Table 4.1 gives a brief profile of the sample villages (the villages are numbered 1,2,3... for convenience).

Table 4.1: Profile of sample villages, Rajasthan

	Cluster 1			Cluster 2				
Village code	1	2	3	4	5	6	7	8
Population Characteristics								
Total population	7280	1811	351	4247	850	228	173	281
%SC	22.76	14.58	NA	33.32	5.76	100	46.24	0
%ST	3.30	27.61	NA	18.04	17.41	0	19.65	97.51
Literacy rate (in %)	46.58	38.10	32.49	39.25	25.67	26.31	20.23	17.4
Males	62.46	52.71	53.98	54.67	41.83	49.55	34.4	30.96
Females	29.28	21.62	5.41	24.00	6.91	4.27	3.75	3.50
Location								
Distance from nearest town	8 km	7 km	5 km	35 km	25 km	4 km	5 km	5 km
Distance from block HQ	8 km	15 km	8 km	35 km	25 km	38 km	40 km	30 km
Socio-economic context								
Caste and community structure	Gujars are the numerically strongest caste. There are also Jats, Brahmins and Yadavs. Muslim and Berwa (SC) are poor. Different castes live in separate localities. Approximately 150 teachers live in the village.	Upper castes, i.e., Brahmins, Rajputs and Gujars live in the main village. Meenas (ST) and Bairwas (SC) live in localities around the main village. Different castes live in separate localities.	98% families are prosperous Yadavs. Bairwa and Meena are poor and hold less agricultural land.	The village has upper caste communities, i.e. Brahmins, Sonis, Mahajans; Cheepas (OBCs), Bairwas (SC), Meenas (ST) as well as Muslims. SC and ST communities live in separate localities while the upper caste communities live together. One upper caste family is extremely dominant and controls the village panchayat.	Gujar community is numerically strong. The poorest group is Bairwas (SC). The Meena community is segregated in few hamlets near the village.	The habitation comprises of people from the Bairwa (SC) community. Houses are <i>katcha</i> and semi- <i>katcha</i> .	The habitation comprises of people from the Bairwa (SC) community. Houses are <i>katcha</i> and semi- <i>katcha</i> .	The largest group is Meena (ST).
Main occupations	Animal husbandry, agriculture, shops, labour work, carpet manufacturing.	Agriculture, animal husbandry, government service (army).	Agriculture, animal husbandry. Prosperous village. Several households own tractors.	Agriculture, tailoring, cycle repair. Many people migrate to the nearest city for work. Several upper caste people have government jobs.	Agriculture	Agriculture, casual labour. Economically poor village. People migrate to Delhi and Jaipur for labour work.	Agriculture. People have tiny land holdings and go to Jaipur in search of work.	Agriculture, labour work. Economically poor village. People migrate to Delhi and Jaipur for labour work.

Amenities available in the village	Pucca road, bus connectivity, bank, post office, television, telephone, public library, hand pump, pipe water supply.	Pucca road, telephone, television, public library but not opened hand pump, open well.	Electricity . telephone, television, hand pump, open well.	Pucca Road, bus connectivity, bank, post office, television., telephone, library in school, hand pump, tube-well, computer.	<i>Katcha</i> road, television, telephone, hand pump.	Pucca road, bus connectivity.	<i>Katcha</i> road, hand pump.	<i>Katcha</i> road, electricity, hand pump.
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As can be seen the villages varied a great deal in terms of population: The largest village (village 1) had a total population of 7,280 and the smallest (village 7) had only 173 people. Large villages, such as 1 and 4, had the maximum physical facilities—*pucca* (tarred) roads, bus service, bank, piped water supply etc. The two smallest villages (7 and 8) had the least physical facilities and lacked even a *pucca* connecting road. The mid-sized habitations fell in between, they were connected by pucca roads and had televisions and telephones but lacked facilities such as banks and post offices. Piped water supply was available only in the largest village.

Notably, the smallest villages had a very high proportion of SC and ST population. Among the three smallest villages, nearly all the population in village 8 were STs and all the population in village 6 were SCs and in village 7 SCs and STs together comprised around two-third of the population. In the larger villages, where there were several castes, different castes lived in separate localities. In village 4, a Brahmin family was found to be dominant.

Agriculture and animal rearing were the major occupations in all the villages. In the larger villages, shop keeping and trades such as bicycle repair were also prevalent. In village 1, carpet weaving was also an occupation and nearly 150 teachers working in schools in the neighbouring villages also stayed in the village. People from around ten families from village 2 worked in the army. In general, occupational opportunities within the villages were very limited., and people migrated to the cities for work, particularly from the three smallest villages

Literacy rates were low. In no village did the female literacy rate touch 30 per cent, while the male literacy rate varied between 30 and 63 per cent. The smaller villages had extremely low literacy rates. In fact, the female literacy rates in all but the three larger villages were below 7 per cent and in the tribal habitation a mere 3.5 per cent.

Educational Facilities

As in the case of physical infrastructure, educational facilities also varied a great deal across villages. Only in the two largest villages in the sample (1 and 4) were there senior secondary schools within the villages. In contrast, in three of the smallest villages (6, 7 and 8), there were no formal schools, but only alternative centres such as Rajiv Gandhi Pathshala, Shiksha Karmi school and a DPEP alternative centre. As noted earlier, these were villages with a very high proportion of SC/ST population, poor connectivity, inadequate physical infrastructure and the lowest literacy rates. Children in the mid-sized villages could access education up to Classes 5 or 8 within the village. For instance, village 2 had an upper primary school, while villages 3 and 5 had formal primary schools. The schools had not been planned well. For example, in village 6 next to the primary school was an upper primary school that belonged to another village. Availability of private schools followed a similar pattern: these were available only in two of the largest villages. In fact, there were nine private schools located in the two large villages. All the villages had alternative centres of various types which catered mainly to the SC and ST communities or to the Muslim community.

Table 4.2: Educational facilities in the sample villages in Rajasthan

	Cluster 1 (C1)			Cluster 2 (C2)				
Village code	1	2	3	4	5	6	7	8
Educational profile								
Highest level of education available in village	Class 12	Class 8	Class 5	Class 12	Class 5	No formal school. Class 5 in alternative school.	No formal school. Class 5 in SK school.	No formal school. Class 5 in Rajeev Gandhi Pathshala.
Formal government schools	Senior secondary school (co-educational), high school (girls), upper primary school (boys), primary school (girls).	Upper primary school	Primary school	Senior secondary school, upper primary school (girls), primary school.	Primary school*	Nil	Nil	Nil
Other types of schools	Madrassa supported by Rajasthan Wakf Board	2 Rajiv Gandhi Pathshalas	Nil	1 Bal Shramik School for Child labourers. 1 madrassa run by DPEP.	1 Rajiv Gandhi Pathshala	Alternative school	Shiksha-karmi school	Rajiv Gandhi Pathshala
Number of private schools available	4 upper primary schools, 1 school for classes 1 and 2.	Nil	Nil	1 private school up to class 10, 2 upper primary schools, 1 primary school.	Nil	Nil	Nil	Nil

*In the campus of this school was situated an upper primary school that belonged to a different village.

Sample Schools

The sample schools illustrate the disparate educational establishments that exist within the state system. In terms of teachers and infrastructure, the schools could be characterized into two categories: schools that had an adequate number of qualified teachers and reasonable infrastructure, and those that did not. In the former category were high schools, situated in the larger villages, which were reasonably well equipped, with a fair number of teachers, though vacancies could be a problem. Upper primary schools, situated in the medium sized villages were less well endowed, but had adequate infrastructure and qualified teachers. The private school had reasonable physical infrastructure, though it lacked a playground or library, but

teachers were a constraint. Posts were vacant and teachers moved to other jobs often, and, in the sample private school, there was not a single trained teacher.

In the second category were formal primary schools and other informal educational establishments. In formal primary schools, physical infrastructure was not too bad and the teachers were qualified, but the number of teachers was too small for the schools to function well. In the single teacher school there was little time for education and even in the two-teacher school, the teachers were busy in management and administrative work. In terms of the number of teachers, the Shiksha Karmi school was actually better-off than one of the formal primary schools, as it had two teachers who, though not formally well qualified, had undergone rigorous training. However, the Shiksha Karmi school lacked even the barest minimum of physical facilities. The Rajiv Gandhi Pathshala and the alternative school had one teacher each, with poor qualifications and training and some bare minimum physical facilities.

Table 4.3: A picture of sample schools in Rajasthan

Rajasthan					
School description	Enrolment in 2003-04	Teachers	Building	School functioning	SDMC/ Community dynamics
Higher secondary school in village 4. Co-educational Grades taught: Classes 6-12. Commerce and Science are not taught in Classes 11 and 12.	The enrolment in classes 6-8 was 220, with 75% boys, and 64% SC/ST children. Many new children from nearby villages enroll in Class 9.	11 regular teachers (with 8 vacancies), 2 guest teachers, 4 non-teaching staff.	23 rooms, separate room for the headmaster, separate toilets for boys and girls, drinking water, playground, no furniture for children who sit on the floor, library books kept in a cupboard.	Observation of class 6 showed an overcrowded classroom where the teacher taught geometry by lecturing to the children without finding out whether they followed or not. Several children in the class could not even do basic arithmetic. In 2003-04, 81 children took the Class 8 examination, 37% scored 33% or less, 58% scored between 33% and 60% and 5% scored above 90%. No midday meal is provided.	SDMC is inactive. Most members belong to neighbouring villages and not the village in which the school is situated. Minutes are usually sent to members to sign. Results are not discussed with parents; children are simply handed report cards.
Upper primary school in village 4. Girls' school. Used to be a primary school, in 2004 Class 6 was added.	There were 164 students (there was no Class 6 in 2003-04), 49% SC/ST, categories and just 4 boys.	1 head teacher, 5 teachers, 1 para-teacher. All women teachers (one ST). All had college degrees and were trained. The PTR was 23.4	5 rooms and 2 covered verandahs, two toilets and one latrine, hand pump, which is also used by villagers, playground, no furniture for children who sit on the floor, there are 400 books but these are kept locked in a box.	Teachers spent a great deal of time chatting amongst themselves and with people who came to fetch water. Two teachers did not come to school on all the three observation days. Teaching time was as little as 10 minutes in a 35-minute period. Teaching was textbook based. During the teaching observed, teachers solved questions given in the textbook on the blackboard. Sometimes children were asked to solve questions on the board. No other activities were observed. Midday meals provided up to Class 5. Earlier <i>ghooghri</i> was provided, which the children often did not eat. At the time of the study <i>dal bati</i> was provided thrice a week, which children eat.	SDMC members, nominated by the head teacher, usually do not attend meetings, but sign papers as needed.
Upper primary school in village 1. The school originally had classes 6-8 (2 sections each), and was a boys' school. In 2004-05, Classes 1 to 5 were added, and	218 boys were enrolled in Classes 6 to 8 and 29% belonged to SC/ST categories.	9 regular teachers, 4 female, one SC. All BA, B.Ed., and one woman para-teacher.	13 rooms, 1 room for headmaster, 3 toilets and 2 latrines, hand pump and pipe line supply, no playground, library books kept in a	During a classroom observation of Class 2, 30 minutes of teaching time was spent marking attendance. The teacher spent another 5 minutes talking on the phone. In the remaining time, she was pleasant and helpful to the children. In 2003-04, 66 children appeared for the Class 8 examination, 64% scored less than 33%, and no child scored more than 60%. Midday meal is provided but many children do	SDMC was formed in after a consultation with the people of the village. Regular meetings are held, and attempts have been made to access funds from the village panchayat for necessary infrastructure.

the school was made co-educational.			trunk, not given to children for fear of being spoilt.	not eat it because they get bored with it.	
Upper primary school in village 2. Co-educational. Classes 1-8.	256 students were enrolled, 66% boys and 58% form SC/ST categories.	5 regular teachers, 4 male and one female, one from ST category. All teachers are trained. One female para-teacher.	5 rooms, 1 room for headmaster, 1 store, separate toilets for boys and girls, hand pump, few books in cupboards.	During classroom observation of Class 6, the teacher came 15 minutes late, spent only 10 minutes teaching and looked out of the window a great deal during this time. Teaching comprised solving problems from the textbook. In 2003-04, 23 children took the Class 8 examination, 35% scored less than 33%, 52% scored between 33% and 60% and a mere 13% scored more than 60%.	The SDMC was dysfunctional. By and large, whatever the headmaster said was accepted.
Primary School in village 3. Co-educational. Classes taught: 1 to 5.	73 children enrolled, 56% boys. 14% children from ST families.	2 regular teachers. Both college educated (one has an MA) and are trained.	3 rooms and 1 office room, separate toilet for boys and girls, hand pump, no playground and library.	School functions regularly. Villagers praised the teachers. There is no timetable. Children are divided in two groups and each teacher tries to spend at least half an hour with each class. A great deal of teachers' time is spent in administrative work and midday meals. Midday meals are served regularly and eaten.	SDMC members do not always attend meetings. In such cases, teachers ask them to sign on minutes, papers etc. One SDMC member is active and takes an interest in the school. The community has contributed for tree plantation within the school.
Primary school in village 6. Co-educational. Classes taught: 1 to 5.	67 students, were enrolled, 73% boys, 10% SC/ST	1 regular teacher, trained.	3 rooms and a hall, 1 toilet not used, hand pump, playground. There are 24 library books, but these are not given to the children.	All children sit together as there is only one teacher. The teacher somehow managed the school and admitted that he did almost no teaching.	SDMC is dysfunctional. SDMC members sign on minutes etc, as needed.
Private school in village 4. Co-educational Classes taught: 1 to 9.	198 student were enrolled , 85% were boys, 19.6% children from SC/ST categories. Children often leave school in the middle of the session.	One headmaster, three teachers. The PTR is 49.5 and there is multi-grade teaching. Teachers either have college degrees or are studying in college. There is no trained teacher in the school. Teachers are paid between Rs. 1,000 to Rs. 1,400 per month.	10 rooms 3 toilets, 1 latrine, tube well for drinking water. There is furniture for the older children to sit on. No playground, no library.	Fees: Classes 1 to 5: Rs. 50 per month Classes 6 to 8: Rs. 100 per month, Rs. 100 as entrance fees. Parents don't pay fees on time. Teacher turnover is high, teachers often leave in the middle of the academic session. Teaching was textbook based. Teachers solved questions from the textbook or asked students to solve the same on the blackboard. There is no midday meal.	All rights are with the school manager. As per the headmaster, there was little contact with parents as they were not educated and did not take an interest.
Rajeev Gandhi Pathshala in village 8. Co-educational. Classes 1 to 5.	46 students are enrolled., all STs.	1 para-teacher. When recruited had passed grade 10. After recruitment, had passed senior secondary and got a teacher	1 room, 2 halls, no toilet, hand pump, playground, no library	The school does not open everyday. Hence villagers do not send their children to school. There is almost no educational activity in the school.	The teacher had nominated his own uncle as the SDMC chairman. People in the village were not aware of the SDMC. The community does

		training diploma.			not complain about the school because they do not want the teacher to lose his job.
Shiksha Karmi School in village 7. Co-educational. Up to Class 4.	37 students were enrolled, 62% boys, all the from SC community.	2 Shiksha Karmis. One SK is educated up to senior secondary and the other up to Grade 8. Both have been given training that is the norm for all Shiksha Karmis.	No building, classes are held under a tree. Some material of the school is kept in the Shiksha Karmi's house. Water is available 500 metres away.	The school functions regularly. During the classroom observation, teaching was found to be conventional, such as the teacher writing numerals on the board and the children copying them.	SDMC is not active. Villagers are poor and are not able to contribute financially to the school.
Alternative school for six hours in village 6. Co-educational Classes 1-5.	51 children enrolled, 47% boys, all from SC category.	1 para-teacher, has a B.A., but no teacher training certificate or degree.	1 room, no toilet, no hand pump, playground, no library.	The school opens only occasionally. There is almost no educational activity in the school.	The school is managed by an SDMC of another village.

In terms of functioning, the Rajiv Gandhi Pathshala and the alternative school could not really be called 'schools', as they simply did not provide regular opportunities for teaching and learning, and in fact functioned only sporadically. Children enrolled in such schools may have technically been 'in school', but in practice did not have access to a regularly functioning school which they could attend on a daily basis. The Shiksha Karmi school did much better and it would appear that a focus on the teacher, i.e., a careful process of selection and training, had had a positive effect. In the two formal primary schools, there was little space for education in the single teacher school. In the two-teacher school, teachers did their best, but each class got around just an hour's teaching every day.

While the category two schools suffered from severe constraints, in terms of functioning, even the category one schools did not perform well. The two important visible constraints here were that on the one hand, teachers did not spend the allotted time teaching, and on the other, when they did, the teaching process was often of a perfunctory nature. In no school could intelligent, inspired teaching and a vibrant classroom be observed. Achievement tests showed learning levels to be extremely poor (see Chapter 6 for more details).

Midday meals were provided to primary school children, including to those in the Rajiv Gandhi Pathshala, Shiksha Karmi and alternative schools. In formal schools these were provided regularly, though in some schools, children did not always eat them because they got bored with the food or for some other reason. As the Rajiv Gandhi Pathshala and alternative school did not function regularly, and children did not go to these schools on a regular basis, it is doubtful whether they had midday meals regularly either. In the Shiksha Karmi school, the midday meal was served regularly, and eaten. SDMCs were largely inert. Members were nominated by the school headmasters and teachers and usually simply signed on papers as required. In fact, only in one village was there any evidence of an even partly active SDMC.

Focus group discussions with students of class 6 in the sample schools revealed that nearly all the children, especially girls, did a fair amount of housework, and also tended to animals. Many children also helped in farms, especially during the harvesting season, or in other family businesses such as shops etc. Children often did not attend school during the busy agricultural season, or during festivals. There was little reading material available to these children at home, and many had illiterate parents. Only a very small number of better-off children

reported that their families bought a daily newspaper. Newspapers were usually available at the local teashops. While some children read these newspapers, many said that they had never read any newspaper, books etc. In one village, comics were available on rent in a shop, and a few children from the private school reported borrowing them.

Andhra Pradesh:

Andhravillage – a profile

Andhravillage was situated five kilometers from its mandal headquarters, and 105 kilometers from the district headquarters. It was a typical Telangana village with a population of 1,800 with more than 50 per cent women. It had a highly stratified social structure. Gowda (BCs) were demographically, politically and socially dominant in the village. A small proportion of Reddys (Other Castes) and Padmashali (BCs) also lived in the village, and were among the well off people. There were Madigas (SC) and Lambadas (ST) as well. Madigas formed the poorest group. Lambadas lived a separate habitation 2 kilometers away from the main village. Telugu Desam Party was dominant in village politics. The seat for the gram panchayat (GP) president was reserved for STs and a college educated (B.A., B.Ed) person of ST category was the president. However, he did not get much support from the other village panchayat members.

Andhravillage had facilities like television and cable reception, telephone, post office and a cooperative bank. An Andhra Pradesh State Road Transport Corporation bus made four trips a day. Agriculture was the main occupation, and women engaged in beedi rolling as well (there were four beedi factories in the surrounding area), while some Gowda families sold toddy. There was a high degree of landlessness in the village. The SC and ST population, largely comprised marginal farmers and landless wage earners. Given the severe drought conditions in the last five years, some small farmers had sold their land to city dwellers for farmhouses and now worked as labourers on the same land. Some people migrated to Hyderabad for work.

Most of the children worked on the farms as well as in local small industries/home based work like bidi rolling. Children also did a fair amount of domestic work. Interaction with children revealed that most worked before and after school and even engaged in wage work during holidays. Girls helped their mothers in augmenting family incomes. They were not paid when they accompanied their mothers, but did get a small wage when they went to work independently. In some families, when girls reached the age of 12 to 13 years, their mothers stopped taking them out for work. Families started looking for suitable marriage partners, and marriage prospects were not good for girls who went out of the house. When a girl was formally affianced, she engaged only in domestic work.

The village had two primary schools (one on the main village and another in the ST habitation) and a high school (classes 6-10). There was one private school in the village, which was English medium, and many better off children from the village attended this school. The fee charged was Rs. 50 per month. Private tuition was also available at Rs. 20 per month, and many BC and OC children accessed it. Several children, especially boys from the dominant castes, also attended schools outside the village and school vans came from a nearby town to pick up the children. According to the enrolment records of the primary school of Andhravillage, it had a child population of about 237 with 97 boys and 140 girls in the school going age group (6-14 years). As per the records, all the school-age children were in school. An in-depth analysis, however, revealed that the data did not clearly represent the total school going children of the village.

The high school in Andhravillage was one of the oldest schools of the district and had the distinction of producing a number of college graduates. Located in the main village, with a green ambience with trees all around, it had a *pucca* building with 5 classrooms, one room for

the headmaster, a staff room, an office room and two small rooms used as a laboratory and a library. It also had an auditorium. There were three toilets, one used by the teachers and the others by girls. There was a bore well and a tap within the school premises. Children had benches to sit on, but they were very high and uncomfortable. Children kept their school bags on the ground and when they wanted to take a pen or pencil, they had to bend in the cramped space between the benches. The width of the benches was inadequate and not conducive for written work, there was no hand rest or back rest.

The high school enjoyed a good academic reputation in the Mandal (meaning good results) and many children from the surrounding villages come to this school, though a few children of high school age of well-to-do OC and BC families attended private English medium schools in the nearby town. However, the school had insufficient subject-wise teachers, as five of the teachers were vidya volunteers. Among them, three had graduated recently and the other two had studied up to the intermediate level. None of them had had any training for teaching. The regular teachers said that the vidya volunteers were enthusiastic but did not know how to teach. The classrooms were overcrowded. Free textbooks were available to the eligible children. Ironically, this meant little because children were encouraged by the teachers to purchase supplementary material in the form of guidebooks and question-answer booklets. Despite instructions from the state government, 'corporal punishment' was still practiced to deal with misbehaviour in the classrooms.

This school had a School Education Committee (SEC). A previous headmaster had created a School Development Committee (SDC) over and above the SEC, in which he had included all the members of the SEC, two donors and a previous chairman of the SEC, former and present GP presidents, a retired Village Development Officer (VDO) and a local leader who had contributed to the school's development. This committee met twice in a year to discuss infrastructure and school development issues as well as funds to be raised. The SEC chairperson and headmaster were the joint account holders of a separate school account. A teacher, who was not a member of this committee, maintained the accounts register.

During the last few years, the formal SEC and the informal SDC had constructed a compound wall and extra classrooms and had improved the sanitation facilities in the school. They had also constructed a hall by accessing funds from Members of Parliament (MPs) and Members of the Legislative Assembly (MLAs). The committees also paid an additional honorarium of Rs. 200 to all vidya volunteers, and had appointed an extra vidya volunteer in the school. They had also sent 42 children for an exposure visit to north India with funds mobilized from the community. In 2004-05, they published a school souvenir and released it on school day celebrations. In the previous year, they had a meeting with the Parent Teacher Association (PTA), in which they addressed school problems. The involvement of the community in the construction of school buildings, in providing furniture, toilet and drinking water facilities and in actively participating in school functions had helped improve the interaction between the community and the school.

Children of this high school also participated in various programmes and competitions. About 60 children participated in a 'one-day one lakh plantation' programme conducted by the National Green Corps in another district in 2003-04. Now, 40 students of classes 8 and 9 (20 from each class) were members of the National Green Corps. They also won a Green School Award and a cash award of Rs. 5,000 from the Forest Department for keeping the school clean and green. Besides this, the children had also participated in *kabaddi* and *kho-kho* competitions and literary competitions conducted at the divisional level.

Every year a couple of students were awarded merit scholarships. For special (physically disabled) children, generous scholarships were available, but the children had to visit government offices regularly to get a certificate that enabled them to apply for scholarships. The children were expected to bring a certificate from an orthopaedic doctor to get the

scholarship, for which they had to go to the district headquarters. This was difficult for many children. As a consequence, in 2004-05 only two children got Rs. 850 per year as stipend. Moreover, the scholarship amount was rarely released in time. The school authorities did not approach the government for the renewal of scholarships to special children. The follow-up was left to the children and their parents.

A vidya volunteer who studied in the same school, said that when she was a student, girls had no chance to play in the ground. They used to remain indoors or in the veranda, playing traditional indoor games while the boys used the ground for outdoor games. According to her this was so because the teachers took no interest in the games that the children were playing. Now, as a teacher, she played with the girls.

Sample Villages

The socioeconomic situation of the sample village is presented in Table 4.4, which shows that these villages enjoyed some similarities and some differences with the sample villages in Rajasthan.

Table 4.4: Profile of sample villages in AP

Village	Mandal 1 (M1)			Mandal 2 (M2)		
	1	2	3	4	5	6
Population characteristics						
Total population	1800	1392	392	2969	2169	1305
% SC	25.06	12.36	17.35	25.46	19.55	18.93
% ST	1.61	18.03	7.14	4.08	3.04	2.53
Literacy % (T)	51.89	43.53	57.91	54.80	44.44	42.01
Literacy (M) %	64.53	52.59	73.60	66.01	56.57	52.10
Literacy (F) %	39.58	36.02	42.05	42.88	33.86	32.33
Distance from nearest town (in km)	25	30	33	25	28	28
Social composition	OBC* politically/socially powerful. SC** and ST**.	OBC Reddy (land owning), SC.	Reddys, OBC, SC.	Reddys (land owners and dominant). SC poorest. OBC small marginal farmers.	OBC majority, Reddy and Vysya powerful. SC poorest. Few Muslims.	Reddy landlords, OBC next. SC and ST poor. Few Muslims and Christians
Main occupations	Rain-fed agriculture.	Rain-fed agriculture, about a 100 villagers work in sand filters.	Mainly wage labour, many sold land and work on other's fields.	Irrigated, 2 crops. Few work in cement factory in neighbouring mandal and rest are engaged in petty business.	Irrigated land, employment available throughout the year. Women go to Guntur for chilly and tobacco plucking.	Irrigated, grow mango and sweet lemon. Other work available throughout the year.
Amenities available in the village	Pucca road, bus connectivity, post office, telephone, nationalized bank, street lights, cable television.	Katcha road, bus service, street lights, television through antenna/cable, drinking water available but not very clean.	Katcha road, bus service, television through antenna/cable, drinking water.	Pucca road, bus connectivity, post office, telephone, cooperative bank, street lights, cable television.	Katcha road, bus service, street lights, television through antenna/cable, drinking water facility.	Katcha road, bus service, street lights, television through antenna/cable, drinking water facility.

Source: Field study data.

Note: * Gowda, Padmashali, Yadavs, Golla, Mutharasi, Kummari, Anusala, Vadla and Mangli, ** Madiga, Mala, *** Lambada, Yerukala.

There was only one very small village while the rest had a population between 1,300 and 3,000 people. This reflects the fact that as compared to Rajasthan, Andhra Pradesh is in general characterized by larger settlements. As in the case of Rajasthan, the two largest villages in the Andhra Pradesh sample, were connected by pucca roads, while the rest had *katcha* roads. The villages were served by a functioning bus service—some of them had buses plying six times a day and streetlights were available in all but the smallest village. In one village, the quality of groundwater was reported to be very poor.

All the villages had a mixed population (unlike the sample in Rajasthan where SC or ST communities only lived in the smallest habitations). Several castes classified as ‘backward’ such as Gowda, Padmashali, Yadavs, Chklis, Vaddera and Gollas along with other castes such as Reddys and Vysalas enjoyed good social status. Reddys, along with other higher castes, were often big landowners. As in the case of Rajasthan, SC communities such as Malas and Madigas, lived in segregated habitations and formed the poorest groups, usually working as wage labourers. In village 2 there was a small hamlet of ST families, mostly landless, who worked as casual labourers.

Three of the sample villages depended on rain-fed agriculture while the others had access to irrigation facilities. The three villages that did not have irrigation facilities also recorded high rates of seasonal migration. Many SC and ST families migrated to Hyderabad city or other districts for work while many migrant families of Orissa come to these mandals to work in the mines, brick-kiln and cement factories.

Educational Facilities

Table 4.5: Educational facilities in sample villages in AP

	Mandal 1 (M1)			Mandal 2 (M2)		
Village	1	2	3	4	5	6
Educational profile						
Highest educational facilities available	Class 10	Class 7	Class 5	Class 10	Class 7	Class 5
Schools	2 PSs, 1 HS	1 UPS	1 PS	1 PS, 1 HS	1 UPS	1 PS
Alternative centre	Nil	Nil	Nil	Nil	1 Alternative School*	Nil
No. of private schools available	1 UPS Private tuition available	Nil	Private tuition available	1 HS Private tuition available	Private tuition available	Nil

* The alternative school was located in a small hamlet of 22 Lambadas (ST) families who are mainly wageworkers and poor.

As in the case of Rajasthan, high schools existed in the largest villages. But while in Rajasthan several hamlets and villages were served only by non-formal school facilities, in Andhra Pradesh an alternative school existed only in one tribal hamlet, along with a formal school in the village. As per the state policy, such centres were not meant to substitute formal schools, but provide education to children who had dropped out or could not attend formal schools.

As in the case of Rajasthan, private schools existed in the larger villages. Better-off villagers from all the villages also sent their children to private schools in nearby towns and school buses or vans were available to transport the children. The fees in such schools ranged from Rs. 100 to Rs. 200 per month per child. Local private schools in the villages charged less, between Rs. 20 to Rs. 50 per month, though some were more expensive. Some families also sent their children to residential schools, paying up to Rs. 12,000 per annum. Private tuition was available in four of the sample villages. In village 5, children paid Rs. 400 per annum for

tuition up to Class 5 and Rs. 600 per annum for tuition for higher classes. Typically, the tuition classes helped children with their homework and in preparing them for examinations.

Sample Schools

Table 4.6 gives a brief picture of the sample schools in Andhra Pradesh.

Table 4.6: A picture of sample schools in Andhra Pradesh

School description	Enrolment	Teacher	Building	School functioning	SEC/Community dynamics
High school in village 1. Grades 1 to 10. Co-educational.	Enrolment in 2003-04 Classes 1 to 7 was 330, 55% students enrolled were girls, 38% belonged to SC or ST categories.	9 regular teachers, 5 vidya volunteers 3 non-teaching staff.	10 classrooms, 1 auditorium and 1 lab room. Separate rooms for headmaster and staff. 1 toilet used by teachers. Hand pump in school premises. Playground. 1,000 library books kept in cupboard in staff room. TV, VCR, radio.	Records available, though not well maintained. School observations of Class 6, taken by a vidya volunteer showed teaching to be unimaginative, with the teacher solving problems on the blackboard with little concern about whether children understood or not.	SEC was extremely active. The previous headmaster had taken a great deal of interest. SEC had improved the school infrastructure, paid extra amount to vidya volunteers, organized exposure visits etc. Parents had also started showing an interest in the school.
High school in village 4. Grades 1 to 10. Co-educational.	Enrolment records could not be obtained for Classes 6 and 7. Enrolment in Classes 1 to 5 was 85. 60% students enrolled were girls, 59% belonged to SC or ST categories.	9 regular teachers, 3 vidya volunteers, 3 non-teaching staff.	7 rooms, 2 toilets used by teachers. Water storage tank with taps. 500 library books. Insufficient playground space.	Records available, though not well maintained. During classroom observation, a physical education teacher taught mathematics to Class 6. The teaching was mechanical and one-sided, the teacher did not show an interest in whether the children were grasping the concepts or not.	The SEC and PTA are not very active.
Upper primary school in village 2. Grades 1 to 7. Co-educational.	Total enrolment in 2003-04 was 240, 47% of those enrolled were girls, 31% were from SC/ST categories.	5 regular teachers, 3 vidya volunteers.	5 pucca rooms, 2 semi-pucca rooms. One room for headmaster and staff. Veranda used for anganwadi. No drinking water facility in school premises, water available at half a kilometre. 2 toilets used by teachers. 300 library books. Playground.	Records available, though not well maintained. Classroom observation in Class 6 showed that though the teacher made an effort to explain algebra to the children, he was unable to. In Class 2, the teacher carried on one-sided teaching, without bothering to check if the children were understanding what he was teaching.	The SEC met three times in the previous year. There was no SC member in the SEC because of lack of awareness that one such member had to be elected. The SEC took an interest in school repairs and midday meals.
Upper primary school in village 5. Grades 1 to 8 (grade 8 started in 2005-06). Co-educational.	Total enrolment in 2003-04 was 283, 47% of those enrolled were girls, 22% were from SC/ST categories.	6 regular teachers, 2 vidya volunteers, 1 teacher paid by panchayat, 1 non-teaching staff.	4 rooms and 4 verandas. 2 toilets 1 for girls, 1 for teachers. Bore well under repair. 125 library books. Playground	Class 2 observations showed that the teacher could not manage the class, and though he encouraged the children, he could not engage them in meaningful learning activities. In Class 6, the teacher wrote problems from the textbook on the blackboard, and explained them. She also gave problems to the children and helped them solve these.	Teachers have a fair degree of interaction with the parents. Regular monthly SEC meetings are held, and the SEC chairperson visits the school every week. Party politics is important in the SEC. TRS* was dominant in the SEC. The SEC had been active in school repairs and selection of vidya volunteers.
Primary	Enrolment in 2003-	2 regular	2 rooms only,	Records available, though not	Members were elected to the SEC

school in village 3. Grades 1 to 5. Co-educational.	04 was 43, 53% of those enrolled were girls, 40% were from SC/ST categories.	teachers.	classes share them. Veranda used for anganwadi. No toilets, tap and bore well. 125 books in a trunk. Playground.	well maintained. Classes 1 to 4 sat in the same classroom. The teacher made considerable efforts to engage their attention, using colourful and attractive teaching-learning materials. However, the teacher taught only Class 2, and could not manage the other classes. In another observation all the students from Classes 1 to 4 participated in the same activity.	on the basis of the donations that they had made to the school.
Primary school in village 6. Grades 1 to 6. Co-educational.	Total enrolment in 2003-04 was 133, 54% of those enrolled were girls, 38% were from SC/ST categories.	3 regular teachers.	3 classrooms, 1 toilet, 1 bore well. Playground. Books in a trunk.	The class was managed casually, though the teacher was friendly with the children. The teacher taught the children addition without much skill. The brighter children grasped the concepts, but others could not.	At the time of elections, there was a conflict between the political parties, so parents of children who were doing well at school were nominated to the SEC. Meetings were held in the previous year, SEC members took part in the enrolment drive, but no special initiatives have been taken.
Private school in village 4. Grades 1 to 10. Co-educational	Enrolment in 2003-04 Classes 1 to 7 was 63. 34% students enrolled were girls. Reliable data regarding SC/ST students was not available.	Several teachers had left the school recently. Only headmaster was available.	Incomplete rented building without doors and windows. 2 classrooms in thatched sheds. No toilet. Public bore well outside school. No playground, no library.	NA	
Alternative school in village 2. Grades 1 to 4. Co-educational.	Enrolment in 2003-04 in Classes 1 and 2 was 27, 63% students enrolled were girls. All the students were from ST category.	One vidya volunteer.	Single room in open shed. No toilet, no drinking water in school premises. No playground, no library.	The teacher attempted to explain numbers to children by writing them on the blackboard, or by doing sums. However, only one or two children could follow her.	

Source: Field study data (September 2005).

Note: * Telangana Rashtriya Samiti

Largely, teacher availability and infrastructure facilities in Rajasthan and Andhra Pradesh are similar. However, unlike Rajasthan in the Andhra Pradesh sample there were no single teacher formal schools, and even the primary schools tended to have a larger number of students as well as teachers. Thus the basic minimum educational facilities provided in the sample villages in Andhra Pradesh were better. All the teachers in the formal school were graduates, and 37 per cent were post-graduates. All were trained. Among the vidya volunteers, out of 13 teachers, nine had passed Class 12 and three had college degrees, while only one had studied less than Class 12. Nine vidya volunteers had also been trained as teachers.

In government schools, girls comprised around 50 per cent or more of the students enrolled, and the percentage of SC/ST children enrolled was higher than their share in the population. In the alternative school, all the children enrolled were STs and 63 per cent were girls. In the private school, girls comprised only 34 per cent of the students enrolled. However, reliable data regarding the SC/ST population was not available. Most children attended school regularly, but a small minority did not. Midday meals were served regularly and we did not hear too many complaints about the quality of the meal.

Classroom observations showed that the situation was a little different in Andhra Pradesh as compared to Rajasthan. Generally, teachers were present in the class during teaching time, and

also attempted to teach. However, while no gross neglect of duty was observed, teaching, as in the case of Rajasthan, tended to be unimaginative. Often the teacher took scant interest in whether or not the students had actually grasped the concepts that he was trying to explain. In some, especially primary classes, the teacher tried but appeared to lack the skills required to create interesting learning situations in which the children could take an active part. In the primary schools, teachers were unable to handle several classes at a time. In almost all the schools, library books (where available) were locked in a staff room cupboard.

As in the case of Rajasthan, Class 6 children enrolled in these schools, especially girls, did a significant amount of housework. Several children, especially those from poorer families, also reported going for wage labour during peak agricultural seasons. As in the case of Rajasthan, many children's parents were illiterate or neo-literates, though in village 5. Only one of the six sample villages had a library run by the panchayat. By and large, children did not have access to much reading material. Only a few families (especially forward and upper castes) got newspapers while some children read newspapers at teashops. Children rarely got any help with their schoolwork. A group of 10 children in one of the villages attend private tuition—spending almost 5 hours attending tuition classes, doing homework and studying. The very poor did not send their children for tuition.

The level of community participation varied from village to village and from school to school. In general, the elected SEC of Andhra Pradesh was far more active than the nominated SDMC of Rajasthan. The flip side was that competition between political parties, as against optimization of school functioning, tended to become important in some SECs.

CHAPTER 5: THE INTERNAL DYNAMIC

Rashmi Sharma

Introduction

The functioning and dynamics *within* the system and the institutions that engage with elementary education in India is related on the one hand to a set of explicit institutional rules and practices, and on the other, to a non-stated but critical dynamic. The importance of stated institutional policies and rules is obvious. The manner in which institutions are structured, i.e. policies vis-à-vis recruitment of personnel, institutional autonomy, rules and procedures, financial support and processes etc., as outlined in various government and institutional documents are without doubt, critical for understanding the key characteristics of the system. However, to get a real grasp on the dynamics, it is important to go beyond formal or stated features and scrutinize unacknowledged practices with equal care. This need becomes clear if we visit, very briefly, recent scholarship on the Indian state.

That the state in India has drifted into turbulent and muddy waters since the late 1960s is well recognized and has been the subject of much comment and debate (Corbridge and Harris, 2001; Kaviraj, 1988; Kohli, 1991; Kothari, 1988; Rudolph and Rudolph, 1987). For instance, Atul Kohli perceives a growing 'crisis of governability' in India, by which he means that '(the) national leadership find it increasingly difficult to put together durable coalitions, to *undertake major policy initiatives*, and to settle political conflicts without violence' (Kohli, 1991: 385, italics added). He locates this crisis in the growing disjuncture between multiplying demands on the one hand and weakening institutions on the other. The nature of politics in India has been such that while various groups have been considerably mobilized and politicized, this politicization has not been matched by a parallel growth in institutions that can respond to the newly mobilized demands. Instead, gaining power through personal appeal and short-term populist measures has become a central dynamic in the Indian polity. This leadership strategy, in turn, has had a de-institutionalizing effect. The Indian polity is highly centralized, but it lacks the capability to bring about socio-economic development because of the weakness of its institutional structures (Kohli, 1991).

Paralleling the above, an important feature of politics in India is the centrality of *competition over state resources* rather than support and opposition to specific *policies*. In the large interventionist post-independent Indian state, access to state resources has been a site for intense electoral competition (Chibber, 1995; Kohli, 1991). Political parties rely on disbursing state resources to mobilize support, rather than building opinion around coherent programmes and policies. Moreover, this competition takes place in a setting described by Kaviraj (1988: 2240) as 'a de-linking of power from ideological and social programmes', which he depicts as the 'structural crisis of the state' where 'a crisis is can be called structural if it arises from inside the basic laws and movements of a system, rather than externalities' (Kaviraj, 1988: 2441). Analysing the period between the late 1960s and the mid-1980s, Rudolph and Rudolph write:

'They (the ruling elite) identified the state with self-aggrandizing public officials who used their power to benefit themselves and their friend and clients...the state...came to be seen...to advance the interests of elected officials and those the state employed and patronized' (Rudolph and Rudolph 1987: 35).

These problems appear to have intensified during the 1990s. Discussing developments after the 1990s, Bardhan says:

‘The diminishing role of elite control and the welcome expansion of democracy to reach the lower rungs of the social hierarchy have been associated with a loosening of the earlier administrative protocols and a steady erosion of institutional insulation in the decision-making process in public administration and economic management. Lower caste politicians, as they come to power, indulge in large-scale transfers. Corruption is seen as the collective entitlement in an amoral game of equity politics’ (Bardhan, 2001).

Current journalistic comment tends to confirm this state of affairs.⁷³ A significant deterioration of political ethos, as observed by scholars has been mirrored in the bureaucracy (Dwivedi et al., 1989; Potter, 1996). To quote:

‘In recent years, Indian administration has deteriorated because of the moral decay that marks the conduct of political leaders...a rigid adherence to procedure combines with a ready susceptibility to personal pressure and intervention’ (Dwivedi et al., 1989: 265-266).

This is a powerful trend, and along with the analysis of stated and formal systemic features, needs to be taken into account. This dynamic is not found in written documents, but is unstated, and has to be ferreted out in practices. This unstated or informal dynamic is rarely considered seriously in policy discussions and analysis. However, as this chapter will show, understanding it is vital to get a true and realistic picture of state institutions. As much as the formal structures and processes of various institutions, these informal practices contribute to the prevailing ethos and ensuing outcomes. In this chapter, an attempt is made to understand the internal dynamic of state institutions related to elementary education as it plays out in Andhra Pradesh and Rajasthan taking into account stated, formal structures and processes, as well as unstated, informal ones.

The chapter is divided in two parts. In the first part, key administrative and management characteristics of the system are delineated. In line with the above argument, two categories of processes within state institutions, termed as the ‘declared rules of the game’ and the ‘unstated dynamic’, are identified separately. The former comprises practices that are open, often available in documents and acknowledged without (or with very little) hesitation by individuals working within the system. For instance, with some exceptions, recruitment and promotion of employees falls within this category. However, the fact that these practices are stated does not mean that they are unproblematic, but merely that they are ‘as per rules’. The latter category constitutes practices that are not acknowledged openly, and are often in contradiction to stated rules and procedures such as rent seeking. After capturing these two dynamics, an attempt is made to depict their interaction and the particular management ethos and patterns that prevail within the system as a consequence.

The second part of the chapter focuses on analyzing the attempts made at changing the system. As Chapter 1 shows, there have been several efforts to bring about a changes in the management and administrative ethos of the system, such as decentralization and community engagement, setting up of project structures, advocacy for privatization and attempts to engage with NGOs. Analyzing how these have fared deepens the understanding of core systemic issues, and also helps in identifying key strategies for change for the future.

The analysis in this chapter is based on a study of various government and institutional documents related to law, rules and orders and detailed semi-structured interviews with people working in various institutions, encompassing their personal experiences as well as opinions on a wide range of issues.

⁷³ See, for instance, Shekhar Gupta in *Indian Express* (3 June 2006) and M.J. Akbar in *The Asian Age* (4 June 2006).

Institutional Characteristics

The declared rules of the game

The formal structures and processes of the various offices and institutions that engage with elementary education are in part embedded in the overall administrative framework of the country and are in part specific to the education department. Across states, government structures and processes are strikingly similar, though there are some differences. We now examine the education departments in Andhra Pradesh and Rajasthan, situating them against the larger administrative system in the country while also looking at their specific features.

The overall administrative context

An important principle of the organization of Indian administration is its structuring along highly fragmented lines with numerous departments (50 or so in a state). Central and state government departments are headed by ministers, who are elected representatives. Executive posts are manned by ‘permanent’ civil servants, recruited when quite young (usually in their twenties), and expected to serve till the age of 60 or so. Civil servants are members of specific ‘services’, which may very broadly be classified into ‘all-India’, ‘provincial’ and ‘subordinate’ with the first enjoying the best salary and status and the last the least. Members of the latter two types of services man most of the executive posts in state governments.

Members of the all-India services are recruited through national level competitive examinations. All-India services include a generalist service, i.e., the Indian Administrative Service (IAS), whose members occupy several top administrative positions in various central and state government departments, as well as separate all-India services for some departments such as the Indian Police Service, the Indian Economic Service and the Indian Forest Service. All-India service members occupy the senior-most posts and are expected to serve tenures with both state and central governments. In state governments, separate provincial services exist for various departments, known as ‘departmental’ services such as services for agriculture, police⁷⁴ and engineering. Members of these services are recruited through state level competitive examinations, man the bulk of the middle management posts, and some are promoted in the later part of their careers to the all-India services. Finally, government personnel are recruited for several central and state subordinate services as supervisors, accountants, clerks etc. Officials become eligible for promotion in terms of ‘batches’, i.e., the year of joining service, and unless a person has an adverse record, the whole ‘batch’ is promoted together. In general, there are few modalities for hiring people from outside government at the middle and senior levels.

For education, there is no all-India service. Though the Education Commission had recommended the creation of such a service,⁷⁵ this service actually never came into being, as almost all the state governments were opposed to it (Naik, 1982). Personnel of the education departments belong to the state services and are mainly school and college teachers. The state elementary education departments are examined in greater detail in the later sections of this chapter. The important point to be noted here is that the most capable people in Indian

⁷⁴ In some departments, such as police and forest, there are all-India service personnel, as well as provincial service personnel, with the former occupying a large number of the top positions.

⁷⁵ Initially, as the directorates of public instruction were established between 1855 and 1865, a wide range of persons, i.e., civil servants, military and medical officers were appointed as inspectors of schools. In 1896, a specialized Indian Education Service was formed. Education personnel were divided into three categories: the highest or superior service whose members were recruited in England, and other two provincial and subordinate services whose members were recruited in India. In 1924, after education was transferred to the provincial governments under diarchy, recruitments to the superior Indian Education Service were stopped (Ghosh 1995; Mishra, 1983).

government are recruited through the 'all-India' services and there is no such dedicated service for education. A small number of all-India service personnel, i.e., IAS officers, who are generalists, work in the education departments, at various points in their careers, at a few top-level posts. At the same time, as the provincial services have lower salaries and poorer promotion avenues than the all-India services, it is difficult to attract the best talent to work in the state education systems.

Within the government, each department maintains a distinct identity of its own. Its field offices are closely linked to, and controlled by, its parent office at the state level which is responsible for formulating and coordinating schemes and activities, allocating budgets and personnel management etc. (Dayal et al., 1974; GoI, 1985). This departmental system has been critiqued as having no locus of accountability as various departments deal with highly fragmented aspects of the same issues. It is incapable of coordinated action and long delays occur because the departmental tasks are interdependent, while the departmental activity is uncoordinated. Moreover, such an arrangement is conducive to the execution of top-down orders rather than the formulation of area specific strategies to address needs and issues that emerge from the ground. Field functionaries are often given unrealistic targets that match the wishes of the top rather than the possibilities on the ground with the result that ground level functionaries sometimes report fudged figures as achievements (Dayal et al., 1974; Hargopal and Balaramulu, 1981; Sambrani, 1978).

The 'district' is the most important administrative unit in India. Nearly all departments have their field offices at the district level. Sub-district units vary with departments, though in most states, including Rajasthan, the 'block' forms an important unit for developmental and social welfare activities. In a few states, including Andhra Pradesh, a smaller unit in the form of a 'mandal' prevails. The district magistrate (DM) has the responsibility of overall supervision at the district level. The DM belongs to the IAS. However, the DM, though often perceived as all-powerful, actually exercises only limited legal authority over the departments. Horizontal linkages through the DM across departments are weak in comparison to the vertical ones within departments. The weakness of horizontal linkages can be gleaned from the fact that at the level of the district an integrated report of all the activities of the district does not exist. This situation gets more aggravated at the sub-district level (Chaturvadi, 1987; Dayal et al., 1974; GoI, 1985; Ray, 1985; Yugandar and Raju, 1992).

An important thrust in Indian administration has been towards democratic decentralization or panchayati raj, i.e., the transfer of power and authority to local elected bodies. This has been considered important for promoting need-based and area specific strategies as well as for encouraging people's participation (e.g. GoI 1957, 1978, 1985). A three-tier system of local government, called panchayats or Panchayati Raj Institutions (PRIs), at the district, block and village level was initiated in the late 1950s and early 1960s. However, during the 1970s PRIs became increasingly marginalized as elections to local elected bodies were not held and these bodies were starved of funds. In the late 1970s and early 1980s, the panchayats were revived in some states (Mathew, 1996). Subsequently, in 1993, there was an important shift: through the enactment of the 73rd and 74th constitutional amendments, regular elections to PRIs as well as reservations for SC, ST and women representatives became mandatory. An 11th schedule, which includes the list of subjects recommended to be transferred from state governments to local governments, has been added to the Constitution. The recommended list includes primary and elementary school education.

Two issues vis-à-vis democratic decentralization need to be recognized. First, the precise functions to be devolved to PRIs remains a matter of state policy and the picture of empowerment of local bodies is highly variable across states. Thus, while in some states local elected governments perform important functions, in others they are marginal players. Second, even within a state, the actual powers of local bodies continue to shift with regime and policy changes. In the case of Rajasthan and Andhra Pradesh, though both states were

among the earliest to adopt the system of panchayati raj during the late 1950s and early 1960s, at the time of the fieldwork, in neither of these states were local elected bodies on the ascendant. In Rajasthan, after an initial euphoria, the role of local elected bodies declined after the mid-1960s and attempts to revive them were not successful (Hooja and Hooja, 2000). In Andhra Pradesh too, panchayats had little autonomy and schemes are decided at the central and state government levels. Additionally, in recent years, the state has seen a proliferation of parallel community based bodies including self-help groups (Rao and Dev, 2003), and a case has been made that with the establishment of these parallel organizations, local elected government has been further sidelined (Powis, 2003). In other words, local government in India is an institution that is still in the making and all policy recommendations and debates about decentralization in education need to contend with this fact.

The transfer of schools to PRIs has been a bit of a dilemma in education. The Education Commission Report (1964-1966) summed up this situation: seeing the engagement of the community with education as desirable on the one hand, but fearing the possible harassment of teachers as local politics plays out in PRIs on the other.⁷⁶ In practice, during the 1970s and 1980s, as administration in general became increasingly centralized and several states withdrew powers from local governments, the role of panchayats in education was marginal. However, the PoA 1992 noted the importance of the 73rd and 74th constitutional amendments for the education sector and envisaged a district level body to plan and implement educational programmes and village education committees to administer educational programmes. The Committee on Decentralized Management of Education 1993 of the Central Advisory Board on Education (CABE) recommended the transfer of administrative control of schools to panchayats, including the appointment and supervision of teachers (GoI 1992, 1993).

The Education Department

The broad institutional structure of the education department in Andhra Pradesh and Rajasthan has been described in Chapter 2. In this chapter, the focus is on the processes and dynamics of these institutions.

Personnel

The overall administrative head, the principal secretary or the secretary of the department, belonged to the IAS in both the states, as did the heads of the directorates and programme offices/societies. All other personnel belonged to the state education departments. Strong departmental and 'service' loyalties were visible among the personnel in both the states. Some departmental officials resented what they perceived as the DM's interference, and also said that the generalist heads did not have enough knowledge of the department, and wanted the department to be more independent. On the other hand, officials of the IAS relied on the DM and sought his support in pushing through their agenda, and some chafed at the lack of managerial abilities of the departmental employees. Officials also reported difficulty in coordinating with other departments that managed schools or had programmes in which collaboration was necessary.

There were some differences in the recruitment patterns in the education departments in the two states. In Andhra Pradesh, recruitment in the education department involved two types of personnel: teachers and educational administrators, i.e., Deputy Education Officers (Dy EOs.). The Dy EOs were recruited through a competitive examination and underwent a six-month induction training upon recruitment during which they worked in several institutions of the department and understood its basic structure and functioning. In contrast, in Rajasthan,

⁷⁶ The Education Commission ultimately favoured protecting teachers from too much intervention from PRIs. Studies of this period indicate that teachers tended to prefer departmental rather than panchayat control (Nuna, 1997; Oad, 1989).

recruitments were made only at the level of teachers and high school headmasters. In 1998, a decision was taken to recruit educational administrators directly, i.e., at the District Education Office (DEO) level, but this had yet to be put into action. In other words, Andhra Pradesh recruited a separate cadre of persons to work in various administrative and other capacities within the department, while Rajasthan did not.

The pattern of teacher recruitment in the two states has been described in Chapter 2 (pages 41-43). In both the states, the qualifications and salaries of teachers varied depending on the grades taught. Elementary school teachers were the least qualified and had the lowest salaries. Both the states hired 'para-teachers' with lower qualifications and salaries than regular teacher. The two states varied in terms of the process of recruitment of regular teachers. In Andhra Pradesh, a well accepted district based system for teacher recruitment, based on a common entrance examination conducted at the district level, was in place. The State Council for Educational Research and Training (SCERT) prepared the papers for this examination. The examination was highly competitive, with only 1-2 per cent of the aspirants being selected. In Rajasthan, recruitment of grade 2 and grade 3 teachers took place at the state level on the basis of qualifications, but the recruitment of grade 3 (primary) teachers had undergone several policy shifts in recent years. Up to 1998, panchayat samitis and zila parishads of rural panchayats recruited (regular) teachers, but between 1998 and 2003, the recruitment of regular teachers had come to a halt and para-teachers were recruited as a substitute. However, since 2004 for the first time in the state's history, grade 3 teachers were recruited based on state level competitive examinations.

In neither state could primary school teachers occupy administrative positions or become teacher trainers. In Rajasthan, people who occupied various administrative positions such as Block Education Officers (BEOs), District Education Officers (DEOs), joint and deputy directors etc. were promoted from among headmasters who had been recruited directly or promoted from among grade 1 and grade 2, i.e., mainly secondary school, teachers. Elementary school teachers did not work in administrative posts in the education department. In Andhra Pradesh administrative posts were filled in two ways. Deputy Education Officers moved up the administrative hierarchy as District Education Officers, Joint Directors and so on. Additionally, school assistants, who taught upper primary grades and high schools, were promoted as headmasters and Mandal Education Officers (MEOs), while headmasters and MEOs could be promoted as Dy EOs.

The recruitment of para-teachers stood apart from the overall recruitment pattern in the government. As has been described in Chapters 1 and 2, para-teachers draw a very small honorarium instead of a salary, their recruitment processes tend to be more informal and they are hired on a year-to-year basis. Like several other states, both Andhra Pradesh and Rajasthan had hired a large number of para-teachers during the 1990s. This process is still to be watched, as it had resulted in extensive litigation.⁷⁷ Some evidence of back-tracking was visible in both the states. Notably, in Rajasthan the recruitment of para-teachers had stopped at the time of the study. In Andhra Pradesh, recruitment of vidya volunteers has always been seen as a temporary measure and the focus at the time of the study was on recruiting regular teachers.

Hierarchy and centralization

Indian administration, like governments across the world, is organized along the lines of a hierarchy, wherein junior officials report to and are supervised by senior ones. Seniority is gained on the basis of merit and experience. Responsibilities and 'powers' are well defined, with senior officials having more of both. In part, this hierarchy functions as a modality for assigning duties and responsibilities clearly, and creating a system of checks and balances. However, the particular pattern of hierarchical functioning in the education departments needs

⁷⁷ Para-teachers alone had instituted 200 writs.

to be seen against two contexts. One, as the education 'system' developed during the colonial regime, a centralized, hierarchical supervision system headed by the director of public instruction, in which the teacher occupied a low place, was put in place (Kumar, 1991; Mishra, 1983). The changes that have come about in this supervision system are discussed in detail later in this chapter. Here, what needs to be noted is that it continues to exist and retain several of its essential features. Moreover, while the salary structure and other benefits for teachers have increased substantially (though reversed in the case of para-teachers), elementary school teachers continue to be the lowest in the hierarchy. Second, Indian the manner in which hierarchy manifests itself in the education system also needs to be seen against the central space that hierarchical ordering occupies in Indian socio-cultural and psychological constructs (Kakar and Kakar, 2007). As the discussion below shows, the hierarchy is not merely a question of being professionally accountable to seniors, but goes much deeper, so that orders from senior officials are seen as more important than the demands of work. Rather than needs and demands from the ground, orders from the top enjoy legitimacy.

This principle of hierarchy was acknowledged and seen as legitimate by a majority of the interviewees, though there were exceptions. For example, a senior official in the commissioner's office in Rajasthan, viewing his rise up the hierarchy with satisfaction, commented:

When I was the headmaster of a high school, I never kept any letters that came from the director pending. When I was district education officer, I used to work in the same mode. Now, I listen first to the state government.

Rather than setting priorities as per broad departmental goals and the needs of the situation, assessing the pros and cons of various strategies and consulting other professionals, the tendency was to simply do what the boss ordered. Importantly, the criteria for promotions supported this stance. An assessment of an officer's performance counted, and this assessment was made by his immediate boss, with those one or two rungs higher up getting a chance to review the evaluation. If the boss made an 'adverse' comment, an official's chances of promotion could be jeopardized. An important repercussion of this 'command-based' system was that the role of scrutiny and deliberation in the decision-making process was small. Critiquing and questioning decisions coming from the top was not seen as part of a good decision-making process. The quality of decisions thus depended mainly on the capacities of individuals in leadership positions and there were few ways of checking poor decisions.

Another way in which the hierarchy unfolded was extreme centralization, i.e., the concentration of decision-making at the state level. For example, an MEO in Andhra Pradesh told the researchers:

'We have not worked out any new plans so far. We want to follow the instructions of our DEO and State Project Director (SPD). No proposals are sent, as there will be no effect.'

In the directorates in both the states, nearly all decisions were concentrated at the level of the director. Even senior officials in the directorate, with more than thirty years of service, could not sanction a couple of days' leave to members of their team. The directorate in turn took orders from the ministry.

This centralization of decision-making, in one of the largest school systems of the world, often caused delays and obstructed actual work. For instance, all the teachers in the sample schools in Rajasthan spent a great deal of time in making information notes (to be fed to central authorities) and doing other non-academic work as directed by some officials. In single teacher schools this meant that hardly any time was left for teaching. Additionally, visits by cluster resource persons, meant to be for academic support to teachers, actually revolved around

collecting information. Moreover, 'command' type, centralized decision-making made the system extremely 'person-centric'. Initiatives developed over several years could be undone with ease with a change in the state level leadership. This was illustrated by the fact that in Andhra Pradesh, new textbooks developed under a particular programme leadership in the DPEP were abandoned by the next.

Notably, hierarchical functioning and centralization were not limited to administrative offices, but extended to academic institutions. For instance, in Andhra Pradesh, teachers and activists who had been involved in pedagogic reform processes complained that the SCERT faculty had overruled their suggestions without adequate consideration. Faculty in academic institutions also reported that they had to get the director's permission for every activity. The DIETs undertook activities as per the orders of the SIERT/ SCERT.

Extension of the hierarchy

The hierarchical ordering did not exist only in terms of the command-style functioning and centralization, but a range of several other processes were viewed through the hierarchical lens too. For instance, facilities were provided not on the basis of the work expectations of a particular office, but on the basis of the rank of the official who headed the particular office. This meant that field offices and centres, which had the primary responsibility for supervision and day-to-day contact with schools, lacked adequate staff and facilities. For instance, at the BEO's office in Rajasthan, the staff was inadequate. It comprised one BEO and two assistant officials, which meant that each official was expected to supervise around 75 schools. Though this office was responsible for monitoring more than 200 schools, it had no telephone or vehicle, two facilities that appear necessary for the role expected. Similarly, the sample district in Andhra Pradesh had 2,879 primary and upper primary schools, 7,070 teachers and more than 3,000 para-teachers. However, the supervisory staff at the district level comprised of only five people.

In the same vein, in Rajasthan very little money was available to pay the travel and medical bills of teachers and these remained unpaid for several years. However, funds were available for the same expenses for senior officials. Even training was dispensed according to status rather than as a necessary perquisite. While the deputy education officers in Andhra Pradesh were considered worthy of six months' induction training, those who were promoted from the ranks got little or no orientation. For example, an interviewee who had joined as a typist and had been promoted to a middle level post in the directorate in Andhra Pradesh had never attended any training programme.

The focus on hierarchy was expressed in a somewhat comical manner wherein secondary education was considered 'superior' to elementary education, even though the latter system served a much larger number of children. As noted above, in both the states, elementary school teachers were required to have lower qualifications and were paid less than high school teachers. In addition, high schools in both the states were best equipped in terms of number of teachers and facilities, while the primary schools were the least well equipped, with the upper primary schools falling somewhere in between. While secondary school teachers were eligible for promotions to various administrative posts as well as to DIETs, elementary school teachers were not.

In Rajasthan, the secondary and elementary education directorates had recently been bifurcated from a single directorate, with the secondary education department getting better facilities and retaining control over all the personnel except grade 3 teachers. Officials of elementary and secondary education continued to comprise a common pool and the cadre controlling authority continued to be the directorate of secondary education. The secondary education department got a bigger budget for medical and travelling allowances. A BEO (for elementary education), who dealt with service matters of 1,000 or so employees had three

clerks, while a secondary school headmaster, who managed a handful of teachers, had two. An employee who had recently shifted from the secondary education department to the elementary education department was surprised at the difference between the two departments and said that secondary education was far better organized and education was taken much more seriously there than it was in the elementary education department.

Notably, this hierarchical ordering corresponds to the fact that students in elementary schools tend to be from a far poorer background than students of high schools, as by the higher school stage, a large mass of poor students drop out of schools. One way of looking at this phenomenon is that it is this differential social composition of elementary and high schools that leads to this particular hierarchical ordering. This connection is undeniable. However, it is the prevalence and acceptance of the principle of hierarchy *within* the system that gives strength to this phenomenon and allows it to go unquestioned on more objective and rational grounds. The chances of these imbalances being looked at more objectively would have been higher if the system itself had functioned on democratic lines.

In this hierarchical ordering, SCERTs and DIETs occupied a place below the administrative structure and were placed firmly in its control. In both the states, leadership in these institutes was junior to the director/commissioner of schools and had no powers of disciplinary action over its own employees. SCERT in Andhra Pradesh was not autonomous. In fact, its director was an additional director of the department. SCERT, an institute expected to provide academic leadership in the state, did not have a budget of its own, and approvals had to be obtained from the director of school education. The SCERT director could not even sanction leave to the faculty. It had only one telephone, and all out going calls had to be made through the director's assistant. The building was cramped and sometimes as many as five faculty members shared a room. The SIERT in Rajasthan had an Advisory Council, was better staffed and had a large campus and building and better facilities. However, as in the case of Andhra Pradesh control over the faculty was retained by the directorate. In both the states, the DIET faculty was under the administrative control of the directorates.

Employee protection

While hierarchical ordering had permeated the system in myriad ways, it was mitigated by a high degree of employee protection. Though the teacher continued to occupy a low place in the hierarchy as in colonial times, in the post-independence period, teachers' salaries, qualifications and service conditions had improved substantially (except in the case of para-teachers, who were recruited on a large scale in the 1990s). In the Indian system, once recruited, an employee usually serves until retirement age and processes for dismissal are long and demanding (the exception here is a process of 'suspension' during which an employee may be barred from working and may receive only a part of his salary while a more rigorous inquiry is underway). Arbitrariness in promotions is contained as the number of years served form the main the eligibility criteria. Promotions can be denied if supervisors give a poor assessment, but it is difficult to promote people 'out of turn', i.e., over people who have put in more years of service and who have not got adverse reports from their supervisors. An employee is entitled to promotions, unless a strong case can be made against him/her.

At times, performance may not figure at all in the criteria for promotion. For instance, in Rajasthan, grade 3 teachers were entitled to automatic promotions every nine to ten years, irrespective of performance. This meant that while promotions were not arbitrary, there was also no space for recognizing exceptional contributions. As one interviewee put it:

'Promotions are fair, but there is no place for performance'.

Additionally, in case of grievances, employees could, and did, approach the courts freely. In fact, court cases, largely concerned with appointments and promotions, were a major activity

in the state and district offices in both the states. For instance, in Rajasthan in the recent teachers' appointments through RPSC, SC and OBC groups as well as the handicapped had gone to court on the question of quotas. In Andhra Pradesh, the importance of legal action to protect employee interests was illustrated in a lengthy litigation whereby all appointments and promotions in the education department had come to a standstill in 2003 resulting in a large number of vacancies. As mentioned in Chapter 2, state had had two types of teacher cadres, i.e., those that belonged to the education department and those that belonged to the zila parishads and panchayat samitis. At one time, government teachers stood better chances of promotion to administrative posts, till in 1998, on the demand of the teachers' unions, the government merged the two teacher cadres with the intention of bringing about uniformity in service conditions. Government teachers had challenged this decision in the courts and a long process of litigation followed. The court had stayed all recruitments, promotions and transfers up to the level of deputy directors, and as a result, there were a large number of vacancies.

Unions also protected employee interests. This was clearly evident in the case of Andhra Pradesh that had 18 teachers' organizations. The Progressive Recognized Teachers' Union (PRTU)⁷⁸ in the state was, according to its office bearers, one of the largest in Asia, with a membership of two lakh teachers, and was extremely active in protecting employee interests.

Rules

An important feature of the formal system was that its functioning was governed by a plethora of rules and procedures. Such rules and procedures are necessary in government (and indeed all organizations) to ensure that decisions are not arbitrary but based on objective criteria. There are several positive features of rule based functioning in the system. For instance, recruitment is based on merit (as assessed by various examinations) and not as per a person's political or other affiliations. Moreover, as argued above, employees cannot be dismissed merely because the boss dislikes them. Financial rules and various audit processes are meant to ensure financial accountability. Audit teams scrutinize departmental records to identify irregularities, which departments have to explain. Usually the assumption is that if someone bent a rule, the intent was malafide, and disciplinary proceedings could follow.

However, many rules were so narrowly defined that they constrained the space for working. Financial rules tended to be especially strict. Budgets were allocated very rigidly as per 'schemes' where activities were pre-defined and uniform across the whole state. Money could not be transferred easily from one scheme to another. So important were these categories that often the mark of 'success' was spending the amount allotted for a particular scheme in full, and not the achievement of the goals of the scheme. This left little room for spending as per the needs on the ground. For instance, during this study, people responsible for teacher training said that fixed unit costs made it difficult to take up training programmes as and when needed. Often, it was quite difficult for even very large government organizations such as the directorates to fund fairly small activities that could be useful, but were not part of some pre-defined scheme. Several officials in the study complained of not being able to take up some needed activity simply because it did not fit into the parameters of a pre-proposed scheme.

Professionalism

The system placed small value on professionalism. This was most obvious from the fact that there was little recognition of elementary education as a field of knowledge and expertise. Neither state hired qualified and trained specialists in elementary education. Besides, the

⁷⁸ PRTU has a monthly magazine, and issues its own calendar, diary, phone book, leather bag etc. The PRTU calendar hangs on the walls of many government offices. The president and the secretary of the union were exempted from teaching duties. The district level presidents and secretaries of recognized unions got 21 extra days of casual leave.

qualification criteria for elementary school teachers did not include university education (though several teachers had university degrees) and their avenues for professional growth were limited. Instead, high school teachers functioned as supervisors and trainers, usually with little or no orientation in elementary education. For instance, in Andhra Pradesh, the promotion structure was such that high school teachers zigzagged through elementary and secondary education (Box 5.1).

Box 5.1: Two possible career paths for school assistants (SAs) in AP

<i>Path 1</i>	<i>Path 2</i>
SA (teaching secondary school).	SA (teaching secondary school).
Promoted as DIET lecturer (training elementary teachers).	Promoted as Mandal Education Officers (managing elementary schools).
Promoted as lecturer in CTE (training secondary teachers).	Deputy Education Officers (supervising secondary schools).

Nowhere was the low regard for professionalism more apparent than in the fact that in both the states, little attempt was made to staff SCERTs and DIETs on the basis of professional logic. DIETs and SCERTs, organizations which make curriculum and textbooks and are expected to be academic leaders, did not have personnel with experience or expertise in elementary education. Instead, officials and high school teachers of appropriate seniority, rather than suitable skills, were posted in DIETs and SCERT/SIERT. A senior faculty member from SCERT Andhra Pradesh said that:

‘Aptitude should be examined before people are posted to SCERT. Then there will be people of appropriate calibre. Perhaps there could be selection tests.’

In Andhra Pradesh, as per rules, two-third of DIET lecturers were to be promoted and a third recruited directly. In practice, recruitments to DIETs did not take place. High school teachers holding a M.Ed degree were promoted as DIET lecturers based on seniority. The director of SCERT was one of the senior-most officials of the education department (additional director level), posted to the institute. There was no direct recruitment of faculty to SCERT. Lecturers and professors from institutes that trained high school teachers were eligible for posting at SCERT and not the DIET faculty. Moreover, the Andhra Pradesh SCERT had several vacancies and had faced a severe financial crunch in the past few years. In Rajasthan, DIETs and SIERT followed similar staffing patterns.

The large-scale recruitment of para teachers in both the states showed the low value given to professionalism. The wide swings in the policy for recruitment of teachers in the past decade in Rajasthan underlined this lack of concern. Notably, in Rajasthan, along with the teacher recruitment policy, the policy vis-à-vis teacher education had also swung in different directions. Till 1998, DIETs conducted a Basic School Training Certificate (BSTC), i.e., the pre-service elementary teacher-training course, but after 1998 this was stopped, and SIERT and DIETs started conducting correspondence courses for para-teachers, Shiksha Karmis etc. (GoR 2004-05). BSTC was subsequently started again, but the number of trainees was reduced. In 1998, DIETs ceased to conduct pre-service training for teachers, though subsequently, it was restarted with a reduced capacity. At the time of the study, Rajasthan trained more secondary school teachers (i.e., teachers with a B.Ed degree) than it needed, while the number of elementary school teachers trained (i.e., teachers with a BSTC diploma) was less than the requirement. As a consequence, in many instances teachers with B.Ed degrees worked in elementary schools. At the time of the fieldwork for this study the situation had been complicated by a high court order to the effect that only teachers with a BSTC diploma could teach in primary schools. Similar, arbitrary shifts of the role of DIETs were

visible in Andhra Pradesh. While DIETs played an important role in APPEP, their role was diminished in DPEP (Rao, 1999).

There was little evidence of a coherent professional development plan for personnel in either of the two states. Neither state followed a training and career development plan for departmental employees, other than promotions based on seniority. Significantly, there was no attempt to systematically develop personnel for SCERT/SIERT or DIETs in either state. The situation was marginally better in Andhra Pradesh, which at least provided orientation training to the personnel recruited as Deputy Director of Educations. Additionally, some of the senior officials interviewed in Andhra Pradesh had been trained in the Andhra Pradesh Primary Education Project (APPEP) or at NIEPA. They uniformly reported learning a great deal through these programmes. In Rajasthan, however, the training of headmasters promoted to administrative positions was extremely inadequate and patchy.

In both the states, there had been some attempt to provide more space for professionalism in project structures. The structuring of these offices itself indicated new concerns. For instance, wings for girls' education and community mobilization had been created, which did not exist in the previous structures. In both the states, at least some of the staff had been placed in these projects on the basis of their past performance and interests. In Andhra Pradesh, efforts had been made to choose personnel on the basis of their work potential and in fact, several personnel of the SSA office had worked in important programmes and projects previously. Some of these people reported that they had been trained specially in APPEP and had been part of a national network in DPEP that engaged in pedagogic and other relevant activities. Resource persons from across the country had invested time in training and exposure and several programme personnel said that they had learned a great deal.

The department and the people

The departmental system described above spread out to the villages through a large number of schools, but its capacity to act in response to the needs on the ground was extremely limited. On the one hand, the grassroots institutions and offices were weak, in terms of capacity, autonomy, personnel and infrastructure. Because of these handicaps, the grassroots or field institutions could barely cope with routine matters. Lack of personnel, expertise and autonomy made it nearly impossible for them to plan effectively for school improvement. They could not use their closeness to the schools and the community to respond to needs, because of their own limitations. This posed severe restrictions on the capacity of the system as a whole to intervene effectively to strengthen the schools.

At the same time, though authority was concentrated at the state level, the functioning of these institutions and offices did not facilitate need based planning either. State level institutions and offices had very weak linkages with schools. In both the states, state level personnel reported that they got very little or no time to visit schools or interact with the community. Some simply did not see school visits or discussions with the community as an important part of their job. Others wanted closer contact with the schools but could not leave their headquarters as orders from above were considered more important, and these led to urgent work at the headquarters on a daily basis. Moreover, the hierarchical, top-down mode of functioning blocked the flow of ideas from field level officials to the apex, decision-making level.

In neither state did panchayats play an important role in education at the time of this study and strong tendencies to marginalize local elected bodies were visible in both the states. In Rajasthan, panchayats had been responsible for recruiting as well as transferring rural primary teachers since 1959. This role was sought to be expanded in 2003, when the entire elementary education system, i.e., upper primary schools as well as primary schools and officials were transferred to panchayats (GoR, 2003a). This arrangement stood formally, but in practice, the role of panchayats vis-à-vis upper primary schools was ambiguous and several state level

officials as well as panchayat representatives confessed to a lack of clarity. Between 1998 and 2003, village panchayats played an important role in recruiting teachers for RGSP schools. However, since 2004 the recruitment of primary school teachers has been centralized and teachers are selected through a competitive examination at the state level (GoR, 2004b). Moreover, in recent years, large-scale transfer orders had been issued through instructions from the state level. In Andhra Pradesh too, elected bodies did not play any significant role vis-à-vis government schools

In both Rajasthan and Andhra Pradesh, community participation was sought through specialized committees for education, i.e., School Education Committee (SEC) in Andhra Pradesh and the School Development and Monitoring Committee (SDMC) in Rajasthan. In Andhra Pradesh, SECs were set up in 1998 and as Chapter 2 (page 45) shows, SEC members were elected from among the parents of the school-going children. All the money for construction was sent to an account operated jointly by the headmaster and the SEC chairperson. However, the role of SECs began to decline with the change of government and their composition was subsequently changed, with representation of the village panchayat and several nominated members. In Rajasthan, as the SDMC was a nominated committee with the headmaster of the school its chairperson, and virtually a departmental body.

Comparing Andhra Pradesh and Rajasthan

The similarities in the formal systems of Andhra Pradesh and Rajasthan were striking. In both the states, several features of the formal system did not promote high quality work. We saw that placement policies did not promote professionalism and efforts made towards the professional development of personnel were inadequate. There was little recognition for exceptional contributions. Neither state developed SCERT and DIETs. Additionally, the hierarchical pattern of functioning, several rules and procedures and the paucity of deliberative processes curbed initiative and did not promote informed decision-making.

However, it can be argued that Andhra Pradesh had a marginally better formal institutional structure than Rajasthan. The most important advantage that Andhra Pradesh had over Rajasthan was with respect to its teacher related policies. Andhra Pradesh had a well developed system of teacher recruitment against the wavering policies in Rajasthan. Moreover, at the upper primary stage, the schools in Andhra Pradesh had teachers with specialization in their subjects, while in Rajasthan they formed a common pool, with teachers without specialization in math and science teaching these subjects at the upper primary level. Additionally, though both the states had hired a large number of para teachers, in Rajasthan, these were concentrated in a few schools such as RGSP and Shiksha Karmi schools, as these schools *substituted* for formal schools, and such schools did not have any trained teachers. On the other hand, in Andhra Pradesh para teachers were spread across all schools along with regular teachers, so that schools without *any* trained teachers were not in existence. Alternative centres were run by para teachers, but these had not been established *in lieu* of formal schools. Related to the was the fact the while in Andhra Pradesh, policies vis-à-vis pre service education of teachers had been consistent, in Rajasthan there had wavered a good deal.

Some advantages were also visible with respect to the administrative and management system. Andhra Pradesh had created a unit called the 'mandal' (with around 25-50 schools), which was much smaller than the block (with around 200 schools), which schools could access more easily and where problems of individual schools could be identified and addressed. As resource persons also operated at the mandal, a much stronger grassroots structure, closer to the people, had come into existence. In contrast, in Rajasthan, the two tiny block level offices were separate. Though the mandal education offices (MEOs), as well as the BEO offices and Block Resource Centres (BRCs) were inadequately staffed, with inadequate infrastructure, the combined MEO office in Andhra Pradesh was better than the block offices the in Rajasthan.

Andhra Pradesh also had an edge over Rajasthan as it recruited educational administrators and provided them with induction training, building up a cadre with some (though not adequate) training and plenty of experience in educational administration. In Rajasthan, high school teachers were promoted with no or very minimal preparation as educational administrators. Moreover, in Rajasthan the splitting of school education into two directorates has resulted in a weaker structure for elementary education, though the intent may have been quite the opposite. Notably, it was not the creation of a separate administrative structure for elementary education *per se* that led to this problem but the lop-sided distribution of facilities and personnel in favour of secondary education. Splitting the directorate required additional personnel and these had not been recruited.

Though Andhra Pradesh had an advantage in recruiting educational administrators, much of its potential was lost as these officials did not become MEOs, i.e., they did not work at the basic supervisory level. Like the BEOs in Rajasthan, MEOs were promoted from among secondary school teachers as they neared the end of their careers, and were provided with no systematic training. Officials in Andhra Pradesh acknowledged that MEOs were not very skilled or effective. The professionalization of educational administrators in Andhra Pradesh, though greater than those in Rajasthan, remained small and the induction training of educational administrators was largely an orientation for day-to-day functioning.

In recent years, neither Andhra Pradesh nor Rajasthan had strengthened their panchayati raj structures. Instead, centralizing tendencies were visible in both the states. At the time of the fieldwork for this study, the two states differed strikingly in the manner in which they had constituted the SDMC and SEC. While members of the Rajasthan SDMC were nominated and the chairperson was the school headmaster or head teacher, in case of SEC in Andhra Pradesh, the members were elected from among the parents of the school. However, the SEC has now been replaced by a body with several nominated members.

The unstated dynamic

The unstated dynamic that permeated the system centred on maintaining power and profiting from it. The two dominant practices, inter-related but not identical, of this unstated or informal dynamic, were the existence of huge and powerful informal patronage networks that operated within the system, as well as outside it, and rent-seeking. The importance of the informal dynamic derived from the fact that it was a *dominant* trend that drove a considerable part of the agenda, and therefore may be conceived as a sub-system within the larger system. It was not confined to sporadic incidents and a handful of lower level functionaries, but influenced the very core of all institutions and activities. As illustrated below, the unstated dynamic operated from the top, just as the formal system did. In the prevalent top-down, centralized mode of functioning, this made it a force to reckon with.

An important point to be noted is that this study found that patronage and rent-seeking played an important role not merely in the manner in which state actors and institutions related to other actors, but were important dynamics *inside* the systems of both the states. For instance, patronage was an important principle not just in say, regulating private schools, but also in the manner in which the state employees were themselves treated. In other words, patronage and rent-seeking were as much a part of the internal dynamic of the system as they were of its interface with society.

The patronage nexus

The large number of employees, especially teachers, that exist within the state education systems make them fair game for manipulation. This manipulation is easily exercised through transfers, which remain beyond the pale of rigid rules and legal intervention. The importance

of transfers in the establishment of a 'counter-system', in which individuals within the system are rewarded and punished for unstated rather than stated goals, is considerable.

Transfers derive significance from the fact that though the salaries and other emoluments of the employees are the same throughout the system, their working conditions can actually be changed significantly through transfers. In government, in a large number of posts of equal salary and status, in some there is more opportunity for exercising leadership, and also for rent-seeking. For instance, an official posted as DEO and another posted as deputy director in the directorate would have the same rank and salary. Nevertheless, the DEO would have much greater autonomy and decision-making powers, and if he is so inclined, more opportunities for rent-seeking. Transfers can therefore be used to assign important and 'lucrative' posts. Additionally, transfers can be used as favours to make an employee comfortable if allowed to work in a place of her choice, where her spouse may work, or where her children's educational aspirations can be met, while an employee can be made extremely uncomfortable by posting her to a place that is personally inconvenient. Notably, in both the states, teachers coveted urban postings.

In Rajasthan, teachers' transfers were regarded as ways of 'obliging' teachers who were close to powerful people, or were doled out as rewards (and punishment) for services rendered such as assistance in political campaigns. In some cases transfers were simply rent-seeking operations with teachers having to pay a sum of money to get a preferred place of posting. During the study, several instances that illustrated this dynamic were observed. Both the major political parties had rewarded their supporters and punished the supporters of their opponents on assuming power. A great deal of the contact between politicians and officials was regarding teacher transfers. Though local elected bodies were technically competent to transfer primary school teachers, there was in fact, consistent intervention from the state level and secret 'camps' had been held for teacher transfers. Several state officials said that teachers helped candidates during political campaigns, in calculating votes, predicting the outcome of elections etc. and that teachers posted in urban schools were often enmeshed in local politics. Subsequently, such teachers were rewarded with 'good' postings or protected even if they neglected their work. Transfers were also rent-seeking operations, where teachers were asked to pay a particular amount of money to get a transfer to their preferred place. During interviews, 7 of the 16 government school teachers in Rajasthan said that transfers were based on relationships with powerful political figures and the exchange of money. Three of these teachers related personal experiences. One wanted a transfer but had not been able to obtain it, and was now in the process of resigning himself to the idea of paying for it. Another teacher admitted to having obtained her current posting through her connections. A third, recently recruited, was planning to pay for a transfer to Jaipur in another six months.

In Andhra Pradesh, a similar situation appears to have existed until 1998, when the Andhra Pradesh government decided to regulate teacher transfers through the process of 'counselling', a process which entails a transparent allotment of postings based on pre-determined criteria (Box 5.2).

Box 5.2: Teacher counselling in AP

All teachers who have served for eight years in a particular place are put on a transfer list. Those who have been posted in a particular place for 2-8 years can apply for transfers if they so choose. Those who have been posted in a particular place for less than two years cannot apply.

Teachers get 'entitlement points' based on several criteria such as length of service in backward areas and seniority. Additional points are given to teachers who are about to retire, to unmarried women teachers etc. Preference is also given to certain categories such as handicapped, those with medical problems and widows. Based on the points obtained, a seniority list is prepared and fed into the computer.

Actual counselling takes place at the mandal headquarters. Vacant posts are displayed on a big screen in the counselling hall as well as outside. Teachers are called in as per the seniority list, and allowed priority in selecting their place of posting. A committee headed by the zila parishad president monitors the process (GoAP, 2005a).

This process functioned well for five years, though since 2003, because of the court case mentioned earlier there had been no transfers. In this study, all the teachers interviewed in Andhra Pradesh had greeted this change with enthusiasm. The personal experiences that they related stood in sharp contrast to those related by teachers in Rajasthan. A lady teacher and her husband had earlier been forced to work in different districts, but after the counselling process started, had got postings in nearby villages. Moreover, teachers said that as staying in a remote posting added to the chances of getting postings of choice at a later date, there was some incentive to work in remote schools for some period of time. Counselling was not confined to teachers, but had been extended to all officials up to the level of deputy directors (see Box 5.3).

Box 5.3: Why AP decided to go in for counselling for teacher transfers

The quiet initiative to limit the arbitrary exercise of power through counselling was an unusual development in Andhra Pradesh. Perhaps it is a forerunner of a decline of the informal system. Whatever the long-term trends, the immediate circumstances of this decision deserve some attention. How had Andhra Pradesh managed to take this step? Why had powerful individuals who benefited from this system agreed to give it up? A combination of circumstances appears to have led to this development.

Rent-seeking in teacher transfers was at its zenith in the state in the 1990s, and the zila parishad was the agency that effected teacher transfers. This was a phase in Andhra Pradesh where panchayats were marginalized and in the political equations that existed at that time, the ruling party was keen to reduce the role and power of local bodies. Moreover, the then chief minister of the state projected a pro-development, modernizing image. Rationalization of teacher transfers was in accordance with this image, and at the same time it cut at the patronage distributing power of the local zila parishads and reduced their importance. As the then chief minister was a dominant figure in his party, it was possible for him to push this agenda through, despite some resistance from other members of his party. Once counselling had been put in place, pressure by strong teacher unions in the state made it difficult to reverse it.

Though Andhra Pradesh had brought about substantial changes in the criteria of posting of teachers and junior officials, as far as transfers and postings of higher level officials were concerned, it was no different from Rajasthan. In both the states, our investigations left us in little doubt that transfers were based on various kinds of personal 'links' rather than work

interests. Nearly all interviewees admitted to this scenario with several relating personal experiences. An official from Rajasthan said:

There is no transfer policy lecturers upwards. If the government is happy, they remain. Otherwise they are transferred. The politicians think of officials as being supporters of one political party or the other. If an official is not able to meet some demand, then he or she is assumed to be supporting the opposite party. I was transferred during the last regime, because the minister wanted very large-scale transfers of teachers and I said that it could not be done. When the party in power changed, I was posted back to the same place.

Interviews showed that a great deal of manoeuvring took place around transfers in both the states. At the time of this study, one official claimed that she was about to be transferred due to pressure from the unions and caste politics, and was trying to get the order cancelled. Another senior official had been transferred, but had then come back to the same post. Though transfers were meant to be secret, some of the officials interviewed knew exactly where their files were, and jockeyed hard to retain their posts or get the posts that they wanted. This form of jockeying was very different from open competition where officials could apply and be assessed in a transparent manner based on their suitability for the post in terms of qualifications and performance. Instead, success or failure in getting a particular post depended on informal networks and the willingness to 'play the game'. A senior official interviewed in Rajasthan thought he was entitled to some amount of choice for a 'good' posting, but watched junior officials get better posts. These were not people whose work was superior, but they could wield more influence because they were close to the power centre.

In this system, while most officials had become 'smart' about getting good postings, others just resigned themselves to senseless movements from post to post. A senior official in Andhra Pradesh said:

There are no problems with politicians except that I may be transferred any time. We are tools in their hands. There was a proposal to shift me, but that was postponed because of the municipal elections. I have told the minister, "keep me where you want". Jealousies play an important role in all this.

Notably, the transfer system is often seen as a modality through which the ruling party ensures that civil servants who are sympathetic to its goals and programmes are placed in key positions so that policy formulation as well implementation gets facilitated. Such a system may work productively in countries such as France, where politics is organized around ideology and policies. However, the politicization of the bureaucracy in developing countries is rarely along ideological or policy lines, because politics in these countries itself gets organized around patronage and rent-seeking, rather than ideology and policy. For instance, in several countries of Africa, where politics and administration have been fused on the grounds of developing a more consistent and coherent governance structure, and where the civil service has been totally politicized, i.e., where recruitments have also been made along political lines, the result has been a bureaucracy committed to supporting a particular party in power and interested in gaining the approval of superiors rather than addressing the needs of the populace (Cameron, 2003). The 'politicization' of transfers appears to have created a similar effect in India, and this is consistent with the characteristics of Indian politics noted in the beginning of this chapter.

In the two states under study, the process of transfers served as a mechanism for keeping the government's personnel policy out of the spheres of public and political scrutiny, as well as its own internal processes of accountability. The transfer process was extremely opaque and no one really took responsibility for it. As some of our interviewees, who explained the formal features of the transfer policy, said, 'transfers are done for administrative reasons', i.e. not as per the stated policy, with the term 'administrative' functioning as an euphemism for

inappropriate and inadmissible reasons. In other words, if a transfer led to an inappropriate selection of an official for an assignment, and subsequent poor performance, no political party would take responsibility for it or be held responsible by the opposition political parties or the electorate. No parliamentary committees or audit teams would scrutinize such decisions. This is not quite (competitive democratic) 'politics', or for that matter governance or public administration. The important point here is that the problem of transfers needs to be seen not in terms of the politics-administration divide and other policy related arguments, but in terms of a process that does not follow *any* publicly stated and owned criteria or policy. This distinction is related to the argument made in Chapter 1 of this book, that a large chunk of the problem with elementary education is not a question of 'which policy', but the fact that actions are divorced from *any* stated public policy. Such actions escape 'policy considerations', but form an important part of the reality all the same.

Though transfers allowed the easiest play for patronage-based functioning, it permeated other activities too. The selection of para-teachers had created new opportunities, as did new programmes. A CRCF told the researchers that though he had received complaints about a dysfunctional alternative school there was pressure from politicians to not take any action. In Rajasthan, even the selection of resource persons for teacher training (where some honorarium was paid) was not based on capability, but was doled out to those who were in favour. Moreover, employees were manipulated not just through transfers, but also through other personnel matters such as sanction of leave and payment of travel and medical bills. The 'establishment' or personnel section was considered extremely important and a great deal of deference was shown to those who held such positions.

Corruption

While allegations of corruption were present in both the states, its existence was starkly revealed in the case of Rajasthan. The dysfunctional schools in Rajasthan (see Chapter 4 and 8) were an outcome of, along with poorly thought out policies, rent-seeking operations within the system. Several members of local elected bodies told the interviewers of this study that supervisors often had a 'setting' in some schools where they overlooked the fact that it did not function regularly. One of the interviewees, a tribal woman chairperson of the block level panchayat, was aware that the block officials were linked to powerful politicians on the one hand, and turned a blind eye to malfunctioning schools where they had such an 'understanding', on the other.

Often, processes were changed and adapted to enable rent-seeking. In a single teacher school in Rajasthan, money was sanctioned for the construction of a hall. There were already three rooms in this school, and in any case, all the children sat in just one room, as there was only teacher school. The contract had formally been awarded by the SDMC, but actually the school teacher had been asked by a supervisor to engage a certain contractor and had acted accordingly. The 'contractor' in turn, was another teacher from a nearby school! Another cluster resource person, who owned a shop and spent a great deal of time there, wanted a greater role in 'monitoring' expenditure. His academic responsibilities were apparently not rewarding enough. A block level functionary claimed that at the district level, all contracts were awarded to a single party. He had asked for a budget for stationery, but it was not sanctioned. Yet, at the end of the financial year, budgets were provided to officials in other blocks upon payment of a certain sum of money to the district office. There was evidence of corruption in teacher training too. Non-residential training programmes were shown as residential ones and the residual money pocketed. Teaching-learning material (TLM) had perished in most schools. The widespread nature of rent-seeking in the system was also apparent from the fact that some interviewees expressed a sense of pride at not being corrupt, obviously seeing it as a not very common achievement.

Allegations of corruption existed in Andhra Pradesh too. NGO representatives pointed to the misuse of the TLM amount and alleged that some MEOs provided textbooks, which were meant for free distribution to children in government schools, to private schools. Corruption also occurred in the process of obtaining permission for getting recognition for private schools. In both the states, corruption existed at all levels, across professional groups (though some had more opportunities than the others) and pervaded all types of activities. That the 'top' was involved was clear from what was tolerated—dysfunctional schools, creation of inappropriate assets and hijacking of teacher training programmes. The supervisory level was clearly implicated (it was the most easily visible). In recent years, with increasing availability of money for elementary education, teachers too had started engaging in unethical practices.

Though in both the states the dispensation of patronage and rent-seeking were important dynamics, from the evidence of this study, Andhra Pradesh appeared to have an edge over Rajasthan in having a less pervasive informal system. As has been noted, in Andhra Pradesh, for postings of teachers and junior officials, the patronage-based system had been replaced by a transparent, rule based process. Though allegations of corruption existed in Andhra Pradesh too, in the village case studies researchers did not come across the more flagrant variety that was visible in Rajasthan. Schools that existed on paper and teachers who did not teach, were phenomena prevalent in Rajasthan, but not in Andhra Pradesh.

The emerging pattern

This stated formal and the unstated informal dynamic can be conceived of as two separate sub-systems within the system. The two have different goals. The formal sub-system is oriented towards carrying out government business, even though its capacity and processes to do so are limited. The informal sub-system is oriented towards personal profit, wherein state actors attempt to maximize personal gains either in terms of retaining political power or in terms of deriving monetary benefits from it. The informal dynamic is better visualized as a sub-system, rather than a few aberrations or practices, because it is highly organized, has its own unstated rules and processes and 'disciplinary mechanisms', and is widely prevalent.

It is important to keep in mind that *both* the stated and non-stated sub-systems operate, and they operate *together*, i.e., at the same time. For instance, while there are many individuals in the system who seek rents, there are also others who do not. The former operate as per the 'rules' of the unstated sub-system, and the latter as per the rules of the stated sub-system. In the same vein, the informal sub-system has to constantly contend with the accountability processes of the formal system such as audits, which as noted above, can be quite powerful. Moreover, both the formal as well as the informal sub-systems draw strength from individuals and institutions outside government. Parts of the media and the judiciary, as well as some activists and political opponents of ruling parties, and in some instances ordinary people, try to expose and attack the informal system. At the same time, individuals who benefit from the informal system, i.e., those who get lucrative contracts or other benefits through it, support it in several ways including helping their collaborators get important postings.

In the two states, the coexistence and interaction of these two sub-systems created a pattern that comprised the management practices of the system. This pattern was quite complex and interesting because at times the practices of the two systems built on each other, while at other times they stood in opposition and in still other instances a totally new process could emerge to accommodate both the sub-systems. This management pattern formed the context and the working conditions in which employees functioned, in which many of their actions, which seem 'stupid' and meaningless at first glance, were explicable. As the following discussion indicates, the new management pattern that had emerged exacerbated and magnified the problems of the formal sub-system.

A new hierarchy

In Rajasthan and Andhra Pradesh, as the formal and the informal sub-systems operated in tandem, they produced a new hierarchy that functioned along two, rather than one, dimensions. The first dimension was constituted by the formal hierarchy, dependent on the stated rank or seniority of a person and the second dimension on the hidden rank or power based on the importance of the individual in the patronage and rent-seeking networks. The hierarchical pattern that emerged is given in Table 5.1.

Table 5.1: The two dimensional hierarchy

Formal hierarchy: High Informal hierarchy: Low These individuals occupied high-ranking posts, but not necessarily the most important ones. They had a fair degree of influence vis-à-vis stated policy, but were unlikely to have much influence over the placement of personnel. Such individuals were transferred frequently, not given their due and even harassed. However, their travel and medical bills were settled, and they were reasonably well paid.	Formal hierarchy: High Informal hierarchy: High These were the most powerful (and feared) individuals in the system. They held high ranking as well as important posts. They had a say in policy and also some authority over the placement of personnel. Such individuals usually got choice postings and were not easily transferred. Sometimes when political regimes changed, such individuals could be harassed, but they were usually capable of making new alliances. Many of them amassed huge wealth.
Formal hierarchy: Low Informal hierarchy: Low Such individuals tended to be powerless. They had no voice in matters of policy, and their experience was usually discounted, even though many of them had daily interactions with schools. They were often ignored and isolated. Harassment occurred in several ways. The extreme case was that of hapless teachers who could not get their medical and TA bills settled, and had to pay money to get transfers to convenient postings.	Formal hierarchy: Low Informal hierarchy: High Such individuals were junior officials. Several supervisors and field personnel fell in this category. They did not formulate policy or take important decisions, but they were treated with a great deal of respect within the system. They often got 'choice' postings and usually could not be removed from a post for non-performance and rent-seeking. Many managed to amass a fair amount of wealth.

Thus, formally junior employees were sometimes quite powerful. For instance, in our sample a block level official who was well linked politically, had been removed from his post by the district collector because of visible and flagrant rent-seeking. However, he had subsequently got himself posted back to the same location. Individuals who were high up in the informal hierarchy were often pace setters and role models, as they led an easier and materially more rewarding life. Several interviewees—teachers as well as officials—said that others often followed their example.

An important characteristic of this management pattern was that almost no one appeared to have the power to push through policies and programmes on the ground. The most powerful individuals in the system, high in the formal and informal hierarchies, were often far more interested in manipulating the system for personal gains than in programmatic issues. Those who ranked high in the formal system but low in the informal one, had only a small say in matters. It was possible for them to push through some *articulations* of policy and programmes, but they had little influence when these were subverted on the ground. This could hold true even of political leaders, whose party members could stymie their plans. People low down in the formal hierarchy had little say in the formulation of policy and

programmes, but those who were high up in the informal hierarchy often engaged in activities which sabotaged or changed policy substantially. The emergence of 'ineffective policy' in such a system is not surprising.

The two hierarchies sometimes built upon each other, as in the case of senior officials who profited the most, and hapless teachers who had to pay for transfers. At other times, they also clashed. One instance of a rather dramatic confrontation between the two hierarchies was the case of an erstwhile director of education in Rajasthan, who had attempted to 'clean up' the system but who had been threatened with violence and as a consequence had stayed on leave for three months. In general, formally top-ranking officials who were not part of the informal network occupied a risk-ridden terrain, especially if they took their formal position and the stated rules and principles too literally.

Such stark confrontations between the two hierarchies and systems were, however, rare. In general, the existence of the informal sub-system exacerbated and magnified the role of the hierarchy within the system. The distance between personnel high up in the formal as well as informal hierarchy and those low down was much wider than the one in the simple formal hierarchy. As has been noted earlier, in the formal sub-system, along with the focus on hierarchy, there was a considerable degree of employee protection in terms of security of service and disciplinary procedures etc. that mitigated top-down functioning. However, with the operation of the informal sub-system through transfers, these redeeming safeguards were destroyed considerably, and those on top of both the hierarchies enjoyed enhanced power to act as per their personal interests and biases, over those below.

Authority and responsibility

In the formal system, along with authority, came responsibility. This responsibility tended to induce some democratic practices as well as reflection, as these promoted good decision-making. For instance, this study found that despite the extreme centralization of the formal system, several second rung officials said that most of their proposals got approved, though this remained tenuous, and depended a great deal on the particular head of the institution. On the other hand, an important characteristic of the new hierarchy was the disassociation of authority and responsibility. The informal system operated through oral orders of persons who did not openly claim responsibility for their decisions, unlike written orders that emanated as a consequence of government procedures. The responsibility instead lay with the official who did the paperwork. For instance, in Rajasthan, unsigned transfer lists were handed over by junior ministry officials (who had obviously received orders from elsewhere) to officials of the directorate, who in turn were asked to merge the lists with 'legitimate' ones while issuing orders. If problems arose subsequently, those who had acted 'under pressure' were liable to be punished, while those who had exerted the pressure were not reprimanded. This disassociation of authority and responsibility made the new hierarchy especially unaccountable.

Increased centralization

The informal system also reinforced centralization. Individuals who were high up in the formal as well as informal hierarchy acquired a special stake in centralization (which included ultimately the centralization of rents) and defended it with vigour. Centralization aids opaqueness, which in turn is essential for the operation of the parallel informal system. Notably, in both the states local elected bodies had little role in elementary education. While this cannot be read simply as a way of aiding the informal system, local elected bodies do pose certain problems for it: they create political competitors who benefit from exposing rent-seeking operations of their opponents and if they have to be co-opted, then they have to be allowed a share of the rents.

The informal dynamic aggravated the weakness of field offices that resulted from the hierarchical ordering of the formal system. As per the principle of the formal hierarchy, the closer an institution was to the ground, the less well equipped it was in terms of qualified and trained personnel and infrastructure. In addition, officials at these levels were subject to constant pressures from the informal system to flout rules. They either caved in, or were constantly engaged in dealing with illegitimate demands from powerful individuals within the system. Consequently, several of them felt harassed and had little time for work. In Andhra Pradesh, one salutary effect of the process of counselling was that district and sub-district officials could no longer be bullied regarding teacher transfers. District officials said that though politicians wanted to be involved in transfers, it was easy to refuse them as the rigidity and transparency of the process made it impossible. In general, in Andhra Pradesh as compared to Rajasthan, the administrative officers were far more engaged with the task of supervision.

Changed working conditions

Government officials and teachers worked in a context created by these two simultaneous and often mutually contradictory sub-systems. This created new 'working conditions', i.e., fresh incentive structures and ethos. The simultaneous operation of the two sub-systems created a system of 'double-bind'. On the one hand, there was constant pressure from powerful figures to flout rules and on the other there was the constant fear of being caught. A district official interviewed in Rajasthan who was nearing retirement, complained of persistent political pressure regarding transfers. Moreover, he felt harassed by the district collector, an enthusiastic and energetic but relatively inexperienced official, trying to make things work. The DEO feared both the politicians and the district collector, as being pushed into doing something wrong by one and being caught by the other, was a possibility that he wanted to avoid, and retire 'safely', i.e., with his retirement benefits intact.

In general, professional relationships had been considerably transformed through the operation of the informal system. Collaborations and disagreements centred not on work related issues, but other extraneous matters. Several supervisory officers, rather than supervising to get better quality of work, were themselves enmeshed in a system of rent-seeking and patronage networks. For instance, a cluster resource person in Rajasthan ran his own private school and encouraged students to leave the government school and join his. The case of a supervisory officer who had been removed from his post but had returned has been noted above. 'Supervision' by such individuals comprised the establishment of networks for rent-seeking.

At the same time, supervisory officers who were not part of the patronage and rent-seeking rackets were not keen to risk taking disciplinary action against those who were, because those who were disciplined could well be more powerful in the informal network and thus capable of stalling disciplinary proceedings and making things difficult for the supervisors in addition. There was a tendency on the part of such officers to maintain 'harmonious' personal relationships, often at the expense of work. A block level official in Rajasthan preferred to 'get along' with everyone, rather than exercise his disciplinary powers. Barring exceptions, local government representatives were not engaged in battles over the functioning of schools. Such an attitude is comprehensible if we take into account the fact that individuals who were low down in the formal hierarchy could be high up in the informal one, and in case of a 'clash', the latter often won.

Within the system, it was fairly well accepted that errant employees could not be held to account. For instance, many employees in the Bikaner directorate owned shops and left the office by 3.30 pm instead of the official time of 5.30 pm. Senior officials who did not agree with this practice felt that they could not intervene. A senior official in the directorate said:

It is best to keep *kamchors* (shirkers) engaged in activities such as gathering information. If you take them on, they will launch agitations. Union pressure and political pressure helps people escape work. Getting into a conflict with such people is like getting into *keechar* (a filthy swamp).

Similarly, as part of a quality improvement programme, an MEO attempted to counsel teachers to develop a more positive attitude towards their work. According to him, while some teachers changed, others had not. The MEO merely ignored those who had not.

Such careful behaviour notwithstanding, relationships within the departments in both Rajasthan and Andhra Pradesh are not harmonious. In fact, during our research, we encountered several instances of hostility among officials of the same department, but these were not focussed on work related issues. With the importance of the informal system, professional relationships had become highly personalized. Professional competition was not mediated and softened through shared professional goals and recognition for genuine contributions. It was not refereed on the basis of performance and achievement. Instead, 'professional' competition centred on capturing various posts through manipulation. The important posts came not because of a person's work, but as a consequence of currying favour with the people at the top. 'Doing well' was increasingly a question of manipulating the system for personal gains rather than performance with respect to the goals of the department. One incident that came to light in this study was a bitter feud between two senior officials in Andhra Pradesh as both jockeyed for the same position using politicians as well as teachers' unions. Who was better in terms of work or contribution, was immaterial. Another incident in Rajasthan seemed to arise from competition over rents. A block level official was bitter because his district supervisor had cut him out of all financial transactions. He said that he was keen to move out as his supervisor made things increasingly 'difficult'.

Hostilities also arose from clashes between the formal and informal practices of the system. The simultaneous operation of the formal as well as the informal sub-systems led to a constant sense of distrust. Several supervisors did not trust those they supervised and vice versa. The conflict between elected representatives and the bureaucracy was also exacerbated. The more straightforward officials resented unreasonable transfers and the constant pressure to accommodate the informal system. Those high up in the informal system saw such officials as 'impractical' and 'negative'. 'Team building' and 'team work' were not possible when teams were split along core values.

Leadership

The operation of the informal system along with the formal system affected the dynamics of institutional leadership. The formal system dictated that individuals become leaders as per their seniority, rather than the quality of their work. In such a system, we may expect leaders of reasonable ability (as recruitment procedures are fair and there is tough competition) and a great deal of experience. However, such a system cannot ensure that specially capable and dynamic individuals will be placed in leadership positions, though some leaders may have these qualities. With the informal system, new handicaps had been added. Often, patronage and rent-seeking formed the basis for selecting leaders, as this was how 'important' posts were obtained. In the case of patronage based selection of leaders, 'selected' leaders were less likely to be interested in the work of the organization and more in milking it for their own interests and for perpetuating the agenda of their mentors.

Patronage based selections, however, could not have their full play because of the importance of seniority and automatic promotions in the formal sub-system. Moreover, sometimes genuinely capable leaders were selected for particular activities or a 'showcase' project, or when there was a crisis. However, the capacity of such leaders to lead was distorted because of the existence of the informal system. In both the states, institutional leaders were highly

mobile and liable to be transferred at any time, often remaining in a post for only a few months. As Table 5.2 shows few leaders had adequate time—say three years—to make an impact. Notably, during the course of this study, the leadership in SIERT/SCERT in both the states was tenuous. The director in one state had recently been removed because of an alleged political controversy and another had not been appointed, while in the other, he had been ‘under transfer’ and was not sure whether or not he would stay.

Table 5.2: Average tenure of state level institutional leaders in the last five years

	Administrative		Teacher Support		Programme	
Rajasthan	Director/ Commissioner	10 months	Director SIERT	2.5 years	State Project Director	7 months
AP	Director	2 years	SCERT Director	1 year	State Project Director	7.5 months

Source: Field study data (April to October 2005).

Note: * One person had stayed for 10 years.

Moreover, when in office, the power of the more straightforward leaders was significantly constrained as they had to contend with the informal power of the employees of their organization. They could not make justified demands vis-à-vis work from many employees. Often, leaders had to invent their own ways of somehow getting the most urgent tasks done. The head of an organization in Andhra Pradesh told the researchers,

‘If someone gives me an application, I keep a photocopy because I cannot trust that it will be followed up. With the staff, one has to appear to be very trusting, but still be very cautious.’

In other words, the very nature of leadership and the capacity to lead has been transformed beyond recognition.

Personnel management

The importance of the informal sub-system was clear in the ‘personnel policy’ that had emerged in practice, where placements were made with scant regard to an individual’s suitability for the work. If the formal sub-system posted SCERT and DIET faculty without much regard for the expertise of the personnel, the informal sub-system made the situation worse by posting well-linked (in the informal system) individuals to these institutes as per their convenience. Those who were well linked in the informal system often felt free to do very little work. Consequently, key institutions were staffed with people who chose to be there for various personal agendas, rather than those who were likely to perform the job best. One faculty of SCERT noted,

‘SCERT is a free zone. People come here for personal reasons, for instance if the spouse is working in the city.’

In Rajasthan, a more merit based system for placing people in SIERT and DIETs had existed at one point of time, but had subsequently been abandoned. Besides this, work assignments within organizations could be changed casually.

As noted earlier, in the formal system, there was no modality for appreciating exceptional contributions made by individuals. In the informal system, reward and punishment took place around transfers for reasons of personal loyalty rather than work. Consequently, individuals who had initiative and approached their work creatively often ended up looking elsewhere for employment. For instance, in Andhra Pradesh one headmaster who had shown an initiative in developing the government school had ultimately set up a private school of his own. In a different type of system, such people may look forward to swifter promotions, more important

job assignments as well as recognition and appreciation. However, the reward system positively promoted and rewarded an individual's capacity to manipulate the informal system, with the result that individuals interested in genuine professional goals sought their rewards elsewhere, either leaving altogether, or developing other professional networks. This depleted the professional capacities of the system further.

Procedures: tied up in knots

An important outcome of the coexistence of the stated and unstated sub-systems were extremely complex and long-drawn processes, which appear senseless if seen in terms of work efficiency, i.e. the 'red tape' for which the government is famous. To some extent because the government departments are large organizations and because all government procedures have to incorporate processes to ensure public accountability, this is inevitable. For example, in Andhra Pradesh, scholarships were often delayed for up to one year as the process was time consuming. The MEO had to take information from the schools and send the proposal to the DEO office. This was then verified by the MRO. The extreme centralization and rigidity in rules and procedures which characterized the formal sub-system further complicated government processes. More interestingly, however, it was the existence of the informal sub-system which leads to a host of sterile, unproductive practices.

Some complicated processes had arisen as a simple outcome of rent-seeking. As has been mentioned earlier, in one instance, allotment for stationery to subordinate offices was made at the end of the year and only after the payment of a commission. Obviously, procedures take more time if commissions and other benefits have to be 'fixed'. For instance, one field functionary in Rajasthan told the interviewers that proposals for new schools had remained pending for a long time because the location of these schools had to suit political expediencies. Notably, in this example, it was not possible to allot schools purely because of patronage, as government norms regarding population and distance had to be observed and the original proposals were made on this basis. However, schools were not sanctioned according to these criteria. Hence, there were delays because narrower concerns had to be 'fitted in' with formal requirements.

Cumbersome processes were also instituted as the formal sub-system attempted to counter the informal one. For instance, in Andhra Pradesh for most of the year there was a ban on transfers, a procedure instituted to stem arbitrariness, particularly as transfers in the middle of the school year can be especially problematic for several employees. In the 'ban' period, transfers could be carried out only with the approval of the chief minister, which meant that the process took a long time. While this provision reduced employee harassment, it introduced rigidity and caused hardship to some people with legitimate needs. Similarly, in Rajasthan, a block official had asked the teachers to note the year (in which it was made) on the teaching-learning materials, a process which did not lead to any benefits for the teacher or the students. The assumption here was that the TLM money would not be used for the stated purpose, and the supervisor was attempting to devise an accountability measure.

Similarly, the extensive legislation that characterizes the education system today is only partly comprehensible in terms of genuine employee protection. Its prevalence is also related to the fact that employees cannot be sure of getting justice in the type of system that we have described. In Rajasthan, there were more than 9,000 court cases in the elementary education directorate and in the words of a senior official, the directorate was 'a client of all the major lawyers and law firms in the state'. At the time of this interview, a senior state level official in Andhra Pradesh had just dispatched a team to the court and said that it was a daily occurrence. He said:

You know how I organize my time? Every day I list out the number of court cases as well as urgent correspondence and requests. First, I try to sort out all the urgent priority items. I leave office at seven or 8 pm. I also take files home.

The simultaneous existence of the two sub-systems increased the (non-productive) workload on the employees as they dealt constantly with 'pressures' and followed cumbersome procedures. One senior official in Rajasthan remarked that he received a 'sack full' of applications everyday and constant visits by employees, politicians and union leaders were a common feature. Officials also reported spending long hours dealing with petitions and demands. Often, what appeared inaction at the end of a day in terms of educational goals, was actually a great deal of hard work that the officials had put in 'fire-fighting'.

Decision-making became increasingly difficult and risk-ridden, and people often avoided taking decisions, even when they could, especially financial ones. A senior official in Andhra Pradesh complained:

'Even if there is a puncture in the tyre of a car, the file comes to me. Everyone is scared of taking decisions. The attitude is, let the boss take the rap.'

A middle level official in a state office in Rajasthan said:

We do have the capacity to take decisions as per rules. However, we want to please the boss and shy away from taking independent decisions so that if there is any problem later on, he is accountable and can answer...I do not want to take any financial decisions. The accounts people always put a spoke in the wheel. When there is real corruption, they are the first ones to get involved.

Several senior officials also wanted to escape responsibility. The director's post at Bikaner was generally unpopular because it involved dealing constantly with pressures regarding transfers, rather than work related to education. As the actual achievement of goals seemed increasingly difficult, the tendency was to shift the blame to the juniors. An NGO representative in Andhra Pradesh commented:

'The reason for reporting fake enrolments is the blame game. The blame is constantly shifted down. The fellow at the bottom then just gives false numbers.'

The marginalization of work

In the management pattern sketched above, the end result was the marginalization of work. We saw that the processes in the formal sub-system of promotions based on seniority rather than merit, several rules and procedures and a rigid hierarchy etc. militated against a productive work ethos. This was considerably exacerbated by the operation of the informal system. As the personnel within the system engaged with the informal sub-system in various ways, the time available for productive work declined. We also noted how work time got utilized in various sterile practices. Moreover, the informal sub-system stood in the way of decisions to improve the education scenario. A very visible example of this was the fact that in Rajasthan, pupil teacher ratios were skewed in favour of urban areas. Yet rationalizing these by transferring teachers away from cities to villages was difficult, as it invariably led to 'pressure', exercised in favour of employees rather than communities. The ultimate result was the existence of a large number of schools without adequate teachers in far flung areas, while several urban schools had more teachers than needed.

The supervision of schools indicated the same dynamic. On the one hand, supervisory officers told interviewers that they were engaged in so many tasks that they could not find time to supervise. On the other hand, discussions with teachers showed that when they did supervise,

they rarely went beyond checking records.⁷⁹ A teacher's comment on the last visit of the supervisor was typical:

'He chatted with the teachers and checked the records regarding midday meals. He did not ask for any information about teaching or children's learning levels.'

We have also noted that several supervisory officers had established networks for rent-seeking, which was highly profitable, and in any case, doing their work well led to no professional recognition beyond personal satisfaction, which sustained only a tiny minority.

The subsidiary place of work was also evident in the marginalization of academic practices, which form the core of the work or 'output' that the system is expected to produce. For instance, grassroots resource persons visited schools to collect information rather than discuss the teaching-learning process. Concerns about TLM centred on the misuse of money, and not on obtaining and using relevant materials in the classroom.⁸⁰ Examination results were manipulated in various ways in both the states. In Andhra Pradesh, a 'handbook' that listed various important questions and answers had been prepared at the district level. This was obviously a signal to the paper setters as well as teachers to focus on certain questions to ensure that too many children did not fail. In the previous year's examination, results in the sample district had deteriorated, not because of any issues related to curricula, teaching or assessment, but because monitoring of malpractices had become stricter.

In such a scenario, it should not come as a surprise that on the ground, the functioning of the schools was not satisfactory. This situation was especially disturbing in the sample villages of Rajasthan, where several RGSP and alternative schools functioned only occasionally. In formal schools too, some teachers were not present in the school at the time of the researchers' visits. In both the states, officials, teacher educators, headmasters, panchayat representatives and even teachers said that teacher motivation was a 'problem'. To quote a teachers' union leader in Rajasthan:

I have been working since 1979. Ethics have really declined in society since then. Teachers have become less dedicated. They have got disassociated from academic activities and don't use the library. Some get involved in other businesses.

Even when teachers were present, teaching time in schools was very limited, especially in the sample schools in Rajasthan. Some teachers struggled to manage more than one class at a time and in the process got little time to teach. This problem was aggravated when teachers had to do non-academic work such as preparing information sheets. Some teachers simply chose to chat. Here, it should be kept in mind that the problem of teacher absenteeism and lack of motivation did not exist independently, but was rooted in the management ethos detailed in this chapter. Teacher motivation needs to be seen against the signals that were being given to the teacher through supervision that had little to do with the teaching-learning process, meagre academic resource support, and in the case of Rajasthan, the threat of transfers. The sample schools studied were too few for us to make any general statements about the entire states of Rajasthan and Andhra Pradesh. However, in our school sample, we did find that schools in Andhra Pradesh functioned better than the schools in Rajasthan. The connections with the management ethos in the two states are obvious.

⁷⁹ By 1966, a Rajasthan government report was already remarking on the deterioration in supervision of schools. One reason was the rapid expansion of schools and the other the fact that supervisors were not trained for the job, but were secondary school teachers. It also talks of harassment of teachers through transfers, where panchayat samitis have abused their powers (GoR, 1966).

⁸⁰ A study of Teacher Centres in DPEP Andhra Pradesh found that politically influential teachers remained absent from teacher training programmes (APDPEP, 1991a).

Crisis and Redress

The systemic characteristics that have been identified lead to outcomes, which while serving the interests of several powerful individuals within the system in the short run, result in problems in the longer run for the government. As noted above, the whittling of the space for productive work leads to dysfunctional schools, poor teaching in schools, and subsequently low achievement levels and high dropout rates. These become problematic because on the one hand they are issues on which the government can be attacked in Parliament, by the media and by various civil society groups, and on the other, because some people within the government are genuinely concerned about this state of affairs. As noted in Chapter 1, in the last decade and a half, domestic as well as international pressures to make elementary education universal have grown, but the poor quality of government schools keeps several children out of school. Additionally, poorly functioning schools in villages present a face of the government that hardly helps the popularity of the ruling party.

The fact that an increasing number of children are choosing to opt out of the government school system ultimately also affects government jobs. For instance, in Rajasthan because of increasing privatization, several schools had become uneconomic. As per state policy, government schools had to be closed down if they attract less than 50 children over three years. This study found that usually, when such a situation developed, teachers were asked to increase enrolment. However, 31 schools which had teachers, but not enough children, were closed down in Bikaner. Aggrieved teachers had filed a court case, but the government had won. More proposals for shutting down schools were reportedly pending decision. In Andhra Pradesh too, schools were closing down because of lack of students.

The concern in parts of the government with out-of-school children and the quality of schools, as well as the problems and embarrassments caused by poor outcomes, had led to attempts to rectify the system. Some of these attempts were aimed at correcting the limitations of the formal system, while others aimed at reducing the space for informal practices. These attempts at change may also be seen as part of the 'system', as they often lead to new institutions, initiatives and practices which subsequently become 'mainstream'. Understanding these attempts is useful for comprehending systemic characteristics in greater depth and also for understanding how *effective* change can be brought about.

Change attempts can be divided into three broad categories: first, there are campaigns and policy shifts through which the government attempts to change the parameters of systemic functioning. Second, increasingly during the 1990s, the government has attempted structural changes within the system through decentralization and community participation and by establishing project structures. One idea underlying these changes was that they would be better than the existing structures in terms of accountability and performance. Third, again increasingly during the 1990s, the government attempted to engage with non-state actors, shifting responsibility away from the public system to various degrees. We now examine how these change attempts have fared in Andhra Pradesh and Rajasthan.

Campaigns and frequent policy shifts

In general, the government system is characterized by a constant search for programmes, schemes and campaigns that are more effective than the previous ones. Consequently, 'new' policy thrusts and programmes are a familiar feature of the system. Many of the programmes initiated are desirable and address previously neglected areas. However, many are simply repetitions of old formulas in new schemes with a few new features added on. Some are symptomatic of the constant search for the 'right' programme or policy thrust, others are more cynical window dressings and several contain both these features.

At the time of this study, a *shiksha maha andolan* (grand drive for education) had been started in Rajasthan with the new school session. Enrolment drives had been a regular feature in the state, and the new feature added with this campaign was that in place of the usual village level enrolment drive, senior officers from the IAS, the police and the education department were allotted districts for enrolling out-of-school children. Similarly, in one of the mandals studied in Andhra Pradesh, in spite of the existence of large-scale programmes to address the quality of education, a new programme called the 'Quality Improvement Programme' was started. In the same vein, in Rajasthan, the policy for the recruitment of rural primary teachers had changed twice since 1998.

However, the success of such policies and programmes tended to be short-lived and programme ineffectiveness was a recurrent feature as all programmes ultimately had to contend with the practices of the formal and informal sub-systems. Thus in the DIETs set up under the teacher education scheme, on the one hand professional expertise had been given scant attention, and on the other, these institutes had become parking slots for those who wanted to stay in a particular place. New policies and programmatic thrusts were simply adapted to suit existing institutional practices, which ultimately undercut them. The enormous importance and power of the informal system in particular, shrank the space for policies and programmes, as these got transformed into patronage dispensing or rent-seeking operations.

Changing structures

Several policy makers and many persons who work in state and non-state organizations and engage closely with schools and other parts of the system are aware of this state of affairs. Consequently, there was some engagement, particularly during the 1990s, to bring about changes in the structures and processes of the system. Two types of attempts were made in this direction. The first was a focus on decentralization and community participation, and the second, the establishment of 'project structures'.

Decentralization and community participation

The idea of systemic improvement and change through decentralization and increased community participation has been a recurrent theme in Indian administration in general, and has also been considered and recommended in the context of elementary education (GoI, 1992, 1993). However, the space for people's participation is intimately related to existing state practices. It has been argued that the formal sub-system promotes centralization and the informal sub-system exacerbates it. Both the sub-systems create propensities for the marginalization of community-based institutions.

In Rajasthan in particular, major decisions had been taken away from the panchayats. A district panchayat representative in Rajasthan, on being asked what the role of the panchayat was vis-à-vis education said,

'There is no role at present. Transfers and postings are the only important things. This was done by a committee of the panchayat earlier, but now DEOs are called and asked to issue lists.'

However, panchayats did continue to review the functioning of schools through meetings and panchayat representatives continued school visits.

At the same time, the SDMC in Rajasthan, expected to institutionalize people's participation, was a nominated body with no independence. SDMC meetings were usually a hoax and members were merely asked to sign on various papers, which they did. In fact, by expending funds through the nominated SDMC, even existing bureaucratic safeguards had been bypassed, while there was little evidence of any accountability to the community. There were

also indications that panchayats were co-opted into the informal system. For instance, panchayat representatives admitted that dispensation of patronage had played an important role in the appointment of para-teachers when it had been the responsibility of the panchayats; school teachers and SDMC members confirmed this.

As in the case of more straightforward officials, panchayat representatives who were not part of this system stood marginalized. For instance, a block panchayat president who was a scheduled tribe woman and a community leader interested in better education rather than personal profit, said that she refused to sign on salary bills of teachers who were not working well. However, teachers were not afraid of her inspections, as they had good links with higher authorities. She told the researchers that supervisory officers followed their own interests and agendas, using political influence. Similarly, a village panchayat president said that he had no solution for making teachers work better.

The panchayats stood marginalized in Andhra Pradesh too, but at the time of the fieldwork, the SEC presented a far more positive picture than the Rajasthan SDMC did. It was an elected body, and participation from SECs and the community in general was far more visible in Andhra Pradesh than in Rajasthan. In three out of the six sample villages in Andhra Pradesh, villagers had contributed to the school. Interestingly, in one village members had been elected to the SEC on the basis of the monetary contribution that they had made to the school. While some amount of community participation in Andhra Pradesh was a positive sign, its limits need to be recognized. SECs had engaged mainly with school buildings and other infrastructure development. Moreover, several SEC members said in their interviews that they favoured beating children to discipline them.

As in the case of panchayats, the role of the SECs had shifted depending on political expediencies. The SEC in Andhra Pradesh was part of a larger thrust towards community based institutions such as self-help groups (and not panchayats) in the state. Some observers have suggested (Powis, 2003) that such institutions were used by the ruling party to bypass the panchayats and extend its influence and build party cadres. This is also indicated by the fact that within the SECs, political party competition was high. In one village, voting had become difficult because of a political conflict at the time of elections. Subsequently, the officials had taken an arbitrary short cut with the result that an unwilling chairperson, a parent whose child was doing well, had been nominated. In fact, as institutions, SECs were even more tenuous than panchayats. At the time of writing this book, elected SECs had already been replaced by bodies with several nominated members.

In other words, decentralization and community participation too were affected by the dynamic outlined in the first part of this chapter. Decentralization of power was least likely when powerful individuals within the system benefited personally from its centralization. Moreover, decentralized bodies too could be co-opted. Decentralization did not *per se* ensure effective state functioning.

Project structures

An intriguing development during the 1990s in the education sector was the creation of 'project structures'. The large-scale use of such project structures since the mid-1990s has been the subject of a great deal of debate as well as criticism (Kumar et al., 2001). Project structures on a large scale, which function somewhat independently of the regular system, emerged in the context of DPEP, an externally funded project, and were retained in SSA. One explanation about the emergence of project structures has been that they facilitate the execution of externally directed policies. From the point of view of this chapter, however, another aspect of the project structures is of greater interest: the perception that project structures deliver more efficiently than government departments. Notably, the projects structures of DPEP have been continued in SSA, which is mainly domestically financed. The

creation of project structures is also an attempt to develop a more viable structure or institution, and it is not surprising that these were used extensively first in DPEP, as externally funded projects place more pressure on the government to show results. This aspect of the project structures is best understood by examining two small, early project structures of Rajasthan (Box 5.4).

Box 5.4: Project Structures in Lok Jumbish and Shiksha Karmi

Two programmes in Rajasthan—Lok Jumbish and Shiksha Karmi—were small, early projects. At their best, these projects evolved management structures and processes that effectively countered several non-productive features of the system and were able to produce outcomes that went well beyond the usual capacities of the system. These project structures differed substantially from the formal departmental system and they also managed to mitigate some of the effects of the informal system.

In both these projects, the composition of the ‘apex’ or decision-making bodies stood in sharp contrast to the simple hierarchical structures and unaccountable informal networks that characterized the regular system. The Lok Jumbish Parishad and the Shiksha Karmi Board, the apex decision-making bodies in these programmes, comprised professionals from within and outside government with varied sets of skills. This composition of the decision-making body served to reduce the extent of hierarchy as well as patronage within the system. A *group* of people, several of whom were not part of the state system, became part of the decision-making process. Consequently, patronage-based decisions became difficult. Instead, intelligent strategies emerged as a variety of individuals with genuine interests and capacity for educational reform came to have a say in major decisions.

Another important feature of both these projects was the talent hunt for personnel from within and outside the government that was undertaken. This contrasted sharply with the basis of personnel deployment in the state system, i.e., seniority in the formal hierarchy and links in informal networks. Importantly, these projects selected the most motivated and capable people from *within the government* and placed them in key positions. These projects were thus able to create a capable, work-oriented leadership.

One more important feature was that grassroots institutions were considerably strengthened. In Lok Jumbish, at least in the original project, each block had a block level body, the Khand Stariya Prabandan Samiti (KSPS), with a distinguished person with an interest in education as the chairperson and the project officer as member secretary. Again, as in the case of state level committees, decision-making by a group of people undercut patronage and rent-seeking. Once these structures with better capabilities, protected from the informal system, had been set up, upgrading technical skills of project personnel and accessing expertise of NGOs and resource agencies further strengthened their capacity. Correspondingly, engaging them in several project activities and allowing them to be part of the decision-making process deepened links with the community.

Finally, project structures with competent and motivated personnel engaged intensively with the community. Community engagement enabled the project structures to address needs emerging on the ground and community support vastly increased their capacity to actually implement programmes. As the patronage and rent-seeking possibilities were countered *and* technical capacities upgraded, it became possible to free the new structures from rigidities that came in the way of creative project design and implementation, introducing more financial flexibility and cutting out the usual red tape. The new flexibility could then be used to take up more context-specific activities (as against using it as an opportunity for seeking rents). Consequently, there was a shift from activities as per pre-set rules to meaningful processes to achieve core goals.

Source: DPEP (1998), Sandhan (1997), SIDA (1996).

The success of these, and other early projects paved the way for project structures of a much larger scale to be established in DPEP, which have continued since in SSA. These consist of a set of institutions described as the ‘programme sub-system’ in this study. The addition of programme structures to the regular structure (i.e., the directorates) has led to three important changes in the system: the availability of personnel freed of routine tasks who can focus specifically on educational development, including personnel for gender, marginalized

communities and community mobilization/participation; creation of grassroots structures for academic resource support, i.e., cluster resource centres, block resource centres and mandal resource centres; and some form of community based organizations such as the PTA, SEC or SDMC. These are logical and needed additions to the directorates. A criticism of these programme structures has been that these have weakened the Directorates and the SCERTs, as the focus of activities has been shifted away from the latter institutions. The same personnel could have been added to the existing institutions. However, as noted, the motivation for creating parallel structures appears to have been improved programme implementation through new institutions.

Our study showed, that while the expectation may have been that the new institutions would be superior to the existing ones, this held only for a short period of time, after which the new institutions tended to become very like the old ones, as, over the longer term, they tended to adopt the same work ethos. At the time of this study, the programme structures in both Andhra Pradesh and Rajasthan showed some attempt to create a new work ethos, but this had been overshadowed by the practices of the larger system. To begin with, some efforts were made to induct capable and motivated people in the state project offices. However, subsequently, personnel had been posted on the basis of their links to powerful people. Several of them had not been trained for their jobs, and performed poorly.

Other problems of the larger system were also visible. In Rajasthan, there had been eight state project directors in the last five years. Centralized and hierarchical functioning was also visible. The focus of project activities had shifted substantially in Andhra Pradesh with changes of SPD. Those who had worked in projects earlier, i.e., in APPEP, Shiksha Karmi and Lok Jumbish said that they were not able to follow the same practices—not spend as much time in the field, or function with the same autonomy. In Rajasthan, the resource person interviewed at the block resource centre merely performed routine tasks such as keeping files and writing letters and several programme personnel were involved in rent-seeking. Even large-scale teacher training had become a rent-seeking operation. In other words, attempts to set up parallel structures within the system and insulate them from the larger system did not seem to work.

The non-state actors

Various groups and individuals often see non-state actors, i.e. NGOs and the private sector, as providing an opportunity for breaking out of the limitations of the existing government system. This study did not delve in any detail into the possibilities and pitfalls of these players, and therefore, the discussion here is limited. The purpose here is only to highlight an interesting pointer, i.e., the relationships between state and non-state actors which impacted and shaped the non-state actors. This was visible in the case of the for-profit (private sector) as well as the not for profit (NGOs) institutions (this does not mean that non-state actors do not influence the state. However, the focus here is the state and the strategy of moving to non-state actors as a solution to the limitations posed by the nature of state institutions).

NGOs

In the course of this study, we encountered three different types of relationships between state actors and NGOs. In both the states, there were NGOs that had a deep concern with education, and accordingly they related to state institutions and attempted to affect them. In general, the state's relationship with such NGOs remained fraught with conflict. Several difficulties arose because of the nature of the system. For instance, if training became a rent-seeking operation, collaboration with actors genuinely interested in training was difficult. If enrolment figures had been cooked up, then working with NGOs to eliminate child labour posed several problems. The representative of an NGO, which had worked in collaboration with the

government on a programme, which had been a failure, said that the reason was that education secretaries had changed every six months and each education secretary had had a different emphasis. Thus collaboration with such NGOs remained prickly and tenuous.

In a second instance, a more businesslike relationship had emerged, as in the case of one NGO in Rajasthan, a multipurpose organization which implemented several state projects with efficiency, without attempting to impact the state institutions or coming into conflict with them. Such relationships were contingent on the containment of rent-seeking proclivities within the state system. A third type of relationship encountered was when an NGO was complicit in the informal network of the state, as in the case of an organization in Rajasthan which ran a fraudulent centre for the education of working children for which it received a grant from the government. The existence of this NGO was contingent on the existence of the informal sub-system, as it required the complicity of supervisors. At the same time, the existence of such practices in some NGOs made the more straightforward government officials wary of collaboration. This introduced another element of mutual suspicion, and further complicated the state-NGO relationship.

The private sector

The dynamic of the burgeoning for-profit private schools was also problematic. On the one hand, private schools in these villages were far from the robust, efficient institutions that they are portrayed to be. On the other, they too were affected by the character of the state.

In Rajasthan, since 1995 (GoR,1995), it was no longer necessary to get recognition from the state level to start a primary school. Our study threw up the dangers of promoting privatization of education without attempting to put adequate safeguards in place against malpractices. In one of the sample villages, a former government teacher who had lost his job when it was discovered that he had obtained his college degree fraudulently, had started a private school in one of the sample villages. This school is not presented as being representative of the private sector, but it is important to realize that practices that are reprehensible in the government system can appear in a more virulent form in an unregulated private sector where there may be no mechanisms for accountability. At the same time, there was little evidence that private schools were institutions that created a facilitative learning environment (Box 5.5).

Box 5.5: Running a private school in Rajasthan

This co-educational school had classes from one to eight and taught in the Hindi medium. There were 198 children in the school, 20 per cent reportedly belonged to SC and ST categories; 85 per cent students were boys. Over the last five years, enrolment had risen by 58 per cent.

The school had a managing committee, but in practice the headmaster, who supervised the teachers closely, took all decisions. It had three teachers (though there were supposed to be eight). There was multi-grade teaching. The teachers in the school, though college educated, were untrained and were paid a monthly salary of Rs. 1,000 to Rs. 1,400. Teachers were not happy with the salary and the lack of job security, but taught because of lack of any other employment. The turnover of teachers was high and they often left in the middle of the school session. One teacher interviewed was keen to get a government job.

The building had ten rooms of around 120 square feet each, there was furniture for children to sit, drinking water and three bathrooms. However, there were no books or a library. Children were not provided midday meals or any other benefits.

Primary school children were charged a fee of Rs. 50 while those in the upper primary grades had to pay Rs. 60. Another Rs.10 was payable as examination fees. New students were

charged an entrance fee of Rs. 250. The headmaster of the school complained that children did not pay fees regularly, resulting in considerable financial difficulties. According to him, there was little contact between the school and the parents since the latter were not well educated, and were not interested in their children's education.

The school functioned regularly, but the timetable was not followed. Teaching was done depending on the attendance of children and teachers on any given day.

In Andhra Pradesh, the government continued to regulate the private sector. The process for obtaining recognition for private schools was demanding, though several officials said during interviews that in recent years, recognition was given more liberally. A sum of Rs. 1,000 has to be paid as inspection fee and Rs. 25,000 as endowment fees and permission had to be renewed every five years. In the sample villages of our study, small private schools tended to shut down and the private school included in our sample (run by a retired government headmaster) was running at a loss as parents did not pay fees regularly. Some teachers had left the school and children had started joining other schools. Larger, 'corporate' private schools were reportedly booming and many big players were entering the field. These schools advertised to attract students and drove the smaller schools out of the market. This had made some of the smaller private schools unviable and existing private schools were asking the government to regulate the entry of new players.

Researchers were told by several sources that on the one hand state politicians and officials had become promoters of such schools and on the other, the state machinery was increasingly incapable of regulating them. A senior state level official was reported to run her own network of such schools. Researchers were also informed that money changed hands in the process of granting permissions. A district official told the researchers that in some cases if various permissions etc. were delayed even a little, there were problems for officials. Several such schools violated norms regarding timings and place of functioning etc., but officials felt that they could not do anything about it. An MEO said that it was no use inspecting private schools as either politicians or people who had money owned them.

For those who view privatization as an answer to the problems of the state system, the fact is that given the present mode of functioning, the government's capacity to regulate the private sector is very limited and a patronage-based, rent-seeking nexus is as likely to emerge in 'public-private partnerships' as within the government system. At the same time, the unregulated private sector presents severe problems.

Unions

The systemic dynamic had impacted the teachers' unions too. Teachers' unions presented different faces in the two states that matched the differences observed in the functioning of the departments. In Andhra Pradesh, the unions were active in genuinely protecting employees. The teachers interviewed in the state said that unions helped them in getting leave sanctioned, in procuring government orders and also in solving community related problems. The main preoccupation of the largest union at the time of this study was the merger of panchayat and government teacher cadres. On the other hand, officials complained that union actions supported unprofessional practices and that disciplinary action against teachers who did not work was impossible. In Rajasthan, the teachers interviewed did not mention the unions very much. Rather, the unions appeared to operate 'from the top', and on a patronage basis. Union leaders pleaded the cases of their favourites. State level officials complained a great about them and claimed that they supported unprofessional practices. Union leaders in both the states did not support an increased role of panchayats in elementary education.

This discussion on non-state actors is admittedly limited, because it does not explore the dynamic of these players. More studies about NGOs, private players etc., and their work are needed to fully understand their contributions and limitations. The limited point to be noted here is that the non-state actors are affected by the manner in which the state relates to them, which, in turn, is affected by the dynamic within the state. Thus the 'alternative' presented by non-state organizations is in fact affected a great deal by the characteristics of the state institutions themselves.

A Last Remark

The attempt in the chapter has been to delineate the management and administrative practices within state institutions. The above discussion shows that the state institutions lack, in Peter Evans' words, 'internal coherence', i.e. their own capacities and processes do not enable to approach goals and tasks effectively. The problems come from within, rather than elsewhere. It is not surprising that a system riddled with so many internal problems should find it difficult to set and reach goals. Change attempts tend to fail, because often they do not address the core issues. In the final chapter of this book, we shall explore what the more effective strategies for change can be.

CHAPTER 6: TEACHING AND LEARNING - THE PRACTICES

Hridaykant Dewan (Hardy)

Background

Approach

The word ‘classroom’ calls up a ‘physical space’ with the teacher and the taught participating in a complex ‘process’ which includes, among other things, teaching and learning. Not only must this process lead to students reading, writing, doing simple mathematics and other subjects, but it must also propel them into becoming good human beings. This poses three issues. One is, of course, of defining and reaching a common understanding of what a ‘good’ human being is. The second is of articulating what it is that needs to be done in the classroom to nurture a good human being. We need to keep in mind that the definitions of a good citizen and of a quality class or school can vary. Against this, the question that emerges is of what we should observe when in a classroom that will help us decide whether students are learning and whether they are becoming good citizens.¹

Classrooms vary in terms of what children do, how they sit, how they are involved in activities. They also vary in terms of relationships between teachers and children, the teacher’s behaviour in class, the explanations given by the teacher, the conceptual clarity shown and the methodology of engaging children and communicating with them. Apart from this, there are physical features and infrastructure details that define classrooms and schools. However, the most important variations are in terms of the worldview, social understanding and the underlying aims of education. Many fundamental beliefs determine classroom processes and assumptions held by the teachers and headmasters impact on the rest of the system. It is important to understand these.

To meaningfully observe what goes on in a classroom, it is imperative to interpret and absorb the value system of the school, the teacher, the students and the community. This is possible only if we have a view of what to expect from and demand in our observations, i.e., a constant or known standard is necessary. Consequently, in attempting to understand the classrooms in the sample schools in Andhra Pradesh and Rajasthan, we shall examine the classroom processes against the ideas and values in the National Curricular Framework and other national policy documents. We shall pose the analysis of classrooms against the statements made in these documents and their underlying principles. We also try to see the broad patterns that have emerged in these documents, and changed over time.

Along with the overall curricular and pedagogic thrusts, it is important to understand two other aspects. The first of these is the relationships that individuals involved in education have with each other. It is clear from the preceding chapters that the government school system provides education largely to poor and marginalized children, while children who are well off attend private schools. Consequently, an overriding factor for understanding the process of teaching and learning is the link between academic performance and equity, and classroom practice and school ethos need to be viewed against the background of children attending specific schools. Second, various actors, i.e., teachers, managers, policy makers, elected representatives, parents

¹ The major challenge is to do so during the limited time spent in the school and/or the classroom. The fact that the classroom process as well as the behaviour of students and teachers can vary from one day to the other only makes the task more difficult.

and others from different social groups and regions, have their own perceptions and feelings about schools and education. These beliefs, which can be very disparate, also influence the quality of education. These numerous perceptions and their variations across different types of people have to be kept in mind while attempting to understand a classroom.

This chapter is divided into five parts. The next section deals with ideas about the aims of education and learners that pervade the system. In the section that follows, the academic institutional structure is examined. With this background, the next section deals with key pedagogic initiatives, i.e., curricula and textbooks, learner evaluation and capacity building which are led from the state level. In turn, these initiatives frame the teaching-learning process in schools. These are described in the final section of this chapter.

Some methodological issues

Let us first briefly look at how part of the data informing this chapter was compiled and how the study was carried out. It is difficult to analyse classrooms for quality because it is not easy to judge if students are becoming good citizens, and virtually impossible to find out in a short span of time if they will eventually be independent learners, capable and confident. Nevertheless, there is something that we can say about their ability to work with language, texts and comprehending mathematical concepts through the study the pedagogic practices of the system through a few observable components including the performance of learners, impact of textbooks, teachers' training or capacity building and classroom processes.

These various processes have been approached in two ways. First, the attempt was to understand the general precepts that were prevalent in the system, such as the aims of education, the learning process, the definition of a good school and a good teacher. Second, the focus was on specific pedagogic processes and practices such as textbooks, learner evaluation and teaching methods. To examine these aspects in detail, only one subject—mathematics—was selected, and perceptions and practices vis-à-vis this subject across the system were traced. The reason for choosing mathematics was that its teaching, unlike that for other subjects, is relatively less dependent on the medium or language in which it is taught, an important consideration for this study which was conducted in two states in different regions with different language families.

For the purpose of our analysis the 'pedagogic framework' was defined to include:

- Statements on the purpose of education, nature of learners and their context,
- Classroom processes that the children experience,
- The nature of mathematics and how to teach it,
- Articulations on how to relate to and treat children,
- The process of developing curriculum and textbooks,
- Capacity building of teachers and other functionaries.

Various national policy statements at different points of times have commented upon in these areas, including National Curriculum Framework (NCF) Documents of 2000 and 2005. We try to analyse some elements of the pedagogic framework in relation to these documents and contrast the observations with reality on the ground.

The findings in this chapter are based on a range of investigative activities. These comprise interviews with teachers and faculty from DIETs and SCERTs as well as managers including their views on textbooks, teacher training, learner assessment, the nature of children, the nature of mathematics teaching and learning processes, a good school and a good teacher; observations of mathematics classes in grades 2, 4 and 6 in the sample schools and administration of a test in these classes; the study of both the mathematics textbooks for these classes as well as the recommended processes for their teaching; the attitude of teachers

towards children, the situation of the latter at home and in school and the opportunities available to them for learning; and capacity building efforts.

It needs to be clarified here that the discussion in this chapter is based on the analysis of the observations and is not a comparative study between states, except where it illustrates a general principle.

Aims of Education, the Child and the Learning Process

What happens in the classrooms is influenced by many tangible and intangible factors. Among the most important of these are the ideas that inform the entire process, which can be discerned from documents that act as edifices in the minds of people influencing these processes. In this section, we shall examine the ideas perceptions about education through the system.

To do this, we first examine briefly some key notions in the major landmark documents in education that have defined national policy. In 2005, NCERT, undertook a major exercise and prepared the National Curricular Framework document i.e. NCF 2005. This document looks at all aspects of education and presents the direction that education in India should take. It has sections on the learner, classroom processes, assessment and textbooks. It also includes, in a separate volume, a section that focuses on the teaching of mathematics in the classroom. We peg our observations against the ideas contained in these chapters in NCF.

However, NCF 2005 also has predecessors, the most recent being the National Curriculum Framework Document of 2000, i.e. NCF 2000. Important ideas have also been expressed in the report of the Education Commission of 1966, the National Policy of Education 1986, the report of the review committee for the National Policy of Education (GoI, 1990) and the Programme of Action prepared in 1992 and. Comparing these documents with NCF 2005 can deepen our understanding of the development of national policy and the issues involved, and consequently, of the situation on the ground. While some broad concerns persist through all these documents, i.e. poor quality of education, providing poor students access to education, the need for deeper commitment from the government system, education as a vehicle for societal functioning and transformation etc., there are also changes in approach which we shall underline.

Aims of education in National Policy

The first premise (stated or unstated) on which any school functions is why education is necessary or desirable, what the role of education is, and what is 'education'. NCF 2005 recognizes school education as a deliberate intervention over and above and distinct from education at home and in the neighbourhood. Education is seen as a process of self-discovery and learning the truth about oneself, while imbibing the values of equity, justice, democracy and rationality. NCF 2005 elaborates on the notion of patriotism and states that children should be a strong visible symbol of the abstract concept of national pride. This, it says, should be complemented by building pride in humanity. The document also campaigns for plurality of culture and the need to bring marginalized cultures to the centre. It advocates a participative classroom in which both the teacher and the children are active and reflective.

If we compare NCF 2005 with the statement of purpose in NCF 2000, we find similar strands of thought, but also important differences. NCF 2000 also underlines equity and the need for education as a vehicle for it. However, at the same time, there is a paradigm shift from the words used in the NCF 2005—cohesion, duties of children, uplifting of a section of people and national goals. These are not the key ideas in NCF 2005. NCF 2000, while recognizing the importance of abolishing exploitation 'handed down by tradition', states that education has to contribute to the uplifting and empowering of weaker sections (SC, STs., women, minorities

and children with impairments), and presents education as the most powerful instrument for achieving social cohesion. It identifies important national goals as secularism, democracy, equality, liberty, fraternity, justice, national integration and patriotism.

NCF 2005 emphasizes the need to make certain rational and human values integral to the process of education and to instil respect for other people, plurality and diversity in the learner. NCF 2005, while emphasizing the paramount need for interweaving education with values, distances itself from a separate programme for value education. Its approach to the ethical development of children suggests having a dialogue with them, letting them choose rationally and encouraging them to question the social structure. NCF 2000 on the other hand, sees religion as a major source of value generation and sees the need for education in relation to different religions, the values inherent therein and a comparative study of the philosophy of all religions. It talks specifically of building value-based education programmes. It believes that it is the responsibility of the school to impart this education to children. At the same time, the document differentiates between religious education and education about religions.

In this discussion, several philosophical differences in defining the purpose of education are visible. But differences also exist in terms of the sociological perspectives which inform the major challenges, constraints and problems of reaching universal quality education. The difference between NCF 2005 and NCF 2000 lies in the way each defines the problem of education. NCF 2000 says:

The problem of education: A sizeable segment of the contemporary Indian society seems to have distanced itself from the religion-philosophic ethos, the awareness of the social design, and the understanding of the heritage of the past. Influenced by the alien technological ethos, the parents and the educational institutions emphasize the acquisition of high-grade techno-informative knowledge alone. However, the impact of westernisation has been limited to only the elite members of the society, leaving the masses unaware of these developments. This has brought into a sharp focus the rural-urban, the agrarian-industrial, the affluent-destitute and the literate-illiterate divides. In this way, the structure of authority in the Indian agrarian society has been disturbed (Page 3 Section 1.1.),

The overall view in this document is of a somewhat reluctant acceptance of transformation and development. On the other hand, NCF 2005 takes a dynamic view of the development of society, valuing the spread of the democratic fabric and promotion of rational thought. It also considers the ability to take independent decisions as valuable for society.

State perceptions about the aims of education

How are the aims of education perceived at the state level? Our most important finding was that the differences between the two national curricula frameworks described above engendered confusion at the level of state institutions, rather than a clear articulation of a position or a stand in favour of one or the other. Interviews with SCERT faculty and other key players showed no sharp arguments about the contrasting purposes of education in the two documents. Instead, SCERT faculty members saw curricular development as an effort to add certain levels of learning to the syllabus along with moral values. When asked about the aims of education, none of them emphasized the development of the child as an independent, concerned and rationally thinking being. Often, they simply provided facts about government programmes in an unanalysed and uncritical way. Thus an important characteristic of state level perceptions about the aims of education was a diffused, unarticulated and little examined idea.

An important idea about the aims of education expressed by interviewees was that children learn to follow norms considered significant by those in positions of authority. This was also

evident from the manner in which one SCERT had formulated the curriculum—based on the recommendations of middle level political representatives and government functionaries from different locations, and not on academic debates and research studies. For teachers, the emphasis was on getting children to understand their duties and know the correct and appropriate behaviour patterns. Most interviewees thought inculcating personal values like honesty, courage, sacrifice and respect for elders and social values such as nationalism, environmental sensitivity and discipline were *de rigueur*. As to the manner in which children could be imparted these values, the participants believed in repeated lecturing and coercion.

Notably, except for a few, most of the teachers and administrators interviewed attributed the lack of interest in education among children to its inability in providing employment. This was mentioned especially for members of the socially backward and poor communities. This belief not only revealed the interviewees' views about the purpose of education, but also suggested that they did not perceive a *systemic* responsibility for involving students from these communities in the process of education. The stated objective of education in fostering an egalitarian society and opening the doors of learning for children, irrespective of their social background, appeared totally lost on the ground. Their 'school' was seen an island unconnected with the world of children and their aspirations, and children as putty to be shaped according to the whims and fancies of the teachers. Contrast this with NCF 2005, which emphasizes dialogue with the child and enabling her/him to evolve value systems based on experience with an understanding of equity and justice.

The nature of learners

We now examine a second set of ideas across the system, i.e., how learning and the learner are perceived. In this context, NCF 2005 lays emphasis on understanding children and their context. It sees an important role for the construction of knowledge by children and argues for space for their ideas. It advocates freedom from stress, irrelevance of what is transacted and non-comprehension. It argues for openness, equity in opportunity and relatedness to what is being transacted. It calls for a classroom process to enable children to be proactive in exploring their own questions, using the teacher as a sounding board. NCF 2005 recommends that textbooks, teaching plans, evaluation and all classroom practices be modified so that children feel valued in the classroom.

However, the disparity between the lofty ideals of the document and value system of the people who work in the state systems was evident from our interviews. As in the case of the aims of education, the ideas of our interviewees about the process of learning were also fairly diffuse. In fact, when asked about the process of learning, most of them (including SCERT faculty and teacher educators) did not present clear views and preferred talking about other things (see Box 6.1). Teachers and administrators expressed sympathy for the children, as the mathematics textbooks and curriculum were 'too much' for young learners, yet had no specific suggestions regarding how children could be helped to learn or what kind of principles could be utilized to teach them or why the curriculum 'load' could not be reduced. Interviewees largely agreed that boys and girls were not intrinsically different with respect to learning, and most of maintained that caste differences were not a determining factor. However, they also pointed out that differences in environment and background affected children's learning.

Box 6.1: Perceptions about Learners

A Senior Academic Administrator:

Exposure, and not community, determines the differences in the quality of the learners. The more exposed they are to different influences, the better. A study says the first-born is better. But interaction time decides whether the child will learn. Boys and girls are in no way different although boys may be better in some situations because of physical strength. In examinations, girls outperform boys.

Emotional competency needs to be kept in mind. An enabling environment has to be built. We need to understand a child's needs. We have to move away from reproducing what is there in the book. A child needs to construct knowledge. This leads to forming of structures in the mind. A child who learns fast prefers to sit in the front of the class and informs the teacher about his work...

A Senior Administrator:

Preparedness for education at home matters. Educational background of parents' matters a lot, so does pre-primary education. SC/ST children are at a disadvantage but only because of preparedness. Otherwise, there is no difference among children.

An Academic Functionary:

Good learners generally possess the following qualities:

- Interest
- Ability to answer as well as question
- Regularity

Students become poor learners because of:

- Ill-health
- There could be some genetic reasons
- Parents' illiteracy/negligence
- Poor educational background of the parents
- Poverty

Good learners have the following qualities:

- Intelligence
- Discipline
- Respect for elders
- Helping the peer group
- Neatness
- Regularity

Teachers:

Mathematics is not difficult for the students who study regularly.

Mathematics is not difficult, but some children do not understand it because their intellect is weak.

Mathematics seems difficult to children, but only to those children whose basics are not clear. Further on, the difficulty keeps on increasing as the subject becomes increasingly difficult.

Overall, the perception across the system seemed to be that children from poor or weaker communities could not learn because of a different belief system. Interviewees seemed to think that the curriculum was such that it required parental support for the child to learn, but this was usually not available to the economically deprived child. Another perception was that children did not understand the language of the teacher, were not able to negotiate textbooks and made no effort to fulfil the expectations of the school. In a few cases, those working in the system visibly resented the presence of these children in schools. One extreme was the following response from a senior person working in SCERT:

‘Children from *jhuggi jhopri* (slums) are cut off from *jeewan mulyas* (life values). When I was a District Education Officer, I saw a school in the slums. Several children were sitting there on a *thela* (hand cart). I asked a woman there if she could teach them something. She said, “You can’t. They will do anything to survive, steal, or anything else.”

The mother is the first teacher...These children know foul words by the age of five. A very sensitive teacher may be able to do something, but it is not easy...

Learning levels are okay, considering the problems of these children. Teachers are trying. However, the family background is very important for learning.

Another comment from a district level administrative official was:

All problems are related to population. This, in turn, is related to poverty. Population in ST areas is more. We did a survey in a block...each woman had 5-6 children. There are health problems... problems of malnutrition. Those who are well off send their children to private schools and are aware. The other group is not aware. They cannot send their children to school even after enrolment.

Some interviewees said that genetic differences or low IQ could be the reasons why some children learned and others did not. An important point here is that most interviewee did not see the curriculum or the method of teaching as issues. Children, they said, did not learn because of their own inabilities. The hostility against children from marginalized and poor backgrounds, however, was tempered by the realization that the very existence of government schools was dependent on these poor children, as those who could afford it sent their children to private schools. In fact, some interviewees were somewhat unhappy about this trend and even suggested that they would be happier if government schools had more children from the same background as the teachers.

The underlying principles of homogeneity and adherence to specific values, and the belief that poor children could not learn because of the differences in their environment, particularly the family background, that emerged during the interviews needs to be counter-posed against NCF 2005:

Cultural diversity is one of our greatest gifts. To respect and do justice to others is also to respect and do justice to their respective cultures or communities. We, therefore, need to radically change the centre versus periphery perspective on inter-cultural relationships in our country. Cultures on the so-called periphery must receive as much attention as cultures in the centre. As for education, its implication is that ways of life other than one’s own must be imaginatively and effectively presented as deserving of as much respect as one’s own (Vol. II, Page 7, Point [iv]).

Notably, contrary to the popular belief, it was not just the teachers who did not want to teach and deal with poor children and were therefore responsible for non-learning but academicians and administrators as well, did not value diversity.

The Teaching-Learning Process in the Classroom

The touchstone of whether an idea on education is valid is in its execution or experimentation in the classroom. In this section we examine the basic characteristics of the classroom process in the schools observed. These processes bear a significant relationship to the perceptions that have been outlined above.

The space for education

While attempting to understand the classroom, the paucity of personnel and resources as well as the problems of school functioning, i.e., teacher absenteeism and inadequate teaching time, highlighted in the previous chapters, need to be kept in mind. Barring a few exceptions, we observed that in most of the elementary schools as little as 30-40 per cent of the school hours were used for actual teaching. Moreover, we found that the younger children in primary schools were the most neglected as the teacher usually devoted his/her time to the older children, especially those about to take board examinations. Often, teachers left their class for various non-academic chores: to prepare midday meals, collect correspondence, or do other sundry non-teaching tasks. The impact of the midday meal on the school has many dimensions, as this observation from an upper primary classroom illustrates:

Around 9 am, a lady came to cook the midday meal. Two girls of the upper primary classes joined her in carrying the utensils from the store. Some time later, two other girls joined in to get water from the hand pump to cook porridge. The teacher was on duty when the children were eating. We were told that irrespective of what was cooked, it took at least one and half to two hours for one teacher to get the meal ready.

This illustrates the general paucity of resources, i.e., inadequate personnel and infrastructure which chips away at teaching time.

The child's 'place' in the classroom

We often heard the word 'activity' during our interaction with the teachers and the need to make classrooms 'activity-based' was emphasized all the time. However, no questioning, exploration, nor any space for creativity and logic—all a part of school activity—found place in the classroom. This huge discrepancy between what people expressed as desirable and its implementation was all too evident in the limited or minimal attempt to involve children in classroom activities. There was no plan to engage them or pay heed to what they were saying. Teachers also showed insensitivity towards the students. In the classes where teachers were present and teaching, most of them used the blackboard for much of the time, with little involvement or effect. Even though they knew that no more than 10–20 per cent students could make any sense of what they were teaching, the teachers either did not, or could not, stop to involve the other children. In most upper primary classes, the presentation was extremely formal and the gap between what was taught and what the children knew or could follow was immense. The problem was compounded by the fact that the number of children in some of the classes was high while the infrastructure was inadequate (see Box 6.2).

Box 6.2: A View from the Classroom

Class VI: First visit—20 September 2005

Out of the 100 enrolled children only 81 are present. The class teacher is a vidya volunteer. The space is inadequate for even the children present, and the class is very crowded. Boys and girls sit separately.

Upon entering the class, the teacher does not announce or introduce the topic. She simply writes the following problem on the blackboard:

$$\begin{array}{r} 308 \\ 1 \end{array} + \begin{array}{r} 6 \\ 10 \end{array} + 5$$

Without asking or waiting for the students' response, she starts solving the sum. Sometimes, she shouts at the students and tells them to look at the blackboard. Many students cannot see what she has written. Nevertheless, they do not ask any questions and scribble whatever they can follow. The teacher frequently instructs them to 'write in your notebooks whatever I have written on the board. Then, I will give you another problem.'

She sends a girl to the office for a duster. The child returns after 15 minutes with the object. The teacher has neither moved in the class nor monitored what the children are writing. Interestingly, all the children are staring at the teacher and watching her activities. After a while, the teacher writes the following sum on the blackboard:

$$285 + \begin{array}{r} 3 \\ 10 \end{array} + \begin{array}{r} 5 \\ 100 \end{array} + \begin{array}{r} 7 \\ 1000 \end{array}$$

The teacher starts asks the children to solve the problem but then, without waiting for any response, says, 'You do not know how to do division' and goes back to writing on the board. The children start chattering loudly and there is a lot of noise in the class. Some students copy from their friends notebooks because they cannot figure out what the teacher has written on the board. The teacher writes 96/36 on the board and asks the students to 'convert' it. The question is vague and the children do not know what to do. However, two girls come forward and show their answers to the teacher. The teacher gets annoyed and says, '*Maatalu ravaddu chetalu raveli*' (Not words, I want action). The children slowly lose interest in her teaching and start playing. One boy is lying on the bench, whom the teacher apparently does not notice. She is busy writing on the board.

In most classes, the textbook was the centre of all the teaching-learning activity with teachers making little attempt to move away from it. Even though classroom observations showed that children had a lot of difficulty in understanding, and the teachers faced many problems in teaching, they could not articulate the nature of their difficulties during interviews. As mentioned earlier, most of the teachers believed that the children's background or lack of previous knowledge was a deterrent to their learning. Notably, authors of mathematics books teacher-trainers and other managers shared this view.

Clearly, the idea of allowing space to children—appreciating their contexts and employing their experiences—and encouraging them to explore their world, which we encountered in NCF, did not translate into classroom reality. The child was the least valuable entity in the school and having her learn was not so important either. This ethos or view fitted in with the

hierarchical functioning of the system described in Chapter 5. The most important thing was impressing supervisors and providing information as demanded by senior authorities. Details of surveys, children in the village, etc., had been compiled, but no plans had been formulated about what was to be taught in the class the next day.

Opportunities for learning

We now take a look at classroom processes in more detail against ideas about learning expressed in NCF 2005. NCF 2005 argues that cognition involves the capacity to make sense of the world. Meaningful learning is a generative process of manipulating concrete things and mental representations. It emphasizes concept development and understanding rather than mere recall and retrieval. Thinking, speaking, writing and doing are intricate processes. Children's abilities to work with the hypothetical and the world of the possible increase only gradually. The document says that (i) Children need to be viewed as naturally motivated people, who need time and space for reflecting and working on their own—two important ingredients of learning, (ii) Children should be able to make concepts meaningful rather than be forced to learn in some predetermined ways, and (iii) Children learn when they are engaged actively, and since learning takes place both inside and outside the school, the document suggests that classroom processes should be linked in some ways with what children do at home.

We found that the classrooms, except in one instance, were contrary to these expectations. In the primary schools studied, in most classes the children were either reciting multiplication tables or copying from the blackboard (or just idled about). The questions handed out were formal sums of column type addition and other such mechanical type operations. In most classes, children were supposed to follow the teacher's solutions where the numbers were to be added column-wise. Many children did not even clearly understand the presentation on the blackboard and gathered around those who could copy properly from the board. There was virtually no dialogue in the classroom and no child was asked to estimate numbers or relate these numbers to his/her life. The handful of classes that were relatively better were the ones in which the teacher tried to get the children to count on their fingers and asked a few of them to try and answer components of textbook questions on the board or orally.

As has been noted earlier, mathematics classes are usually not interesting. Factors such as the size of the classroom, air, light and the availability of teaching-learning material (TLM) makes the classroom physically and visually unattractive. Teachers failed to involve students in positing and solving problems. Forced by circumstances or their own temperament, they were rude and insensitive to the children. At the same time, as Chapter 4 shows, a majority of the children come from homes which cannot provide any academic support and the school is their one and only source of learning. Some children do not come regularly to school, which also affects their learning.

Academic Institutions

The ideas and perceptions that have emerged in the discussions above, as well as the pedagogic processes are seeded in the institutional structure. In this section, we look at the academic institutions in Andhra Pradesh and Rajasthan, to get a firmer grasp on the institutional framework in which concepts and issues are debated and formulated. As already discussed earlier in this book, SCERTs, DIETs and block and cluster centres have been set up to intervene and strengthen pedagogical programmes. However, as discussed in Chapter 5, the faculty is 'posted' to these institutions without much consideration about the ability and interest of individuals. Consequently, these institutions remain academically weak. It is against this background that the lack of a real dialogue about learning and education that we have discussed earlier needs to be seen.

SCERTs

SCERT was intended as a key institution to execute curriculum reforms and function as the hub of academic research, innovation, inspiration and motivation as well as a symbol of quality. It could logically be expected that this apex body would have not only an academic vision, but also a philosophical and sociological understanding of education and its implications for the transformation of society. How the SIERT/SCERT faculty perceives these institutions is illustrated in Box 6.3:

Box 6.3: Perceptions about SIERT/SCERT

‘SIERT plays the same role at the state level as the NCERT at the national level. SIERT is responsible for curriculum, textbooks and teachers’ training (pre-service and in-service). The institute has academic control over DIETs and fixes the latter’s academic calendar. All DIET faculties come here. We prepare an academic calendar in consultation with DIETs. Then, each of them is free to modify it as per local needs.

...We made the curriculum in 2000 with NCF 2000 as the base. Then we did book writing. These days we are working on a teacher’s guide that is being prepared for the first time in the state. The Sarva Shiksha Abhiyan conducts teachers’ training, while SIERT prepares resource persons. Then there are science fairs and quiz programmes for high schools. For elementary schools, we take up life skill development programmes. Children of elementary classes have competitions at various levels for art etc. We analyse the books every year. We have two programmes for book review in which we get reactions from schools and the field. Then we have a community science centre where we keep various types of models etc. Many tourists see it.

We need to inculcate values in our people. SIERT had made a programme for this through EduSat. Discussions are going on in the government about this programme.’

‘SCERT should be a guiding light for DIETs, CTEs and IASEs. It also makes textbooks and provides resource support to various departments. The council is already doing the common examination for the Diploma in Education. We also do counselling for teacher education admissions. We provide model question papers to the Director of Government Examinations. We do training and hold workshops for the social and tribal welfare departments’.

The views outline the role of the apex institution in an apparently comprehensive way, yet they fail to capture the sense of SCERTs as academic hubs in the role of vision makers and contributors to the transformation of society. Some faculty in SCERT were even more restricted in their role definition. For example, a professor in charge of curriculum and material development defined the role as:

The main role of SCERT is to take up curriculum revision as decided by the minister and the secretary. Every five years, the curriculum has to be revised. The curriculum has not been revised since 1998. The budget was not sanctioned. This government has given permission for curriculum revision. We have consulted all teacher educators, teachers, parents and government departments. We have sent letters to people, held meetings and invited opinion from the public. We have letters from the traffic control board, pollution control department and the women and child welfare department. All this will be submitted to the concerned authority and the curriculum will be revised.

It is evident that the performance of SCERTs at present is dependent on what bureaucratic and political pressures dictate. Processes like textbook development and teacher training are not institutionalized. SCERT faculty had taken up such activities as and when directed without

much thought. As a result, instead of confronting the ideas and responses of the administrators, the SCERT faculty merely echoed them. And if SCERTs suffered from hierarchical oppression at the hands of NCERT and the state government, they in turn ‘transmitted’ this to DIETs, adopting a similar ‘order based’ approach. Moreover, it was also clear that SCERTs’ plans and programmes did not include a relationship with the processes at the school level, except by way of what may be called monitoring inspection. The following were the concepts of a ‘good school’ of the faculty (Box 6.4):

Box 6.4: Perceptions of SIERT/SCERT Faculty about ‘A Good School’

‘The quality part is okay. The main problem is monitoring. Teachers should be there and they should teach something. No one monitors to see if all students are achieving the standards. Teachers should be appraised and every teacher should be checked. Simple monitoring is enough for the elementary stage.’

‘A good school is one where there is good rapport between the head of the institution and the teachers; the teachers are of good calibre and the head is capable and can control everything. A good school should have good practices: the community should make financial contributions to the institution, PTA and SEC meetings should be held regularly and the school should get suggestions from the parents and the community.’

These perceptions appear no different from those of an inspector of schools or an official without any academic background. The advocacy by a teacher educator of policing and monitoring as the answer is a good indicator of the vacuum in education. Yet, the expectation from these institutes is that they will intervene and speak to teachers about possible changes in schools! Predictably, these institutions did not engage in any quality research.

Within the institutions, there was a lack of autonomy and freedom to take any initiatives. To quote one senior faculty member:

I take no independent decisions. We obey officers. For doing a workshop etc., we have to get approval from the Director, SCERT. For financial matters, approval has to be obtained from Director of School Education. There is not separate budget for the SCERT. Every activity has to be planned in advance. We should have the freedom to take up research. Right now we have to plan in advance and get things approved. Even to give a lecture somewhere, I have to take the Director’s permission. If I don’t take permission, there will be a problem.

This discussion earlier in this chapter shows that in terms of understanding and beliefs, the faculty and processes at SCERT were still steeped in folk beliefs of pedagogy and behaviourism, and this affected how they perceived their own role and performed it. In response to questions about their work, the faculty often gave answers that had little relationship to their area of specialization. For example, a professor dealing with learner evaluation described the work of her department as:

We have conducted programmes and workshops. There is a quality improvement programme currently running in the whole state. My department looks after evaluation. We are also doing a survey of the midday meal scheme in three districts. We have also participated in educational television programmes. The purpose is to see that there is uniformity in all the districts.

and:

Assessment is a big issue. Often, the questions are not specific. For instance, five marks may be assigned to a question that may not be a scoring question. The instructions given in the

question papers are not clear. It is not indicated how many lines there should be in an answer. Therefore, the children's response is to keep writing reams and reams.

DIETs

The DIET is supposed to function as the hub of academic activity in the district for training and as a resource centre, expected to:

- Undertake pre- and in-service teacher education, set
- Set up evaluation processes and mechanisms,
- Conduct research,
- Understand educational issues in the district
- Provide support to schools to make the school more meaningful.

The institute should be conducting and coordinating multi-dimensional work. Its faculty should have the time, inclination and capability to think out of the box and set up systems that allow them as well as the teacher to be more flexible and reflective. DIETs must function autonomously, explore new ideas and develop their own plans of action. The reality, however, is vastly different.

The DIETs that we visited were poorly staffed with many vacant posts. There did not appear to be room for creative faculty members. There were no innovative academic activities or plans to network with schools or even with Block Resource Centres (BRCs) and Cluster Resource Centres (CRCs). These institutes did not bring in context-specific local elements in the class or attempt to understand the district in a way that would make this possible. DIETs seemed to be waiting for directions from the top, with no institutional energy or initiative.

In general, the structures expected to lead the academic process showed a singular lack of spirit, capacity and keenness to fulfil individual and collective roles. However, it was not clear whether the lack of autonomy arose out of cynicism, levels of ability, lack of initiative or the lack of space in the system.

The Pedagogic Edifice: Learning Mathematics

In this section we explore in greater detail the gamut of pedagogic ideas and strategies in the system, the textbooks, teacher training, the learner evaluation system and learner achievement levels along with perceptions, ideas and classroom practices. We do so by focussing on how mathematics is approached and taught, and how it is learned. To begin with, we explore some basic ideas about the aims of teaching mathematics to children.

Why Mathematics?

National Perceptions

The Education Commission Report 1964 emphasized the study of mathematics in order to understand, among other things, scientific data expressed in quantitative measures. The Commission noted that math played an important role in the development of physics, which was increasingly used in the advancement of the life sciences. Since this, together with automation and cybernetics, would lead to an industrial revolution of sorts, it was necessary to pay special attention to the study of mathematics. The right foundation had to be laid in schools (Page 347, Chapter VIII, Section 8.62). Thus, the Commission looked at mathematics as an important medium of learning other subjects in modern education. When it considered the curriculum for mathematics, the focus was on the underlying logical formulations and the common principles in various sub-disciplines in mathematics (Page 347, Section 8.63). It did

not consider the division of mathematics into arithmetic, algebra and geometry important, and in fact, suggested clubbing arithmetic and algebra, laying emphasis on the rules and principles of mathematics and a rational thinking system. The 1986 policy document also looked at mathematics as the vehicle to train the child to think, reason, analyse and to articulate logically. It expected mathematics, apart from being a specific subject, to be treated as a concomitant to any subject involving analysis and reasoning.

The NCF 2005 description focuses on an, ‘ability for mathematicization’ and a two-fold objective for teaching mathematics in school (Box 6.5). The first objective is linked to quantitative knowledge required for science, with the ‘narrow’ aim being to develop useful capabilities related to numeracy, number operations, measurement, decimals and percentage. The second, and ‘higher’, purpose is to develop the child’s resources to think and reason mathematically, take assumptions to their logical conclusion and handle abstractions. This includes developing the ability and the attitude to formulate and solve problems. NCF 2005 calls for an ambitious mathematics curriculum and seeks to achieve the higher aim. It argues that the curriculum needs to be coherent so that the variety of methods and skills available for teaching arithmetic, algebra and geometry come together to address the problems of other domains such as science and social science in senior classes. It also emphatically argues that every child must learn mathematics as it helps develop the mind and enriches the ability to analyse experiences and the world around us (Page 38, Section 3.2).

Box 6.5: The vision for school mathematics in NCF 2005

Vision for School Mathematics

- Children learn to enjoy mathematics rather than fear it.
- Children learn important mathematics: mathematics is more than formulae and mechanical procedures.
- Children see mathematics as something to talk about, to communicate, to discuss among themselves, to work together on.
- Children pose and solve meaningful problems.
- Children use abstractions to perceive relationships, to see structure, to reason out things, to argue the truth or falsity of statements.
- Children understand the basic structure of mathematics: Arithmetic, algebra, geometry and trigonometry, the basic content areas of school mathematics, all offer a methodology for abstraction, structuring and generalization.
- Teachers engage every child in class with the conviction that everyone can learn mathematics.

It should be pointed out here that in contrast, NCF 2000 lays a far greater emphasis on the utilitarian goal (the ‘narrow’ goal of NCF 2005) of mathematics. It says:

One of the basic aims of teaching mathematics in schools is to inculcate the skill of quantification of experiences around the learners. Toward this, carrying out experiments with numbers and forms of geometry, framing hypotheses and verifying these with further observations form an inherent part of mathematics learning. It would also include generalizing these findings with proof and developing competence to solve problems. Mathematics helps in the process of decision-making through its application to real-life situations in familiar as well as non-familiar situations. It contributes in the development of precision, rational and analytical thinking, reasoning, positive attitudes and aesthetic sense (Page 55, Section 2.8.6).

It goes on to add:

The mathematics curricula must develop an appreciation and understanding of the contribution of Indian mathematicians, along with that of others. This would develop a sense of self-esteem and self-confidence amongst the learners.
(Pages 55-56, Section 2.8.6.).

Thus mathematics can be seen on the one hand as laying the foundation of such conceptual structures in the child's mind that help him/her comprehend abstract ideas, build logical arguments and appreciate the methodology for abstraction, structuring and generalization, and on the other hand as a subject concerned with procedures and algorithm and facts, formulae and mechanical procedures. There is also the deeper relationship of mathematics to the development of society as well as the human mind. For example, the development of natural sciences has obviously depended a great deal on capabilities of generalization, abstraction and formulation of logical arguments. The visualization of spatial placements of atoms in a molecule or of the structure of a crystal has made possible a deeper appreciation of how the world can be understood. Similar examples can be got from economics, architecture, linguistics, designing, painting etc. These uses and applications of mathematics are not related to the need for quantification, but to what has been defined as mathematicization.

Perceptions at the state level

What do the state level academicians and managers feel about why and the ways of teaching mathematics? Box 6.6 gives some excerpts from interviews of people who are responsible for providing good mathematics education to children.

Box 6.6: Perceptions: Why Teach Math in Schools?

Senior Academic Administrator:

(Math) improves logical thinking and helps understand step-wise treatment, sequential order, measurement of quality and quantity (in day-to-day use). For teaching, the presentation should be good. For example, the Pythagoras theorem can be proved using railway tickets (squares).

Teacher Training Coordinator:

Math should be taught for skill development. Other skills are related to math.

Teachers:

- Math helps children to know about numbers. It is very useful in daily life.
- Mathematics sharpens the IQ. It is useful.
- Mental abilities improve. It helps in developing the ability to think.
- It is useful for quantifying daily needs, for high scientific calculations and for (all kinds of accuracy in) measurement.
- It is a part of the children's context and extremely useful in daily life.
- Math is necessary for learning numbers and counting. That is the only way children can do business when they grow up.

A study of current textbooks and the articulations of the people in the system, including those responsible for conceptualizing teacher training and textbooks, reveals a heavy emphasis on what can be described as algorithms, shortcuts and methods. The other emphasis is on learning facts including definitions. The organization of space, spatial relations, logic, the idea of generalization and of variables did not emerge in any articulation. The overwhelming emphasis was on working with arithmetic in daily life, and that meant numbers and operations. A meaningful understanding of mathematics and its teaching had not yet emerged. The described manner of good teaching was giving children solutions to difficult problems and tutoring them to look for clues and for specific algorithms. Concepts and general principles, that are actually the basis of mathematics, were not recognized. This is in stark contrast to the expectation of NCF 2005 which expects the child to develop her own new strategies and

appreciate the power of generalization of mathematics based on her own conceptual formulations.

Ideas about curriculum and textbook development

In this section we look at the curriculum and textbooks. Let us first examine what key people in the system say about these before moving on to how curricula and textbooks are changed.

National views

The 1968 policy document recognized the difficulties that children have in learning mathematics and science, the importance of learning these subjects in the Indian context and the effort needed to make it interesting. It emphasized the need to free learners of primary level mathematics from the pressures of higher education. Subsequently, the Yashpal Committee set up in 1992 also spoke of how the child was weighed down by the burden of non-comprehension, besides having to carry a school bag heavier than her/himself⁸¹. Recently, serious questions have arisen regarding the mathematics curriculum and its relationship with other aspects of learning, particularly in primary classes. The initiative of an NGO, Eklavya, in developing an alternative primary school programme in Madhya Pradesh merits a mention. Eklavya categorized the syllabus into areas of learning that were coherent and holistic rather than abstract disciplines. This brought interesting insights to the mathematics programme. The Centre for Science Education in Delhi has contributed significantly towards analyzing mathematics curriculum for primary classes and in evolving material for the classroom.

However, despite the fact that the opinion on how to teach mathematics in schools has come a long way since 1968, in practice the mathematics curriculum remains by and large the same. To find out why things have not changed on the ground, let us look at the process of developing the curriculum. Beginning with the policy statement, there are several steps in the actual formulation of the curriculum and textbooks. To begin with, NCERT formulates the document. Following this, it sets up committees that convert the curriculum framework into syllabi for each subject. NCF 2005 admits that these committees have often excluded those who had alternative ideas and approaches. Consequently, the syllabi have been roughly the same even though they have been worded differently. Little surprise then, that in 1991 we spoke of activity based child-centred education in the same breath as itemising learning through minimum levels of learning (MLL)s!⁴ Individual states then take up the process of curriculum development. Let us examine this in detail in Andhra Pradesh and Rajasthan.

⁸¹ In general, it can be said that the documents authored from as far back as over the period of India's independence reveal mutually conflicting concerns. The process also reflects a lack of sufficient sharing and respect for the rationale in previous documents. The involvement of school experience and of teachers has been perfunctory and those key to the penning of these documents were largely unaware of existing government schools and how they functioned, or of recent research in the area. The documents reflect swings of views rather than evolution through dialogue and mutual learning.

⁴ The minimum levels of learning were standards in terms of competencies that were set for the nation as the levels to be expected in all schools. These were to be a progress on the information and rote based teaching and assessment. The document was published and advocated by the MHRD all over the country. The MLLs were to be the cornerstones of all aspects of primary education including teaching-learning materials, activities, teaching process, classroom organization, assessment etc. They were set for each class in an itemized and linearly hierarchical manner and reflected a view of education, of learning, of the child and of the nature of subjects that was contradictory to the concepts already being discussed in the country.

The state level process

Each state customarily evolves its own curriculum and textbooks. They can do this keeping in view the NCERT as curriculum, which is discussed with the states in various forums. The logical expectation here would be that state institutions would enrich discussions on the issues of the mathematics syllabus, teaching-learning methodology suggested and presentation in textbooks. Box 6.7 gives some views of personnel responsible for mathematics curriculum and textbooks at the state level.

Box 6.7: Ideas About the Mathematics Curriculum

Teacher Training Coordinator:

Many children fail in mathematics. The syllabus has too much content. Children do not need all that. They need to learn simple operations of plus, minus, multiplication and division. But there are many other things as well. We do not have competent teachers. There is a problem in teacher selection. Teachers are selected on the basis of overall marks. They are not selected for specific subjects. Teaching of mathematics should be more activity-based, the syllabus should be reduced and the competency of teachers upgraded.

SCERT faculty in Curriculum and Textbook branch:

Children can solve problems. Telugu-medium students are better at solving problems because they use their mother tongue. Mathematics is not required for all students. For instance, algebra is not necessary for everyone. Only general math is enough.

Slow learners find it difficult to solve problems. Only basic numeracy and simple additions and subtractions should be taught up to Class 2. Current syllabus up to Class 4 is all right. Mathematics for Class 5 is somewhat difficult and for Class 6 even more so.

Sums that use borrowing, fractions, decimals and business mathematics are difficult for children.

These views show a basic lack of comprehension about the process of teaching and learning mathematics and the place of curriculum and textbooks in this process. When these topics were discussed, often, interviewees simply sidestepped the issue and talked about something else. The types of views that came across were: school mathematics was difficult and unnecessary, the seeming implication being that it was best to drop mathematics teaching as early in the school cycle as possible; or that teachers were not competent and there was little that could be done about it.

While NCF 2005 envisages that textbooks be interactive and should try and draw from the experiences and the environment of the child, none of our interviewees appeared to view textbooks in this light. For most, books formed the complete teaching kit prepared by experts. This view, coupled with a lack of awareness about the need to engage children in active learning, would have us believe that the only complaint, if any, of teachers, teacher educators and officials was that textbooks had too much or too little content and that teachers were competent or incompetent.

At best, interviewees discussed issues like whether a book was ‘Minimum Level Learning (MLL) based’ or not. The experiences that people remembered most clearly about curriculum appeared to be periods where there was talk of ‘MLL-based’, and then during DPEP, around ‘child centred’. And these, though quite disparate and even contradictory, could *co-exist* as valuable ideas. However, these ideas did not stay long enough, or were not pursued long enough, to permeate the system permanently. Their usage lacked clarity and depth, and many times they were only labels to describe a host of often contradictory activities. It was clear that

project based quality initiatives such as in DPEP could only trigger a process, but if it was not sustained and continued with patience, openness and confidence, it was likely to disappear, leaving only faint memories behind.

In general, people working in SCERTs, DIETs or senior education officers had not thought about the curriculum and its relationship to learning with any seriousness. Neither the SCERTs nor DIETs had any follow-up studies or research on how the curriculum/textbooks fared in the classroom. No faculty interviewed had spent time in any classrooms to observe the processes and learn from them. Consequently, there was no continuity of engagement with the curriculum in the system. A review of the curriculum and the syllabus was a mechanical job that had to be performed every five or ten years (there was no clarity even on the frequency). There was no framework to serve as a basis for curriculum choices. The changes were based on the trends and buzzwords of the time. The effort to understand, debate and absorb the crucial differences in the various frameworks was singularly lacking.

In Rajasthan, the ‘Jansammat’ curriculum (titled Shikshakram, Rajasthan Rajya Shaikshik Anusandhan Ayavn Prakshishan Sansthan (SIERT, Rajasthan: , 2001 Udaipur) was developed largely from inputs given by elected representatives like MLAs and presidents of panchayats and government officials. It relied heavily on Minimum Levels of Learning (MLLs) and the development of human values, including spirituality, as suggested by the National Curriculum Document of 2000. In our interaction with people at SCERT, although there was a mention of inspiring leadership of the then SIERT Director, there was very little knowledge of what this document had sought to do. The memory that remained of that period was a flurry of movement and document production, and not ideas about the philosophical, sociological, pedagogical or curricular content principles.

However, some ideas were enmeshed in the system much more than the others. Though ‘activity’, ‘engagement with the child’, ‘making a child an independent learner’ etc. have been a part of the pedagogic vocabulary since the 1986 policy, it is the idea of competency and minimum levels of learning that seemed to have struck a chord in the minds of teachers and educators. This occurred in spite of the serious criticism of MLLs in academic discourses soon after its launch and NCERT’s subsequent introspection on this issue. Notably, the Minimum Levels of Learning contain itemized expectations at the end of each year in basic subjects. The critique of MLLs is that these are based on the behaviourist output sensitive pedagogy, which considers learning to be a linear progression of acquiring items. Although stated to be competency based, the document takes a flawed view of the term and is not able to separate information retrieval from any other notion of competency. The document is largely silent on concept acquisition. However, for most personnel engaged with curricula and textbooks at the state level, both the textbooks in use and the curriculum statements were MLL based. Importantly, MLLs give a kind of broken up and itemized understanding of learning that could be tracked and monitored and presented a basis for a centralized mechanical review and a handle to make so-called objective comparisons.

State level practices

So far we have focused on perceptions of key state level actors and the institutional practices in leading academic institutes. Let us now examine some of the state level pedagogic practices that consequently ensue. We examine with reference to mathematics, the curricula and textbooks, the learner evaluation system and teacher training. These practices ‘frame’ the classroom as they provide the structure in which teachers function.

Curriculum and textbooks

Processes of formulating curriculum and textbooks

Though curriculum changes had been attempted in both Andhra Pradesh and Rajasthan in recent years, it was hard to discern any real shifts. In light of the above discussion, it is not hard to understand why. The SCERT faculty responsible for carrying out this task generally lack the confidence, exposure and material to bring in new ideas. For example, one SCERT had initiated discussions on the NCF 2005 document without involving crucial participants even from within the state. The entire exercise had been taken up like a routine activity. The senior people in the administration were not aware about how people had been selected for the discussion, and nor did they know of the issues on the agenda. The curriculum material and development cell, which organized the process was not clear why these discussions were needed or how they should proceed. No groundwork had been done. In fact, no one appeared to have read the NCF text—partly or wholly. The other SCERT did not show any awareness of the existence of NCF 2005.

This illustrates clearly that the exercises of curriculum development and textbook revision were not well organized or given priority, and there was a lack of appreciation of the importance and the need to build a system and a structure for these tasks. In Andhra Pradesh the curriculum document and the textbooks remained unaltered because of the confusion in work division among the officers ⁶ (see Box 6.8).

Box 6.8: The Process of Curriculum Development

‘The old curriculum was made in 1989. Then it was decided that a new curriculum would be made on the basis of what people want. First, we invited people’s representatives at the block and district level and allowed them to speak on what type of education they wanted. We simply noted down what they said. Then learning areas were marked out. SCERT then went to two panchayat samitis where they met public representatives as well as government officials from different departments. That’s how curriculum development started. This process took two to three years. In the new curriculum, some of the concepts that had been introduced in Class 1 (such as carry over) were introduced in Class 2. As the new curriculum was tested with the people, new learning areas emerged. The curriculum had to be competency-based and made as per the MLLs.’

The NCF document 2005 elaborates the need to involve a variety of people in the production of textbooks and ensure access to a variety of materials for those engaged in the process. However, some managers reported that teachers and SCERT faculty often resist, the involvement of persons from outside the government in such processes. Importantly, curricular change was rarely informed by actual experience with children. For example, opinions on the need to modify the content and reduce the load on the child were sharply divided. A case in point here is that the introduction of 4-digit or 6-digit numbers and operations is done too fast and early. This means that the child does not get the required time and practice to understand basic concepts and be confident of the ideas and of playing with numbers and relationships. Instead, with larger numbers, the emphasis shifts to computation and making the child a calculator who can solve a given problem. This is not mathematics and reduces the expectation from students to a mechanical performance, rote learning of answers and/or shortcuts and methods. For similar reasons, though the mathematics syllabus was difficult for a majority of the children and also the teachers, neither teachers, nor trainers or administrators thought this

⁶ The Director of SCERT told us that the task of curriculum development was transferred from the Council to the Directorate of Education following the transfer of one person from the SCERT to the Directorate. This had resulted in some confusion regarding responsibilities as it was then shifted to the Director DPEP’s office located in the same building. Since the modification of the syllabus and the textbook was not the responsibility of the Project Director, DPEP, there was no system left to take it forward when priorities of the individuals posted as SPD, DPEP changed.

was a good enough reason to cut back on the irrelevant and excessive load in the syllabus. Instead, the overriding concern was that diluting the syllabus would lead to an adverse impact on the brighter children and make the state appear backward!

The selection of authors and coordinators for producing textbooks also showed the lack of a system that could produce good results. For example, one coordinator for the mathematics textbook depended on the abilities of two people recommended by the Director. They were the backbone of her 'team'. But in fact, there was hardly a 'team' to speak of, as the other members in the group were not important. She did not feel the need to develop a library of material of her own. In our discussions on textbook development, she did not refer to the national or state curriculum documents. The principles involved in the framework documents, even when developed within the state, did not effect any significant change in the material available to schools. There was also no mechanism for orienting or for the capacity enhancement of the people engaged in these tasks.

Some innovative work with textbooks had been taken up during the Andhra Pradesh Primary Education Project (APPEP). A Mandal Resource Person (at the time of the study placed in the SSA office), who had been involved in the participative exercise, described the books as:

Math books are inclusive— they see the need to build language and alongside have activities that allow children to engage and explore and have units instead of fixed chapters with many components being taken together. We believe textbooks need not be linear. The spiral approach allows for the gaps in a child's learning to be filled when those ideas come up again. Books follow a process in which the contexts and concepts are linked. For example, in closed and open shapes we use the tiger and goat example. We also use modified *bolo bhai kitne*. (A mathematics grouping game. This involves learners moving around to the rhyme of *bolo bhai kitne, aap bolo jitney* (say how many, as many as you say). Then quickly getting together in to groups of number size that is given. Those not being able to get in to groups of correct size are out and the game continues with the rest.) We have de-emphasized one correct method and have given the opportunity to the child to develop many possible correct methods and have also included questions that are open ended. The focus is on concept development. A group that works on textbooks must be mixed. It must have many teachers, experts and officers. But the new approach to math was not accepted by the senior brass in SCERT and, hence, has not continued after class 3.

However, as indicated earlier, this new approach was abandoned within two-three years. Even when attempts are made to develop different, more carefully thought out textbooks, these do not necessarily sustain, as the system is basically not oriented to working with such ideas.

To conclude, curriculum and textbook development require sustained and long-term intervention with an extensive statement of purpose assuring continuity of learning. Institutional memory of the processes and the reasons for undertaking them is essential. A more open participation is required to widen horizons (as was the case for some time in Andhra Pradesh). A strategy needs to be evolved to empower those responsible for syllabus and textbook formulation to act autonomously and logically, and be able to defend their choices and assumptions with their superior officers as well as with external academicians.

Production and distribution

The types of management related problems described in Chapter 5 also bog the production and distribution of textbooks, and this too affects the quality and eventual use of material by children. For example, a person involved in textbook production said:

Production is a big headache. Computer typing is a problem. Two or three softwares have to be used together in the math textbook. Tenders are floated for typing. The tender given was low,

and then people who quoted the lowest rate, walked out of the project. In the first typing, the book came to 413 pages. The rate is fixed per page. So, those who do the setting of the book tend to increase the number of pages.

Notably, textbook production revolved around finding writers, typists and computers and getting on with the job. No time was factored in for review, reflection or discussion. This trend did not encourage an open-ended exploration of the design, presentation style etc. but aimed merely at a feedback on typographic errors and minor suggestions.

The nature of textbooks

We now briefly discuss the broad characteristics of the textbooks. The vision statement of NCF 2005 suggests that mathematics should be activity-oriented and enjoyable. Instead of being constantly plagued by a fear of failure, learners should look at mathematics as a challenge, even joy. The language should be such that the child can understand it. The material/textbooks should emerge from the context of the children. There should be space for them to reason and provide their own logical arguments. Textbooks should be geared towards the higher aim of learning mathematics and conceptual development.

Our analysis of the mathematics books of class 2 in Rajasthan and Andhra Pradesh showed that they shared some common points. Both were like workbooks (the child was expected to answer the questions in the book), and there were instructions for the teacher at the bottom of some pages. In the Rajasthan textbook, some exercises with concrete materials were suggested. The language, however, was reflective of the pedagogy. An example, was ‘Keep 10 objects of different kinds in front of the children. Ask them orally which the fourth object is. Then ask the specific place of some other object.’ Although there was an attempt to include some verbal problems and poems, the book did not have many illustrations. Questions in the books rarely reflected the contexts and reference points of the children. The language was simple but flat, perhaps uninteresting. The competency (MLL) to be facilitated by each page was stated at the top. The assumption seemed to be that one kind of explanation or exercise held good for all children and would produce the same specific competencies in all. The book did not ask the teacher to add or improvise on the set of exercises to be completed in the classroom or at home. There were no games or puzzles that children could discuss with each other after the mathematics period is over. The class 2 book in Andhra Pradesh had much less content on each sheet as compared to the textbook used in Rajasthan. It had some puzzles and games and addressed itself to the children.

The class 4 books in both the states were regular, conventional textbooks. They looked formidable and dealt with extremely large numbers. They were oppressive in terms of both content and language, and inspired neither independence nor accomplishment in their users. Most chapters had virtually no illustrations, there were no suggestions to use concrete materials and the language was fairly complex. The contents page for both the books was similar. The exercises involved extremely long and often complicated manipulations with numbers. None of the books even mentioned, let alone nurtured, the sense of exploration in a child or the need to articulate ideas in her/his own way. Rules regarding numbers and shapes and their properties had just been stated and the children were not expected to attempt to find generalized rules and any patterns independently. Overall, the books lacked sympathy with the children or their milieu, in terms of both content and form. One may argue that the syllabus given to our team did not have enough scope for bringing in the experiences of children. However, we perceived in the classrooms that no such effort was made even when there was enough opportunity.

In the Rajasthan class 4 textbook, explanations and definitions used technical words and terse language. Technical words were used to explain concepts of factor, multiples, profit and loss

etc. Class 4 students were expected to understand the abstract notion of line, line segment and ray, angles and their types. All kinds of fractions, with complicated procedures, were part of the books. The supposition of the writer(s) was that children would already have learnt a fair amount regarding fractions in Class 3. While the chapter on fractions in the Class 4 book for Andhra Pradesh book was less complicated, the one on whole numbers had some extremely difficult and tedious exercises. There were very few opportunities for children to go beyond the chapter. The book trod the beaten path of collective classroom work and rarely encouraged children to go beyond the text and work in groups or individually. The exercises were aimed at making children 'learn' the method and the technique rather than at understanding the concept. Neither the class 4 nor class 6 books in the two states had exercises that the children could do on their own.

The Rajasthan textbook of class 4 asked the teachers to ensure that learning did not degenerate into rote, and be geared towards developing expected competencies in children. The book emphasized that experimental work and measurement be taken up to test mathematical formulations. This is in line with NCF 2000, which also talks about the importance of experimental proofs for mathematical statements. However, in the context of mathematics, this strategy is limited, because a proof has to be based on axioms and previously proved statements. The ideas in mathematics are abstract and ideal and hence cannot be proved or disproved by experiments. They can only be verified or demonstrated by using concrete materials. It may be pointed out that NCF 2005 specifically makes a distinction between verification and proof and also advocates the use of concrete materials in a very different sense than as experiments. It suggests the use of materials as a scaffold and a temporary device to keep the information visible while the child learns to abstract and understand concepts.

As in the book for class 4, the language used in the class 6 book for both states was extremely formal⁸². The emphasis in both the books was on children learning definitions and remembering rules. They were expected to follow the suggested methods to solve the problems given after every chapter. Only an occasional stray exercise in a chapter indicated that children solve similar problems other than the ones in the book. However, the exercises where these examples occurred did not subsequently build on them and there was no illustration of such attempts.

The Assessment System

Assessment of children has been a major area of concern in the country, particularly at the elementary stage. Many believe that assessment and evaluation through examinations is a major reason for high dropout rates and failure. On various occasions, policies of non-detention have been taken up, and some of these are still in place. Elaborate exercises have been developed on how to assess children. Apart from this, organized efforts to help teachers understand and diagnose the students' problems and use assessment to initiate remedial teaching have been emphasized. There has been talk of comprehensive and continual assessment and of assessment as an aid to help the teacher understand what is going on in the classroom. NCF 2000 affirms that:

Teaching for successful learning cannot occur without high quality evaluation. Evaluation, therefore, needs to be integrated with the process of teaching and learning..... Hence, evaluation has to be so designed that it can be used as a powerful means of influencing the quality of what teachers teach and what students learn. But while doing so, special care must be taken to ensure that it is humane and it enables the learner to grow into a responsible and productive citizen. Not only this, evaluation has also to provide constant feedback regarding the effectiveness of the course, its contents, classroom processes and the growth of individual learners besides the appropriateness of the evaluation procedures. It must, however, be flexible

⁸² We saw only seen the English edition of the Andhra Pradesh book

enough to the extent that it can be experimented with and adapted according to the specific situations and needs of the learner groups.....Evaluation is a systematic process of collecting, analysing and interpreting evidences of students' progress and achievement both in cognitive and non-cognitive areas of learning for the purpose of taking a variety of decisions. (Page 97, Chapter 4, Evaluation).

The 1986 policy sees assessment of performance is an integral part of any process of learning and teaching. It suggests that examinations be employed to bring about qualitative improvements in education, and the examination system re-cast to ensure a method of assessment that is a valid and reliable measure of students' development and a powerful instrument for improving teaching and learning.

The ground realities in schools against the aspiration of NCF 2000 and the 1986 policy to make evaluation an integral part of the teaching and learning process are given in Box 6.9.

Box 6.9: Views about the Learner Evaluation Process

Teachers:

- We write the test for Classes 1 and 2 on the blackboard for the half yearly and annual exams. We bring question papers from the market for Classes 3 and 4 and bring it from the CRCF office for Class 5. Monthly test papers in all the classes are written on the board. Since the papers for Classes 3 and 4 are bought from the market, we do not know who composes them.
- Test papers for Class 6 had come from the department. We do not know from where the test papers of Classes 3 to 5 had come. We copied the test papers of Classes 1 and 2 on the board.
- Question papers had come from the nodal centre, the same question papers had been distributed in the entire district.

Most teachers did not frame their own papers. The papers were either bought from shops or obtained from the block office/teacher centres. This clearly indicates that the question papers were not related to the manner in which students were taught or to the specific competence level the students. The major purpose of the assessment, i.e., giving teachers an idea of what her students are capable of doing, was therefore lost. Notably, though examinations were held at regular intervals and the test papers were available at some source or the other, grading was not a favourite task with teachers and others. Nor was there an understanding of the fact that performance or marks in a test were not a permanent measure of a student being weak or strong in a particular subject. In fact, many believed that children made an effort to learn only under the stress of examinations or evaluation. The major concerns expressed here, with a few exceptions, were towards ranking children accurately and properly. Such an evaluation system seems to provide a handle to the system to monitor the teacher, rather than to provide a set of learning tasks to children to encourage them.

Moreover, officials told us that teachers were also under a lot of pressure from senior officials to pass children, and the result was often mass copying. Additionally, children's answer papers and procedures of assessment were rarely shown to their parents, and therefore they could not show interest in the children's progress. Thus the teacher was obliged to pass the students on the basis of externally produced question papers so that she could convince herself and others that she had excelled at her task. Naturally, teaching and learning took a back seat and assessment became all-important for a reason quite different from actually 'assessing' the child.

Let us contrast the above situation with NCF 2005, which says that:

The purpose of evaluation is not:

- To motivate children to study, under threat.
- To identify or label children as ‘slow learners’, ‘bright’, ‘problem children’. Such categories segregate, place the onus of learning solely on the child, and distract from the role of pedagogy.
- To identify children who need remedial teaching (this need not wait for formal assessment, it can be detected by the teacher in the course of teaching and attended to as a part of pedagogic planning, through individualised attention).
- To diagnose learning difficulties and problem areas, while broad indications on conceptual difficulties can be found out via evaluation and formal testing. Diagnosis requires special testing instruments and training. It is also specific to foundational areas of literacy and numeracy, and not for subject areas (Page 65, Section 3.10.1).

We do not explore in detail what NCF 2005 says about the deeper meaning of evaluation and assessment. However, we look briefly at the following paragraph on the teaching of mathematics:

We talked of fear and failure. While what happens in class may alienate many, it never evokes panic, as does the examination. Most of the problems cited above relate to the tyranny of procedure and memorisation of formulas in school mathematics, and the central reason for the ascendancy of procedure is the nature of assessment and evaluation. Tests are designed (only) for assessing a student’s knowledge of procedure and memory of formulas and facts, and given the criticality of examination performance in school life; concept learning is replaced by procedural memory. Those children who cannot do such replacement successfully experience panic, and suffer failure.

...While mathematics is the major ground for formal problem solving in school, it is also the only arena where children see little room for play in answering questions. Every question in mathematics is seen to have a unique answer, and either you know it or you do not. In language, social studies, or even in science, you may try to demonstrate partial knowledge, but (as the students see it), there is no scope for doing so in mathematics. Obviously, such a perception is easily coupled to anxiety (Page 55).

It is clear that teachers, teachers educators and managers must be better trained to understand the concept of assessment.

Capacity Building

A system becomes what its component of people makes of it. To improve the school and its processes, capacities of those who have the power and the authority to initiate change in the system need to be built. In this section, we examine how capacities of individuals placed in various positions within the system have been built.

Capacity building of leaders

In our interactions with some of the decision-making people within the system, we found that none of them had gone through any orientation or capacity-building exercises on pedagogy (see Box 6.10).

Box 6.10: Capacity Building Experiences of Leaders

A sub district level manager: ‘I was not given any rigorous or professional training. I have learnt everything on my own and from my experiences. However, I was able to attend a number of workshops conducted in accounts, administrative matters, midday meal management, child labour and communication skills.’

Director of SCERT: ‘I have learnt mostly on my own. I have not participated in any capacity-building interactions in the last few years. The programmes I attended included a six-month training when I joined and another three months of training from NIEPA for district planning. Subsequently, there was a three-week administration management training, also from NIEPA. I have also been through a school-mapping training at NIEPA.’

Senior Officials in Elementary Education: ‘I think of myself as a student. I read rules, government circulars and make notes. I find doing this helpful. I have a collection of all the rules. This gives me confidence. I received training for 21 days on becoming a headmaster but none thereafter. That training was good. I believe training is very necessary.’

‘Trained at the State Administrative Institute, I learnt a great deal about land-related issues in the initial years. I have also worked in urban development and in administration. But there has been no training in education.’

While decision makers had trained in planning, management and administration, they had had little or no opportunity to receive training or interact on what constitutes a good classroom or other education-related aspects. They may have had their own notions of what constitutes good education or a good classroom, but they had never been required to analyse their thoughts in a framework. Apparently, to be familiar with the rules of management, accounting and administration, enmeshed with some ambiguous theories of education, was considered adequate...pedagogy, philosophy, psychology and the sociology of education be damned!

This was despite the fact that the senior officials quoted in Box 6.11 were expected to (as part of their core duty) decide how education could be improved. They were required to motivate trainee-teachers, textbook writers and developers of training modules etc. Yet they had had no opportunity to review their understanding of issues that influence the quality of education. It was not surprising that most of them believed that their primary obligation was providing reports at the right time and ensuring tabulation of information. This was also a possible reason why institutions had no memory of what they had done and how specific methodologies and ideas had fared. Each attempt was a new effort, often repeating the key mistakes and limitations of previous efforts.

Teacher training

Structures such as SIERT, DIETs, BRCs and CRCs have been set up to ensure the effective capacity-building of teachers. Many programmes have been initiated, including in DPEP and SSA, to train teachers at the elementary school level. However, despite these institutions and programmes, most teachers have not had the necessary interaction to be able to stay in touch with their subject. Many teacher trainers and officials recognize the need for more effective capacity building. While observing teacher training programmes was not in the ambit of this study, we interacted those who had been/were involved in teachers’ training (see Box 6.11).

Box 6.11: Teachers' Training – Some Comments

SSA coordinator for teachers' training: 'We are trying to improve quality through training. Most of my time is spent in office work and I am not as involved with the field. But I can say that we have tried to improve training and have looked at the module. We examined what the teachers wanted and reduced the number of days of training from 15 to nine. We then got the DIET and SIERT faculty to attend, increased the honorarium of the resource person, made the training residential and introduced a monitoring system through BRCs and DIETs. All these initiatives were eventually manipulated. We tried to improve things but could not. Rules are very complicated and there is no punishment for flouting them.'

Highly Placed Person in SIERT: 'SIERT is responsible for curriculum development, teachers' training (both pre-service and in-service). SSA does the training and SIERT prepares the resource persons. Decisions about training are taken at the government level. There was an earlier decision to involve SIERT in training but the government said training had to be done within a month in the summer holidays. We had made a six-month programme which was manageable but that was not accepted. Therefore, the training is now being done through SSA.'

Teacher Educator: 'We train teachers under various programmes as per goals. However, what they learn in training is not implemented in the class. A teacher who undergoes training should tell everyone in the school about what he/she has learned. It is the duty of the headmaster to see that teachers use the techniques imparted during training in the class. But the problem is that 2nd grade teachers are per force made headmasters and they don't seem to bother about it.'

Teacher Educator: 'During training, teachers should be encouraged to talk openly about their problems and share their experiences; some do this to a certain extent. But most of them don't. If only teachers would speak up, the trainers will be better placed to solve their problems.'

People in the system at one level displayed lack of deep understanding on the other hand felt frustrated about the response to the attempts at change and improvements made by them. Before examining the software of training, we shall look at the hardware, i.e. the way training was organized.

Preparation, planning and implementation

Our investigations showed that as part of the 'cascade model' that was used for training, a few individuals were trained at the state level. They in turn trained people at the district level who then trained masters-trainers. The masters-trainers would in due course train teachers at the block or cluster level. Importantly, crucial issues regarding the duration, content and process of training were decided by an arbitrary process. Speediness, rather than quality, was the criterion for deciding on the trainers, the argument being that unhurried training did not guarantee quality. Field-level personnel could not decide the pace of their training. Our discussions showed that while they were not sure of the areas that needed to be chosen for interaction or the content during training sessions, they were convinced that the current modules were not appropriate. One could sense the constant conflict between cynicism and resignation towards status quo and the hope that the structure would allow honest choice, review and reflection.

Focus in training programmes

While analyzing the quality of training, some of the functionaries also mentioned tangible components such as teaching-learning material. These, however, were the only areas mentioned to accomplish quality training. For example, a mandal resource person said:

‘The quality of the training was very good. Teachers participated and a lot of TLM was made.’

However, there was no mention of the points that were discussed during the training or the manner in which it was executed. What remained in people’s minds was the concrete output, not how it could relate to learning or how it could help children. This attitude may be posited against the fact that TLM is an aid to help teachers involve and engage children. If there is no clarity about what the children should be involved in and for what purpose and in what manner, then making TLM is meaningless. Cynicism tinged with disappointment about the problem areas in training found its way into the comments of an SSA coordinator for teachers’ training:

At present we cannot even properly focus on competency development among teachers. We work on content, methodology etc. but more work is needed. We also need to work on attitudinal issues. We do not have the type of educated trainers in the numbers that we need. To some extent, training modules are responsible for the situation. Long-term training is needed for building conceptual understanding as well as attitudes. But there isn’t enough time. I do not feel that we have been able to make any changes.

An Assistant Programme Officer told the researchers: ‘

‘We were called to the headquarters for three-four days. We were told about the objectives of SSA, how to ensure cent per cent enrolment and help out-of-school children join the mainstream. We were also advised to execute the programmes in such a manner that the community got a sense of ownership and participation.’

This was passed off as ‘training’.

Who wants to be trained?

A problem that came up was the unwillingness of teachers to train. The trainers that we spoke with said that only those teachers who did not have ‘big’ degrees were amenable to training. One teacher training coordinator who had experience of several programmes said:

‘As far as training of teachers is concerned, the clients for Shiksha Karmi (SK), Lok Jumbish (LJ) and the SSA were different. SK training was the best because the type of individual who came to be trained was not permanent and not formally trained.....the problem is the “know all” attitude of teachers. Teachers are so arrogant...we need to go beyond this arrogance and work at the child’s level...Selection of teachers for the SK project was different and so was the training. We had teachers who had just passed Classes 8 or 10 but had a good attitude. And now we also have teachers who have done MSc, but they don’t want to stay in the village and teach.

This view has two aspects, one the fact that the teachers who do not have high degrees and are new to the system are more inclined to exert and learn, and the other is that the trainers are themselves not equipped to work with and dominate those who are academically better qualified and experienced. The notion of learning together, identifying new problems to explore and generating excitement for the trainees does not exist.

A key goal of training is change in the classrooms. As mentioned earlier, teachers did not remember the content and purpose of the training. The trainers and those who planned the exercise had very little to say on this issue and were happier talking about numbers and logistics. There was also little concern expressed on the content of training or about ideas not

being shared or followed up. At the state level, the feeling was that both the quality of training and follow-up programmes needed to be more effective and efficient. But the idea of monitoring was in the 'inspectorial' mode, rather than in attempting to understand the needs and effectiveness of the programme and improving it.

Trainers

This discussion shows that while there are concerns about the motivation, technical ability, attitude and understanding of people charged with the responsibility of guiding, counselling, supporting and ideating with teachers, there are no programmes for preparing supervisors or master trainers in a manner that gives them time, ability and confidence to be open on pedagogical issues. Their competence in the subjects is also suspect.

It is clear that the preparation and selection of people as trainers needs another look. If we could have a mechanism of giving long-term training to trainers at the district level and placing them in DIETs, cluster or block support offices for the purpose of supporting schools and understanding the processes in the classrooms, we could have a cadre that is capable of providing quality inputs. Currently, however, there is a paucity of quality people in education. To quote a person from SCERT:

'At present the Mandal Resource Persons (MRPs) are available but the system is defective. We are now making subject-wise MRPs. When the MRP goes to the school and is not recognized as a suitable person then they will not listen. We are planning an exam and selecting MRPs subject-wise.'

The important point here is that the role of the people who are providing support and inputs to the schools is crucial. They need to think about the requirements of teachers in terms of suggesting activities and tasks to make classrooms interesting for children. They must be able to conceptualize the processes that ensure retention and learning in the classroom and help children become confident and comfortable while learning. This is not the understanding of the present role that the system has for these persons.

The teacher training 'bind'

Given the above scenario, teachers did not appear to benefit a great deal from the training programmes. An officer said:

'Teachers think it is the same old module, the same old trainer, the same place, the same five, seven or nine days of training and they are bored at the thought of it. It becomes a burden. Ironically, they also feel that earlier training was interesting but have no clear idea as to how and why it was useful!'

So, on the one hand teachers find training coming out of their ears and on the other, government officials feel that they are so caught up in organizing endless trainings. At the same time, all documents and discussions see the capacity building of teachers as the most critical element in improving schools. Notably, several new players have entered the training fray. The SCERT Director of one of the states said that they were planning to get Microsoft to organize training for teacher educators and prepare lessons on computers for this purpose. Others would perhaps follow.

The way forward

From our discussions about the ideas regarding education and learners that pervade the system, it is clear that there is a need for training. However, what is also clear is that the system does not know how to organize and conduct training programmes. This discussion makes the

reasons for the ineffectiveness of training clear. But these reasons are rarely explored when actually planning teacher training. Reviews of teacher training strategies are not rooted in the ideas of equity or the other main defining features of meaningful thought for social and public spaces as reflected in the preamble to the Constitution and in the context of what we understand about the construction, evolution and development of educational thought in any human mind. Instead, all kinds of ideas regarding organizing different kind of training, which do not emerge from a rational analysis, float around. For example, an SIERT official felt that spiritual education of teachers was necessary for training to be a success. Another training coordinator told us of plans to develop literature to train teachers through correspondence. There was no clarity on the nature of the content of training or on the kind of material that the teachers could be interested in. The need for correspondence training had arisen to fulfil targets within the allocated budget.

Our analysis shows that some basic issues need to be addressed for training to be meaningful. Officials in charge need to understand education better and a process to orient them needs to be put in place. They need to have a background in philosophy, sociology, psychology, teaching-learning methodology and in educational management. Often, those in charge of training just have a fleeting experience of schools, classrooms and of families in villages. These officials need to undergo training as teachers do on various issues, spend some time as teachers or academic observers in schools and have a constant and patient interaction with children in the schools. Moreover, team building among those planning and conducting the training also needs to be ensured. The orientation of teachers cannot be treated as a task to be accomplished in 5-10 days. The relationship of the training team with each other, their understanding of the strengths of individuals in a team and of building a climate of partnership with interest and support would help the quality of training.

We also know that quality follow-up and support is a critical factor in improving teacher understanding. Each experience should feed into the next attempt. This can be possible if trainers follow-up programmes and training designers share their experiences regularly and feel responsible for improving quality in schools. They need to appreciate each other's work and present a common understanding to the school. It is difficult to do this while following the cascade model of training done at breakneck speed. Hence, a more comprehensive capacity-building programme needs to be set in place.

While overall teacher training has tended to disappoint, yet there have also been effective training programmes from time to time, and it is possible to learn from these. The following description from an MRP provides a direction that the training can probably take:

In 1996, the DPEP studied training started with Special Orientation for Primary Teachers (SOPT). Teachers were not happy with lecturing. We went to observe and participate in workshops.

Change came with inputs from outside...The state team sat hearing and absorbing ideas on the 'Nature of Child, Nature of Learning, Activity, Nature of Subjects'... these were the important components of the new training design. These inputs were of many dimensions and from multiple sources. The State Resource Group (SRG) was set up—SCERT/DIET/MRP were involved as resource persons and many of them were SRG members who underwent a series of training from experienced people and were from outside the system. MRPs were appointed. The interaction moved from one-sided lecturing to a participative discussion. Teachers demanded change in textbooks.

In the School

We have explored ideas about education and learners, the institutional structure that provides leaders and manages the academic process and also pedagogic practices and inputs such as curricula, textbooks, learner evaluation and capacity building. Against this background we now try to understand what happens in the school.

Overall, the schools visited presented an ambience of a system without internal dialogue, reflection, a sense of purpose or a curiosity to learn. The people lacked hope, their confidence in using or creating space to explore ideas was low and they were unsure of what was expected of them. The situation in private schools was not much better in terms of what went on in the classrooms, yet the impression created on the surface was of regularity and seriousness.

The reasons for this disinterest at the school level are many. As mentioned earlier, the space and time for the teacher and children to be engaged in meaningful learning tasks was extremely limited. In this section, we try to understand how the various perceptions about the teaching- learning process and pedagogic processes contribute in making the school what it is.

What is a good school, who is a good teacher

The creation of a 'good' school depends on the vision of what constitutes a good school. We posed a question, i.e. what is a good school, at several levels across the system. Cutting across categories of administrators, teacher educators and teachers, a good school was seen as one that had enough teachers, adequate rooms, blackboards, chalk, some kind of toilet and drinking water facility and that opened on time and continued for the full duration. Thus largely, the vision itself was limited and showed a system attempting to cope with basic issues of resources and management. Only a few respondents mentioned the school as a place where children were happy learning. However, no one said a good school must be sensitive towards its children and ensure their participation in the classroom. Not one interviewee mentioned that sharing and learning among teachers and a responsible community were the cornerstones of a fine, flourishing school.

Similarly, for most people a teacher was good as long as she was sincere, regular, punctual and seen to be teaching the class. How she taught and engaged the children or what material and activities she used to make learning a fun experience was immaterial! This good teacher could get away with being rude if only she abstained from unnecessarily beating up the students. While we did not come across any instances of corporal punishment in the classrooms, very often teachers would humiliate children and speak rudely to them. Supervisors and resource persons lamented the decline of the teacher's '*dharma*':

Teachers are not self-motivated and only want their salaries. The emphasis is on demands, but there is no dedication. Their morale has plummeted. No awards, no punishments seem to have an effect. They do not heed instructions or do what is expected of them. They do not keep records or diaries and do not take up remedial teaching or make TLM...

Sometimes, amid all the distress and difficulty, we heard voices of verve and vigour. Some teachers differed from the administrators on what was a good school and articulated the need for more space, freedom and cohesive functioning as being important for a fine, flourishing school.

However, while articulating personal qualities of teachers, almost all the respondents including the teachers, tended to visualize the ideal teacher as a god-like being. The image of an ideal teacher was one who was sensitive, caring as well as firm, and disciplined. The teacher was expected to transcend material demands and aspire only to sacrifice. While admitting that the

administrative system requires many corrections and is far from being ethically sound, and that the social and political systems that influence schools and their administration are equally devoid of strong ethical moorings, the expectation remained that the teacher should be ethical and self-sacrificing. Whatever the constraints, he/she was expected to be dedicated, sensitive, full of verve and initiative, creative, obedient and compliant. Unable to identify, or incapable of addressing, the larger problems within the system, most respondents saw the teacher as finally responsible. The problems of textbooks, curriculum expectations, systemic conflicts, societal problems, administrative mismanagement and all other factors were disregarded. Interestingly, most teachers also accepted this interpretation.

With all these constraints, teachers somehow ‘managed’ their work. How they did this, is illustrated below in their approach to teaching math, and to ‘weak’ children.

Teaching mathematics

NCF 2005 emphasizes designing learning tasks that match the capabilities of children. It says that there can be no real learning if the learner is not taught to value her own ability to think and reason. Tasks that are difficult, repetitive or mechanical, or assume the passivity/obedience of children are not meaningful. This places the onus of holding meaningful classes on teachers, textbooks and curriculum. However, the classrooms we observed showed that neither teachers, nor books and material were capable of drawing up tasks in keeping with the child’s competence. Instead, they were mechanical and dull. In the classrooms, children were passive receivers; they were neither expected to construct new problems nor solve existing ones on their own. They were not given any tasks that gave them a sense of achievement or enhanced their confidence. Some of the methods that the teachers used to teach math are given in Box 6.13

Box 6.13: Teachers Talk about Teaching Math

- I need to get more time to teach the children. I think, then, they will not have any difficulty in understanding mathematics. Time is a major issue. We are not able to get teaching work done due to lack of sufficient teachers.
- We get children to memorize formulae, we give them repeated practice of examples in the lesson done. We get various exercises done. This can be made interesting through aids/TLM, charts etc.
- Many a time, they face difficulties due to lack of information about mathematical rules. We give information about these rules. If there are many example and exercises given in the book, then we solve these exercises.
- The subject can be made interesting by giving information about the usefulness of mathematics in daily life.
- Usually we solve most of the examples ourselves and explain the solutions to the children. But sometimes we face difficulty with the math taught in class 5.
- While teaching mathematics, one should ask about numbers. Through this, the children’s knowledge of numbers increases.
- The difficulty is that there are 48 children in class 6. We can pay attention to only seven or eight intelligent children. We are not able to pay attention to the weak children. The course is too much and the time not enough to cover it.
- The students are able to answer questions that are straightforward and direct. But they are not able to do indirect questions.
- If the topics are not related to each other, then they seem to become more difficult as we move further. The subject gets more difficult to understand if the child is not inclined or interested.
- Difficulties are faced in getting new topics covered. We have to go prepared for topics such as equations and trigonometry because children find them difficult to understand.

It is important to mention here that the classrooms in private schools were no different from those in the government schools. The teachers did the explaining and writing on the blackboard without paying much attention to the children or their responses. The content and

the manner of teaching were similar and the students were barely interested. The only materials used were the blackboard and chalk and the only way to solicit children's involvement was to ask them to respond to the teacher's questions.

Clearly, teachers believed that children needed to be explained questions and memorise formulae and rules. They did not feel the need to examine their teaching methods but also did not know how to deal with some children learning faster than the others. While all the curriculum documents have expressed concerns about children failing in mathematics and have focused on the need to teach at the level of the learner, most teachers did not admit to any problems in teaching the subject, though they grudgingly mentioned that some of the topics were not related to each other. They also felt that the course content could be a bit too much for the children even though they also suggested that it is a problem of the 'weak' children (Box 6.14).

Box 6.14: Teachers' Views about the Math Curriculum and Textbook

- Questions in junior classes are usually easy. Children manage to do them. However, as they move to higher classes, the questions become increasingly difficult.
- Most of the children have passbooks (key/guide books), and copy from them. But they face difficulties in answering oral questions. Even if a minor change is made in the questions given in the book, they cannot solve the question.
- Whichever textbook is prescribed, we teach it. We have never thought about whether the textbook is good or bad.
- Change has come about in the textbooks for classes 6, 7 and 8. No change has come about in teaching.
- The mathematics textbook for class 2 is found to be all right by an intelligent student but a child who is weak, does find it difficult.

One teacher very succinctly expressed the general opinion about why children failed in mathematic:

‘Children who work, learn. It is only lazy children who are unable to learn.’

In addition, an officer at the block level said,

‘Teachers try their best to teach mathematics but students are not interested. Some of them have low IQ and their parents are also not bothered.’

For all the hype about pedagogical renewal including textbooks and training, most teachers seemed ignorant about the changes in the textbooks or when these had taken place. Most had not compared the old and the new textbooks and were not even sure if the books had changed. Those who did know about the changes were indifferent to them and did not think that they had affected teaching in any way. The only thing that they considered significant was that the books had more pictures and colour. These remarks reveal the confusion in teachers' minds as to why mathematics puts a blight on children. Teachers swung between terming the books difficult or blaming the conditions and/or the children for their failure to learn. This is relevant because when we look at the books for class 6 of both the states, we see them filled with mathematical expressions and explanations.

However, in Andhra Pradesh people seemed to remember the changes in the Telugu books of Classes 2, 3 and 4. It may be useful to consider a process for textbook writing that was sustained for a while in Andhra Pradesh. *Notably, this process involved selection of a team from volunteers from across the state through an open advertisement. The mixed group picked from this process went through a facilitated engagement with the task and a sustained interaction with each other. The team developed a framework for the books and were exposed*

to other materials and interacted with other teams and also with children to get feed-back on their ideas.

Though officials, teacher educators and teachers believed that the onus for poor performance was on the children and their parents, the classroom processes belied this claim. We observed teachers writing and solving complicated problems without attempting to dialogue with the children and/or engage them actively in the solution process. The authors and the proponents of the National Curricular Framework may cry hoarse about the need for multiple answers, engaging children to formulate their own problems and evolving individual strategies to approach solutions but such an education in mathematics remained a pipedream. Meanwhile, the buzzwords—‘level of children’, ‘alternative play way methods’ etc.—continued to do the rounds without meaning anything on the ground. That the situation of mathematics in the school has always caused concern is evident from this remark in the Education Commission in 1964:

The modernization of mathematics teaching is vital to any programme of reform in school education. But the new curricula and the modern approach can be introduced into our schools only gradually. The pace in this matter will be determined by what can be done for the training of new mathematics teachers, the retraining of the teachers already in schools (through refresher and correspondence courses) and for the preparation of new text materials. (Page 348, Section 8.66)

A reading of NCF 2005 suggests that not much has happened on these lines. Since all policy documents and programmes of action have talked about this, the question is why it does not change on the ground, in the school and in the classroom?

Clearly, teachers need to know more mathematics. They need to enjoy mathematics and be capable of exploring new problems in it. Also, they need to have the capability of understanding the learning processes and the kind of stages that the children are going through. It is only then that classrooms will change. Given the present scenario, this requires massive capacity building at all levels through processes that are not mechanical and aimed at showing large achievement targets, but aimed at making a real difference in the capacities of the people engaged with the tasks. They need to understand why mathematics needs to be taught, why everybody should learn it, and what learning math means. As has been stated earlier, knowledge of mathematics is necessary not merely to solve problems using algorithms and resolving long calculations, but to be able to reason and explore new answers. This appreciation was lacking and mathematics teaching remained focused on shorter and better ways of memorizing algorithms and rules.

Accommodating ‘weaker’ children

Policy and education documents after 1986 have backed a ‘child-centred approach’ to learning and the more recent ones also speak of the ‘child as the constructor of her own knowledge’. The National Policy of Education 1986 defines child-centred approach as:

Child-Centred Approach—A warm, welcoming and encouraging approach, in which all concerned share a solicitude for the needs of the child, is the best motivation for the child to attend school and learn. A child-centred and activity-based process of learning should be adopted at the primary stage. First generation learners should be allowed to set their own pace and be given supplementary remedial instruction. As the child grows, the component of cognitive learning will be increased and skills organized through practice. The policy of non-detention at the primary stage will be retained, making evaluation as disaggregated as feasible. Corporal punishment will be firmly excluded from the educational system and school timings as well as vacations adjusted to the convenience of children (**Page 11, Section 5.6**).

NCF 2005 reflects on this and says similar things with a greater emphasis. It has been 20 years since the National Policy on Education (NPE) 1986. Since then, the educational system has been through many programmes of quality improvement. However, schools and classrooms remain the way they were. In fact, there seems to be ignorance of even the most commonsensical ideas of dealing with children. We did not come across a single process in a classroom that took into account multiple rates of learning and appreciated children who were slow to learn or could not pick up things in the first instance.

The views of the administrators revolved around key ideas including that of teachers' lack of knowledge, perception of TLM as magic and lack of awareness about the incompetence of trainers. To quote:

Mathematics is not tough, if teachers teach it well. Children are told at home that mathematics is tough and, therefore, find it tough. They should be told that it is simple. The concept should be clear. It should be made clear through teaching aids and activities. Earlier, there used to be many problems in this regard. There was no money for such things. Now there is money because of DPEP and SSA. More TLM is used now. If this process continues, there should be some improvement. Training is also needed. Sometimes, trainers are also not competent and teachers' concepts are not good.

Some who had taught senior classes in mathematics said:

The main difficulties in mathematics are because of the teachers. I have had to teach basics like division when teaching class 10. Most teachers don't understand basic things. We need subject teachers in primary classes. Teachers don't read the textbook. A teacher just looks at the example given in the lesson, the question in the exercise, and solves it without explaining to the children. For instance, the degrees in a triangle can be measured and then explained, but teachers don't do that.

Of the people we spoke to, only one MRP placed in the SSA office talked about the need for strengthening concepts. He also mentioned the need to form groups in classrooms and use them to attempt activity-based learning. He emphasized involving and waiting for 'weaker' children. But as a rule, administrators, teacher educators and teachers blamed the children's deprived backgrounds or lack of exposure for their inability to learn. Many of them admitted that children found mathematics difficult but they did not have any way of ensuring their involvement. In fact, we found no one in the system willing to understand children or sensitive to their difficulties. *The problem with mathematics learning was largely seen as children not exerting, not being keen to learn, being genetically deficient, or with the teachers being lazy and not wanting to teach.*

What do children learn?

We now come to the question of what the children who experience the classrooms and the system described above actually learn. As mentioned earlier, we gave a mathematic test to the children of Classes 2, 4 and 6. The test in mathematics was set in Hindi and Telugu and the questions were simple. The pattern of questions for all the three classes was the same. The paper was not linked to the chapters that were being taught but was a basic mathematics ability test. Box 6.15 provides described these tests:

Box 6.15: The tests for classes 2,4, and 6

Class 2: There were 18 simple questions involving addition, subtraction and multiplication; three word problems; and two questions that required the child to count the number of objects in pictures.

Class 4: There were 21 questions involving the four operations as well as some fractions; a question on identifying the fraction depicted in a picture;

Class 6: There were 25 questions involving numbers and fractions; 4 questions involved multiplication or division with decimal numbers; several verbal problems.

Larger numbers were used for higher classes, though at the class 4 level, these were within a couple of hundreds. Slightly more complex division problems were included in the test for class 6 as compared to that for class 4. For class 2 there were no questions involving division.

Below are given some highlights from the class 2 results (see Box 6.16):

Box 6.16: Highlights of Class 2 Results:

- Questions on counting objects pictures were done reasonably well, but not by all children.
- Question related to finding missing numbers was attempted reasonably well in Andhra Pradesh, but poorly in Rajasthan, and the average score obtained by a respondent in Andhra Pradesh was 60 per cent and around 10 percent in Rajasthan.
- Questions that were solved the best in Rajasthan were those written in the format of columns where the numbers had already been positioned. Even within these, the success rate in Rajasthan was mainly in additions without carry over. The average student score for these was just 33 per cent. In questions with carry over, the average score was 19 per cent. While the same scores for Andhra Pradesh were 67 per cent and 26 per cent respectively.
- The overall average student score obtained in Andhra Pradesh was around 50 per cent while in Rajasthan it was around 12 per cent.
- Out of the nine schools in Rajasthan children in six, scored zero in all the questions. It was only in the remaining schools that the children were able to do something.

In general, in class 2, students did worse whenever slightly large numbers were involved, i.e. wherever the numbers were unknown by sight. Classroom observation data was consistent with this performance. In all questions, the performance of children in Andhra Pradesh was better than in Rajasthan and in some Andhra Pradesh outperformed Rajasthan by several times. The students of the two private schools in both the states did marginally better.

The class 4 results showed a similar pattern. Here again, the Andhra Pradesh schools did far better than schools in Rajasthan. Students in Rajasthan were only able to answer questions given in the standard format that required them to just add columns or follow the algorithm. Children found questions written in the horizontal format far more difficult than those written in the column format. Wherever there were zeros involved in multiplication, particularly if it was in the lower number, the students found it problematic.

However, the comparative performance of students in class 4 in the schools in Rajasthan seemed to be better than the performance in class 2. This was in contrast to Andhra Pradesh where the performance was roughly the same. Notably, classroom observations showed that in Rajasthan much more attention was paid to teaching and learning in class 4 than to class 2. The schools that had only a few teachers generally concentrated on classes 4 and 5, leaving classes 1 and 2 to fend for themselves. This was also reflected in the results.

The performance of class 6 students in the two states also showed an interesting change. Even though students from Andhra Pradesh a little did better, the gap had reduced and become quite small. Students in Andhra Pradesh were 27 per cent marks on an average, while those in Rajasthan obtained 22 per cent. Children from Rajasthan were a notch better in division and multiplication questions. Although students from both the states did quite poorly in fractions, students in Andhra Pradesh were a shade better. The results here also broadly matched classroom observations. Though the number of classrooms was very small and there was only

one test, the correlation suggests a relationship between test performance and the context of school and classroom processes.

The fact that schools in Rajasthan had caught up with schools in Andhra Pradesh by class 6 needs to be analysed. It is clear that the achievement scores in classes 2 and 4 in both the states were mainly due to sums given in columns where no understanding was necessary. It can be argued that even when children move to class 6, they do not move beyond that initial competency. Scores in word problems, too, were very poor. This is proof enough that the schools focus more on getting children to practise algorithms instead of helping them learn to read or enable better understanding of concepts. In class 6 we observed teachers writing on the board while less than half the class followed what was being taught or was even interested in the proceedings. We also saw that teachers in practically every school addressed and seemed to teach only a couple of children in their class.

The fact that students in Rajasthan had caught up with students from Andhra Pradesh, can perhaps be ascribed to the expected level of abstraction that has to be engaged with and learnt in class 6. Here, both sets of children did poorly. Clearly by this stage, it is not enough to rote learn formulae or just learn the mechanism of solving problems. Notable, in Andhra Pradesh, students in class 6 could only do the kind of questions that they were able to do in class 2 and 4. They had not progressed much. However, the reducing gap in scores can also be due to the fact that more children drop out in Rajasthan than in Andhra Pradesh. Children who are in the school in the early primary classes and have scored a zero are unlikely to remain in school through the upper primary classes. This is something that can be investigated in a follow-up study.

Overall, the evaluation results of the three classes indicated extremely low levels of ability in mathematics. This was consistent with the manner in which mathematics was taught in the classroom, with an emphasis on memorization and/or following procedures mindlessly. The attitudes and understanding of administrators, teacher educators and teachers also matched classroom processes.

Concluding Remark

The classroom processes, the views of the teachers, teacher educators and officers and children's performance on tests, indicate a wide gap between what exists, and what would ordinarily be considered a reasonable mathematics education. The expectations of NCF 2005 are still higher in terms of the goals set out. A crucial requirement for bridging this gap is inculcation of confidence and capability in subjects among teachers and teacher educators, because their attitude to the subject and its transaction are influenced by their own confidence and conceptual clarity. The task here therefore extends to the improvement of university education as also the senior secondary education, to make those entering the profession confident of their knowledge. It also encompasses changes in the notion of knowledge, children, their minds and learning. This has wide implications for teacher training and teaching-learning materials. Many of them have been stated in NCF 2005, but the question of making all this possible still hangs.

The institutions that are responsible for bringing about academic quality need to be strengthened adequately, provided with resources, space and autonomy to function with their own agenda. They need to be made responsible for developing, planning and executing efforts to reach education to all students, particularly the poor and disadvantaged. It is clear that in spite of interventions like DPEP and SSA there is no significant change on the ground. These efforts at the macro level have been directed centrally and have not been pursued with adequate decentralization of responsibility and authority. They have left no space for the

system to think on its own and have placed no demand on it to improve itself qualitatively in terms of its learning.

It is also evident that participants and stakeholders at all levels need to have a clear view about the need for education and the role of the education system. Without this clarity, the system gets lost in rituals and logistics, forgetting the core of teaching-learning. At present, the system as a whole gives the impression of being uncommitted to the goal enshrined in the Constitution and in recent policy documents. It seems to be cynically holding the victims as being responsible for their own plight. Present exercises like trainings, material production, publications and the rest of the wherewithal in SSA and predecessors, and those of the Rajeev Gandhi Shalas and the predecessors, or the vidya volunteers etc., only add to the cynicism and confusion in the system, particularly among the teachers. The notion, that spreads is '*Sab Chalta Hai*' (anything goes) rather than the need to reach and then raise the bar of quality at present.

The system also seems to be developing for each strategy each idea proposed and the emergent plan formulated to improve the learning of mathematics in our schools, a counter strategy that simultaneously makes its implementation difficult and builds resistance to the idea so that there is no effective change. The rush to bring change without doing the ground work to prepare capabilities, the hurry to be able to say we have done it, the overlooking of inaction and pilferage in the system is perhaps both a part of and the result of that the counter-strategy. The centralisation of power and resources, the hierarchy that assumes that one on a higher position is more knowledgeable, with one-way communication and decision-making are fertile conditions for the evolution of counter strategies. The system has to satisfy the ambitions and compulsions of those at the so-called helm and ignore the actual reality that suggests patience and perseverance as the essential for change. There is a need to recognise that sustained efforts for a couple of decades are needed to carry a process of change towards education as described in the NCF 2005. If the system gets in to a hurry to declare that NCF 2005 has been implemented and the country is now ready to move forward to the next set of ideas then we would have blunted the concepts and ideas in the NCF 2005 as well. Change requires resources, time and commitment and the present need of the system to hear the refrain 'sab thik thak hai' can only result in the conclusion 'public system me kuch thik nahin ho sakta'

NCF 2005 is far ahead of the present possibilities in the system and a sustained effort is required to make people running the system engage with it. This needs to include people at the level of the school as well as at the level of the secretariat. It includes both headmasters and officers including directors and secretaries. The areas of intervention in the system are numerous. The intervention, however, has to be consistent and conceived over a long term with constant inputs from people with commitment, experience and understanding.

CHAPTER 7: SYSTEMIC ISSUES FRAMING EQUITY

Vimala Ramachandran

Framing the Equity Debate

Democracy, especially in today's unipolar world has become synonymous with equality and justice—even though history is replete with examples of functioning democracies that are inherently unequal and unjust. At a formal level, democracy is about exercising choices. It is about franchise and it is about one-person one-vote and one-vote one-value. At a little deeper level we also associate democracy with rule of law and institutions. The world expects democratic societies to respect civil liberties and human rights. It places great value in an impartial judicial process, a mechanism through which people can get justice. Democratic societies take great pride in a legislature that makes laws and an executive and administrative system that follows these laws. They also give importance to the right to free speech and a media that functions within a competitive environment. Most importantly, democratic societies value the right to equality. And finally democratic societies are defined by the quality and vibrancy of democratic institutions.

Notwithstanding the popular perception of democracy it is also quite well known that the mere existence of democratic institutions does not ensure democratic practice. In India the Constituent Assembly understood that rights could be realized only if we create a level playing field. As noted in the prophetic words of Dr. B. R. Ambedkar: 'On the 26th of January 1950, we are going to enter a life of contradictions. In politics we will have equality and in social and economic life we will have inequality. In politics we will be recognising the principle of one-man one vote and one vote one value. In our social and economic life, we shall, by reason of our social and economic structure, continue to deny the principle of one-man one value. How long shall we continue to deny equality in our social and economic life? If we continue to deny it for long, we will do so only by putting our political democracy in peril' (Dr. B. R. Ambedkar, 1949, quoted in Sunil Khilnani, 1997: 35). Acknowledging the centuries of discrimination based on caste, we initiated policies of affirmative action and introduced reservations in educational institutions and in government jobs to counter the historical baggage of social exclusion and economic exploitation. There was consensus that (at least at the formal level) affirmative action was in keeping with the spirit of democracy and equal opportunity.

Education was positioned as a central tool for the realization of the constitutional obligation of equal opportunity. It was hoped that education would enable communities that had faced the brunt of social discrimination under a rigid caste system to be able to break out of the vicious cycle of low social status, poverty and illiteracy. Chapter 4.2 of the National Policy on Education (NPE 1986) states: 'Education will be used as an agent of basic change in the status of woman. In order to neutralize the accumulated distortions of the past, there will be a well-conceived edge in favour of women. The National Education System will play a positive, interventionist role in the empowerment of women... This will be an act of faith and social engineering' (Page 6, Chapter IV, Paragraph 4.2 and 4.3). While the approach was not quite as radical when it came to the education of socially disadvantaged communities, the policy nevertheless made an unambiguous commitment to use education as a tool to ensure equal opportunities (see Box 7.1):

Box 7.1: The National Education Policy on Education of SC and ST Children

THE EDUCATION OF SCHEDULED CASTES

4.4 The central focus in the SCs educational development is their equalization with the non-SC population at all stages and levels of education, in all areas and in all the four dimensions—rural male, rural female, urban male and urban female.

4.5 The measures contemplated for this purpose include:

- i) Incentives to indigent families to send their children to school regularly till they reach the age of 14;
- ii) Pre-metric scholarship scheme for children of families engaged in occupations such as scavenging, flaying and training to be made applicable from Class 1 onwards. All children of such families, regardless of incomes, will be covered by this scheme and time-bound programmes targeted on them will be undertaken;
- iii) Constant micro-planning and verification to ensure that the enrolment, retention and successful completion of courses by SC students do not fall at any stage, and provision of remedial courses to improve their prospects for further education and employment.
- iv) Recruitment of teachers from Scheduled Castes;
- v) Provision of facilities for SC students in student hostels at district headquarters, according to a phased programme;
- vi) Location of school buildings, balwadis and Adult Education Centres in such a way as to facilitate full participation of the Scheduled Castes;
- vii) The utilisation of Jawahar Rozgar Yojana resources so as to make substantial educational facilities available to the Scheduled Castes; and
- viii) Constant innovation in finding new methods to increase the participation of the Scheduled Castes in the educational process.

THE EDUCATION OF SCHEDULED TRIBES

4.6 The following measures will be taken urgently to bring the Scheduled Tribes on par with others:

- i) Priority will be accorded to opening primary schools in tribal areas. The construction of school buildings will be undertaken in these areas on a priority basis under the normal funds for education, as well as under the Jawahar Rozgar Yojana, Tribal Welfare schemes, etc.
- ii) The socio-cultural milieu of the STs has its distinctive characteristics including, in many cases, their own spoken languages. This underlines the need to develop the curricula and devise instructional materials in tribal languages at the initial stages, with arrangements for switching over to the regional language.
- iii) Educated and promising Scheduled Tribe youths will be encouraged and trained to take up teaching in tribal areas.
- iv) Residential schools, including Ashram Schools, will be established on a large scale.
- v) Incentive schemes will be formulated for the Scheduled Tribes, keeping in view their special needs and life styles. Scholarships for higher education will emphasize technical, professional and para-professional courses. Special remedial courses and other programmes to remove psycho-social impediments will be provided to improve their performance in various courses.
- vi) Anganwadis, non-formal and Adult Education Centres will be opened on a priority basis in areas predominantly inhabited by the Scheduled Tribes.
- vii) The curriculum at all stages of education will be designed to create an awareness of the rich cultural identity of the tribal people as also of their enormous creative talent. (see Annexure for details).

Source: Chapter IV, Pages 6-8, NPE-1986, Government of India

Yet, if we scan the history of independent India, one very uncomfortable fact stares us in the face: we have failed to ensure that every child has a right to education. Right from the beginning in 1947 planners agreed that universal elementary education and basic literacy are indeed desirable in the practice of democracy. But the founding fathers of democratic India did

not think it was important to make it a fundamental right. Five and a half decades after we declared ourselves a sovereign democratic republic, we have still to redeem the pledge made in 1950. Almost two decades of sustained advocacy finally led to an amendment to the Constitution in 2003, making universal education a fundamental right. The rules and regulations necessary to actualize this amendment are yet to be enacted and the draft bill is being debated across the country. While one section of the political spectrum is comfortable with different kinds of schools for children living in different circumstances, a vocal lobby of social activists is demanding a common school system whereby the government is made responsible for providing equal quality formal schools for children living in diverse situations and locations. This has, once again, put the equity and education debate on the national and international radar with the world keenly watching how India will go about ensuring that its children actually realize this right (Sadgopal, 2005).

Education may have no direct one-to-one relationship with the formal process of adult franchise. However, being illiterate influences the ability of citizens to engage with the institutions associated with the practice of democracy. Being illiterate affects the ability of people to transcend the situation in which they find themselves at birth, impairs their ability to negotiate the maze of institutions that surround them, robs them of self-esteem and confidence and silences the voice of the marginalized and the dispossessed. Ultimately, its absence pushes 'people without voice' into ghettos, resurrects primordial identities and creates new forms of social exclusion and social segregation. Education, despite the power to work as an agent of change, to 'neutralize the accumulated distortions of the past' could actually end up reinforcing existing social and community identities (NPE 1986, Page 6, Chapter IV, Paragraph 4.2 and 4.3).

The relationship between education and democracy is like an invisible spiral that helps those in control remain there and those at the bottom alienated and voiceless. Equally, education has the potential of enhancing the capability of people. As Dreze and Sen convincingly argue: "'Capability' refers to the alternative combinations of functioning from which a person can choose. Thus the notion of capability is essentially one of freedom—a range of options a person has in deciding what kind of life to lead. Poverty, in this view, lies not merely in the impoverished state in which the person actually lives, but also in the lack of real opportunities—given by social constraints as well as personal circumstances—to choose other types of living' (Dreze and Sen, 2002: 35-36). Lack of access to education of acceptable quality actively inhibits the development of the capability of citizens to engage with democratic institutions, thereby denuding the practice of democracy.

Equity and Access

For over five decades now the equity debate in education has revolved around the issue of access to schooling for all children. As discussed in the introductory chapter, the government essentially focused on enrolment to the exclusion of other process and outcome indicators. Even the most recent Sarva Shiksha Abhiyan (SSA) set a universal enrolment target for 2003.

The last 17 years have been particularly important because of a sustained advocacy to make elementary education a fundamental right of every child. Notwithstanding significant progress made on the legal front, as well on the ground, the hard reality is that social status, economic situation and location continue to frame educational access as well as influence the experience of education (Nambissan, 1996; Ramachandran 2003). Children from Scheduled Castes (erstwhile untouchable communities) and Scheduled Tribes—particularly if they live in rural habitations or in peripheral settlements in urban areas—have access to schools with one or at best two teachers, a single classroom school, a multi-grade (where more than one class is taught by a single teacher) classroom or an alternative learning centre. Typically, it is the very poor from illiterate and poor families and communities who send their children to the worst endowed schools, which in many regions of the country function far below the expected norm. And even when such children do go to a regular primary school, they get little attention and

experience schooling from the margins of the classroom. They experience exclusion in different ways, from unfamiliarity with the mainstream language to the treatment meted out to them by teachers and other children. As Nambissan argues, ‘while the large majority of Dalit children are now being included in schools at the point of entry, the terms of their inclusion in relation to institutional structures and processes are discriminatory...’ (Nambissan, 2007).

If we approach the issue of equity and education from a rights perspective, then merely declaring that every child has a fundamental right to schooling is not enough. The government’s ability to provide schools is a first step. The big question is how distinctions made between formal schools and alternative schools/transitional schools affect children from poorer communities and whether they provide basic minimum facilities, teachers and teaching-learning experiences. Even within formal schools, the distribution of single teacher and two teacher schools between differently located villages and habitations, the availability of a basic learning environment by way of books, teaching and learning material, library, the training of teachers and academic support available to them make a big difference. Neutralizing accumulated discrimination of the past demands a proactive approach to not only make schools available but also to ensure meaningful access, i.e., to ensure that schools function effectively and make efforts to address the constraints that prevent children from attending them regularly, and learning.

The question we ask in this chapter is whether such an effort has been made to realize the goal of equal access and non-discrimination enshrined in our Constitution and what are the systemic barriers to equitable access for all children, regardless of their location or social group.

Locating education and equity in social and economic context

While the Indian Constitution is based on the idea of equality of all citizens, Indian society is stratified, hierarchical and iniquitous, forms the context in which the elementary education system in India is situated. Notably, though at the macro level, a common perception is that Rajasthan is far below Andhra Pradesh in most social and economic indicators, Chapter 4 shows striking commonalities among the sample villages of the two states, though there were also some differences. (It may be recalled from Chapter 2 that sample districts and blocks represented the middle ground in both the states.). The most critical dimension of inequity in India are caste and economic status, which usually, though not always, complement each other. In the sample villages of both the states, the village community was stratified along caste lines. Both sets of villages had a mixed population of OBCs, SCs and STs and a few upper castes, and sometimes muslims. While rich forward caste families retained their homes and controlled the land and trade in the villages, many also had houses in nearby towns, where they lived. SCs and STs were the poorest people, and usually worked as farm labourers or marginal farmers. Among the OBCs there were families living in abject poverty and also families that were fairly well off. Interestingly, in both the states, the OBC community owned cattle. Migration to cities/other areas was fairly common among the landless poor, especially in Rajasthan and also in an arid part of the selected district in Andhra Pradesh .

In both the states, most of the forward and middle caste families lived in contiguous settlements, while the SC community lived in separate colonies/settlements, which were located slightly away from the main village. While higher castes had some social contact with the SC and ST families working as wage labourer or as farm hands/domestic servants in their households, this contact was limited. Social discrimination added to the burden of poverty among these groups. Chapter 6 in this book examines the attitudes and behaviour of the personnel across the system, i.e. officers, resource persons and teachers, who were usually upper caste, middle class and urban, toward the mainly poor and lower caste children who studied in government schools. This attitudes ranged from outright hostility, indifference and

empathy tinged with discomfort at the very different life experience of the children who attend government schools.

To this may be added one more example of a school where the social distance played itself out in the midday meal programme. In one village in Andhra Pradesh, forward caste and even OBC children did not eat the midday meal cooked by an SC woman. This issue came to the notice of the Mandal Education Officer, District Education Office and even the Director of Education. The community was formally warned that people who discouraged or disrupted the midday meal programme would be punished. Yet, field visits showed that some children ran home during the lunch break. When parents were asked about this, they said that dogs and pigs roamed around when the children ate, and since the school did not have a boundary wall, the teachers could not keep the animals away.

The facilities/amenities available in the villages have been described in chapter 4. These showed the second important dimension of inequity: size and location, whereby that smaller and 'remote' habitations had fewer amenities. Consequently, children living in small villages had less opportunity and less contact with the outside world. As Chapter 4 shows, this held for educational facilities too: the level of education available in smaller habitations was lower, and children in such habitations had to travel some distance to access even upper primary education. In Rajasthan, this fact coincided neatly with the fact that the population of very small habitations wholly or largely comprised SCs and STs.

Private schools exacerbated both these dimensions of inequity. Private schools existed only in larger villages, and children in smaller villages did not have easy access to these. At the same time, only better off children could afford to go to fee charging schools. The best off families in these villages sent their children to schools in nearby towns. This was especially the case in Andhra Pradesh, where matador vans of private schools came to the villages to pick up children.

The situation in the villages did not promote literacy among the children. As Chapter 4 shows, literacy levels among men and women were quite low, and those among women in Rajasthan were the lowest. There was hardly any reading material at home and some children reported that they saw the newspaper only occasionally. The only books they had were school textbooks. Their families did not subscribe to newspapers or magazines. While the relatively better-off did have access to some kind of written material at home, the poor had almost no access. Unlike in Kerala and Tamil Nadu, we did not come across village libraries, except in one village in Andhra Pradesh where the school had over 1,000 books. The local panchayat office and the tea shops did get newspapers and men accessed these. The government schools had library books in Andhra Pradesh, but in most cases it remained locked.

One of the striking commonalities was the work burden of children, both boys and girls. It is important to note that in both areas it was the very poor who sent their children to government schools; the relatively better-off sent their children to private schools or a better-endowed government school in a nearby village or town. Discussions with children in Class 6 revealed that they woke up early and put in a few hours of work at home before going to school. In both the states boys, especially of the SC community, cleaned the cattle shed, fed the cows or took them out grazing and helped their parents in the field during the peak agricultural season. Girls cooked and cleaned, worked on the family land and took care of a range of domestic chores. In both the states the children told us that they often missed school during the peak agricultural season. In Andhra Pradesh, the daily wage during the peak season in one village went up to Rs. 40 per day for girls. Therefore, parents pulled them out of school during this period. In Rajasthan the older girls went with their parents for wage work even when they were formally enrolled in school.

Social assessment studies conducted by the government of Andhra Pradesh in 1999, 2001 and 2002 reveal that non-enrolment and non-participation (even when enrolled) remained big issues in the state.⁸³ It was the very poor—landless and among them the most deprived social groups—who were still alienated from schooling. The situation of the tribal children was far worse than that of SC children, even though smaller sub-groups of SCs reported high rates of non-enrolment and attendance as evident from the key finding of one study (Box 7.2).

Box 7.2: Key findings of Social-Assessment Survey of Government of AP, 2002⁸⁴

The study surveyed 3,040 families in 76 revenue mandals of 19 districts in Andhra Pradesh covering a population of 12,510 persons and 6,430 school age child population among 65 backward caste communities:

- 72.75 per cent of the children in the sample are enrolled in school and 24.03 per cent of the children were never enrolled in school (26.47 per cent boys and 21.58 per cent girls). More boys were never enrolled! Given the enrolment campaigns many more girls were enrolled but were not attending school. A sizeable percentage of girls (20.37 per cent) were enrolled but not attending; the percentage for boys was 12.43 per cent. Sudden disasters (within the family) and drought leads to withdrawal from schools.
- Communities that are numerically small (nomadic and semi-nomadic) shy away from schools. Communities with a majority of non-school going children include Naga Vaddilu (88.24 per cent), Vanjara (77.78 per cent), Pedamma Vandlu (75 per cent), Uppara (75.42 per cent), Bestha (59.43 per cent), Kuraba (55.88 per cent), Rajaka (55.04 per cent) and Ediga (50.94 per cent). They are marginalized and not covered by most ongoing economic and social development programmes, including self-help groups, mother's committees, village education committees etc. They find that demand for birth certificates and other official identification documents difficult to comply with.
- The sex ratio in the sample is 989 girls for 1,000 boys. School going families showed a lower sex ratio (939) than non-school going families (1,040). Equally, the percentage of girl children is higher among first born or the first two children (higher birth parity) but it goes down in lower birth order especially among families of school going children.
- Families of school going children live in pucca houses or tiled houses and 55.43 per cent families living in huts do not send their children to school (non-enrolled as well as not attending); 56.94 per cent of the sample households were landless, 16.35 per cent were marginal land holdings and 20.19 per cent were small farmers. 52.40 per cent of landless and 64.59 per cent of marginal land holding families do not send their children to school. Only 5.23 per cent of medium size land holding families did not send their children to school. 59 per cent of non-school going children came from families of landless and manual wage labourers.
- Poverty and poor economic conditions emerged as compelling factors for non-enrolment or non-participation in schools. Close to 15 per cent of the families sent their children for wage labour and cattle grazing. Commercial crops like cottonseed cultivation, cotton picking, sunflower and groundnut provided wage employment to children. Region-wise, the Rayalaseema region (19.38 per cent) is followed by Telangana (15.97 per cent) as being the most affected by poverty compulsions that prevent children from attending school.
- 73.16 per cent of the families of non-school going children were illiterate—meaning that there was not a single literate person in the family. 22.89 per cent non-school going children were from families that are literate or neo-literate. Alternatively, only 2.11 per cent non-school going children were from families with at least one member with upper-primary education.
- There is a sense of insecurity with respect to girl children where parents are apprehensive of educating girls beyond the primary level for fear of additional expenditure for marriage, as they would have to find educated boys. This is more pronounced among some communities and is most palpable in Rayalaseema (14.06 per cent) followed by Coastal Andhra (13.96 per cent).
- Parents wish to equip their children with whatever traditional skills they can, as they are concerned about the future of their children. They feel education does not contribute to better livelihood opportunities and education beyond the primary level leads to refusal to work in agriculture. Among the reasons given for non-enrolment are learning the family trade (1.90 per cent), domestic work (11.43 per cent), wage labour (33.62 per cent), too poor to send children to school (44.13 per cent).

⁸³ See K S Chalam (1999) and Reddy (1999 and 2001).

⁸⁴ Professor Ch. Umamohan (2002): 'Constraints to schooling: Social Assessment Survey of Backward Caste Communities in Andhra Pradesh', Report submitted to DPEP. Government of Andhra Pradesh (October).

cent), sibling care (0.95 per cent), no birth certificate (4.44 per cent) and no incentives (4.44 per cent).

- Fathers and mothers jointly take decisions on enrolment, but for the girls the mother's have a bigger say.
- Parents of children who were not attending or had dropped out said that they were not learning anything and schooling was a waste of time. Some said that the kind of education given was not relevant in their lives and children who study beyond a point refuse to do any manual labour. Rigid and unattractive curriculum was cited as a reason for not attending schools.
- Incentives like giving rice (before the midday meal was introduced) and textbooks have played a very important role in enrolment, but not so in regular participation. Parents still have to spend on notebooks and proper clothes. Lack of proper clothing is an inhibiting factor in school attendance.

Source: Prof. Ch Umamohan, Anantapur, October 2002.

Government Initiatives for Equity

If we examine the national programmes described in Chapter 1, there has been some focus through the NFE scheme and its successor, the EGS and AIE scheme, to enrol out of school children, who are largely from poor communities. How such initiatives have actually played out, is discussed in greater detail below. The attempt in OB, to provide basic minimum facilities in all schools, was a move in the same direction. DPEP focused specifically on low literacy (and mostly poor) districts and encompassed a range of initiatives, which included attempts to enrol all children, reduce drop out and improve quality of schools. These initiatives are also part of SSA. Moreover, in DPEP and SSA, the national government increased its attempts to reach out directly to the most deprived children, such as by funding free textbooks.

Both the states were identified as high 'child labour' employment states with a persisting problem of out-of-school children. As Govinda and Biswal point out: 'Estimates from the 7th All India Education Survey (AIES) showed that in 2002-03, most of the out-of-school children in the age group 6-11 were accounted for by a few selected states, namely, Andhra Pradesh, Bihar, Madhya Pradesh, Rajasthan, Uttar Pradesh and West Bengal. It should be mentioned that, way back in 1978 while preparing the Sixth Five Year Plan, the Planning Commission identified these states, along with a couple of others, as educationally backward states. To find the same cluster of states appearing as low UPE states after implementing development plans for nearly 25 years and going through at least four full cycle Five Year Plans raises serious questions on the strategy being adopted for overcoming inter-state disparities in the country'⁸⁵.

Certain aspects of state policies described in Chapter 2 such as provision of free textbooks, scholarships to students from deprived communities, and especially the midday meal in recent years, (for which the central as well as state governments contribute), were important initiatives. However, there were some differences here between the two states. As Chapter 2 shows, in Rajasthan, children still paid fees in government schools. Girls were exempt, and SC and ST boys paid half the amount, and the amounts charged in government schools were much lower than those in private schools, but several very poor children continued to pay fees. In Andhra Pradesh, no fees was charged. Moreover, Andhra Pradesh provided a higher amount per child for midday meals than did Rajasthan.

Two programmes of Rajasthan need to be highlighted in the context of equity: The Shiksha Karmi Project (SKP) and Lok Jumbish. The Shiksha Karmi Project has been described in Chapter 1 (Box 1.4). Lok Jumbish was another innovative government programme launched in 1992 which folded up in June 2004. The project sought to 'develop, demonstrate, catalyse and

⁸⁵ At the upper primary stage, even relatively better performing states such as Karnataka and Maharashtra show serious problems with a large proportion of children remaining outside the fold of schooling' (Govinda and Biswal, 2005)

transform the mainstream education system with the objective of ensuring that every child has access to basic education (Grades I to VIII).’ The project evolved interesting models for community involvement and was instrumental in designing and running effective alternative schools for children from very deprived communities. However, at the time of the fieldwork there was little institutional memory in the government about the strategies adopted in Lok Jumbish.

Equity and Education in the Sample Villages

From Chapter 4, which provides a glimpse of the sample villages in the two states, gives a general idea of how the equity-education plays out. This equation, as it played out in the sample villages, is examined in greater detail below.

The Rajasthan Villages

In the largest village of our sample, i.e. village 1, educational segregation was visible through the existence of five private schools along with four government schools. The private schools were considered better, and those who could afford it sent their children to private schools. Poorer people sent their children to government schools, but were dissatisfied with the quality of education, and children, especially girls dropped out early. The richest people in the village sent their children to schools in nearby towns. However, many children from smaller surrounding villages came to village 1 to study. The school that was considered the best was run by the Bhartiya Shiksha Parishad, affiliated to the R.S.S. The teachers of this school maintained consistent contact with the parents, informing them about children’s progress and conducting monthly PTA meetings, and ensured that children did not drop out. Another private school was managed by the CRCF who lived in a neighbouring village, who used his influence to enrol students to the school.

Village 2, also a large village, presented a different pattern of inequity. It had no private schools, but the government school system itself provided differential access to schooling to children from different social categories. As Table 4.1 in Chapter 4 shows, this was a village where upper castes lived in the main village and SCs and STs lived in separate habitations/localities a little away from the village. The main village had an upper primary school with five regular teachers, one para teacher and a five roomed school building. This school functioned regularly. Each of the satellite settlements had a Rajiv Gandhi Pathshala up to grade 5, with one teacher each. These Rajeev Gandhi Pathshalas had reasonable buildings. Teachers in both had college degrees and teacher training diplomas. The Rajeev Gandhi Pathshala in the Meena (ST) habitation functioned regularly. However, in the other habitation, where a large number of Bairwa (SC) families lived, the school functioned for around 15 days a month. While a few boys from these habitations went to the upper primary school in the main village, no girls did.

Let us now examine the situation from the equity angle in the village case study of Rajasthan described in Chapter 4 (village 4). In this village, there were seven schools, four of which were private. The three government schools included one coeducational government primary school, one girls’ upper primary school and one co-educational higher secondary school (classes 6-12). We saw that the primary school became dysfunctional when a RGSJP was opened in 1999 and a Shiksha Karmi school (SKP) was opened in 2001 in nearby habitations. This meant that in the main village, boys in the 6 to 11 years age group *had to* attend private schools. The village had a Madrassa and a school for working children, which were dysfunctional. This combination, i.e. dysfunctional primary school and alternative centres ensured that the most marginalized children got no meaningful education. The situation of the marginalized communities in the neighbouring habitations was similar, or worse (see Box 7.3):

Box 7.3: Scenes from marginal habitations surrounding a large village

- Habitation one: There are 40-45 households of the Meena (ST) community, who are all small and marginal farmers, and go to Jaipur and Delhi as casual labourers. The village has a handpump and electricity, but no school. Boys go the main village to study. The government primary school is dysfunctional, so 8-10 boys study in the private school. Rest stay at home. No girl goes to school. No one has come to the habitation to persuade parents to send children to school.
- Habitation two: The population is around 400, 90% of whom are from the Meena (ST) community. The occupations are agriculture, casual labour goat and cattle rearing. Girls tend to goats, and rarely study for more than 2-3 years. There is a DPEP alternative school (upto class 5) in the habitation, in which around 57 children are enrolled. It has no building, and functions under a tree. The teacher has made his own house into an 'office', for which he gets Rs. 300 per month from DPEP.

In the main village, the upper primary school that catered to girls opened regularly and teachers were also present, but little time was spent on actual teaching and learning activity, which was less than 20 minutes per period of 45 minutes. Thus parents who preferred to send their daughters to a girls' school, also sent them to a school with casual academic processes. The school that functioned best in this village was the government higher secondary school and even forward caste children moved on to this school after class 5. *However, the nature of elementary schooling facilities had ensured that only better off children had access to the higher secondary school.* Children from the RGSJPs, madrassa and other alternative centres were unlikely to stay in the school system long enough to access this school, and given the casual teaching in the girls' school it could be expected that many would experience academic failures. Finally, the better off children, studying in private schools at the elementary level, especially boys, would enrol in the better functioning high school.

Near village 4 was village 8, connected by *kaccha* road, where 30-35 families of the Meena (ST) community and one family of the OBC community lived. There was one Rajeev Gandhi Pathshala (upto class 5), with one teacher to teach five classes. It had a building with three rooms, and also a handpump. However, the school functioned for less than ten days a month, as and when the teacher felt like. There was hardly any teaching activity. Children went to the school, but came back because there was no activity. Villagers did not complain because they did not want the teacher to lose his job. The Rajeev Gandhi School was affiliated to the SDMC of the primary school in village 4 and neither the teacher nor the parents were aware of what went on there. Around 5-7 boys went to the main village to study. Girls did not study beyond class 5. It is important to understand here that the new Rajeev Gandhi Pathshala in this village had actually *worsened* the educational situation for many poor children: the formal primary school in village 4 had become dysfunctional, and the new Rajeev Gandhi school did not provide any meaningful education.

Village 6, where people belonged to an SC community, was a similar case study of indifferent education handed out to the poorest children. It had one alternative school (AS), with five classes, supported by DPEP. It had only one room and was managed by a para-teacher of the same community. The school had one room. The research team visited the school 10 times, and found it closed 9 times. Upon checking, it was found that on these days, the school had been shown as functioning. On the one day that it was found to be functioning, a very small number of children attended. The teacher simply hung a chart and asked the children to copy.

However, village 7 in our sample, also with a high SC population, was different. It had one Shiksha Karmi school and a highly motivated male teacher in charge, and also a woman teacher. Discussions with the community revealed that the selection of the teacher had been

done with care. He participated in in-service training and continued to get academic inputs from the Shiksha Karmi Board. The teachers were exceptional and they seemed to be dedicated to their work. The community admitted the shiksha karmis taught regularly. The test results in this school were by far the best in our sample. The interesting part was that this school did not have a building, yet it was the sheer dedication of the teachers that the school functioned and the midday meal was cooked with care, the children sat in an orderly fashion and ate their meal. This school illustrated that investment in teachers in recruitment as well as training could lead to much better education for marginalized children. However, why such schools could not be provided buildings and facilities, while some schools had buildings they did not use, was a question that remained.

Village 3, though not much larger than village 6 and 8 in terms of population, was also quite different. This was inhabited largely by a prosperous Yadav (BC) community, and had had a government primary school since 1963. There were no private schools or Rajeev Gandhi Pathshalas. The primary school had two well-qualified teachers, and a new building had been provided recently by DPEP. The school functioned regularly, teachers worked hard, and children also attended regularly. The teaching learning process did suffer because the two teachers spent a great deal of time in office work, or the pulse polio drive, but there was no gross negligence. The community had contributed to plant trees around the school. Midday meals were served and eaten regularly. Notably, though village 3 suffered from the limitations of a small village, i.e. it had only a two teacher primary school, it had a far better school than habitations of similar size with SC and ST populations.

The situation in village 5 illustrated how poor planning and rent seeking behaviour led to the squandering of resources that could be used for better quality education for poor children. This was a small village where two formal schools functioned in the same campus! One belonged to another village technically, but in practical terms, where one large school with adequate teachers and infrastructure could have been created, resources had been used to create two small schools. The government primary school that belonged to the village had only one teacher. Another RGSJP in the village essentially catered to children from the Meena (ST) community, while the formal primary school (PS) catered to the Gujar (OBC) community. Poor use of resources was illustrated by the fact that there were three rooms and one hall for the formal PS, but only one teacher, and all the children sat in one room. Notably, the existing rooms had been constructed by a contractor who was also a teacher in a neighbouring government school, and now an additional room had been approved. The teacher admitted that he did not find time to teach. The research team went to study the RGSJP three times and invariably found it closed. The room in which the RGSJP was supposed to function was run down and did not look as if it had been used in the recent past.

The Villages in Andhra Pradesh

Chapter 4 provides a detailed account of the educational scenario in village 1. Let us now take a more detailed look at three more villages, beginning with village 4. This village had a large Madiga (SC) community, who were mainly agricultural labourers. Reddys (OCs) were the dominant group, and owned land and held government jobs. Social segregation was pronounced in the village. The village had good irrigation facilities, which had two important outcomes for inequity. First, as the well off people were busy in agricultural work and earned well, they preferred to send their children to private schools and tuitions. Second, children from poor families, especially from among SCs, worked in the fields. Girls from poor families worked in chilli and tobacco fields/processing during the summer vacations and on holidays. During peak agricultural seasons, poor upper caste children also worked in the fields. Children from poor families tended to drop out after grade 5 or 7 and began to earn instead. Some villagers said that a few boys worked as bonded labourers as well.

There was one government primary school and one government high school in the village. The schools were crowded: children from the neighbouring villages were also enrolled in both the

schools. The upper caste children did not eat the midday meal served in the government primary school, which was cooked by a woman from the SC category. The Congress and Communist (CPM) parties were fairly active in the gram panchayat. There was an SEC but it did not meet regularly, even though it supervised the midday meal. Apparently the SEC had been active during the tenure of the Telegu Desam Party but did not have the political support of either the Congress or the CPM.

Village 3 was a small village. The seat for the sarpanch was reserved for a woman. However, the elected lady sarpanch had little say, and her husband managed the show. There was a great deal of tension between the sarpanch (husband) and ward members over corruption charges in handling panchayat funds. The sarpanch's husband was critical of the SEC and believed that the village panchayat should play a more active role. While his children studied in a private school, he was well informed about the government high school in the village, and informed us that 10 out of the 50 children in high school were also enrolled in a private school. The chairman of the SEC, a forward caste leader (Reddy) from a poor family, who had studied till class 10, was active. His children studied in the government school. The midday meal was cooked by a forward caste person and was regular. Some children came to school hungry and had their first meal in school. In the last academic year there had been no teachers for two months—the SEC had been pro-active and had lobbied the gram panchayat and the mandal panchayat to get a teacher.

There was a small ST community in the village. While all the children from this community were enrolled in school, they were absent for long periods when their parents migrated in search of work. When they were in the village, the children attended school. According to the teacher they came for the midday meal! The teachers admitted that the learning levels of all the children were poor and that 'the standard of the textbooks was so high that the children could not cope with it'. The school had two rooms and two verandas but only one room was used as only one teacher was present at any given time. Both the teachers were highly qualified (double MA, M.Ed, M.Phil and one was doing his Ph.D). They were unhappy about teaching in a primary school and were therefore indifferent. The parents said that they assumed that their children were learning something in the school but given their workload they did not check what and how much the children were learning. The teachers admitted that examinations were a farce and all the children were promoted.

Village 5 was located in an irrigated area with big landlords who grew paddy. Traders come to the village to purchase paddy and the sarpanch collected Rs. 20 from every trader and donated it for school development. The village panchayat had built a water tank, one veranda, a bore well a compound wall and levelled the playground in the school. However, as the motor was under repair, children had to go out of the school campus to drink water. There were benches in the classroom though they were high and uncomfortable. The children were fairly regular and the school functioned regularly. The sarpanch visited the school regularly, and enquired about classes and academic achievements. Interestingly, the SEC chairperson and sarpanch talked to the male teachers, but not to the women. The SEC chairperson said that SEC members needed orientation on what the SEC was empowered to do, and felt that greater clarity about the role of the GP and the SEC vis-à-vis the school was needed.

Children from poor families helped their families during the peak agricultural season. As adults got work throughout the year, the children had to do a lot of housework. Discussions with the children in Class 6 revealed that they did not get time to have a bath in the morning. Many children woke up early to go for tuitions between 5.30 and 8 am. Then they (boys as well as girls) did some housework and reached the school by 9.30 am. They faced the double burden of a prosperous agricultural society in which their labour had monetary value, and an industrializing society where academic achievements was important, and led what appeared to the researchers an overworked life. The children also complained that the tuition masters beat them. In this village, running tuition classes was big business.

There was an alternative school in the section where the ST (Lambada) community lived 3 kilometers from the main village. This school was established to enroll out-of-school children and was run by a vidya volunteer in a shed⁸⁶. Most children from the community attended this school, though five attended the formal school in the main village. The teacher came to the school every day from the main village. She took classes but admitted that language was a problem. Many children did not have notebooks or pencils. Some children came for midday meals and then disappeared. The teacher visited children's homes to ask them to attend school. The teacher was concerned about learning levels and scared of losing her job. But given that she was not trained and did not get any academic support from the MRPs, she found it difficult to manage the school.

The Complex Framework of Equity and Education

The equation between equity and education is a shifting one. At the time of India's independence, an important ingredient of this equation was the difference between those who had been close enough to a primary school to enrol in it, and those who were not. While this still holds true for some children living in very tiny habitations, for the vast majority, the equity debate has shifted, to access to upper primary and higher levels of education, to the differential access to schooling, and the quality of schools.

Access to upper primary education

In both Rajasthan and Andhra Pradesh official records revealed a negligible number of out-of-school children in the 6 to 11 years age group. A cursory look at schooling facilities may give the impression that the government has provided for schools and that the problem of access has been taken care of. Most government sponsored as well as independent studies done in the last 15 years reveal a sharp increase in the number of recognized as well as unrecognized primary schools. However this was not the case for upper primary education, where access and formal enrolment were still important issues. This is illustrated by the case on Andhra Pradesh.

The recent SSA survey in the sample villages of Andhra Pradesh by the government showed that the number of out of school children in the 6-11 age group was negligible (Table 7.1).

Table 7.1: Enrolled and out-of-school children in the sample villages, Andhra Pradesh

Andhra Pradesh	Total	Boys	Girls	SC		ST		BC		General	
6-11 age group				Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls
Village 1											
Total	117	54	63	20	27	0	1	25	32	9	3
School going	117	54	63	20	27	0	1	25	32	9	3
Village 2											
Total	173	93	80	20	12	-	-	62	55	10	12
School going	173	93	80	20	12	-	-	62	55	10	12
Village 3											
Total	54	27	27	5	9	3	1	10	6	9	11
School going	54	27	27	5	9	3	1	10	6	9	11
Village 4											
Total	149	75	74	29	27	-	1	30	28	16	18
School going	147	73	74	27	27	-	1	30	28	16	18
Out of school	2	2	-	2	-	-	-	-	-	-	-

⁸⁶ Five children from this community attended the formal school in the main village.

Village 5											
Total	217	115	102	19	15	17	13	60	49	19	25
School going	207	111	96	16	13	17	13	59	45	19	25
Out of school	10	4	6	3	2	-	-	1	4	-	-
Village 6											
Total	351	247	104	177	25	4	5	52	14	60	14
School going	351	247	104	177	25	4	5	52	14	60	14

Source: SSA Survey data kept in the village (September 2005).

Annual school enrolment drives (Chaduvula Panduga) had been fairly effective in ensuring enrolment in the local school.

However, the situation of children in the 11+ years age group was somewhat different. Notably, access to upper primary education was much poorer than access to primary education. In the two sample mandals, all the habitations/villages had a primary school within one kilometre (only 2 villages of mandal 1 did not have a school within the habitation. However, to access upper primary schools, children had to walk 2 to 4 kilometers to reach their school. As a result, twice as many children in the 11-14 years age group were out of school than in the 6-11 years age group (113 in mandal 1 and 94 in mandal 2). Table 7.2 provides the number of children in the age group 11 to 14 years who were out of school in the sample villages of Andhra Pradesh.

Table 7.2: Out-of-school children in 11 to 14 age group, Andhra Pradesh

	Total	Boys	Girls	SC		ST		BC		General	
				Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls
Village 1	0	0	0								
Village 2	4	1	3								
Village 3	0										
Village 4	20	8	12	4	5	0	0	4	7	0	0
Village 5	4	2	2		1	1		1	1		
Village 6	5	5				5					

Source: SSA Survey data kept in the village (September 2005).

Since primary education, and enrolment of all children at least at the primary stage, has been the focus of attention in recent years, the obligation in the Constitution to provide free and compulsory education to children *up to the age of fourteen* has not received the emphasis that it deserves. Moreover, as more and more children are enrolling in primary schools, their ability to go beyond this stage is inhibited by the availability of upper primary or middle schools. This is now an important dimension of equity and education in India, as each child needs to access at least eight years of elementary education, and not just five. This is not the case yet, and many children remain deprived of this right.

Gap between enrolment and attendance

Physical access to some kind of school, and the registration of a child's name in the school register, which is often called 'enrolment', is not a meaningful interpretation of the right to education. This type of technical interpretation leads to poor children being 'enrolled' while they actually continue to labour in fields and household chores. At the very minimum, children's attendance in school, as against mere enrolment, needs to become the yardstick.

Taking the case of the sample villages of Rajasthan, there was a clear gap between formal enrolment, attendance as reported in the school records, and what was observed during our field visit (Table 7.3). In all the schools, there was a progressive reduction as we moved from

formal enrolment to attendance as marked and what we actually observed. This gap was more pronounced in a few schools. If we add the number of children who are just formally enrolled and those who are irregular, to the non-enrolled, then the number of children who do not access education is quite high.

Table 7.3: Attendance on day of visit against total enrolment and official records, Rajasthan

Village	Enrolled students						Attendance according to Register						Present in class on the Day of Visit					
	Class - 2			Class - 4/6			Class - 2			Class -4/ 6			Class - 2			Class -4/ 6		
	B	G	T	B	G	T	B	G	T	B	G	T	B	G	T	B	G	T
UP1(i)	34	0	34	60	0	60	17	0	17	54	0	54	16	0	16	45	0	45
UP1(ii)	15	10	25	18	13	31	9	7	16	14	13	27	5	4	9	14	12	26
PS1	8	9	17	7	5	12	7	6	13	5	5	10	6	6	12	5	4	9
HS2	NA	NA	NA	57	0	57	NA	NA	NA	52	0	52	NA	NA	NA	52	0	52
UP2	0	45	45	2	45	47	0	42	42	2	39	41	0	35	35	2	39	41
PS2	12	5	17	11	5	16	10	3	13	4	2	6	10	3	13	3	2	5
PV-C2	15	1	16	23	0	23	14	1	15	21	0	21	14	1	15	21	0	21
AS-C2	2	6	8	8	1	9	0	4	4	0	2	2	0	3	3	0	1	1
RG-C2	7	2	9	5	3	8	6	1	7	4	3	7	6	1	7	4	3	7
Total	93	78	171	191	72	263	63	64	127	156	64	220	57	53	110	146	61	207

Source: Survey data in the nine sample schools (September-October 2005).

In Andhra Pradesh, a social assessment survey commissioned by the government (October 2002) revealed that close to 22 per cent of the enrolled children did not attend school for a variety of reasons, the most significant being agriculture and wage work (Table 7.4). Surprisingly, while official data claimed near universal enrolment, 24 per cent children in the survey sample were reported as being never enrolled. It may be noted that this survey was done among the socially and economically disadvantaged groups in 19 districts across the state and covered children in the 6 to 14 years age group before the introduction of the midday meal programme.

Table 7.4 School age children and status of schooling (as reported by parents), percentage

Status of schooling	Boys	Girls	Total
Attending Primary	29.82	28.69	29.25
Attending Upper primary	4.13	4.45	4.29
Attending Secondary	4.54	2.52	3.53
Studying (unspecified)	4.19	2.24	3.22
Irregular in attending	12.43	9.59	11.01
Drop outs	1.86	1.56	1.71
Enrolled but not attending	16.56	29.37	22.95
Never enrolled	26.47	21.58	24.03
Total number of children	3219	3211	6430

Source: Professor Ch Umamohan, Anantapur (October 2002).

In the sample schools on Andhra Pradesh, we found close to 80 per cent of the children officially enrolled present in the school on the day of our visit. Additionally, teachers knew about the irregular children and the reasons for their not being regular, and were able to identify them.

Differential Access to differential schooling

The biggest initiative to counter inequity since independence has been the expansion of schooling itself, as education has become available to a much larger number of children,

especially those from deprived groups. But the manner of this expansion has several iniquitous features. On the one hand, it has tended to reproduce existing inequalities, and on the other introduce new ones. Almost all the studies done in the last ten years reveal that the social status, location and economic situation of families influence the kind of educational facility that a child can access (Aggarwal, 2000; Jha and Jhingran 2005; Gopinathan Nair et al., 2000; Ramachandran, 2003). Differential access to schooling is a complex dynamic in India, and for simplicity, we may look at it in two parts: the three broad classes of schools in existence, i.e., private schools, regular government schools and alternative schools; and differences within the regular government schools.

Alternative, government and private schools

The description on the sample villages of Rajasthan and Andhra Pradesh presented above shows clearly that children from different better off social groups tended to shift to private schools. While this differentiation in schooling was a fact in both the states, critical differences existed between the two states in their approach to alternative schools and para teachers. In Rajasthan, Rajeev Gandhi Pathshalas *substituted* for 'regular' government schools. These schools had poorly paid and untrained para teachers and few physical facilities. There was no such policy to substitute formal schools with alternative centers in Andhra Pradesh. Though Andhra Pradesh too hired poorly paid and untrained para teachers, these were distributed across all schools. Thus the attempt was to provide regular schools for all children, and all schools has a mix of regular and para teachers, unlike Rajasthan where some children had access to schools with para teachers only.

As the description of the sample villages above indicate, RGSJPs often did not function and there was little any evidence of regular teaching and learning taking place in them. This was the case for DPEP supported and other alternative centres too. The notable exception here were the Shiksha Karmi schools. The community was wary of complaining as the teachers were well connected and in most cases belonged to the same caste. It was also clear that local politics influenced the location of new schools, leading to poor use of resources. The opening of RGSJPs had actually worsened the educational situation in several villages, by making the formal schools dysfunctional, and shifting the children to RGSPs that existed mainly on paper. Children of SC and ST communities who earlier had access to a formal school were now forced to make do with an alternative school or RGSJP.

The existence of RGSJPs led to a false sense of security about universal enrolment. For instance, on paper, only 71 children in Raj Block were out of school (Table 7.7). Yet, being enrolled in a RGSJP or a madrassa meant little. But if we add the number of children enrolled in the alternate school, the madrassa and Shiksha Mitra schools, the number comes to 4,04, nearly 10% of children.

Table 7.5: Children in 6-14 age group enrolled and out-of-school, Rajasthan

Name of Block	No. of children of 6 & 14 Age-group			No. of children Enrolled			Out of School Children		
	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total
Raj Block	24098	18140	42238	24073	18094	42167	25	46	71

Source: Annual Action Plan (2005-06), Block Resource Centre, Raj Block.

If we take Raj Block (Table 7.6) as a whole, the distribution of children across different categories of schools is quite revealing. Many more SC, ST and OBC children study in government schools of all kinds and the proportion of forward caste and general category (also including Muslim children) children in private schools is quite marked. On the other hand, SC/ST children were disproportionately represented in the informal schools.

Table 7.6: Raj Block, Raj district enrolment status of children in classes 1 to 8

Category of School	SC			ST			OBC			General			Total		
	B	G	T	B	G	T	B	G	T	B	G	T	B	G	T
Panchayat Samiti	1524	1306	2830	1301	1173	2474	2849	2826	5675	585	702	1287	6259	6007	12266
Elementary Edu.	1619	1078	2697	1023	569	1592	3510	2349	5859	910	974	1884	7062	4970	12032
RGSJP	691	695	1386	700	649	1349	784	707	1491	92	92	184	2267	2143	4410
SKS	131	106	237	86	50	136	208	126	334	10	3	13	435	285	720
Sanskrit	16	13	29	0	0	0	11	12	23	17	21	38	44	46	90
AS DPEP (6Hr.)	419	440	859	290	307	597	326	400	726	128	119	247	1163	1266	2429
AS DPEP (4Hr.)	83	146	229	29	92	121	138	478	616	13	32	45	263	748	1011
Madrassa*	0	3	3	0	0	0	54	63	117	64	70	134	118	136	254
Shiksha Mitra	13	92	105	1	93	94	12	67	79	0	11	11	26	263	289
Bridge Course	0	30	30	0	0	0	0	25	25	0	5	5	0	60	60
Private	1055	515	1570	894	290	1184	3144	1193	4337	2138	1469	3607	7231	3467	10698
Other	120	35	155	109	127	236	176	98	274	144	61	205	549	321	870
Total	5671	4459	10130	4433	3350	7783	11212	8344	19556	4101	3559	7660	25417	19712	45129

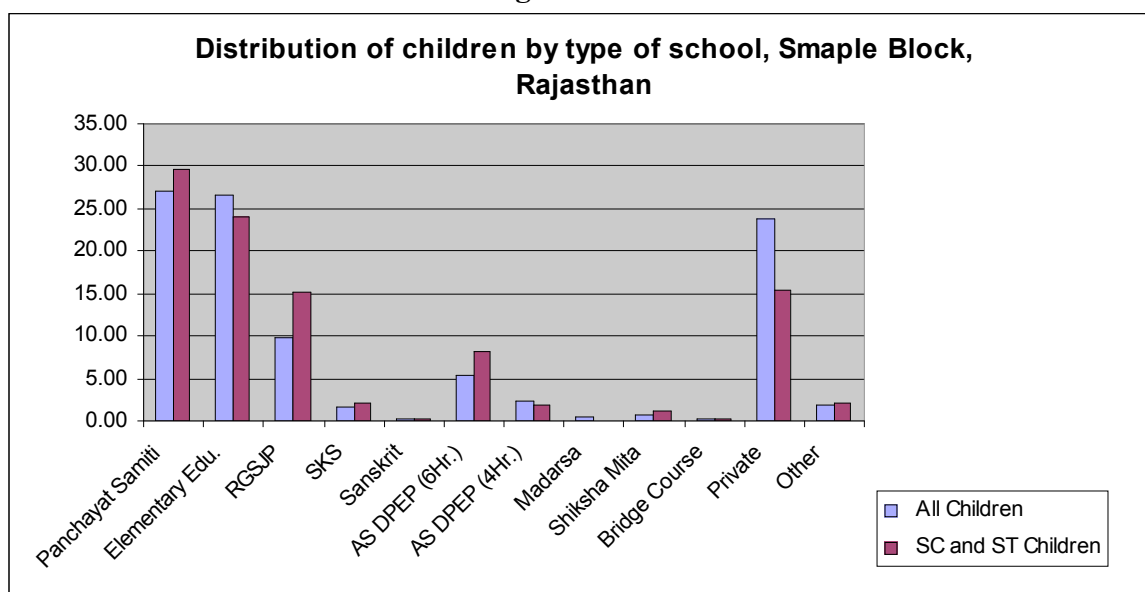
Source: Annual Action Plan (2005-06), Block Resource Centre, Raj Block.

Note: This figure includes under age and over age children enrolled.

* The general category children in madrasa and alternative schools were mostly Muslim.

As is evident from Figure 7.1, 23.7 per cent of all children were studying in private schools as compared to 15 per cent SC and ST children. As the village accounts in the sections above show, these accounted for the more influential sections and those who had a voice in society and in government. While there may be a lot of variation in, and issues with, the quality of private schools, the moot point is that those who can pay have opted out of the government school system. As a result there is little pressure on the government system to ensure quality education—53.8 per cent of all children in Raj Block studied in formal government schools while the proportion of SC and ST children was 43.55 per cent. Equally, while close to 10 per cent of all children were in RGSJP the proportion for SC and ST children was 15 per cent.

Figure 7.1



The gender dimension on the above issues was quite disturbing. Alarm bells rang when the Census of 2001 released the juvenile sex ratio in Rajasthan, which stood at 909 girls for 1,000 boys. The male literacy rate was 75.70 per cent and the female literacy rate was 43.85 per cent—a 31.9 percentage point difference between men and women! The discrimination against the girl child was also evident in the choice of schools. There were more boys in all social groups enrolled in private schools (Table 7.7). This is significant because there was a general perception that private schools are better. At the same time, the ratio of girls was high across all social groups in alternative schools of various types.

Table 7.7: Ratio of girls to boys enrolled in different types of schools, Raj Block Rajasthan

Category of School	SC	ST	OBC	General
Panchayat Samiti	0.86	0.90	0.99	1.20
Elementary Education.	0.67	0.56	0.67	1.07
RGSJP	1.01	0.93	0.90	1.00
SKS	0.81	0.58	0.61	0.30
Sanskrit	0.81		1.09	1.24
AS DPEP (6Hr.)	1.05	1.06	1.23	0.93
AS DPEP (4Hr.)	1.76	3.17	3.46	2.46
Madarsa	-	-	1.17	1.09
Private	0.49	0.32	0.38	0.69
Other	0.29	1.17	0.56	0.42
All schools taken together	0.79	0.76	0.74	0.87

Source: Calculated from Annual Action Plan (2005-06), Block Resource Centre, Raj Block.

As has been stated earlier, stratification and inequity are important aspects of the Indian social structure. The fact that children from different social strata access very different types of schools, heightens these features. The pervading apathy not only towards formal schools but more importantly towards RGSJP and alternative schools, is explicable by the fact that these are accessed by poorer children, and by girls. Given that alternative schools and other transitional arrangements have an overwhelming number of very poor children from disadvantaged social groups and given that many more girls are sent to these schools and centres, there is almost no administrative or social interest in ensuring that these centres run

with any degree of regularity. This is perhaps the most formidable systemic barrier to equity, and critical for reproducing and exacerbating existing inequities.

Differences within the 'regular' government schools

Within the government school system, the schools varied widely in terms of their basic facilities. Notably, many schools did not have even the bare minimum teachers needed for any meaningful teaching and learning. For example, there were 473 single teacher schools in the sample district of Rajasthan, accounting for 29 per cent of all schools, while in the sample district in Andhra Pradesh only 4 per cent of the primary schools were managed by single teachers ([Table 7.8](#)).

Table 7.8: Schools by number of teachers, AP and Rajasthan

	Andhra Pradesh			Rajasthan	
	AP District	AP Mandal 1	AP Mandal 2	Raj District	Raj Block
Primary					
No teacher	12	1	0	1	0
Single teacher	63	2	1	473	103
Two teacher	1267	24	26	831	119
More than two teachers	1537	20	33	285	84
Total	2879	47	60	1590	306
Upper Primary					
No teacher	5	0	0	1	0
Single teacher	0	0	0	12	2
Two teacher	13	0	0	39	7
More than two teachers	902	11	11	414	66
Total	920	11	11	466	75

Source: DISE data (November 2005).

As [Table 7.9](#), 2002-03 7th AIES data shows, single and two teacher schools were usually located in rural areas. Interviews and field visits showed these to be predominantly small and far-flung habitations.

Table 7.9: Single teacher primary schools in rural and urban, AP 2002-03

State	Rural		Urban		Total	
	% of Single Teachers Schools	Total Schools	% of Single Teachers Schools	Total Schools	% of Single Teachers Schools	Total Schools
Andhra Pradesh	19.87	53916	8.78	7251	18.55	61167
All India	16.41	573085	4.63	78290	14.99	651375

Source: Govinda and Biswal (2005).

The data on teacher-pupil ratios was also quite revealing. Teacher-pupil ratios in schools in the more backward and remote areas were far worse than those closer to the road or in bigger villages. As is evident from [Table 7.10](#) the situation in Rajasthan was far more worrisome than that in Andhra Pradesh.

Table 7.10: Schools with PTR (primary and upper primary separately) numbers

	State	District	Mandal		State	District	Block
Government Schools Primary	AP	AP District	AP Mandal 1	AP Mandal 2	RAJ	Raj Dist.	Raj Block
>100 PTR	851	42	0	2	4186	82	17
75 to 100	902	29	0	0	4688	74	11
50 to 75	3906	141	0	2	12455	282	75
40 to 50	5865	240	4	3	9712	275	50
< 40 PTR	53586	2392	43	53	24176	783	135
Total	65110	2844	47	60	55217	1496	288
Upper Primary							
>100 PTR	795	50	2	2	501	9	1
75 to 100	375	26	1	0	462	12	5
50 to 75	814	53	1	0	1098	26	6
40 to 50	712	27	1	0	1020	28	7
< 40 PTR	15697	614	3	5	10821	285	45
Total	18393	770	8	7	13902	360	64

Source: DISE data (November 2005).

While the shortage of teachers in Andhra Pradesh was less severe than in Rajasthan, it existed, and as of September 2004, 15.7 per cent primary schools in Andhra Pradesh were single teacher schools with 6.8 per cent children enrolled in them.

Infrastructure in schools too remains uneven, and in many schools, grossly inadequate. Recent data from Andhra Pradesh reveal that 34.3 per cent of primary schools function in a single classroom, even though they may be more than one teacher in such schools. Equally, 15.2 per cent of the schools had a pupil-classroom ratio of more than 60 (NIEPA, 2004). In Andhra Pradesh, the last ten years have witnessed a steady increase in the enrolment of children in primary schools but the infrastructure development has not kept pace. Discussions with people engaged with education in the state revealed that while a large number of primary schools have been formally upgraded to upper primary schools, infrastructure, facilities and availability of trained teachers remains problematic (discussion with Dr. Shanta Sinha, November 2005).

The above discussion shows that children in rural schools and within rural areas, from habitations or settlements that are away from the main village had a disproportionate share of both single and two teacher schools and they also had high teacher-pupil ratios.

School functioning

As has already been indicated in the previous chapters, and the village descriptions above, there were significant school functioning. Teacher absenteeism was a persistent problem, especially in Rajasthan, and particularly in smaller schools. Even when there were two teachers, the normal practice (at least in the remote settlements) was that one of the two teachers was invariably not present. They left behind a letter, and in the case of an unexpected visit by education officials the other teacher submitted that letter for casual leave, and one teacher was left to manage the entire school. It is important to note that teacher absenteeism is indeed a big issue and is closely linked to larger governance issues. Moreover, even when teachers and children were present, classroom observations revealed that very little time was

spent on actual teaching and learning. The sample schools of Rajasthan were worse than the sample schools of Andhra Pradesh in this respect. While negligible teaching activity was visible in the sample schools in Rajasthan, in Andhra Pradesh, teachers were regular and we observed at least 25 to 30 minutes of actual teaching happening in each period.

A recent World Bank study (Kremer et al., 2004) found 25.3 per cent of the teachers absent and what was more worrying was that 57 per cent of the time there was no teaching activity, even when the teachers were formally present (Table 7.11).

Table 7.11: Teacher absence and teaching activity in government schools, 2004

State	Teacher Absence %	Non teaching activity %
Rajasthan	23.7	63.4
Andhra Pradesh	25.3	57.0
Weighted average India	24.8	55.3

Source: Kremer et al.,(2004).

Chapter 5 shows the administrative and management ethos, particularly the existence of patronage and rent-seeking networks which gave rise to this state of affairs. From the equity point of view, it is important to note that poorly functioning schools hurt the marginalized children most. The better off children moved to private schools, and poor children, especially girls, SCs and STs were at the receiving end of ineffective schools.

Teaching and learning

The teaching learning processes in the classrooms were not satisfactory. As Chapters 4 and 6 show, classroom processes were mechanical and did not stimulate the children, and the tests conducted in the sample schools showed children's learning levels to be poor. Chapter 6 shows how this problem pervaded the whole system: the lack of a meaningful discourse on education within the system, the lack of understanding about the process of learning, ineffective textbooks, teacher training and so on. Importantly, officers, resource persons and teachers tended to see the problem of poor achievement and learning in terms of children's background, their personal characteristics such as IQ, rather than in terms of curricula and teaching strategies. At worst, some were hostile to the underprivileged children who accessed government schools. Even when not hostile, officers, resource persons and teachers appeared to have no plans for addressing the needs of specific children. Often, the teacher focused her attention on the few bright children, ignoring the rest.

Here, the problem was as serious in Andhra Pradesh as in Rajasthan. Though teachers were more regular and spent more time teaching, their knowledge of their subject and teaching skills were very unsatisfactory. Corporal punishment was common. Discussions with SEC and village leaders in Andhra Pradesh revealed that they felt that some amount of punishment was necessary to discipline children and this attitude emboldened the teachers to use the cane freely. In Andhra Pradesh, as little learning seemed to happen in schools, children, especially in the larger villages, went for tuition. In one village, the children spent several hours every day in the tuition class. As a result, children from very poor families or those who could not afford tuition learnt little. Surprisingly, children were beaten far more severely in tuition classes than in schools. The children we interviewed admitted that the beating was quite severe but they went for tuition classes because their parents forced them to do so.

Unlike Rajasthan, alternative schools and bridge courses functioned regularly in Andhra Pradesh. However, as the example of the alternative school in one of the Andhra Pradesh villages presented above shows, the teacher was not able to deal with the learning issues. Notably, the alternative school was set up with good intentions, but, in spite of the sincerity of

the teacher, did not serve the educational needs of the children. Though it was not an outright fraud as in the case of several such schools of Rajasthan, it was a mere token. It was too weak an intervention to be effective: what was needed was teachers trained specially to deal with first generation children and also those in terribly deprived situations.

In other words, even if a child from a socially deprived community did somehow remain in a 'regular' government school, even if she attended school regularly, and even if the teacher also taught regularly, there was little possibility that the teaching and learning process would actually help her learn meaningfully. The government school system did not provide real educational opportunity to the largely less well-off children that it served. Chances of such children dropping out at an early stage were extremely high.

Midday meals and free textbooks

Two important initiatives to counter inequity need to be noted. In both the states we found a functioning midday meal programme. In both the states, teachers said that these meals helped very poor children attend school. Similarly, free textbooks were provided in both the states. These initiatives need to be viewed against the fact that villages and schools described in this study showed a high degree of poverty and the engagement of children in domestic as well as non domestic labour, and the provision of noon meals as well as free textbooks appears essential in this context. Notably, in Andhra Pradesh the noon meal was a huge incentive during the two years of drought. These strategies countered the iniquitous social background of the children who attended government schools.

Comparing the Two States on Equity and Education

Even though there were a large number of commonalities between the two states vis-à-vis the situation on the ground, and in neither state had the situation been address adequately with respect to equity, there were also some differences. Social and gender differences in society were being reinforced to different degrees in both the states. While the most marginalized communities were left to deal with single teacher schools and dysfunctional alternative schools in Rajasthan⁸⁷, this was not the case in Andhra Pradesh, where the attempt was to provide access to formal schools, and at least two teachers in a school, to all children. In Andhra Pradesh the government had also done a household survey to identify the number of children who are not in school. In 1995-96 (during DPEP) the government re-defined out of school children to include those who may be enrolled but did not attend school regularly, and teachers in the sample schools were aware of, and dealt with this issue. The midday meal programme ran better in the Andhra Pradesh and the state government had contributed a larger amount towards it.

In both the states, teacher availability and teacher-pupil ratios varied across schools according to their location and categorization. The situation in Rajasthan was characterized by excessive political interference in these processes. The situation in Andhra Pradesh was much better

⁸⁷ 'The situation seems to be the result of special moves made by several states to establish schools in small habitations and thereby making schooling facilities accessible to remotely located and un-served groups. While one cannot question the motive behind such measures, it is important to carefully examine their long-term viability and quality of provisions. Creation of temporary measures—including Education Guarantee Scheme (EGS) centres or enrolling children in residential camps for mainstreaming should be viewed distinctly from developing a system according to acceptable standards and that are sustainable. Temporary measures are acceptable only if they are seen so and are not counted as part of the progress in building a national system of education. A roll back plan for such measures without disturbing the long-term plan developments is critical. Utmost effort should be made to avoid creation of unviable and substandard schooling infra structure as upgrading at a later date would prove more difficult than creating afresh' (Govinda and Biswal, 2005).

with little evidence of rent-seeking/corruption in teacher deployment. The persistence of corruption/rent-seeking had a negative impact on the more deprived areas and the most deprived communities in Rajasthan. It led to a larger number of single teacher schools and more skewed TPR, and children in smaller, more remote villages, often from SC and ST communities, who could not go to private schools, bore the brunt of this state of affairs. Even among teachers, those who came from relatively well-connected or resourceful families were able to exercise greater leverage than the others.

Schools that existed only on paper were a fact in the Rajasthan sample but not in Andhra Pradesh sample. This question was directly relevant for equity, as it was the schools that catered to the poorest children that were more likely to be dysfunctional. As mentioned, teaching-learning processes in both the states were highly unsatisfactory. But the greater amount of teaching time in the sample schools in Andhra Pradesh meant that children who accessed government schools in the state were more likely to stay in the system and learn. The achievement levels of children reflected this difference.

Systemic Issues and their Impact on Equity

The overall impact of systemic functioning on equity

The picture sketched above indicates clearly that the elementary education system does not provide equitable education for all children. On the one hand, the system does not enable high quality schools, so that the mass of poor children who access the government school system get an inadequate, discouraging education. On the other hand, the system breeds inequities by creating various types of schools with the most marginalized accessing the schools most ill equipped in terms of teachers and infrastructure. Thus it is a system that reproduces and even exacerbates inequalities, rather than countering them.

The problems within the system that lead to the type of schools that have been described have been delineated in detail in the previous chapters. Here, it is important to point out that our study has shown clearly that the poorest children bear the brunt of the most severe problems of the system, i.e. rent-seeking, the shrinking space for work, poor classroom processes and others. Rent-seeking in transfers and postings in Rajasthan and the power wielded by the informal power brokers ensured that schools in remote and inaccessible habitations are the worst hit. On the other hand, greater transparency in the transfer posting regime in Andhra Pradesh has helped in improving teacher availability. The schools meant for the most deprived functioned the worst—whether it is teacher availability or schools managed by one or two teachers; or the dysfunctional RGSJP or the non-existence of Shramik Vidyalayas. Again, lower teaching time affected the poor adversely as they could not access private tuitions. The dice then gets loaded not only against the very poor but also against girls even in relatively well of families. Essentially the analysis of the data from an equity perspective reinforces the widely held belief that a dysfunctional or poorly functioning government school system has a greater impact on the poor and among the socially disadvantaged.

Who are the children who make it through the system? The answer is obvious: those who attend poor quality primary schools and alternative schools are the ones who dropout of the system. An attrition policy, both in terms of numbers as well as in terms of quality is built into the very structure. Only an exceptionally gifted child from a rural remote primary school can hope to make it to a secondary school. Inequality is inherent in the system, right from the time a child is born till s/he becomes an adult. What implication does this have for a democratic society? Do growing differences threaten its democratic fabric? If the last 15 years provides any indication, the consequences are frightening.

This situation remains in spite of stated policy thrusts to the contrary. The Government of India and the Government of Rajasthan documents pertaining to SSA highlight the need to enhance access and bridge gender and well as social gaps. There is official recognition of the problem of differential access to different kinds of schools. Yet, for every stated strategy to provide schools for all children a number of unspoken practices frustrate the efforts. Rent seeking (for transfers and postings of teachers) and patronage (location of RGSJP, alternative schools) negate the very strategy. The poor in Rajasthan are the worst affected as they are neither able to pay, nor are they part of the high and middle caste patronage networks. These hidden strategies that work behind the scene effectively reinforce existing social and economic inequities thereby negating the potential of any well intentioned programme or strategy seeking to promote social equity and justice. Every year a large number of children are officially mainstreamed or enrolled in schools. The number of teachers and the number of schools are justified on the basis of official enrolment data. Yet, teachers with influence or those willing to buy transfers invariably relocate themselves to more convenient locations. Official teacher-pupil ratios have little meaning on the ground. Remote, inaccessible and in many cases just far off villages and habitations end up with one or two teachers who end up taking turns to come to school. While all is well at the policy level and even at the level of district plans, the interplay of a range of political and administrative forces bring these plans to naught.

For reasons described in Chapters 5 and 6, our study showed that as one programme or project came to a close and another took its place, there was a break in the flow with the attention of administrators and teachers shifting to the new goals and targets. This meant that several equity oriented interventions did not get the needed time to get absorbed in the system. Notably, DPEP had paid a lot of attention to joyful learning and the importance of treating children with love and kindness. But the fieldwork showed that old attitudes and practices were creeping back into the system. At the time of the study corporal punishment was practiced in many schools and often also supported by parents. Messages regarding gender equality too had similarly got diluted with a change of programmes. In general, the system was not able to sustain and reinforce the good practices adopted. Achieving cumulative change necessitates the carrying forward of the lessons of one set of interventions into the next round of programmes. Starting every programme as ‘new’ and on a clean slate does not help the government carry forward the gains and consolidate them.

Notably, Rajasthan is quite unique in having developed a range of very effective strategies to provide meaningful access to education, promoting community participation and involvement and also in micro-planning and school mapping. These innovative strategies were reported to have been quite effective (Ramachandran and Sethi, 2000). Yet the mainstream education system just brushed them aside. Each new initiative was started afresh—Shiksha Karmi, Lok Jumbish, DPEP, Rajiv Gandhi Pathshalas and now SSA. Institutional memory in the state is weak and it is believed that factionalism in the administration and also the political system effectively ensure that a new incumbent starts by negating or undoing what his or her predecessor had done.

Planning for marginalized children

An important point to be noted is that planning for the most marginalized children has proven to be a hard task for the system. Given the hierarchical style of functioning, the rigidities of rules and procedures, this is not surprising. However, it means that the system is unable to address the needs of the most marginalized in any meaningful way. The situation as it obtains on the ground in Rajasthan was succinctly captured by a district level official who said:

We need to work the hardest to reach parents, those who are resigned to their fate. We are not able to reach the labourer class and the migrating

population. There is lack of data about the migrating population and they are not part of a district. We need to work on educational satellites, but cannot do so because of lack of manpower...at present nothing much is being done. We think that just getting children to school is enough. There are no tangible results. We are told: do this in a week. We ought to be able to focus on strategies and do things from the heart...We are generally bound by the Annual Work Plan and are not working in an optimal way...Most of the time is spent on file processing, reading documents. The higher ups, i.e., the secretary etc. decide the direction and we tow the line...if we had some teeth, we could work the hardest to reach the parents, i.e., those who are labourers etc.....I had organized these Kala Jathas, in which the Director had told me to take the decisions. But the tribals who participated in those Kala Jathas have not been paid. The big shots, the media people etc. have been paid. The next time the tribals will not participate.

Notwithstanding the rhetoric of district based and bottom-up planning, there is little evidence of such context and group specific planning and strategy formulation. A major systemic barrier to achieving equity goals lies in the inability of the government to plan for specific groups of children. What is worse is that there is little recognition of the importance of such micro planning at the state level. The elementary education systems in both the states were caught in the game of showing impressive figures. Block and district officials were more concerned about collecting and forwarding data to the higher-ups.

This inability to plan coherently as per the needs of children from different social groups is visible most clearly in the creation RGSJP schools, an intervention that was applied evenly across the state, irrespective of the reasons of the particular children being out of school. What is even more remarkable is that within Rajasthan, more varied strategies with an attention to the context had been tried out in Lok Jumbish, and processes for teacher selection and community engagement had been developed in the Shiksha Karmi Project. Yet none of these were used in setting up the RGSJP schools. The reasons for this are located in the huge problems of systemic functioning described in Chapter 5. From the point of view of equity, these systemic constraints do not allow the problems of the most marginalized to be addressed.

Context specific planning to address the educational needs of the most deprived was not visible in either of the two states. In stead, there was a tendency to adopt piecemeal approaches. There was not much comprehensive planning to address infrastructure, teacher availability, training, incentives and monitoring attendance. Equally, a broad state level template was followed to generate district plans. The priorities, targets and budget allocations were decided at the state level. In the last few years there has also been considerable pressure from the Government of India to meet certain targets (teacher training) and for introducing certain models (National Programme of Education of Girls at Elementary Level or NPEGEL). Even the so-called innovations are often 'pre-designed' in the SSA programme. The state governments are caught in a trap, they have to conform to national guidelines to access SSA resources, which are being disbursed in the name of district specific plans. In Rajasthan, close to 36,000 teachers were to be appointed through the Rajasthan Public Service Commission and these appointments were expected to solve the problem of single teacher schools. Yet teachers recruited through a centralized process were more likely to be urban based and even less willing to stay in rural areas.

To take the example of Andhra Pradesh, there is a general feeling that the state is doing quite well and almost all the problems of non-enrolment and dropouts has been addressed through bridge courses. Yet, the interventions designed to bridge gender and social equity gaps were not strong enough to neutralize the devastating impact of growing agrarian distress in the

state.⁸⁸ According to P. Sainath, the situation in Andhra Pradesh, which witnessed a spate of farmer suicides, is quite alarming and the sample district was a part of this belt of agrarian distress⁸⁹. While rapid growth of commercial crops, mounting indebtedness, and a spate of suicides by farmers has been reported extensively by the media, what has not been reported is the steady increase in the number of children, especially girls, working in cottonseed farms, sunflower harvesting and cotton farms. The impact of multinational and indigenous cottonseed cultivation on growing incidence of girl child labour is well documented (D. Venkateshwarlu and Lucia da Corta, 2005). Equally, the increased employment of children as domestic workers in both rural and urban areas is another issue that merits more focused attention. While we have not examined the linkage between economic policies and education in this study, it is obvious that this needs to be examined in greater detail if educational strategies are to make a positive impact on meaningful access to good quality education.

Similarly, in Rajasthan, officials claimed that the problem of access had been solved and all children have access to a school within walking distance. However, official DISE data reveal that the Gross Enrolment Ratio (GER) for Rajasthan is 100.20 at the primary and only 50.36 at the upper primary levels. The Net Enrolment Ratio is even more alarming; it is 72.95 at the primary and only 34.63 at the upper primary level (NIEPA, 2005). Discussions with officials revealed that they thought that this data did not reflect the real picture because it only accounted for children enrolled in formal and recognised schools. When the issue of access to a properly functioning school was raised, reactions varied from denial to it being a 'different matter'. Denying that the problem exists is an accepted mode to respond to challenges.

In the same vein, the educational needs of physically handicapped children were not even mentioned by the teachers and administrators interviewed. In Rajasthan, we did not come across even one physically challenged child in the schools that we visited. Pointed questions led to a resigned response about the difficult situation on the ground in Rajasthan. We were informed that the Lok Jumbish project made concerted efforts to mainstream physically challenged children but the current leadership was not aware of the efforts made in the innovative programmes of the past.

The Question of Small Schools

Parallel to the inability to plan for children with specific types of problems in attending school, the system is unable to plan adequately for different types of schools too. In both Andhra Pradesh and Rajasthan, as in the rest of the country, there are a large number of small primary schools, which have only one or two teachers. In Andhra Pradesh 46.3 per cent of the primary

⁸⁸ It is evident that while aggregate SDP per capita has increased moderately since 1993, SDP in agriculture per capita shows no such increase, and has actually declined. In fact, between the triennium 1994-96 to 1995-96 and the triennium 2001-02 to 2003-04, per capita agricultural product actually declined by around 12 per cent... What is even more significant is that per capita consumption fell after 1993-94 in all the regions of rural Andhra Pradesh barring the coastal Andhra region. This fall was particularly marked for Rayalaseema (comprising the Southwest and Inland Southern regions). So, in most of the rural areas of the state, average consumption expenditure actually declined in real terms in the period 1993-94 to 1999-2000... But in addition to the agricultural patterns, the general stagnation of the rural economy, and the absence of non-agricultural income generation possibilities, contributed further to the deterioration of living standards in the countryside. Part of the problem in employment generation stemmed from agriculture itself —not only was this sector depressed, but the increasing mechanisation implied falling labour use per hectare of cultivation (Ghosh and Chandrashekar, 2004).

⁸⁹ Sainath's article on farmer suicides in June 2004 and other articles published in *The Hindu* in 2004 and 2005 (downloaded from Indiatogether.org in November 2005). In the August 2004 article Sainath reports, 'The farm crisis has also wrecked the education of many girls. In Kurugunta in Anantapur, the bright younger daughter of G. Hanumantha Reddy has had to quit school. Her family cannot afford it. This district has seen some young girls committing suicide after being pulled out of school.'

schools have less than 50 students while this figure stands at 32.9 per cent in Rajasthan. It is noteworthy that the Rajasthan data did not include RGSJP and alternative schools (NIEPA, 2004).⁹⁰ Therefore, the actual percentage of small single teacher schools may be higher in the state.

As Chapter 4 shows, the teaching learning process in these small schools tends to be the most disorganized, and teachers in such schools get no or very little time to teach. Consequently, children who attend such schools are at a considerable disadvantage. Yet in neither of the two states was there any evidence for special strategies to improve the functioning of such schools. There are obviously no easy answers to these problems, but a host of possibilities exist. It would be useful to study school functioning and postulate a minimum number of needed teachers, to have specially trained teachers in such schools, have special teaching learning materials etc. This is an issue that needs to be addressed with seriousness for equitable education.

Local Bodies, participation and civil society

As has been argued in the previous chapters, in both Andhra Pradesh and Rajasthan, the role of local elected government vis-à-vis education had reduced in recent years. The situation was somewhat better in Andhra Pradesh, and local gram panchayat and other local leaders were vocal and in some cases involved with the school even though many of them sent their children to private schools. In Rajasthan, panchayat leaders appeared more apathetic, and the SDMC existed only on paper, so that community participation in the school processes was minimal. In Andhra Pradesh, at the time of the fieldwork greater participation from the SECs, elected from among parents of the school-going children, was visible. However, subsequently, the composition of the SEC was changed to include nominated, rather than elected members. The point here is that neither state followed a *consistent* policy for community participation. Consequently, the institutions for such participation were not robust.

At the same time, the passivity of community leaders was baffling. In Rajasthan, leaders of Dalits, tribal communities, minorities and most ironically the women's movement had not raised their voice against iniquitous strategies and poor quality education. Dysfunctional rural schools rarely attract the attention of Dalit and tribal leaders in the state. Equity in education is not an election issue. Perhaps the answer to this all-pervading apathy lies in the fact that most people in positions of power, even those who belong to the disadvantaged communities, do not have a stake in government schools and even less in alternative schools and transitional arrangements. While the content of textbooks and writing of history seem to generate a lot of passion, the systematic marginalization of the poor in education does not elicit the same kind of response.

Yet community action has a potential for addressing the needs of the most marginalized children. Notably, a sustained anti-child labour movement spearheaded by the MV Foundation in Andhra Pradesh turned the spotlight on the most vulnerable section of children in the state. They led a state-wide campaign against child labour and mobilized NGOs, social activists and ordinary citizens to demand universal education. This led to the introduction of 'bridge courses' to get out-of-school children back into formal schools. In the early 1990s the bridge courses were primarily run by NGOs. However, from 1996 the government has also started running these courses. As a result of a sustained campaign against child labour, access to formal schooling in the state became non-negotiable and the government used the alternative mode purely as a bridging strategy. The IT industry and its associated boom in employment opportunities for the educated middle class also had an effect on education. The demand for

⁹⁰ We were informed that from 2006 RGSJP data has been integrated into DISE and that many of them had been 'converted' into formal single teacher schools. Equally, many primary schools had been upgraded into upper primary schools.

education up to the secondary level and demand for English medium also increased. From the 1990s, sending children to school emerged as a social norm in most of Andhra Pradesh, especially among people who aspired to jump onto the IT and the service sector boom.

Rajasthan went through a very different trajectory. At one level the state was home to a range of innovations in elementary education like the Shiksha Karmi Project and the Lok Jumbish Programme. These projects were introduced by civil servants with vision. However, despite encouraging results, the mainstream political and administrative system remained indifferent. During fieldwork in the sample district and villages, the officials in charge of elementary education hesitated to talk about Lok Jumbish and shrugged off questions regarding the difference between Shiksha Karmi and RGSJP schools. Most of the well known NGOs in the state had not only participated in both the programmes but continued to run residential camps for girls and other interesting models for accelerated learning. Yet unlike Andhra Pradesh, the issue of children's right to education was never taken up as a full-fledged campaign in Rajasthan. There was little social mobilization on the issue of child labour or right to education. A few interesting and well-run programmes that dot the state and education did not become a rallying point for social activists. The lack of a social movement against child labour or for universal education and the apathy of the non-government sector could perhaps offer us some insights into the situation as it obtained in Rajasthan. This perhaps also explains to some extent the fact that a less iniquitous situation prevails in Andhra Pradesh than in Rajasthan.

Finally, notwithstanding the prevalent scenario, the parents we spoke to were eager to send their children to school and the children themselves were full of hope. Their aspirations, however, were tempered by the lack of opportunities. The initial enthusiasm to participate gradually turns to resignation and apathy and the children become irregular, hang around at home or on the streets and often start pitching into family work, in most cases dropping out without learning anything. The inter-generational cycle is perpetuated, with the new generation being (at best) semi-literate, weak and apathetic or in many instances angry at a system that has treated them with such indifference.

CHAPTER 8: IMPLICATIONS FOR THE FUTURE

Rashmi Sharma and Vimala Ramachandran

We began this book by demonstrating that severe problems in elementary education, i.e., a substantial number of children out of school, iniquitous educational opportunities and inadequate quality of education, have persisted in India across decades and have survived a fair range of policy thrusts as well as shifting socio-economic and political scenarios. In parallel, recent studies have shown basic school functioning itself to be a problem area, while initiatives for systemic reform have been few. We argued, subsequently, that neither the economic and socio-political scenario, nor education policy, fully explained the shortfalls of elementary education in India (though both these are important). Instead, there was a need to examine the 'system' i.e. institutional structures, capacities, processes and dynamics.

We focused on two states, i.e. Rajasthan and Andhra Pradesh, in the north and south of India. Rajasthan is characterized as a low income state, and Andhra Pradesh a middle income state, and the two states are broadly representative of a large part of the country. Both these states share certain pan Indian features, i.e. a caste-based society, a mainly agrarian economy and high incidence of poverty etc. In both the states, literacy rates remained low for five decades after independence and had only recently started accelerating. However, while following the south Indian regional trend, Andhra Pradesh was on the verge of a demographic shift, and also had better human development indicators. While the Congress Party has been a major political force in both the states, Andhra Pradesh had seen the rise of a regional party with a large number of backward caste supporters, as well as a strong communist movement in the Telangana region while in Rajasthan the RSS began to gain support during the 1960s and the BJP is the other major political force in the state.

In these two states we examined a range of institutions, i.e., administrative, academic and community-based, across various levels, from grassroots structures to apex organizations. We also included an investigation of a small number of schools and villages for the reader to get a 'feel' or 'texture' of the system as it functioned on the ground. While scrutinizing the system and its institutions in detail, we examined them from two points of view—their internal organization and links to society. On the one hand, we attempted to comprehend the structure, capacities, processes and dynamics of various institutions, and on the other, explored how with their particular organization and dynamics, the various arms of the system interacted with the larger social system.

In this chapter we shall briefly summarize the systemic dynamic as delineated in the previous chapters and then attempt to spell out the implications for the future. The chapter is divided into three parts. In the first part we summarize the broad systemic dynamic. In the second, we visit important strategies that have been attempted to bring about change, and in the light of the systemic characteristics, why they have succeeded or failed. Finally, learning from these two sections, we examine the implications for the future.

The systemic dynamic

The systemic dynamic is summarized in four parts. The first part addresses the question of resources. The second part describes the system and its institutions, in terms of its management ethos and the manner in which it perceives and practices 'education'. In the third part are summarized briefly, the 'outcomes' on the ground, i.e., the schools. The last part summarizes the larger social impact of the Indian elementary education system.

Resource base

As Chapter 3 shows, elementary education in India is chronically under-funded, and this has been the case in both Andhra Pradesh and Rajasthan, which stood 24th and 21st respectively in terms of per capita expenditure on education among 28 major states, Rajasthan in 2004-05. Though an allocation of 6 per cent of the GDP for education is a policy goal, in 2004-05, Rajasthan allocated 4.4 per cent, while Andhra Pradesh allocated 2.6 per cent GSDP for education. The percentage of SDP spent on elementary education was 2.6% and 1.4% in the two states respectively. Notably, till the 1980s, expenditure on education in Andhra Pradesh had been higher than in Rajasthan. But from 1993 onwards, as Andhra Pradesh aggressively followed economic reform policies, there was a significant decline in expenditure on education. Now, despite the fact that Andhra Pradesh is a middle income state and Rajasthan a low income one, on education, Rajasthan outspends Andhra Pradesh.

In both the states, government expenditure accounted for more than 95% of all expenditure on education, which illustrates the very central role that the government plays. More than 90 per cent of this expenditure was on salaries and allowances of the teaching and other staff. Of the total government expenditure, state governments incurred around 90 per cent, and the central government, the rest. The union government met a large proportion of the plan expenditure, as much as 50 percent, through centrally sponsored schemes. At the same time, state plan expenditure fluctuated a great deal over the years, which is not conducive to a healthy growth of the education system. Notably, since most of the state governments' funds went towards salaries, states depended a good deal on the central government schemes for up-gradation of infrastructure and quality improvement. However, central funding as per state needs is not *assured*. In this context, the practice in DPEP and subsequently in the SSA to provide central funding as per state plans to make elementary education universal, rather than schemes that are limited in scope, is a positive development. However, while in recent years SSA has provided a large quantum of funds for teachers and infrastructure, there is no guarantee that this funding will continue as long as it is needed by different states.

The household expenditure on elementary education was between 25 to 30 percent of all expenditure in education in the two states. At the primary level, the annual average household expenditure in Andhra Pradesh and Rajasthan was estimated to be Rs. 430 and Rs. 518, and for middle school Rs. 820 and Rs. 831 per annum. This indicates substantial under-funding of education in the two states. Notably, in spite of recent attempts to increase funding in education in Rajasthan, household expenditure remains quite in the state. Household expenditure strengthens inequities, as sending children to school entails substantial hardship for the very poor, and many poor children may not have access to education.

It is estimated that substantial financial investments are essential to make elementary education universal. To this end, it was estimated that in 1999, Andhra Pradesh required an additional financial requirement of Rs. 8,697 crores and Rajasthan Rs. 6,508 crores.

The system and its institutions

Our scrutiny showed that there existed severe problems with the system and its institutions which whittled its capacity to deliver as per policy goals. In this context, a critical dynamic within the system was that along with its formal and stated processes, i.e., practices which were available in rules and codes and acknowledged without hesitation by the personnel, a set of informal and unstated practices, i.e., patronage dispensation and rent seeking were prevalent. These were widespread in the system, their scale was huge, and they permeated all activities. Significantly, these also played a critical role in the *internal* management of the system, and consequently, transformed the formal and stated features of the system. consequently, the functioning of the system could be comprehend in terms of its unstated practices along with its formal features.

The important *formal* features of the system comprised an extreme hierarchical functioning coupled with a high degree of employee protection, along with extreme rigidity in schemes and rules and a lack of attention to professionalism. Hierarchical functioning was manifest in various ways such as the following of orders from above rather than seeking situation specific solutions, extreme centralization of decision-making, lack of real deliberation on issues because of 'the boss is always right' ethos, etc. Hierarchical functioning was not limited to management practices but extended to placing organizations and offices within the system in a hierarchy, rather than understanding their separate roles and significance. Thus there was a concentration of facilities and personnel in the state level offices, while institutions close to the grassroots remained weak and deprived of basic facilities. Secondary education was considered 'superior' to elementary education, and the latter suffers as a consequence. Academic institutions were placed below administrative ones.

For the personnel working within the system the hierarchy was mitigated by strong employee protection. Promotions were more or less guaranteed. Dismissing and even punishing employees required complex processes of enquiry and proof, which were often unsuccessful. Employees also took frequent recourse to litigation. This level of employee protection tended to impart a rigidity to the system, whereby the system was unable to recognize the special contributions made by individuals, as such recognition could be seen as being 'unfair' to the others. This rigidity in personnel matters was matched by a style of functioning guided by rigid rules and procedures, especially pertaining to finances. It can be quite difficult sometimes for even apex offices with huge budgets to fund small relevant activities not included in the budget.

The lack of professionalism within the system ensued partly from the characteristics described above and partly from a lack of focus on professional development. Hierarchy oriented functioning blocked reflection and the emergence new ideas. The weakness of grassroots processes limited the systemic capacity to actually deliver on the ground. The subordination of academic institutions to administrative ones reduced the focus on educational issues and processes. At the same time, professionalism was, not emphasized in the managerial as well as academic sense. In the two states studied, the recruitment system did not make place for inducting individuals of high calibre, serious plans for the professional development of available personnel had not been developed and there were few mechanisms for supporting the professional practices of learning, deliberating and researching. The system did not develop professional managers and individuals were often placed in managerial positions with only very minimal training, though Andhra Pradesh was somewhat better than Rajasthan in this respect.

The institutions that provided academic leadership, i.e., SIERT/SCERT and DIETs, were characterized by neglect. The placement of faculty did not take into account the skills and capacities needed in either state. The people responsible for making curriculum and textbooks lacked rigorous training, confidence, exposure and access to new ideas and materials. At the same time, SCERTs as organizations were not sufficiently strengthened and empowered for leadership and lacked autonomy and freedom to take initiative. The faculty did not engage in research of any quality and had few linkages with other resource institutions. Instead, the institutions followed a fairly ad hoc agenda which varied as per the directives given from time to time.

In general, issues related to teaching, learning and pedagogic practices were simply not perceived as central, even by those responsible for academic leadership. For example, large-scale teacher training programmes had taken place in both the states. However, in these it was not the content of training but the numbers and logistics that became critical elements. Perceptions of a 'good school' most frequently revolved around adequacy of infrastructure and teachers. Similarly, perceptions of a 'good teacher' were concerned with qualities such as being sincere, regular and punctual, and not how the teacher taught, the type of classroom

activities she encouraged and the manner in which she engaged with children. were considered appropriate.

An important characteristic of the system was that a meaningful discourse education and learning simply did not exist, even at the state level. For instance, though there had been considerable debates and shifts regarding the desirable curriculum and the teaching-learning process in recent years at the national level, at the state level the ideas expressed by academic leaders, i.e., curriculum makers and textbook designers were diffuse and patchy. The essence of the debates at the national level were not reflected in terms of positions taken at the state level, and in fact, few state level leaders had well thought out, well argued positions on the aims of education, the process of learning and the pedagogic techniques etc.

This lack of a genuine reflection about education was constantly visible in the pedagogic practices across the system. Textbooks were seen as major means of teaching and learning in the classroom and a curriculum based on competencies, an ideal goal. The idea of the child's active engagement to make her an independent learner, though part of the national vocabulary, was rarely voiced. For example, the underlying belief about mathematics was that it was something to be explained and given to children and that specific exercises would produce the same specific competencies in all the children. The math textbooks dealt with extremely large numbers and did not attempt to relate mathematics to the experience and environment of children. The main idea about examinations and assessment was that these were necessary to assess children and to make them study. That learner assessment was an important ingredient of the teaching strategy itself was rarely understood. Teachers often bought question papers from the market and board examination papers were textbook focused.

To these formal structures and processes were added several unstated, but pervasive and influential practices. The unstated practices of the system centred on the creation of networks for patronage and rent seeking. Rent seeking was pervasive. All opportunities available, i.e., school construction, teacher training and provision of teaching-learning materials were 'up for grabs'. In these processes, the manipulation of government servants through transfers and postings, which were left to the discretion of those in the highest positions of authority, played a critical role.⁹¹ Transfers formed the fulcrum of the reward and punishment system as per unstated rather than stated rules. Those who stood in the way of the unstated goals of patronage dispensation and rent seeking were marginalized and sometimes threatened. Those who helped in such activities were rewarded with important charges and also benefited financially from rent seeking. In Rajasthan, teachers reported 'paying' to get a posting in a place of their choice. In Andhra Pradesh, significant changes had come about at the lower levels and the transfer system had been made rule based. At the higher levels, the problem persisted.

The stated and unstated practices co-existed and produced a new management and work ethos. A new hierarchy of the informal system was superimposed on the existing formal hierarchy, i.e., persons who managed rent-seeking networks were also powerful, irrespective of the formal positions that they occupied. At the same time, the power of individuals who were high up in the formal hierarchy, but not in the informal one, was diminished. Some mitigating characteristics of the formal system such as employee protection were vitiated. Though an employee could not be dismissed, she could be given an unimportant or personally inconvenient posting and this chipped away at employee welfare. Teachers having to pay for

⁹¹ As this is written, a dramatic raid was conducted at the house of the Director, Public Health Services in Bhopal, Madhya Pradesh, where stashes of cash were discovered. Hindi newspaper *Dainik Bhaskar* reported the sources of 'regular income': District chief Medical Officers pay Rs. 15 to 20 lakhs for choice postings, lower staff pays Rs. 10,000 to 50,000 and private hospitals, nursing homes etc. pay up for 'recognition' (9 September 2007, page 7).

postings was another case in point. Significantly, authority and responsibility were de-linked. 'Orders' in the informal system were oral, and no records existed about where they emanated from. At the same time, junior officials stood at the risk of being victimized, both when they went along with the unstated practices and when they resisted them.

The interaction of the formal and the informal system whittled down the place of work and professionalism. The reward structure worked against professionalism as the attainment of the 'best' postings was contingent on complicity in the informal system. In fact, individuals who chose to exercise initiative vis-à-vis work, often found few 'takers' within the system and frequently transferred their dynamism elsewhere. Working relationships within organizations were transformed into client-patron complicities and personal rivalries that centred on capturing various posts and seeking rents, rather than on shared goals and competition over performance vis-à-vis state goals. The selection of organizational leaders was not based on their capacity to meet the formal goals of the organization, but instead on their willingness to either further, or tolerate, the informal system. Field level offices were further weakened as their energies and time were tied up in the activities of the informal system. The supervision system was ineffective, as supervisors were not well trained and often engaged in rent-seeking themselves.

The coexistence of accountability mechanisms of the formal system along with the informal system led to an increase in rigidities and in long and cumbersome processes. This happened as new rules and processes were constantly invented to counter the informal system. Several distortion of practices to suit the priorities of the informal system and as attempts to check rent seeking. These ultimately resulted in delays. Decision-making became a site of new conflicts between the formal and informal systems and the energies of individuals in the system were often tied up in these conflicts. The space for meaningful action was whittled further.

The schools

Given the above scenario, the poor functioning of state schools which has also been documented in several other studies, was not surprising. The chronic under-funding of in elementary education was visible in the paucity of teachers in Rajasthan and in poor building infrastructure in Andhra Pradesh. In addition, both these states relied on para teachers, who were paid very poorly and were often not trained as teachers. The paucity of teachers in several schools meant that one or two teachers teaching five classes were largely involved in basic management, and got little time to teach. Single teacher schools existed in the form of Rajiv Gandhi Swarnajayanti Pathshalas (RGSPs) and alternative centres of various type. In such schools, even the most sincere teachers found it difficult to engage in any meaningful teaching.

The impact of the patronage based and rent seeking system was visible in several schools, especially the RGSPs of Rajasthan. These usually functioned sporadically, so that they could not be called 'schools' in any meaningful sense, i.e., they were not institutions where students could go regularly to learn. This was made possible because well-connected teachers could not be taken to task and because deals could be struck with supervisors. Moreover, as some people got away with professional neglect, demanding it of others became more difficult. Many teachers had a lackadaisical attitude towards their work.

Several schools functioned regularly but the amount of time devoted to teaching and learning was often shockingly low. Some lacked teachers. In others, teachers were often involved in non-educational tasks. And even when these problems were not severe, the actual teaching time remained brief. Untrained teachers simply did not know how to handle the class (as in the case of one teacher who spent the entire class time marking attendance). Some teachers did not care to teach and the supervision system, because of its own problems could not address this issue.

Even where a reasonable number of teachers and basic infrastructure was available and the teachers addressed their work with a degree of sincerity (as in several schools in Andhra Pradesh), the state schools did not present a facilitative learning environment. Classroom

had to teach and their teaching was unimaginative. In the classrooms, children remained passive receivers and listeners. They had no occasion that required them to think of new problems or solve problems on their own. Their voices did not find place in the classroom. This situation within the schools was comprehensible in the context of the general absence of educational discourse across the system. The rigid syllabus and examination pattern further blocked creativity among teachers and the inability of the system to spot and support excellence was a de-motivating factor. Not surprisingly, achievement levels in both the states were extremely low.

The system and equity in society

The education system in India exists in the context of an iniquitous society. While the Constitution envisages education as one modality of reducing unequal opportunities, in actual fact, while the elementary education system built on and even strengthened the social cleavages within society. On the positive side, the expansion of schooling post independence had provided access to education to a very large number of children, and the emphasis during the 1990s on universal elementary education had further widened the outreach of the system. Through midday meals, scholarships, free textbooks etc., some interventions to address the needs of the least well off children had been made.

However, these initiatives had been vitiated to a very large extent because of the poor functioning of government schools. For one, the fact that government schools did not function well pushed the privatization of schooling. The better-off children tended to move to fee-charging private schools, the large poor majority studied in government schools. If these government schools had provided high quality education, the system would have empowered the poor children it catered to. However, in the existing school system, most children left the school system after very few years of education. And even those who remained for a longer time period were ill prepared, to say the least, for the opportunities and challenges offered by the growing economy. Thus because of the poor quality of government schools, social and gender inequities were preserved and even exacerbated rather than mitigated.

The perceptions of key actors within the system also contributed to the inability to address inequity. It is important to remember here that the larger social system impinges constantly on the system, as personnel within the system are drawn from the same society and share its social, cultural and intellectual constructs. Our study found that several people placed in key positions within the system believed that children from less privileged backgrounds were unlikely to become as good students and that difference in the capacity to learn arose largely because of genetic make up and social background. Notably, curriculum and teaching methodologies were not perceived as important instruments for creating a learning environment for all children. A consistent finding in classroom observations was that teachers often taught without bothering to find out if the students had understood or not, betraying a lack of concern along with a lack of skills. A greater degree of professionalism, especially a keen understanding and discourse about learning and education could have shifted these perceptions and the teaching style, at least to some degree. As it was, traditional beliefs and ideas were carried into the professional sphere.

The inherent acceptance of inequity was visible sharply in the fact that within the government school system, schools varied widely in terms of facilities. Government schools in large and politically influential villages were better endowed while those in smaller villages and with a

larger share of marginalized communities were one or two teacher schools with the most rudimentary facilities. The poorest children attended 'alternative' school systems. In Rajasthan, the scope of the alternative school system had been expanded (rather than confined to the bare necessary minimum) through the RGSP schools, which were the worst in terms of facilities and teachers and were often dysfunctional. The most marginalized children were thus offered the least effective education.

The lack of a professional ethos within the system resulted not only in poor quality schools, but the system also had limited professional capacity to evolve effective strategies to address the needs of the most marginalized communities. Strategies such as RGSPs were easy to push through where the educational needs of children from marginalized communities were not well understood and articulated. Had these been considered carefully, a pressure to provide better than average and special facilities for such children would also have been built. Poor professional capacities and rigidities in functioning also meant that need based planning remained in the background. There were few mechanisms for addressing the varying needs of different children such as those who did not speak the mainstream language, girls and working children. Where such interventions were attempted they were often not effectively implemented because of a lack of a complete understanding of the issues involved, or the weakness of the grassroots implementing structures.

The paucity of funding for elementary education meant that children incurred substantial costs for education. While this was not significant for the better-off children, it a significant deterrent for those whose families were poor. In both Rajasthan and Andhra Pradesh, household expenditure on education was substantial. This was be an additional contributor along with the unsatisfactory quality of schools, to children from marginalized communities dropping out.

It is important to note here that even as the elementary education system reproduced existing inequities, it also did not promote excellence in any way. The children who studied in Indian schools were not empowered to fulfil their human potential. The functioning of the school system did not highlight the value of democracy and of a humane approach to others. Nor did it promote independent thinking, problem solving or even the acquisition of skills. Thus the system did not contribute to a humane society, an intelligent and thinking polity and did not maximize the economic potential either because of the low level of skills. The overall social costs of the type of education system that exists today are huge.

Change attempts

The need for change in this scenario is clear. However, the dilemma here is, that a move 'out' of this type of system is difficult. Since the 1970s, there have been several attempts to bring about change from outside government, as well as from within it. We now briefly sum up the dynamic of some major change attempts.

Change through policy

Attempts to bring about change have been attempted most often by introduction of new policies and programmes. However, at best, these shifts have had a very limited impact, and often, no impact at all. Firstly, several characteristics of the system are not conducive to the formulation of well thought out policies. Hierarchical functioning and the lack of professionalism obstruct the process of thinking through problems and rigorous strategy formulation. And secondly, even if a potentially effective policy somehow gets formulated, it is easily stymied in 'implementation', i.e. becomes the victim of the weakness of the institutions that translate them into action, and the play of patronage dispensation and rent-seeking. This is not to deny the importance of well-thought out policies and programmes, but

to stress that institutional weakness blocks both appropriate strategizing and intelligent implementation.

Curricular and pedagogic reform

Curricular and pedagogic reform shared the same limitations as any other policy or programme. Our study showed that major changes in curricular statements at the national level were perceived in a very diffused manner at the state level, with little comprehension of what these changes implied, as a meaningful understanding and discourse about education was absent *across* the system. This lack of understanding was visible not only among administrators but also among those who occupied leadership positions in academic institutions, resource persons and teacher trainers.

The most sophisticated and well thought out interventions have to ultimately deal with an institutional structure which may not even be capable of comprehending the initiative, let alone adapting it to the local context in a creative manner. In such a scenario, resources spent on teacher training may have little impact, as the quality of training itself may be highly inadequate and trainers themselves not understand various concepts. At the same time, even if carefully thought out curricula, textbooks and teacher training modules are developed on the basis of a discourse among the most informed persons in the country, these may not be understood by key players in the states. What is striking in this scenario is, as in the case of new programmes, the very limited impact that national level attempts at academic reform tend to have.

Shifting activities to non-state actors

Another type of solution that has been attempted, especially in recent years, is providing increasing space for non-state actors. Many market enthusiasts have particularly seen privatization as a solution to the poorly functioning government school system. There are, however, several problems with this idea. The first of these is the equation between privatization and equity. If a child's quality of education is linked to the capacity of the parents to pay, then existing inequities are strengthened. On the other hand, a school system where all children can access a high quality of education, irrespective of their social background, lays the foundation of an informed and equitable society.

At the same time, an important point that emerged in this study was that privatization too cannot be seen independent of state functioning. First of all, there is little evidence that rural private schools provide substantially better quality of education in terms of teaching methodologies and the learning environment etc., though they may claim to teach English, open regularly, make children wear uniforms etc. The lack of understanding about the educational process appears to extend to these schools.⁹² In fact, the quality of education imparted by private schools is as yet a highly under-explored area which needs to be researched and discussed in greater detail.

Additionally, privatization of education needs regulation. Our study showed that unregulated privatization, as had been attempted in Rajasthan, has the potential of creating a new set of serious problems. At the same time, regulating privatization invariably involves state agencies. This means that the problems of state functioning cannot be sidestepped, but are likely to be reproduced.

⁹² As this is written, a newspaper in Madhya Pradesh reports that no student could pass the eighth grade examination from 36 private fees paying schools in Bhopal district alone (*Dainik Bhaskar*, 12 May 2007, page 2).

An argument that has emerged in recent years revolves around state funding of education with the privatization of the delivery of services. The proponents of the 'voucher system' argue that the delivery of education should be privatized totally. As per this system, the government would simply provide each child with a voucher which would enable her to attend any (private) school and the government would simply reimburse a certain stipulated sum to the school on the basis of the vouchers presented. It is argued that children will opt for better functioning schools and ultimately only such schools will survive. In other words, schools would be set up and managed by private players, but funded by the government. Each child would be able to choose the school she wishes

The privatization of delivery of services as envisaged in the 'voucher system' requires strong regulation by the state. There would be the need to intervene in the case of schools that offer short-term gains such as organized cheating during examinations or an education oriented narrowly to cracking exams, as well as schools promoted by undesirable players. Moreover, the state machinery would need to be efficient enough to ensure that each child actually gets a voucher (and only one voucher). Fraud schools, where no or minimal teaching takes place and the benefits of the voucher are simply divided among school authorities and students with indigent parents co-opted, cannot be ruled out. These types of malpractices would need to be prevented. However, such regulation is impossible unless state practices improve.

In fact, privatization along with unreformed practices within state institutions could increase rent-seeking rather than reduce it, as accountability would become diffused. For instance, if a government school runs badly, the government is obliged to take responsibility if parents, civil society or the media protest. But if a school to which students provide government vouchers runs badly, the responsibility is ill-defined and public accountability is reduced. At the present juncture, where the need to increase public accountability is paramount, the creation of mechanisms that reduce such accountability is likely to compound, rather than solve the problem.

Changing state institutional structures

Finally, there have been some attempts at changing the system itself and the two major thrusts in this direction have been decentralization and people's participation on the one hand, and the creation of project structures on the other.

Decentralization and people's participation

Decentralization and people's participation have been seen as antidotes to many problems of the state, not merely in elementary education but in governance in general. A scrutiny of policy documents since the mid-1980s in India shows that decentralization and people's participation, especially through local government, have been envisaged as increasing accountability, making public institutions more responsive and creating space for the voice of the poor (Sharma, forthcoming). Decentralization of education in particular has been seen as a modality on the one hand for gaining the support of the community and on the other, improving school management through greater accountability to the people on the other.

There are, however, several critical issues here. First, local elected government is still a contested institution in India. Though since 1993 holding local government elections has become mandatory, in actual fact these bodies have been empowered to different degrees across states, and within a state, this scenario shifts across time. Local government is in a state of flux and sometimes other types of community-based bodies are used to weaken them. In this study at the time of field work, local government in both Andhra Pradesh and Rajasthan had been weakened. In Rajasthan, several powers had been delegated to School Development and Management Committees which were *nominated* bodies. This study showed them to be defunct with members merely signing on various papers as instructed by the school

headmaster. The elected School Education Committees in Andhra Pradesh were more active and people's participation was more visible in the sample villages. However, the encouragement provided to these bodies by the previous government had ceased as the government of the day saw them as agents of the previous one. As this book is written, the composition of these bodies too had been changed to include several nominated members.

It is also important to recognize here that in actual fact local governments often reflect the macro-governance scenario for various reasons. Local governments are shaped by the larger socio-political and governance structures, and the governance practices within them are often similar to those of the macro-structure (Sharma, forthcoming). In this study too, local government representatives were caught up in patronage based and rent-seeking practices as were the centralized agencies. Moreover, the understanding that local governments or other local agencies bring to elementary education is extremely limited. For instance, the SECs in Andhra Pradesh had limited themselves to building and infrastructure related issues and several SEC members were in favour of beating children to make them learn.

To sum up, local government is as yet a contested institution in India, and 'decentralization' usually implies the transfer of power to rather tentative structures. Though decentralization of education is seen as a solution, in actual fact there are no strong, viable grassroots institutions to which education can be decentralized across most of the country. Moreover, local governments tend to reflect the characteristics of the macro-governance structure and their capacities vis-à-vis education tend to be limited. While decentralization and building strong grassroots structures may still remain desirable goals for the future, given the present institutional structure and ethos, decentralization is unlikely to bring about rapid change.

Project structures

In recent years there has been a tendency to create 'parallel' structures to implement projects, i.e., within government agencies separate from the usual departmental structure have been set up. Such agencies have then attempted to induct more competent personnel (often from within the government) and make greater space for tapping professional expertise. However, the study showed that though some project structures have been able to function more professionally and vibrantly than regular departmental structures for short periods of time, such attempts have not succeeded over the longer term. Patronage based placements and the practices of the formal as well as informal systems tend to be adopted in these projects too and unprofessional practices follow.

The Way Forward

The scenario painted in this book shows that while change is essential, changing the system is difficult. However, this does not imply the inevitability of a sub-standard education system. Rather, it implies the need for a carefully thought out and nuanced strategy which addresses core, rather than peripheral, issues. The findings of our study bring out the most critical systemic constraints. We now sketch the contours of a more effective strategy to counter these problems.

The centrality of systemic reform

The results of this study indicate that questions of systemic capacity and functioning need to be brought centre-stage for the provision of high quality elementary education for Indian children. Policy changes and new programmes are not enough, as systemic problems permeate every policy and programme to the point that policy interventions have become increasingly marginalized in the system. In other words, *the state system has to be reformed*. This may appear to be an 'obvious' conclusion. Yet, policy discussions and debates about elementary

education in India rarely recognize the *centrality* of systemic reform and its overriding importance. New programmes, rather than changes in the core practices within the system, tend to dominate the discourse. To begin with, it is important that the centrality of systemic reform and its overriding importance be recognized.

As stated earlier, systemic reforms need to *address the core problems identified within the system*. Some of the solutions that are being offered at present simply do not appear adequate. For instance, privatization or a shift in agencies does not solve the problem, as practices within the state affect the private sector, just as they affect all other agencies. In fact, privatization or shifting agencies away from government can often offer new opportunities for rent-seeking, and can thus exacerbate, rather than solve problems. Similarly, new structures within the state apparatus may succeed for some time, but tend to adopt the mainstream practices in the long run. More monitoring in various forms is not likely to yield results either, as the monitoring system too becomes part of the problem. Reforms that focus only on technical up-gradation, management techniques etc. tend to run into similar problems too.

The approach to systemic reform consequently, has to be more focused. This study shows that problems arise from the stated management practices of the system, as well as from the existence of a powerful 'informal system' based on patronage-based functioning and rent-seeking. There are huge gaps of professional expertise, and of resources. Uneven availability of teachers and infrastructure instead of an appropriate provision of the same across the system, hurts the poorest children the most. These core problems need to be addressed for any real shifts to come about in the elementary school education system.

Addressing the informal system

The importance of the informal system

An issue that needs to be considered carefully is the importance of the informal sub-system, i.e. patronage and rent-seeking networks within the system, and the fact that the existence of this sub-system stymies change. For instance, attempts to introduce flexibility deteriorate all too easily into corruption, and rigidities are then re-introduced. Teacher training, meant to upgrade professionalism, can get taken over by rent-seeking. In other words, systemic reform is not possible without addressing the unstated practices within the system. Yet, questions of patronage and rent-seeking are often seen as issues subsidiary to the more important questions of policy thrusts and management-oriented reforms. Their capacity to jettison almost any initiative does not get due recognition. In fact, attempts at change, i.e., major policy shifts and the restructuring of the formal system will be meaningful only if the informal system too is brought into view and interrogated. Effective systemic reform needs to address rather than ignore this issue.

Addressing the problems of the informal system is not an easy task for it is a prevailing concern across government institutions in India. However, the breadth of the problem does not reduce the need to address it in specific sectors. Because various government departments deal with different groups of people as well as activities, the dynamic of rent-seeking behaviour and its consequences also shift. For example, there has been some success in controlling rent-seeking in the telecom sector in recent years, and the fact that in this sector the government interfaces with some of the most powerful individuals and companies in the country is of no mean significance. In contrast, the interface of the elementary education system is with the least powerful members of socially disadvantaged groups, i.e., poor children. On the one hand, rent-seeking behaviour tends to go unchecked because of the powerlessness of the social groups involved and on the other, rent-seeking affects the most vulnerable children and shapes not only their present but also their future. The social consequences of rent-seeking in elementary education are enormous, and very long term.

Rent-seeking poses some critical questions about the use of public funds. First, there is the simple question as to whether or not those in positions of power can appropriate public funds for their own use. In theory, the answer to this question may be obvious, but in practice it is far from so. Rent-seeking implies a subversion of democracy because it means that funds are not used as voted by the parliament, but arbitrarily by those in positions of power. Rent-seeking lays a strong basis for perpetuating inequity because those who can afford to pay, or have clout, can subvert the system in their favour. In other words, rent-seeking vitiates democracy itself and needs to be viewed in this light. At the present juncture, sections within the government and also outside, are aware of the magnitude of the issue, but there are also many well-meaning people who have not come to terms with its full impact and significance.

An important point that is often missed is that the informal system is made effective through certain practices *within* the government, notably the placement of personnel. Consequently, it is important to recognize that government personnel policy is not merely a problem of those within the government, but is a *public* issue. Thus the 'transfer and posting' processes and the selection of institutional leaders are not merely 'internal' management processes, or the professional successes and failures of individual government employees, but are important decisions about whether or not the *public funded* education system is going to be managed well. Government personnel policies need to maximize efficiency and effectiveness so that public funds can be use optimally. The 'internal' personnel management policies of the government are, consequently, issues of public concern. We have argued that the personnel placement in the government revolves around the development of patronage networks for the maintenance of power and rent-seeking. The point being made here is that the effects of this practice are not limited to a few people who work within the system but have much wider consequences. It has become a powerful instrument for appropriating public funds and for blocking and vitiating quality and equity in the school system.

Possible strategies

The informal system can be whittled be whittled by increasing accountability. This has been recognized by several civil society groups. Yet, the type of accountability that these groups demand is often 'output' oriented, i.e. in terms of what the government does or does not do, school results etc. The processes that lead to poor outcomes are often seen as the 'internal' practices of no interest to outsiders. Yet output oriented accountability can be self-defeating, for it rarely reaches the source of the problem. For example, in recent debates and studies teacher absenteeism and other such issues, visible at the level of the school, have become the focus of attention. Yet such scrutiny often hides more than it reveals, as the reasons for poor school functioning remain obscure, and therefore un-addressable. What is needed is a greater scrutiny of intermediating, 'internal' practices such as personnel management or financial dealings, which promote the poor functioning of schools.

Accountability can be strengthened in two ways: by fortifying internal accountability mechanisms, and by increasing outside scrutiny and pressures. Increasing internal accountability has limited, but definite, potential. The potential for strengthening internal accountability mechanisms is limited because it involves confrontation with extremely powerful individuals and institutions that can 'capture' the internal accountability mechanisms created. However, in spite of this limitation, internal accountability mechanisms need to be stressed since it is through various practices within the system that the patronage based, rent-seeking proclivities are supported. An important point here is that attempts, often made by well-meaning individuals, to make rules more rigid and processes more complex in order to ensure accountability usually do not work. They merely serve to make government processes more cumbersome and ways are found around them to promote rent-seeking anyway.

Internal accountability is difficult to ensure precisely because it is 'internal'. Here it needs to be noted that far as teacher placement is concerned, the 'counselling' system introduced in Andhra Pradesh, through internal accountability mechanisms, is a breakthrough. However, it is also quite rigid, and is not a viable solution at all levels, as at higher levels, placements need some understanding of an individual's expertise and experience. But the Andhra Pradesh experience indicates that the involvement of a larger number of people in the placement process, as well as spelling out the criteria for placement clearly, may address the issue to some degree. Thus one solution would be to shift to group decision-making in which people outside government may also be nominated, as against processes that are heavily dependent on the decisions of one or two individuals. However, even this would have its limits, as given the widespread incidence of rent-seeking, collusion among a fairly large number of people including alliances with individuals outside government, are possible. Indeed, rent-seeking in government is often made possible with the support of individuals outside. We saw in our study the collusions with private actors as well as so-called NGOs.

One possible solution here is the creation and strengthening of independent bodies with specialized knowledge of various processes that can act as watchdogs. The Indian experience shows that in certain cases, the creation of independent bodies, which scrutinize practices within the system, and which ensure that the rules of the game are followed, can be successful. The higher judiciary and the Election Commission are two institutions that have intervened successfully to reduce arbitrariness in public life. However, both these institutions have a high degree of autonomy and stature. Institutions with less autonomy and stature, created to curb rent-seeking, have too often been ineffective. It would appear that the creation of genuinely independent bodies of a certain stature, or strengthening of existing ones, which can constantly point out lacunae in various processes, is one way in which internal accountability can be increased.

To be effective, internal accountability mechanisms would need to be supported by the activities of groups outside the system that are genuinely interested in curbing patronage based, rent-seeking practices. As yet, interest in these aspects of those outside the system has been small. While 'policy issues' such as the nature of the curriculum, or accessing external funding for elementary education, are often debated hotly, the fact that some small tribal child may not get her textbook at all (no matter what the curriculum), because someone has appropriated the funds for it, is generally not scrutinized. This lack of scrutiny smoothens the operation of the informal system, for no one questions it. It subverts and vitiates Indian democracy, as the capability of its institutions to deliver as per policy, i.e., decisions taken through the democratic process, is reduced and affects its poorest citizens adversely. Yet, this phenomenon goes largely unquestioned.

The two exceptions here are the highlighting of more dramatic cases by the media and the work of some NGOs engaged with the Right to Information. However, these are not adequate to counter the powerful patronage and rent-seeking networks that exist within the government. Much of the informal system operates through 'routine' activities, which are unlikely to be newsworthy. Right to Information has its limits, partly because it is usually quite difficult to obtain documents that actually illustrate or prove illicit activity (paper work is usually in order), and partly because even when some cases of unlawful or dishonest activities are well known, it can be quite difficult to go further than getting information, i.e., get an active response in terms of measures to prevent such incidents in the future. Media interest as well as Right to Information are valuable, but are not enough.

What is additionally needed is a much greater discourse on the processes within government, their transparency and accountability. It is not enough to unearth specific cases of corruption but to constantly attack the mechanisms through which corruption routinely takes place. For this to happen the scrutiny of government processes may require greater attention to detail than is the practice at present. In fact, it may require a large shift in orientation for civil society

organizations interested in this issue, from a broad criticism of the 'government' as a whole, to a more nuanced scrutiny of government processes and a more focused and accurate critique, directed at specific practices. It may also require vocal *support* from civil society organizations for some types of government processes, such as counselling for teachers' postings in Andhra Pradesh, which cut away at the scope for patronage.

Additionally, in public discourse we may have to learn to identify various actors in public life not only in terms of professional groups, e.g., 'academics', 'activists', 'journalists' and 'bureaucrats' and in terms of their political affiliation, i.e., as left, middle or right wing, but also in terms of their integrity. This may seem 'obvious', but in practice is far from the case in public life, where ethical lapses can be quite easily overlooked if there is professional or political affinity. That a journalist and a bureaucrat who believe in an ethical government and practice their profession ethically, have more in common than two journalists who differ in terms of ethical practices, is often not recognized. Individuals in public life are rarely categorized as 'corrupt', or 'not corrupt'. Given the scope and importance of the issue, it may become necessary to develop a new lexicon.

The formal system

As this study shows, the restructuring of the formal system is necessary. However, it needs to be underlined that such restructuring, when attempted without a simultaneous diminution of the informal system, can simply become a part of the informal system. It may then become meaningless. For example, technology up-gradation and modern management techniques, superimposed on a rent-seeking apparatus are unlikely to bring about anything more than very superficial alterations. Reform of the formal system, without addressing the informal one, can even backfire. For instance, attempts to make the system more flexible and capable of responding to emerging needs in a context where patronage based functioning and rent-seeking remain dominant, may merely promote the informal system, for example in the form of 'flexible' patronage based promotions on the pattern of transfers and postings. To the extent that the informal sub-system remains a significant presence, the space for reform of the formal sub-system is made correspondingly smaller.

Within the formal system, the question of professionalism is the most critical. A very important constraint of the elementary education system in India is a lack of comprehension and an informed professional discourse about teaching, learning and pedagogic practices. This lack of concern and comprehension about the process of learning and education exists not only at the school level, but also pervades the apex institutions that engage with these issues. The reasons for this situation are complex.

The first structural constraint in creating professional rigour in elementary education is that this area occupies almost no space in the higher education system. Training of elementary school teachers is carried out in DIETs, which are very weak institutions rather than in universities. Moreover, at the university level there are few programmes that address education as a discipline, which acts as a constraint in the creation of a systematic body of knowledge. Moreover, the system simply does not have mechanisms for developing well trained educationists, i.e., curriculum and textbook developers, specialists in gender or even language and science teaching and educational managers for elementary education. Some exceptions to this rule are emerging. The Delhi University runs a Bachelor of Elementary Education course, which imparts undergraduate level training for elementary school teachers. Recently, the Tata Institute for Social Sciences has started a Masters programme to train professional educators. But the need is far greater. Supporting programmes that focus on elementary education in universities and other resource institutes is very important for sustainable professionalization. Long-term planning to lay the basis for greater professionalism in school education is clearly essential.

The second structural constraint is that state systems also do not recognize the need for increasing professionalism within the system. A major contributor is the lack of a large enough body of professionals in this area. Since there are very few trained professionals, the whole idea that a more professional approach to elementary education is needed is rarely even discussed. SCERT and DIETs remain neglected as a consequence of this lack of awareness, as well as the vested interests of the informal sub-system.

While to some extent the professional up-gradation of SCERTs and DIETs is dependent on the longer-term development of overall professional expertise in school education, it is also possible to bring about significant shifts in this situation in the short-term. Rationalization of the existing placement policy, along with intensive capacity building can yield significant results. Capacity building is essential not just at the level of the teachers but also at the apex levels, i.e., the directorates and SCERTs. Such an initiative can, in a relatively short time, have a large impact. Second, within the government system space can be created for the induction of available professionals in apex institutions. Once professional practices are established in apex state level organizations, these institutions can support such practices in schools. However, it is important to note here, that the professionalism has to be genuine, i.e. well conceptualized and context-specific. Given the complex Indian context, the simple borrowing of ideas and practices from elsewhere is not likely to work. Professional experience and knowledge generated through work in Indian schools will be essential.

The management processes within the system need to shift. The extreme hierarchical and authoritarian patterns of functioning are a barrier to organizational learning and capacity for problem solving and clearly need to be addressed. The hierarchical patterns of functioning need to be replaced by deliberative ones and the question of institutional restructuring and leadership needs to be addressed in detail and with care. This would require work at two levels. The first would be a detailed scrutiny of existing government processes to increase the space for deliberation and informed decision-making. At present, the decision-making processes are structured so that authoritarian, rather than deliberative decision-making patterns are facilitated. Secondly, it would be necessary to orient officials and manager towards new ways of working, i.e of acting after understanding the situation, consulting with other professionals, rather than as per the 'command-style' system. Notably, some types of management reforms, such as 'hire and fire' policies, may be counter-productive, as these can not only increase the play of the patronage, but also strengthen the hierarchical, authoritarian tendencies within the system.

Strengthening grassroots offices and institutions is another key area for reform. The structures and processes of these institutions are a critical in determining the manner in which the system relates to society and the type of capacity that it has to act on its decisions. Understaffed, under-equipped offices and poorly trained and disempowered personnel in grassroots institutions pretty much jinx the chances of effective action. Grassroots institutions need to be viewed not as the lowest step in the hierarchy ladder, but as critical interfaces of the government with the people. In a multifaceted and changing society, the task of such institutions tends to be extremely complex. And it is in these terms that grassroots institutions need to be re-envisioned and developed. The personnel in such institutions need to be selected keeping in mind the needs of the work expected, and not in terms of their suitability to function at the bottom of a hierarchy. These institutions also need to be equipped accordingly.

Yet another important area of work within the system is of increasing its responsiveness to the community. Greater responsiveness demands a two-pronged approach. First, there needs to be consistency in policy towards community based institutions as well as local governments, so that such institutions can become genuine forums for the articulation of community concerns. Some attention also needs to be paid to the institutional design of these forums. Notably, for some types of community institutions in India, such as self help groups, the development of

rigorous processes has enabled the development of strong, sustainable community based organizations. A similar attention to the processes of community based institutions that deal with education would be necessary. Second, teachers and headmasters need to be trained to establish relationships and work with such organizations and create a dialogue about education with such bodies. At present, the assumption often is that community based bodies would somehow automatically be able to act to improve elementary education. In fact, this can happen only if a dialogue between the schools and such bodies has been established.

Here, it is necessary to understand the relationship of the process of community engagement with an improved internal functioning of the system. At present, community linkage has been attempted largely without internal reform. This leads to the community based organizations becoming part of the problem by joining rent-seeking nexuses, or becoming indifferent defunct, as the school cannot improve because of the larger systemic dynamic. A genuine strengthening of community-school linkages is possible when the internal dynamic of the system has shifted towards professional and societal, rather than ulterior, goals.

Redefining the space for policy

As mentioned earlier, because of the existence of a strong informal system and the constraints of the formal system, the space for 'policy' itself has become small. This does not imply that policy *per se* has become meaningless, but instead entails the need to take into account institutional characteristics and dynamics described in this book while formulating policies or planning. The present practice is of conducting policy and planning discussions and exercises without taking systemic characteristics into account, and subsequently laying the blame of policy failure at the door of 'implementation'. The naïve statement, that 'the policy/ plan was good, but the implementation was poor', is made liberally in policy and planning circles in India.

The fact is that any policy or plan that is oblivious to the existing reality is no more than a wish list. A policy to train teachers on a large scale which ignores the limited ideas about education within the apex system, or one to appoint para-teachers through informal mechanisms while ignoring the likelihood of patronage based selections, runs into the inevitable and foreseeable problems of 'implementation'. Teacher training tends to be below standard and not as originally envisaged and para-teachers, rather than being people who are willing to work voluntarily for the community, tend to be people connected to various powerful actors. The problem here is that policy makers as well as their critics often assume some characteristics of the government which do not exist in reality. Policies are often debated and introduced with the assumption that the institutions function optimally and the informal sub-system does not exist. In fact, when policies are conceived without keeping in view the institutional and organizational characteristics in which they play out, they can sometimes unwittingly widen the scope for the informal system. The recruitment of para-teachers is a case in point, where the ideas of recruiting teachers from the community have resulted in the de-formalization of recruitment processes and provided a greater scope for patronage based appointments.

A redefinition of 'policy' is clearly required in the Indian context. First of all, policy needs to address some core issues that confront the government today. A basic question that has not been addressed is whether people who are part of the system will appropriate public funds for their own use or not. As stated earlier, while the answer to this question may seem obvious in theory, in practice public funds are *widely* appropriated by individuals across the system. Consequently, an appropriate policy to ensure that public funds are not misused becomes a priority. Similarly, instead of an assumption that the institutional structures and processes are satisfactory, what are needed are policies for institutional reform.

While on the one hand, the areas of policy concern have to be redefined, on the other, all policies need to keep the existing characteristics of the system in mind. For instance, at the present juncture many times it may be preferable to tolerate rigidities because ‘flexibility’ creates the space for misuse. Teacher transfers in Andhra Pradesh have become extremely rigid, but this rigidity is preferable to patronage-based transfers. Ideally, processes that combine flexibility with accountability should be developed, but at this point of time, flexibility gained at the cost of accountability would be expensive.

The need for financial investment

The need for greater financial investment remains. Learning from within the country, the early success in elementary education in Kerala as well as the more recent one in Himachal Pradesh, have been backed by a much higher levels of investment in elementary education than in the two states under study in this book. In this case, the declining trend of expenditure on education in Andhra Pradesh, is a matter of concern.

Attempts were made in the previous decade to resolve the investment gap in education by reducing the cost of education, especially by hiring para-teachers. In fact, as Chapter 3 of this book shows, a major problem with the elementary education system in India has been chronic under-funding. Attempts to reduce the cost of education only compound the problem. If the goal is to make elementary education universal and equitable as per the constitutional mandate, then what is needed universally is a vastly improved classroom. In the present scenario, a school with multi-grade teaching, a bare overcrowded building, no furniture and paltry teaching learning material is the norm. This ‘standard’ needs to change, and this is not possible without increased investment. Our study showed low cost schools to be highly problematic. In fact, low cost schools not only go against the principles of equitable education, but are also a poor financial investment since many of them do not impart any education at all.

However, an important point here is that as in the case of making changes in the formal processes, the returns from an increase in investment are contingent on the diminution of rent-seeking. As Chapter 3 shows, Rajasthan has increased its expenditure on education as a proportion of GDP as well as state income in the last decade. However, the new RGSJP schools, which have been funded with these increases, do not impart meaningful education. Up-gradation of the physical infrastructure in schools, when teaching time is very low, is of little use. Over the long run, large scale rent-seeking cuts away at support for increased investment in education as the money is seen as having little or no impact. With low investment, the quality of education deteriorates further, and a vicious cycle sets in.

Addressing inequity

The elementary education system in India is situated in the context of stark social inequities, a fact which needs to be appropriately recognized. At present, the hierarchical ordering of the system on the one hand, and the illegitimate exercise of power by privileged individuals through the informal system on the other, aligns the system to serve, rather than counter these inequities. To a great degree, systemic reform and a better functioning state-run education system will address the question of inequity, because, as Chapter 7 shows, the inadequacies of the system affect the poor the most. Children from better-off families move to private schools, or supplement schooling with tuition, and it is the poor who bear the brunt of the poor functioning of the system. If government schools improve, even poor children will be able to access good education, like their better-off counterparts.

However, general improvement across the system is unlikely to be enough and the system will need to move towards countering social inequities in more specific ways. At present, a dual approach is visible across the system. On the one hand there are measures that counter

inequities, such as midday meals, free textbooks, scholarships for SC/ST and other poor students etc. Such provisions are essential. A further reduction in household expenditure, i.e., a more complete provision in the form of notebooks and other teaching-learning material to poorer children would be an important aspect of equity. Better implementation of such programmes is also necessary.

On the other hand however, these measures to counter inequity are negated by the fact that the quality of schools tends to build upon existing inequities by providing lesser quality schools to poorer children. This negates the student benefits provided. In fact, scholarships provided by a school which does not function well amounts to a general subsidy to the family and not an increase in educational opportunity. At the same time, the education of children from marginalized communities requires a greater degree of skill, and teachers teaching such communities need to be trained rigorously. Children from such communities may need more support in terms of teaching learning materials etc. Thus schools from children from marginalized communities may need more, and not less than average funding. This is in stark contrast to the present situation where the quality of the school tends to 'match' the socioeconomic status of students, rather than ensuring that the social deprivations faced by these students are countered with their education.

A Last Remark

The Constitution may envisage elementary education as playing a certain kind of role in social transformation, but the extent to which it actually does so, is dependent on the characteristics of the system itself.

We have attempted in this book to sketch the systemic dynamic. We have argued that if the elementary education system is in fact to play the role that it is envisaged to play in the Constitution then institutional and systemic reform is a key concern for the future. We have argued further for the need for reform that addresses *core*, rather than peripheral, issues. These core issues take us into complex terrains, where no easy, 'quick-fix' solutions are possible. New programmes, technical up-gradation, or training, are not adequate. The issues that have to be addressed are larger: a basic acceptance and promotion of inequity within the system, the question of protecting the public sphere from individual greed, and the whole issue of professionalism. These problems are intertwined, and one cannot be addressed without making some headway in the other. Change is possible only by going to the heart of the matter in different ways, i.e., by authentic reform of government processes, by developing ideas about education and learning across the system and by measures based on a genuine comprehension of the needs of the marginalized.

Bibliography and references

1. Aggarwal, Yash. (1999). 'Trends in Access and Retention', National Institute of Educational Planning and Administration (NIEPA), New Delhi.
2. Aggarwal, Yash. (2000a). 'An Assessment of Trends in Access and Retention', NIEPA, New Delhi.
3. Aggarwal, Yash. (2000b). 'How Many Pupils Complete Primary Education in Five Years?' NIEPA, New Delhi.
4. Aggarwal, Yash. (2001). 'Trends in Access and Retention', NIEPA, New Delhi.
5. Aggarwal, Yash. (2002). 'An Assessment of Trends in Access and Retention', NIEPA, New Delhi.
6. Akbar, M.J. (2006). 'The Culture of Money', The Asian Age, 4 June.
7. Andhra Pradesh School Education Society (2005-06): Sarva Shiksha Abhiyan, State Component Plan.
8. AP Gazette (Extraordinary). (1994). Andhra Pradesh Educational Institutions (Establishment, Recognition, Administration and Control of Schools under Private Management)- Rules 1993 (dated 3.1.1994).
9. APDPEP (1999a) Teachers' Centres a Qualitative Study by K. Laxma Reddy
10. APDPEP. (1999b). School and Teacher Grants: A Qualitative Study by G, Narsimha Reddy (Unpublished Mimeo)
11. APDPEP (2003), Annual Report, Sarva Shiksha Abhiyan, 2002-03.
12. Ayyar, RVV. (1995). 'Let Not the Dead Seize the Living: The DPEP Saga', DPEP Calling. (December).
13. Bajpai, Nirupama and Sangeeta Goyal. (2004). 'Primary Education in India: Quality and Coverage Issues', CGSD (Columbia University) Working Paper No 11, (February).
14. Bardhan, P. (2001). 'Sharing the Spoils: Group Equity, Development, and Democracy', in Atul Kohli (ed.) The Success of India's Democracy, Cambridge University Press, Cambridge.
15. Bashir, Sajita. (2000). Government Expenditure on Elementary Education in the Nineties. European Commission, New Delhi.
16. Basu, Anuradha. (1995). Public Expenditure Decision Making: The Indian Experience. Sage. Publications, New Delhi.
17. Behrman, Jere R., and Nancy Birdsall. 1983. The Quality of Schooling: Quantity Alone is Misleading, *American Economic Review* 73(5):928-46.
18. Bharat Patni (2006). From QIP to CLIP in Kameshwari Jandhyala and Vimala Ramachandran: Enabling most deprived children to learn: Lessons from promising practices. Report submitted to ILO, New Delhi (Unpublished).
19. Bhargava P. and Sharma, R. (2004). Assessing the Functioning of Block Resource Centre, Cluster Resource Centre and School Development and Management Committee in DPEP, Institute of Development Studies (Unpublished report), Jaipur.
20. Blaug, Mark. (1972). An Introduction to Economics of Education. Allen Lane the Penguin, London.
21. Cameron, Robert (2003), Politics-Administration Interface: the Case of the City of Cape Town in *International Review of Administrative Sciences*, Volume 69 (1), March.
22. Census of India. (2001). Provisional Population Totals, India Paper 1, Web edition.
23. Centre for Policies Studies. (1999). DPEP Social Assessment Study – Working Children, Hyderabad 1999

24. Chalam, KS. (1999). 'Promoting schooling among scheduled caste children in Andhra Pradesh', study commissioned by DPEP Andhra Pradesh, Vishakapatnam (October). Chaturvedi, A. (1987), Inter-Departmental Dynamics: Relations between Four State Government Departments at the District Level. *Economic and Political Weekly*, Vol. 22 (9), February 28, pp M9-M19.
25. Chibber, P. (1995). 'Political Parties, Electoral Competition, Government Expenditures and Economic Reform in India', *The Journal of Development Studies*, Vol. 32, No.1, October. PAGE NOS?
26. Choudhary, S. (2000). UEE in Rajasthan : A study with Focus on Innovative Strategies. MHRD & NIEPA. New Delhi.
27. Clarke, P. (2001). Teaching and learning – the culture of pedagogy. New Delhi: Sage Publications.
28. Commissioner and Director of School Education, Andhra Pradesh. (2000). Selected Educational Statistics 2000-01, 2004-05, Hyderabad
29. Commissioner and Director of School Education, Andhra Pradesh. (2003). Recruitment of Teachers, District Selection Committee 2003. Hyderabad
30. Corbridge, S. and John Harris. (2001). Reinventing India: Liberalization, Hindu Nationalism and Popular Democracy. Polity Press, United Kingdom.
31. CSO: Central Statistical Organisation (2005). National Accounts Statistics 2005. New Delhi.
32. Dayal, I., Kuldeep Mathur and Mohit Bhattacharya. (1976). District Administration: A Survey for Reorganization. The MacMillan Company of India Limited, Delhi.
33. Dayaram (2002). Parateachers in Primary Education: A Status Report, Unpublished Mimeo. Educational Consultants India Ltd. New Delhi 2002
34. De, A., C. Narhona and M. Samson. (2001). 'Private Schools and Universal Elementary Education', South Asia Education Sector Technical Working Paper No. 3. World Bank.
35. De, Anuradha et.al., (2000). Role of Private Schools in Basic Education, NIEPA and MHRD, Government of India. NAMES OF ALL AUTHORS?
36. De, Anuradha, Claire Noronha and Meera Sampson. (2001). 'India: Primary Schools and Universal Elementary Education', India Education Team Report No. 3. World Bank, New Delhi.
37. Dev, S. Mahendra and J. Mooij. (2002). 'Social Sector Expenditures in the 1990s: Analysis of Centre and State Budgets', *Economic and Political Weekly* 37 (9) (March 2-8): 853-66.
38. Directorate of School Education. (1994), Report of Main Survey 4, Andhra Pradesh Primary Education Project on Implementation of the Project in Schools. (Unpublished Mimeo)
39. DPEP. (1999a). Meeting Challenges: Documentation of Positive Practices in Four DIETs. GoI and Gender Unit of Ed.CIL, New Delhi.
40. DPEP. (1999b). Reaching Out Further: Parateachers in Primary Education. Delhi:
41. DPEP. (1999c). Glimpses from the Grassroots. Educational Consultants India Ltd.
42. DPEP. (2001). Inside the School. Educational Consultants India Ltd.
43. DPEP. (2003). Shifts in Learner Evaluation. Educational Consultants India Ltd.
44. Dreze, Jean and Amartya Sen. (1995). India: Economic Development and Social Opportunity OUP, New Delhi.
45. Dreze, Jean and Amartya Sen. (2002). 'India: Development and Participation', Oxford University Press, New Delhi, pp35-36.
46. Dreze, Jean and Amartya Sen. (eds). (1996). Indian Development: Selected Regional Perspectives OUP, New Delhi.

47. Dreze, Jean and Geeta Gandhi Kingdon. (1999). 'School Participation in Rural India', The Development Economics Discussion Paper Series, No. 18. London School of Economics. London.
48. Dreze, Jean and Haris Gazdar. (1996). 'Uttar Pradesh: The Burden of Inertia', in Jean Dreze and Amartya Sen (eds), Indian Development: Selected Regional Perspectives. OUP, New Delhi. PAGE NOS?
49. Dwivedi, O.P., R.B. Jain and B.D. Dua. (1989). 'Imperial Legacy, Bureaucracy and Administrative Changes', Public Administration and Development, Vol. 9, pp253-269. NO?
50. Education Commission. (1966). Education and National Development: Report of the Education Commission 1964-66. NCERT, New Delhi. (Reprint 1971).
51. EPW Research Foundation. (2004). Domestic Product of States of India. Sameeksha Trust, Mumbai.
52. Firdausi, A.A., Raju, B.Y. Study and G.M. Reddi. (1999). Study of the District Level Support for DPEP in Andhra Pradesh (Unpublished study).
53. Fuller, Bruce (1990) *What Investments Raise Achievement in the Third World*, in David W. Chapman and Carol A. Carrier (eds.) *Improving Educational Quality A Global Perspective*. Portsmouth, NH, USA: Greenwood Press.
54. Fuller, Bruce; John H. Y. Edwards; and Kathleen Gorman (1986) When Does Education Boost Economic Growth? School Expansion and School Quality in Mexico, *Sociology of Education*, 59 (3) (July): 167-181
55. Ghosh, Suresh Chandra (1995). The History of Education in Modern India 1757-1986. Orient Longman Ltd. New Delhi
56. GOI, 2007. Select Education Statistics, Department of Education, MHRD, New Delhi
57. Government of Andhra Pradesh (2005b). Plan Schemes 2005-06, Department of School Education, GoAP).
58. Government of Andhra Pradesh. (1962). The Andhra Pradesh Educational Service Rules. G.O. Ms. No. 359, General Administration (Rules) Department (dated 9.2.1962).
59. Government of Andhra Pradesh. (1971). G.O. No. 1781 dtd. 27-11-197, G.O. No. 162, dated 20-2-1976, Memo No. 262-K2/77-1 (dated 17.1.77).
60. Government of Andhra Pradesh. (1978). Scheme for 7th Class Common Examination. G.O. Ms.No. 234, Education (K) Department (dated 14.3.1978).
61. Government of Andhra Pradesh. (1992). The Andhra Pradesh School Educational Subordinate Service Rules. G.O. Ms.No. 40, Edn. (Ser II), Dept. (dated 7.2.1992). Government of Andhra Pradesh (1994), AP Gazette (Extraordinary) dated January 3, 1994: Andhra Pradesh Educational Institutions (Establishment, Recognition, Administration and Control of Schools under Private Management)- Rules 1993
62. Government of Andhra Pradesh. (1995). Reforming School Education: Report of the Ramakrishna Rao Committee. Hyderabad.
63. Government of Andhra Pradesh. (1999). Andhra Pradesh: Vision 2020. McKinsey & Company for Government of Andhra Pradesh, Hyderabad.
64. Government of Andhra Pradesh. (2000-01). SCERT: Structure, Functions and Achievements. 2000-01.
65. Government of Andhra Pradesh. (2001). The Andhra Pradesh Direct Recruitment of Teachers (Scheme of Selection) Rules 2000. Published in the AP Gazette, Part1, Extraordinary (dated 23.1.2001).
66. Government of Andhra Pradesh. (2004). Performance Budget 2004-05. Hyderabad.
67. Government of Andhra Pradesh. (2005). Proceedings of the Director of School Education. Rc. No. 398/Mdm-1/2005 (dated 19.5.2005).

68. Government of Andhra Pradesh. (2005a). Rules to Regulate the transfer of teachers of AP School Educational Subordinate Service. G.O. MS. No. 102 (dated 5.8.2005).
69. Government of Andhra Pradesh. 1982. The Andhra Pradesh Education Act 1982 The Andhra Pradesh Education Code Vo.1 and Vol. 2. Panchayat Publications. Hyderabad.
70. Government of Andhra Pradesh. 1988. The Andhra Pradesh Educational Institutions (Inspection and Visits) Rules, 1988 in Reddi, PR. The Andhra Pradesh Education Code Vo.1 and Vol. 2. Panchayat Publications. Hyderabad.
71. Government of Andhra Pradesh. 1998. The Andhra Pradesh Education Code Vo.1 and Vol. 2. Panchayat Publications. Hyderabad
72. Government of Andhra Pradesh. 1998. The Andhra Pradesh School Education (Community Participation Act), 1998.
73. Government of Andhra Pradesh. The Andhra Pradesh School Education Society (2005-06): Sarva Shiksha Abhiyan, State Component.
74. Government of India (1997a). Report of the Committee of State Education Ministers on Implications of the Proposal to Make Elementary Education A Fundamental Right. MHRD. New Delhi.
75. Government of India (1999). Expert Group on Financial Requirements for Making Elementary Education A Fundamental Right. MHRD> New Delhi.
76. Government of India (2005a); *Report of the Committee on National Common Minimum Programme's Commitment of Six per cent of GDP to Education*. MHRD. New Delhi: National Institute of Educational Planning and Administration. [<http://niepa.org/download/mhrd%20report.pdf>]
77. Government of India (2005c) *Report of the CABE Committee on Free and Compulsory Elementary Education*. New Delhi: Central Advisory Board of Education, Ministry of Human Resource Development, Government of India, 2005 [Free and Compulsory Education Bill (Draft). MHRD. New Delhi. [http://www.education.nic.in/htmlweb/%20ssa/free_compulsory_edu_bill_2004_intro.htm]
78. Government of India (2005b) *Report of the CABE Committee on Financing of Higher and Technical Education*. New Delhi: Central Advisory Board of Education, Ministry of Human Resource Development, MHRD (June). <http://www.education.nic.in/cabe/Report%20CABE%20Committee%20on%20Financing%20Higher%20and%20Technical%20EducationL.pdf>
79. Government of India (a). (various years). Selected Educational Statistics. MHRD New Delhi.
80. Government of India (b). (various years). Education in India. MHRD. New Delhi.
81. Government of India (c). (various years). Analysis of Budget Expenditure on Education. MHRD New Delhi.
82. Government of India (d).(various years). Education Profile of States and Union Territories (1997 and 1999). MHRD. New Delhi
83. Government of India (e) (various years), *Economic Survey (2004-05, 2005-06, 2006-07, 2007-08)*, Ministry of Finance (MOF), New Delhi.
84. Government of India, 1990 Towards an Enlightened and Humane Society – Ramamurty Committee Report, New Delhi
85. Government of India, Department of Education, Annual Report (1997-98), New Delhi.
86. Government of India. (1957). Report of the Study Team for Community Development and National Extension Service. Planning Commission, New Delhi.
87. Government of India. (1966). Report of the Education Commission 1964-66: Education and National Development. Ministry of Education, New Delhi.
88. Government of India. (1978). Report of the Committee on Panchayati Raj Institutions, Ministry if Agriculture and Irrigation, New Delhi.

89. Government of India. (1985). Report of the Committee to Review Existing Administrative Arrangements for Rural Development and Poverty Alleviation Programmes (CAARD), Ministry of Agriculture, New Delhi.
90. Government of India. (1986). National Policy on Education 1986. Ministry of Human Resource Development, New Delhi. (Revised 1992).
91. Government of India. (1986). National Policy on Education and Programme of Action, New Delhi.
92. Government of India. (1987). Project Document: The Shiksha Karmi Project, Rajasthan. Ministry of Human Resource Development, New Delhi.
93. Government of India. (1989). District Institutes of Education and Training: Guidelines.
94. Government of India. (1991). 'The Central Advisory Board of Education', Proceedings of the Forty-Sixth Meeting, 8-9 March. Department of Education, Ministry of Human Resource Development.
95. Government of India. (1992). National Policy on Education: Programme of Action, New Delhi.
96. Government of India. (1994). Report of the CABE Committee on Decentralized Management of Education. Ministry of Human Resource Development, New Delhi.
97. Government of India. (1996). Report of the Working Group on Elementary Education, Non-Formal Education, Early Childhood Education and Teacher Education for the Ninth Five Year Plan. Ministry of Human Resource Development, New Delhi.
98. Government of India. (1997). Selected Educational Statistics. Ministry of Human Resource Development, New Delhi.
99. Government of India. (1998). Education in India, 1992-93, Vol. I (S). Ministry of Human Resource Development, New Delhi.
100. Government of India. (2001). Census of India 2001, Series 29-AP, Paper 1, New Delhi.
101. Government of India. (2001). Education for All – Developments since Dakar – India Country Paper to the E-9 Ministerial Review Meeting, Beijing, China. Ministry of Human Resource Development, New Delhi.
102. Government of India. (2001). Education for All – Draft National Plan of Action. India Country Report to Sub regional Ministerial Review Meeting, Kathmandu 10-12 April 2001. MHRD, New Delhi.
103. Government of India. (2001). Registrar General of India: Census of India – Provisional Population Totals (Paper 1 of 2001 Census), New Delhi.
Government of India. (2001). Working Group Report on Elementary and Adult Education: Tenth Five-Year Plan. Ministry of Human Resource Development, New Delhi.
104. Government of India. (2004). Select Education Statistics As of September 2003. MHRD, New Delhi.
105. Government of India. (2007). Select Education Statistics As of September 2004. MHRD, New Delhi.
106. Government of Rajasthan (2004-05). Citizen's Charter. Directorate of Elementary Education, Bikaner.
107. Government of Rajasthan (2004b) Rajasthan Gazette (Extraordinary) , Panchayati Raj Department Notification, dtd. February 28, 2004 .
108. Government of Rajasthan. (1966). Primary Education Committee Report. Jaipur.
109. Government of Rajasthan. (1995). GO P 3 (4), Shiksha 5/ 94 (dated 18.3.1995).
110. Government of Rajasthan. (1997). Ninth Five Year Plan: Draft Report. Jaipur.
111. Government of Rajasthan. (2001). Bulletin of Rajasthan Government, Department of Education.

112. Government of Rajasthan. (2001). G.O. No. F. 139(5)/ Panchayati Raj Prarambhik Shiksha/ 2001/816, Jaipur (dated 30.4.2001).
113. Government of Rajasthan. (2002). GO. No. F. 18/ Panchayati Raj Prarambhik Shiksha/ Vidhi Prakoshta/ 1713, Jaipur (dated 5.8.2002).
114. Government of Rajasthan. (2002). Rajasthan Human Development Report. Rajasthan State Cooperative Press, Jaipur.
115. Government of Rajasthan. (2003). Constitution of School Management Committee in government primary and upper primary schools. G.O. No. Shivira/ Shaikshik/ G/1477/2003 (dated 3.7.2003).
116. Government of Rajasthan. (2003a). G.O. No. p.2 (5) Shiksha-1/ PS/ 2003, Jaipur (dated 30.6.2003).
117. Government of Rajasthan. (2004). Government order P 4 (6) Shiksha-1/ 2004 (dated 20.2. 2004).
118. Government of Rajasthan. (2004a). Government Order P.21 (1)/ PS/2003 (dated 1.12.2004).
119. Government of Rajasthan. (2005). Implementation of the Transfer Policy of the Education Department (G.O. 17(1) Shiksha-2/2002/ Part 3).
120. Government of Rajasthan. (2005-06). Nishpadan Aay Vyayak. Directorate of Elementary Education, Bikaner.
121. Government of Rajasthan. (2005a). GO No. F4/ Panchayati Raj Vibhag/ PC/ Disha Niradesh/ o5/ 549, Jaipur (dated 31.3.2005).
122. Government of Rajasthan. (no date) Directorate of Elementary education (n.d.) Prabandhan Vyavastha. Unpublished document.
123. Government of Rajasthan. Rajasthan Subordinate Service Rules 1971.
124. Govinda, R. (1997). Decentralization of Educational Management: Experiences from South Asia. International Institute of Educational Planning, Paris.
125. Govinda, R. and K. Biswal. (2004). Elementary Education in the Tenth Plan: Promise, Performance and Prospects – A background paper for Mid Term Assessment, Unpublished Mimeo. NIEPA, New Delhi.
126. Govinda, R. and K. Biswal. (2005). Mapping Literacy in India – Who are the illiterates and where do we find them, Mimeo. NIEPA, New Delhi.
127. Govinda, R. and N.V. Verghese. (1993). Quality of Primary Schooling in India: A Case Study of Madhya Pradesh. International Institute of Educational Planning, Paris.
128. Govinda, R. and Y. Josephine. (2004). Parateachers in India – A Review Draft. NIEPA, New Delhi.
129. Govinda, R., Ed., (2002). India Basic Education Report. Oxford University Press, New Delhi.
130. Gupta, Shekhar. (2006). ‘Who Needs Checks and Balances?’ The Indian Express, 3 June. Hargopal, G. and C.H. Balaramulu. (1981). ‘Bureaucratic Culture in Panchayati Raj Institutions’, Quarterly Journal of Local Self Government, Vol. 11 (3), January-March. Pp 164-175.
131. Hooja, R. and Meenakhi Hooja. (2000). ‘Chapter on Rajasthan’, Status of Panchayati Raj in States and Union Territories of India. Institute of Social Sciences. Concept Publishing Company??, New Delhi.
132. International Institute for Population Sciences and Marco International. 2007, National Family Health Survey (NFHS-3) 2005-06: India, Volume I, Mumbai IIPA
133. Jayati Ghosh and C P Chandrashekar, Macroscan, Alternative Economic Webcentre, 4 October 2004
134. Jha, Jyotsna and Dhir Jhingran. (2002). Elementary Education for the Poorest and other Deprived Groups. Centre for Policy Research, New Delhi.
135. Jha, Praveen K. (2005). Withering Commitments and Weakening Progress: State and Education in the Era of Neoliberal Reforms in India, *Economic and Political Weekly*, August 13.

136. Joseph, M. (2004). 'Performance of Northern States: A Comparative Analysis', Economic and Political Weekly, Vol. XXXIX No.6, pp 564-579.
137. Kakar, Sudhir and Katharina Kakar. (2007), The Indians. Oxford University Press. New Delhi.
138. Kaviraj, Sudipta. "A Critique of the Passive Revolution. ... India: 'Subalterns' and Beyond," Economic and Political Weekly 23,41 (8 October 1988) pp 2249-43
139. Khilnani, Sunil. (1997). The Idea of India. Farrar Straus Giroux, New York.
140. Kothari, R. (1988). State Against Democracy: In Search of a Humane India, Volume 1. Ajanta Publications, Delhi.
141. Krishnaji, N. (2001). Poverty, Gender and Schooling: A Study of Two Districts in Andhra Pradesh, in Vaidyanathan and P.R.G. Nair (eds.), Elementary Education in Rural India: A Grassroots View. Sage, New Delhi, pp. 131-65.
142. Kumar, K. (1991). Political Agenda of Education: A Study of Colonialist and Nationalist Ideas. Sage Publications, New Delhi.
143. Kumar, Krishna, M. Priyam and S. Saxena. Looking beyond the smokescreen: DPEP and primary education in India. 'Economic and Political Weekly' 36(7): 2001: 560-568
144. Kumar, R.Y. (1998). 'Community Involvement in Management of School Education: An Appraisal of the Proposed Reforms', in Yash Aggarwal and Kusum K. Premi (eds) Andhra Pradesh in Reforming School Education: Issues in Policy Planning and Implementation. Vikas Publishing House Pvt. Ltd., New Delhi.
145. Kurukshetra (2003), Panchayati Raj in Rajasthan: Scope for Improvement, March 2003
146. Lok Sabha Secretariat (), National Policy Studies edited by S.C. Kashyap. Tatal MacGraw-Hill Publishing Company Limited. New Delhi. YEAR?
147. Mahajan, B., R.S. Tyagi and Shanta Agarwal. (1996). Educational Administration in Rajasthan. Vikas Publications, New Delhi. for NIEPA.
148. Majumbar, Manabi. (2001). 'Educational Opportunities in Rajasthan and Tamil Nadu: Despair and Hope', in A Vaidyanathan and P R Gopinathan Nair (eds) Elementary Education in Rural India – A Grassroots View. Sage Publications, New Delhi. Pp 320-394
149. Mathew, George. (1994). Panchayati Raj: From Legislation to Movement. Concept Publishing Company, New Delhi.
150. Mathew, George. (1996). 'Panchayati Raj in India', in Kuldeep Mathur (ed.) Development Policy and Administration: Readings in Indian Government and Politics-1. Sage Publications, New Delhi.
151. Mehrotra, Santosh, P.R. Panchamukhi, Ranjana Srivastava and Ravi Srivastava. (2005). Universalising Elementary Education in India: Uncaging the 'Tiger' Economy. Oxford University Press, New Delhi.
152. Mehrotra, Santosh. (ed.) (2005). The Economics of Elementary Education in India: The Challenge of Public Finance, Private Provision and Household Costs. Sage Publications, New Delhi.
153. Mehta, Arun C. (2005). Elementary Education in India – Where do we stand – State Report Cards, New Delhi.
154. Mehta, Arun C. (2006). Elementary Education India – Progress Towards UEE. Analytical Report 2006, NIEPA, New Delhi
155. Migdal, J.S. (1994). The State in Society: An Approach to the Struggles of Domination in *State Power and Social Forces: Domination and Transformation in the Third World* (Cambridge Studies in Comparative Politics) edited by Joel Samuel Migdal, Atul Kohli and Vivienne Shue. Cambridge University Press. Cambridge.
156. Mishra, B.B. (1983). *District Administration and Rural Development in India*. Oxford University Press. Delhi.

157. Naik, J.P. (1982). The Education Commission and After. Allied Publishers Private Limited, New Delhi.
158. Nambissan, forthcoming 2006, in Ravi Kumar (Ed), The Challenge of Education in India, Sage Publications forthcoming 2006. PAGE NOS?
159. Nambissan, Geetha, Dealing with deprivation, Seminar, Issue number 493, September 2000
160. Nambissan, Geetha. (1996). 'Equity in Education? Schooling of Dalit Children in India', Economic and Political Weekly XXXI (16&17), PP. 1011-24.
161. Narasimhulu, M., 2006; State Level Fiscal Reforms: A Case Study of Andhra Pradesh, in Srivastava D.K. and M. Narasimhulu eds., 2006 State Level Fiscal Reforms in the Indian Economy, Deep & Deep Publications, New Delhi, pp. 757-75
162. NIEPA (National Institute of Educational Planning and Administration) (2000), Implementation of Operation Blackboard Scheme in India: Ground Realities in 1999-2000. (Unpublished).
163. NIEPA. (2005). Elementary Education in India: Where Do We Stand? Analytical Report 2004. New Delhi.
164. NIEPA. (1979). Administration of Elementary Education in Andhra Pradesh: A Study in Relation to Universalisation of Elementary Education.
165. NCAER. (1999). India Human Development Report. Oxford University Press, Delhi.
166. NCERT (2000). National Curriculum Framework 2000. New Delhi
167. NCERT (2005). National Curriculum Framework 2005. New Delhi
168. NCERT 1998. The Primary Years: Towards a Curriculum Framework. New Delhi
169. NCERT. (1969). Future Development of National Council for Educational Research and Training: Report of the Review Committee
170. NCERT. (1993). 6th Educational Survey. New Delhi.
171. NCERT. (1998). The Primary Years: Towards a Curriculum Framework.
172. NCERT. (2003). Compendium of Educational Statistics (School Education). New Delhi.
173. NCERT. (2005). Seventh All India Education Survey: Provisional Statistics. New Delhi.
174. NSSO, Government of India, 2006. Employment and Unemployment Situation Among Social Groups in India 2004-05, Report No 516(61/10/2), New Delhi
175. Nuna, S.N. (1997). Decentralized Management in Primary Education in Maharashtra in *Decentralization of Educational Management: Experiences of South Asia* edited by R. Govinda. International Institute of Educational Planning. Paris.
176. Oad (1989). Primary Education under Panchayati Raj in *Studies in Educational Reform in India, Volume 2*, edited by P.R. Panchamukhi. Himalaya Publishing House. Bombay.
177. P Sudhakar Reddy. (2001). DPEP in Andhra Pradesh – A social assessment study among tribal groups, Tirupati PUBLISHER?
178. Panchamukhi, P.R. (2005). Household Expenditure on Elementary Education, in Mehrotra, Santosh, P.R. Panchamukhi, Ranjana Srivastava, and Ravi Srivastava, (Eds), *Universalising Elementary Education in India: Uncaging the 'Tiger' Economy*, Oxford, New Delhi, pp. 291-339.
179. Planning Commission, GOI (2002b) Financial Progress of Education Sector during Ninth (1997-2002) & Tenth Plan Period. Education Division, Planning Commission, Government of India, New Delhi (Unpublished).
180. Planning Commission (2005) *Mid-Term Appraisal of 10th Five Year Plan 2002-07*. New Delhi: Planning Commission.
181. Planning Commission, 2007; Poverty Estimates for 2004-05. <http://pib.nic.in/release/release.asp?relid=26316>. (March 21)

182. Planning Commission. (2002a). National Human Development Report 2001. New Delhi.
183. Potter, D. (1996). 'The Prime Minister and the Bureaucracy', in Kuldeep Mathur (ed.) Development Policy and Administration: Readings in Indian Government and Politics-1. Sage Publications, New Delhi.
184. Powis, B. (2003). 'Grass Roots Politics and the Second Wave of Decentralization in Andhra Pradesh', Economic and Political Weekly, June 28. Number 38 (26): pp 2617-2622.
185. PROBE Report. (1999). Public Report on Basic Education in India. Oxford University Press, Delhi.
186. Rajasthan Council for Primary Education, Annual Report 2003-04.
187. Rajasthan Education Service Rules 1970, Government of Rajasthan, Jaipur
188. Rajasthan Gazette (Extraordinary) (2004). Panchayati Raj Department Notification, (28 February).
189. Ramachandran, Vimala and Harsh Sethi. (2000). Rajasthan Shiksha Karmi Project – An Overall Appraisal. Swedish International Development Cooperation Agency (SIDA), Stockholm.
190. Ramachandran, Vimala, Madhumita Pal, Sharada Jain, Sunil Shekar and Jitendra Sharma (2005). 'Teacher Motivation in India', Mimeo, Educational Resource Unit, New Delhi.
191. Ramachandran, Vimala. (2003). Gender and Social Equity in Primary Education – Hierarchies of Access. Sage Publications, New Delhi.
192. Ramachandran, Vimala. (ed.) (2004). Snakes and Ladders: Factors Influencing Successful Primary School Completion for Children in Poverty Contexts. South Asian Human Development Sector Report No. 6, World Bank, New Delhi.
193. Rampal, Anita (2003) 'Texts in Context: Developing Curricula, Textbooks and Teaching Learning Materials', in R. Govinda (ed.) India Education Report. Oxford University Press, New Delhi.
194. Rampal, Anita. (2001). Curriculum Change for Quality Education: A Study of Schools in DPEP and non DPEP districts of Kerala. UNICEF. New Delhi.
195. Rao K. (1999). Study of Support Systems and Processes that Underpin DPEP's Pedagogical Strategy in Andhra Pradesh. Educational consultants of India Ltd. (Unpublished document).
196. Rao, C.H. Hanumantha and S. Mahendra Dev. (eds) (2003). *Andhra Pradesh Development: Economic Reforms and Challenges Ahead*. Centre for Economic and Social Studies/Manohar Publishers & Distributors, Hyderabad.
197. Rao. C.H. Hanumantha and S. Mahendra Dev (2003). 'Economic Reforms and Challenges Ahead: An Overview', IN C. H. Hanumantha Rao and S. Mahendra Dev (eds) *Andhra Pradesh Development: Economic Reforms and Challenged Ahead*. Centre for Economic and Social Studies. Hyderabad. pp. 1-38
198. Ray, A. (1985). 'Delivery Systems for Rural Development: A Field View of Institutional Linkages', Public Administration and Development, Vol. 5, No. 4. pp 353-362.
199. Ray, Sunil, 2005; 'The Cost and Financing of Universalising Elementary Education: A Silver Lining in Rajasthan', in Mehrotra, Santosh, P.R. Panchamukhi, Ranjana Srivastava, and Ravi Srivastava, (Eds), *Universalising Elementary Education in India: Uncaging the 'Tiger' Economy*, Oxford, New Delhi,
200. Reddi, PR. 1998. The Andhra Pradesh Panchayat Raj Manual Vo.1 and Vol. 2. Panchayat Publications. Hyderabad. 1998
201. Reddy, V. Ratna and R. Nageswara Rao, 2003; Primary Education: Causes and Constraints, in Hanumantha Rao C.H., and S. Mahendra Dev, Eds, *Andhra Pradesh Development: Economic Reforms and Challenges Ahead*. Centre for Economic and Social Studies/Manohar Publishers & Distributors, Hyderabad. pp. 373-97.

202. Rudolph, Loyd I. and Susanne Hoeber Rudolph. (1987). *In Pursuit of Lakshmi: The Political Economy of the Indian State*, University of Chicago Press, Chicago.
203. Sadgopal, Anil. (2005) in Ravi Kumar (Ed), *The Challenge of Education in India*, Sage Publications forthcoming 2006. DETAILS?
204. Sambrani, S. (1978). 'Mythologies of Area Planning', *Economic and Political Weekly*. Review of Agriculture. December pp A135-A139.
205. Sanker, Deepa and Venkatesh Sundararaman. (2003). *Equity in enrolment and completion in Elementary Schooling in India*. South Asia Human Development Sector Report No 4, World Bank, New Delhi.
206. Sarma, J.V.M. (2003) *Fiscal Management: A Review*, in Hanumantha Rao C.H., and S. Mahendra Dev, Eds, *Andhra Pradesh Development: Economic Reforms and Challenges Ahead*. Centre for Economic and Social Studies/Manohar Publishers & Distributors, Hyderabad, pp. 81-113.
207. SCERT Hyderabad (2005): *Proceedings of the Director, State Council for Educational Research and Training* dated 28-2-2005. NOT CLEAR
208. Schultz, Theodore W. (1961). 'Investment in Human Capital', *American Economic Review* 51(1) (March): 1-17.
209. Sharif, A. and P.K. Ghosh. (2000). 'Indian Education Scene and the Public Expenditure Gap', *Economic and Political Weekly* Vol 35(15):1396-1406, April 15, 2000
210. Sharma, G.D. (1989). 'NIEPA: Instrumentality of Change', in P.R. Panchamukhi (ed.) *Educational Reform in India, Volume 2: Educational Reforms at Different Levels*. Himalaya Publishing House, Delhi. PAGE NOS?
211. Sharma, R. (1997). 'The Dynamics of Learning the Three Rs in Madhya Pradesh', *Economic and Political Weekly*, Vol. XXXII, No. 17, April 26, pp 891-902.
212. Sharma, R. (1998). 'Universal Elementary Education: The Question of "How"', *Economic and Political Weekly*, Vol. XXXIII, No. 26, June 27. Pp 1640-1647.
213. Sharma, R. (1999). 'What Manner of Teacher: Some Lessons from Madhya Pradesh', *Economic and Political Weekly*, Vol. XXXIV, No. 25, June 19Pp 1597-1607
214. Sharma, R. (2000). 'Decentralization, Professionalism and the School System in India', *Economic and Political Weekly*, Vol. XXXV, No. 12, October 14, pp 3767-3774.
215. Sharma, R. (2007). 'Kerala's Decentralization: The Idea in Practice', in Satyajit Singh and Pradeep K. Sharma (eds) *Decentralization: Institutions and Politics in Rural India*. Oxford University Press, New Delhi. Pp 37-81
216. Sharma, R. (forthcoming). *Local Government in India: Policy and Practice*. Manohar Publications, New Delhi.
217. Shukla, P.D. (1989). 'Central Board of Secondary Education', in P.R. Panchamukhi (ed.) *Studies in Educational Reform in India, Volume 2: Educational Reforms at Different Levels*. Himalaya Publishing House, Delhi.
218. Shukla, Snehlata et al. (1994). *Attainment of Primary School Children in Various States*, National Council for Educational Research and Training New Delhi?
219. SIERT. (2003). *Rajasthan Rajya Ke Kaksha 8 Board Pattern Ka Pareeksha Parinam*, (April).
220. Sinha, Narain, R. Sinha and A.K.S. Rathore, (n.d) *Educational Development and State Expenditure on Education: An Empirical Study with Special Reference to Rajasthan*, University of Rajasthan Jaipur.
221. Sipahimalini Rao, Vandana. (2000). *India: Financing of Elementary Education in India in the 1990s*. South Asia Report No. 2, World Bank, New Delhi.

222. Srinivasulu, K. (1994), 'Andhra Pradesh: BSP and Caste Politics', EPW, 29(40): 2583. ... Economic and Political Weekly, October 13, 2001, pp. 3892-94
223. Srivastava, D.K., and M. Narasimhulu. (eds) (2006). State Level Fiscal Reforms in the Indian Economy. Deep & Deep Publications, New Delhi.
224. Srivastava, Ravi. (2005). Public Expenditure on Elementary Education, in Mehrotra, Santosh, P.R. Panchamukhi, Ranjana Srivastava, and Ravi Srivastava, (Eds) *Universalising Elementary Education in India: Uncaging the 'Tiger' Economy*, Oxford, New Delhi. 127-72
225. State Institute of Educational Research and Training Udaipur. (2004). A Report on the Utilization of Teachers Grant for Teaching Learning Material (Unpublished).
226. State Institute of Educational Research and Training Udaipur. (2004). Panoramic Scenario of SIERT, (April) 2005. 2004 or 2005?
227. Sudhakar Rao, N., 2006; 'State Finances in India with Reference to Andhra Pradesh, in Srivastava and Narasimhulu (eds.), *State Level Fiscal Reforms in the Indian Economy*, Deep & Deep Publications, New Delhi.
228. Thomas, Jayan Jose. (2002). 'A Review of Indian Manufacturing in India Development Report 2002', Indira Gandhi Institute of Development Research, Oxford University Press, New Delhi. PAGE NOS?
229. Tilak, J.B. G. (1989) Center-State Relations in Financing Education in India, *Comparative Educational Review* 33 (4) (November): 450-80.
230. Tilak, J.B.G. (1996a). 'How Free is "Free" Primary Education in India?' Economic and Political Weekly 31 (5 & 6) (3 & 10 February 1996): 275-82; and 355-66.
231. Tilak, J.B.G. (1996b). 'Higher Education Under Structural Adjustment', Journal of Indian School of Political Economy 8 (2) (April-June 1996): 266-93.
232. Tilak, J.B.G. (1998). Public Expenditures on Education in Andhra Pradesh: A Review of Trends, Issues and Problems. Government of Andhra Pradesh, District Primary Education Programme, Hyderabad (August).
233. Tilak, J.B.G. (1999a). 'National Human Development Initiative: Education in the Union Budget', Economic and Political Weekly 34 (10-11) (6 March): 614-20.
234. Tilak, J.B.G. (1999b). 'Development Assistance to Primary Education in India: Transformation of Enthusiastic Donors and Reluctant Recipients', in Kenneth King and Lene Buchert (eds) *Changing International Aid to Education: global Patterns and National Contexts*. UNESCO in cooperation with NORRAG, pp. 307-17. The Hague, The Netherlands
235. Tilak, J.B.G. (2000a). Household Expenditure on Education in India: A Preliminary Examination of the 52nd Round of the National Sample Survey. National Institute of Educational Planning and Administration (July), New Delhi.
236. Tilak, J.B.G. (2000b). 'Investment in Human Capital in India: An Inter-State Analysis of Stock and Flow of Human Capital', in Perspectives on Indian Development: Economy, Polity and Society: Essays in Honour of Prof B Sarveswara Rao. Sterling, New Delhi, pp.275-346.
237. Tilak, J.B.G. (2002a). 'Education Poverty in India', Review of Development and Change 7 (1) (January-June 2002): 1-44.
238. Tilak, J.B.G. (2002b). 'Household Expenditure on Elementary Education in Rural India: A Few Stylized Facts', Indian Journal of Social Development 2 (2) (December): 231-67.
239. Tilak, J.B.G. (2002c) Determinants of Household Expenditure on Education in Rural India. Working Paper No. 88, National Council of Applied Economic Research, New Delhi.

240. Tilak, J.B.G. (2003a). *Financing in India: Current Issues and Changing Perspectives*. Ravi Books for National Institute of Educational Planning and Administration, New Delhi.
241. Tilak, J.B.G. (2004). 'Public Subsidies in the Education Sector in India', *Economic and Political Weekly* 39 (4) (24-30 January): 343-59.
242. Tilak, J.B.G. (2006a) Trends in Public Expenditure on Education: A Contrast between Two Educationally Backward States -- Andhra Pradesh and Rajasthan. *Journal of Indian School of Political Economy* 18 (1-2) (January-June): 1-34
243. Tilak, J.B.G. (ed.) (2003b). 'Public Expenditure on Education in India: A Review of Trends and Emerging Issues', in Jandhyala B G Tilak (ed.) *Financing in India: Current Issues and Changing Perspectives* Ravi Books/National Institute of Educational Planning and Administration, New Delhi, pp. 3-54.
244. Tilak, J.B.G. 1994; *Resource Requirements of Education in India: Implications for the Tenth Finance Commission*. (Report prepared for the Government of India). New Delhi: National Institute of Educational Planning and Administration.
245. Tilak, J.B.G.(2008) Political Economy of External Aid for Education in India, *Journal of Asian Public Policy* (Routledge, UK) 1 (1) (March): 32-51
246. Tilak, J.B.G., (2006b) On Allocating 6% GDP to Education, *Economic and Political Weekly*, Vol. 41, No. 7, 18 February.
247. Tilak, J.B.G., (2006c); Cess-Driven Allocations for Education, *Economic and Political Weekly* Vol. 41, No.14, 8 April.
248. Tilak, Jandhyala B. G. (1987). 'Educational Finances in India', *Journal of Educational Planning and Administration* 1 (3-4) (July-October): 132-99
249. Tilak, J.B.G. and Ratna M. Sudarshan. (2001). *Private Schooling in Rural India*. Working Paper Series No. 76. National Council for Applied Economic Research, New Delhi.
250. Tribal Cultural Research and Training Institute: Study of problems of tribal children with respect to enrolment, retention and quality; Hyderabad June 1999.
251. Umamohan. Ch (2002). Constraints to schooling: Social Assessment Survey of Backward Caste Communities in Andhra Pradesh, A report submitted to DPEP, Government of Andhra Pradesh, Anantapur, (October).
252. Vaidyanathan, A. and P.R. Gopinathan Nair. (eds) (2001). *Elementary Education in Rural India: A Grassroots View*. Sage Publications, New Delhi.
253. Vasavi A R, Chand, Vijay Sherry, R Shailesh Shukla. 1997. "Blueprint for Rural Primary Education, How Viable? *Economic and Political Weekly*, 13 December 1997. pp 2053-2063
254. Visaria, Leela. (2003). 'Primary Education in India: What do we know' in Vimala Ramachandran (ed) *Hierarchies of Access - Gender and Social Equity in primary education*, Sage Publications, 2003.
255. World Bank, 1996b; *Andhra Pradesh: Agenda for Economic Reform*. Country Operations, Industry & Finance Division. Report No. 15901-IN, World Bank, New Delhi.
256. World Bank, 2006; *India: Rajasthan Closing the Development Gap*. Poverty Reduction and Economic Development. South Asia. Economic Report No. 32585-IN, World Bank, New Delhi.
257. World Bank. (1995). *Priorities and Strategies for Education: A World Bank Review*. World Bank, Washington D.C.
258. World Bank. (1996). *Andhra Pradesh: Agenda for Economic Reform*. World Bank, Country Operations, Industry & Finance Division, New Delhi (Report No. 15901-IN).
259. World Bank. (1996a). *India: Primary Education: Achievement and Challenges*. World Bank, Washington D.C.

- 260. World Bank. (1997). Primary Education in India. World Bank and Allied Publishers, Delhi.
- 261. World Bank. (2001). India Expanding and Improving Upper Primary Education in India. World Bank, Washington, D.C.
- 262. World Bank. (2003). A Review of Educational Progress and Reform in the District Primary Education programme (Phase 1 and Phase 2). New Delhi
- 263. Yugadhar, B.N. and B.Y. Raju. (1992). 'Government Delivery Systems for Rural Development', Economic and Political Weekly, August 29. pp M123-M127.